



**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 4: Enhancements for Positioning

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.11az™-2022

(Amendment to IEEE Std 802.11™-2020
as amended by IEEE Std 802.11ax™-2021,
IEEE Std 802.11ay™-2021,
IEEE Std 802.11ba™-2021,
and IEEE Std 802.11™-2020/Cor 1-2022)

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Approved 3 December 2022

IEEE SA Standards Board

Abstract: This amendment defines modifications to both the IEEE 802.11 medium access control layer (MAC) and physical layers (PHY). Compared to IEEE Std 802.11™-2020 it enables absolute and relative position estimation with higher accuracy, and reduces wireless medium utilization and power consumption, is scalable to dense deployments, and includes security features.

Keywords: direction measurement, Fine Timing Measurement (FTM), IEEE 802.11™, IEEE 802.11az™, positioning, ranging

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Introduction

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This amendment defines modifications to both the IEEE 802.11 medium access control layer (MAC) and physical layers (PHY). It enables absolute and relative position estimation with higher accuracy executing on the same PHY-type. Further, it reduces existing wireless medium utilization and power consumption, is scalable to dense deployments, and includes security features.

Contents

1.	Overview	24
1.3	Supplementary information on purpose	24
3.	Definitions, acronyms, and abbreviations	25
3.2	Definitions specific to IEEE Std 802.11	25
3.4	Acronyms and abbreviations	26
4.	General description	27
4.3	Components of the IEEE 802.11 architecture	27
4.3.19	Wireless network management	27
4.3.19.19	Fine timing measurement (FTM)	27
4.5	Overview of the services	28
4.5.4	Access control and data confidentiality services	28
4.5.4.2	Authentication	28
6.	Layer management	29
6.3	MLME SAP interface	29
6.3.5	Authenticate	29
6.3.5.2	MLME-AUTHENTICATE.request	29
6.3.5.3	MLME-AUTHENTICATE.confirm	29
6.3.5.4	MLME-AUTHENTICATE.indication	30
6.3.5.5	MLME-AUTHENTICATE.response	31
6.3.56	Fine timing measurement (FTM)	32
6.3.56.1	General	32
6.3.56.2	MLME-FINETIMINGMSMTRQ.request	34
6.3.56.3	MLME-FINETIMINGMSMTRQ.indication	35
6.3.56.4	MLME-FINETIMINGMSMT.request	35
6.3.56.5	MLME-FINETIMINGMSMT.confirm	38
6.3.56.6	MLME-FINETIMINGMSMT.indication	39
8.	PHY service specification	40
8.3	Detailed PHY service specifications	40
8.3.4	Basic service and options	40
8.3.4.2	PHY SAP inter-(sub)layer service primitives	40
8.3.4.3	PHY SAP service primitives parameters	40
8.3.4.4	Vector descriptions	41
8.3.5	PHY SAP detailed service specification	41
8.3.5.14	PHY-RXEND.indication	41
8.3.5.18	PHY-RXLFSEQUENCE.request	41
8.3.5.19	PHY-RXTRNSEQUENCE.request	42
9.	Frame formats	43
9.3	Format of individual frame types	43
9.3.1	Control frames	43
9.3.1.19	VHT/HE/Ranging NDP Announcement frame format	43

	9.3.1.22	Trigger frame format	47
	9.3.1.22a	Subvariants of Ranging Trigger variant.....	49
9.3.3		(PV0) Management frames	51
	9.3.3.2	Beacon frame format	51
	9.3.3.11	Authentication frame format.....	52
9.4		Management and Extension frame body components	53
	9.4.1	Fields that are not elements	53
	9.4.1.1	Authentication Algorithm Number field.....	53
	9.4.1.9	Status Code field	53
	9.4.1.11	Action field	54
	9.4.2	Elements.....	55
	9.4.2.1	General.....	55
	9.4.2.20	Measurement Request element	56
	9.4.2.21	Measurement Report element	56
	9.4.2.24	RSNE	59
	9.4.2.26	Extended Capabilities element.....	59
	9.4.2.36	Neighbor Report element.....	60
	9.4.2.118	MIC element	61
	9.4.2.127	DMG Capabilities element	61
	9.4.2.167	Fine Timing Measurement FTM Parameters element	63
	9.4.2.187	FILS Wrapped Data element	67
	9.4.2.241	RSN Extension element (RSNXXE)	68
	9.4.2.265	EDMG Capabilities element.....	69
	9.4.2.296	ISTA Availability Window element.....	70
	9.4.2.297	RSTA Availability Window element.....	70
	9.4.2.298	Ranging Parameters element	72
	9.4.2.299	Secure HE-LTF Parameters element	79
	9.4.2.300	Direction Measurement Results element	80
	9.4.2.301	Multiple Best AWV ID element	81
	9.4.2.302	Multiple AOD Feedback element	82
	9.4.2.303	PASN Parameters element.....	83
	9.4.2.304	ISTA Passive TB Ranging Measurement Report element.....	84
	9.4.2.305	RSTA Passive TB Ranging Measurement Report element	86
	9.4.2.306	Passive TB Ranging LCI Table element.....	87
	9.4.2.307	LOS Likelihood element.....	88
9.6		Action frame format details	89
	9.6.6	Radio Measurement action details	89
	9.6.6.6	Neighbor Report Request frame format.....	89
	9.6.7	Public Action details	89
	9.6.7.1	Public Action frames	89
	9.6.7.32	Fine Timing Measurement FTM Request frame format.....	90
	9.6.7.33	Fine Timing Measurement (FTM) frame format	91
	9.6.7.49	Location Measurement Report (LMR) frame format	94
	9.6.7.50	ISTA Passive TB Ranging Measurement Report frame format	96
	9.6.7.51	Primary RSTA Broadcast Passive TB Ranging Measurement Report frame format	96
	9.6.7.52	Secondary RSTA Broadcast Passive TB Ranging Measurement Report frame format.....	97
	9.6.34	Protected Fine Timing frame details.....	98
	9.6.34.1	Protected Fine Timing Action field	98
10.		MAC sublayer functional description.....	99
10.3		DCF.....	99

10.3.2	Procedures common to the DCF and EDCAF	99
10.3.2.14	Duplicate detection and recovery	99
10.12	A-MPDU operation.....	100
10.12.3	Minimum MPDU start spacing rules	100
10.23	HCF.....	100
10.23.2	HCF contention based channel access (EDCA)	100
10.23.2.8	Multiple frame transmission in an EDCA TXOP.....	100
10.42	DMG beamforming.....	101
10.42.10	EDMG beamforming.....	101
10.42.10.6	First path beamforming training	101
11.	MLME	103
11.3	STA authentication and association.....	103
11.3.1	State variables	103
11.3.2	State transition diagram for nonmesh STAs	103
11.3.3	Frame filtering based on STA state	104
11.3.4	Authentication and deauthentication	105
11.3.4.1	General.....	105
11.3.4.2	Authentication—originating STA.....	105
11.3.4.3	Authentication—destination STA.....	105
11.3.4.4	Deauthentication—originating STA	106
11.3.4.5	Deauthentication—destination STA	106
11.3.5	Association, reassociation, and disassociation	106
11.3.5.1	General.....	106
11.3.5.2	Non-AP and non-PCP STA association initiation procedures.....	107
11.3.5.3	AP or PCP association receipt procedures.....	107
11.3.5.4	Non-AP and non-PCP STA reassociation initiation procedures	107
11.3.5.5	AP or PCP reassociation receipt procedures.....	108
11.3.5.6	Non-AP and non-PCP STA disassociation initiation procedures.....	108
11.3.5.8	AP or PCP disassociation initiation procedure	108
11.10	Radio measurement procedures	108
11.10.10	Usage of the neighbor report	108
11.10.10.2	Requesting a neighbor report.....	108
11.10.10.3	Responding to a neighbor report request	108
11.13	SA Query procedures.....	109
11.21	Wireless network management procedures	109
11.21.6	Fine timing measurement (FTM) procedure.....	109
11.21.6.1	OverviewGeneral	109
11.21.6.2	FTM capabilities	113
11.21.6.3	Fine timing measurement FTM procedure negotiation	115
11.21.6.4	Measurement exchange.....	127
11.21.6.5	Fine timing measurement FTM parameter modification.....	176
11.21.6.6	Fine timing measurement FTM session termination	177
11.21.6.7	LCI and Location Civic retrieval using FTM procedure	178
11.24	Quality-of-service Management frame (QMF).....	179
11.24.1	General.....	179
11.24.1.2	Default QMF policy	179
12.	Security	180
12.2	Framework	180
12.2.4	RSNA establishment.....	180
12.2.7	Requirements for management frame protection.....	180

12.2.11	EDMG secure ranging sequences	180
12.5	RSNA confidentiality and integrity protocols	181
12.5.3	CTR with CBC-MAC protocol (CCMP)	181
12.5.3.2	CCMP MPDU format	181
12.5.3.4	CCMP decapsulation	182
12.5.5	GCM protocol (GCMP)	183
12.5.5.2	GCMP MPDU format	183
12.5.5.4	GCMP decapsulation	183
12.6	RSNA security association management	184
12.6.1	Security associations	184
12.6.1.1	Security association definitions	184
12.6.7	RSNA policy selection in an MBSS	185
12.6.10	RSNA authentication in an infrastructure BSS	185
12.6.10.1	General	185
12.6.19	Protection of robust Management frames	185
12.7	Keys and key distribution	186
12.7.1	Key hierarchy	186
12.7.1.3	Pairwise key hierarchy	186
12.7.1.6	FT key hierarchy	188
12.7.3	EAPOL-Key frame construction and processing	188
12.11	Authentication for FILS	189
12.11.2	FILS authentication protocol	189
12.11.2.5	Key establishment with FILS authentication	189
12.13	Preassociation security negotiation	190
12.13.1	General	190
12.13.2	Discovery of a PASN capable AP	190
12.13.3	Key establishment with PASN authentication	191
12.13.3.1	Overview	191
12.13.3.2	PASN frame construction and processing	192
12.13.4	PASN authentication with FILS shared key	196
12.13.5	PASN authentication with SAE	197
12.13.6	PASN authentication with FT	198
12.13.7	PTKSA derivation with PASN authentication	198
12.13.8	MIC computation with PASN authentication	199
12.13.8.1	MIC computation for PASN second frame	199
12.13.8.2	MIC computation for PASN third frame	199
12.13.9	Comeback cookies	200
26.	High-efficiency (HE) MAC specification	201
26.5	MU operation	201
26.5.2	UL MU operation	201
26.5.2.2	Rules for soliciting UL MU frames	201
26.5.2.5	UL MU CS mechanism	201
26.12	HE PPDU post-FEC padding and packet extension	201
26.15	PPDU format, BW, MCS, NSS, and DCM selection rules	202
26.15.2	PPDU format selection	202
26.17	HE BSS operation	202
26.17.2	HE BSS operation in the 6 GHz band	202
26.17.2.1	General	202
27.	High-efficiency (HE) PHY specification	203
27.1	Introduction	203

27.1.1	Introduction to the HE PHY	203
27.2	HE PHY service interface.....	204
27.2.2	TXVECTOR and RXVECTOR parameters	204
27.2.3a	LTFVECTOR parameters.....	206
27.3	HE PHY	207
27.3.4	HE PPDU formats.....	207
27.3.11	HE preamble	207
27.3.11.1	Introduction.....	207
27.3.18a	HE Ranging NDP and HE TB Ranging NDP	207
27.3.18a.1	HE Ranging NDP.....	207
27.3.18a.2	HE TB Ranging NDP	210
27.3.18b	HE-LTF field using secure HE-LTF	211
27.3.18b.1	Introduction.....	211
27.3.18b.2	Generation of a randomized secure HE-LTF sequence	212
27.3.18b.3	Pseudorandom and deterministic per spatial stream phase rotations..	218
27.3.18b.4	Frequency domain windowing in HE-LTF field using secure HE-LTF	219
27.3.18b.5	Mathematical description of HE-LTF field using secure HE-LTF.....	221
27.3.18b.6	Construction of secure HE-LTF symbols	222
27.3.19	Transmit specification.....	223
27.3.19.5	Time of departure accuracy	223
27.3.21	HE transmit procedure	223
27.3.22	HE receive procedure.....	224
27.4	HE PLME	224
27.4.3	TXTIME and PSDU_LENGTH calculation	224
28.	Enhanced directional multi-gigabit (EDMG) PHY specification.....	226
28.1	Introduction.....	226
28.1.1	Introduction to EDMG PHY	226
28.2	EDMG PHY service interface	226
28.2.2	TXVECTOR and RXVECTOR parameters	226
28.3	Common parameters	227
28.3.2	EDMG PPDU format.....	227
28.3.2.1	General.....	227
28.3.3	EDMG preamble.....	227
28.3.3.3	EDMG portion of EDMG format preamble	227
28.5	EDMG and non-EDMG SC mode	228
28.5.9	Data field.....	228
28.5.9.2	Symbol blocking and guard insertion	228
28.9	EDMG beamforming	228
28.9.3	EDMG secure ranging PPDU	228
28.9.3.1	General.....	228
28.9.3.2	EDMG secure ranging PPDU structure	228
28.9.3.3	EDMG secure ranging PPDU header fields	228
28.9.3.4	EDMG secure ranging PPDU duration.....	228
28.9.3.5	TRN field definition with Secure TRN subfields	228
28.9.3.6	Secure TRN subfield definition for EDMG secure ranging PPDU	229
28.9.3.7	Transmission of an EDMG secure ranging PPDU	230
	Annex A (informative) Bibliography	231
	Annex B (normative) Protocol Implementation Conformance Statement (PICS) proforma.....	232

B.4	PICS proforma—IEEE Std 802.11-2020.....	232
B.4.3	IUT configuration.....	232
B.4.4	MAC protocol	232
B.4.37	Enhanced positioning (Ranging) features	234
Annex C (normative)	ASN.1 encoding of the MAC and PHY MIB	237
C.3	MIB detail	237
Annex J (informative)	RSNA reference implementations and test vectors.....	242
J.12	PASN Test Vectors	242
J.13	KDK Test Vectors when PASN authentication is not used.....	242
J.14	LTF Sequence Generation Test Vectors	243
Annex AD (informative)	Phase shift feedback calculation	244
Annex AE (informative)	SIR-based attack detection	245

Tables

Table 8-2—PHY SAP inter-(sub)layer service primitives	40
Table 8-3—PHY SAP service primitive parameters	40
Table 8-4—Vector descriptions.....	41
Table 9-28f—NDP Announcement frame variant encoding	44
Table 9-29c—Trigger Type subfield encoding	47
Table 9-29j—UL Target Receive Power subfield in Trigger frame.....	47
Table 9-29l—Ranging Trigger Subtype subfield encoding.....	49
Table 9-32—Beacon frame body.....	51
Table 9-40—Authentication frame body	52
Table 9-41—Presence of fields and elements in Authentication frames.....	52
Table 9-50—Status codes	53
Table 9-51—Category values	54
Table 9-92—Element IDs.....	55
Table 9-98—Measurement type definitions for measurement requests	56
Table 9-134—Subelement IDs for LCI Report	56
Table 9-151—AKM suite selectors	59
Table 9-153—Extended Capabilities field	59
Table 9-279a—Definition of the EDMG Ranging Priority subfield when included in the IFTMR frame	64
Table 9-279b—Definition of the EDMG Ranging Priority subfield when included in the IFTM frame	64
Table 9-280—Format And Bandwidth subfield	65
Table 9-280a—Optional Subelement IDs for FTM Parameters.....	66
Table 9-280b—Direction Measurement Density field interpretation.....	67
Table 9-321—Extended RSN Capabilities field.....	68
Table 9-322a1—Format And Bandwidth subfield	74
Table 9-322am—Max R2I/I2R LTF Total subfields.....	75
Table 9-322an—Ranging Subelement IDs for Ranging Parameters	76
Table 9-322ao—Secure HE-LTF protocol section identifier	79
Table 9-322ap—Type subfield	85
Table 9-364—Public Action field values	89
Table 9-526m—Protected Fine Timing Action field values	98
Table 10-5—Transmitter sequence number spaces	99
Table 10-6—Receiver caches	100
Table 11-11a—Valid combinations of direction measurements requests and the required capabilities at the initiator and responder	126
Table 11-11b—TXVECTOR parameter setting for first path AWW FTM exchange	136
Table 11-16—Default QMF policy	179
Table 12-10—Integrity and key wrap algorithms.....	188
Table 27-1—TXVECTOR and RXVECTOR parameters	204
Table 27-2a—LTFVECTOR parameters.....	206
Table 27-47a—Pseudorandom octet index for each nonzero subcarrier index in the nth 20 MHz secure HE-LTF sequence.....	212
Table 27-47b—Pseudorandom octet index for each nonzero subcarrier index in the nth 40 MHz secure HE-LTF sequence.....	214
Table 27-47c—Pseudorandom octet index for each nonzero subcarrier index in the nth 80 MHz secure HE-LTF sequence.....	216
Table 27-47d—Pseudorandom octet index for each nonzero subcarrier index in the nth pair of lower and upper 80 MHz segments	217
Table 27-47e—Pseudorandom octets for generating per stream pseudorandom phase rotations	219
Table 27-47f—Deterministic phase rotation $\beta_{r,s}$ (radians)	220

Table 28-1—TXVECTOR and RXVECTOR parameters 226

Table 28-13—EDMG-MCS field definition when the Number of SS field is 0 227

Table 28-74a—EDMG-A header fields setting for secure DMG TRNs 229

Figures

Figure 6-17—Fine timing measurement FTM primitives and timestamps capture for EDCA based ranging measurement exchange	32
Figure 6-17a—FTM primitives and timestamps reporting capture for non-TB ranging measurement exchange	32
Figure 6-17b—FTM primitives and timestamps reporting capture for TB ranging measurement exchange	33
Figure 6-17c—FTM primitives and timestamps reporting capture for passive TB ranging measurement exchange	33
Figure 9-59—Sounding Dialog Token field format	43
Figure 9-61e—STA Info field format in a Ranging NDP Announcement frame when the AID11 subfield is equal or less than 2007	44
Figure 9-61f—STA Info field format in a Ranging NDP Announcement frame if the AID11 subfield is 2043	45
Figure 9-61h—STA Info field format in a Ranging NDP Announcement frame if the AID11 subfield is 2045	46
Figure 9-61g—STA Info field format in a Ranging NDP Announcement frame (if the AID11 subfield is equal to 2044)	46
Figure 9-64m—Trigger Dependent Common Info subfield format for the Ranging Trigger subvariants Poll, Sounding, Secure Sounding, and Report	48
Figure 9-64n—Trigger Dependent Common Info subfield format of the Passive Sounding Ranging Trigger frame	48
Figure 9-64o—User Info field format for the Poll and Report Ranging Trigger	49
Figure 9-64p—User Info field format for Sounding subvariant	50
Figure 9-64q—User Info field for Secure Sounding subvariant	50
Figure 9-256a—Relative Compact LCI field format	57
Figure 9-256b—Antenna Placement And Calibration subelement format	57
Figure 9-256c—Antenna Information field format	58
Figure 9-256d—Antenna Placement And Calibration field format	58
Figure 9-337—BSSID Information field format	60
Figure 9-535—MIC element format	61
Figure 9-549—DMG Capabilities element format	61
Figure 9-554e—DMG Direction Measurement Capabilities field format	62
Figure 9-554f—DMG Fine Timing And Range Measurement Capability Information field format	63
Figure 9-624—Fine Timing Measurement FTM Parameters element format	63
Figure 9-625—Fine Timing Measurement FTM Parameters field format	63
Figure 9-626a—DMG Direction Measurement Parameters subelement format	66
Figure 9-626b—DMG Direction Measurement Parameters field format	66
Figure 9-626c—EDMG Specific Parameters subelement format	67
Figure 9-626d—Secure Ranging Operation Parameters field format	67
Figure 9-664—Wrapped Data element format	68
Figure 9-788aq—Data field of the Beamforming Capability subelement format	69
Figure 9-788ec—ISTA Availability Window element format	70
Figure 9-788ef—ISTA Availability Information field format	70
Figure 9-788eh—RSTA Availability Information field format	71
Figure 9-788ei—Header subfield format	71
Figure 9-788ej—Availability Window Information field format when the Availability Window Broadcast Format subfield is set to 0	71
Figure 9-788eg—RSTA Availability Window element format	71
Figure 9-788ek—Availability Window Information field format when the Availability Window Broadcast Format subfield is set to 1	72
Figure 9-788el—Passive TB Ranging Parameters subfield format	72

Figure 9-788em—Ranging Parameters element format	72
Figure 9-788en—Ranging Parameters field format.....	73
Figure 9-788eo—Non-TB Specific subelement format.....	76
Figure 9-788ep—TB Specific subelement format.....	77
Figure 9-788eq—Example of a bitmap with 200 TU periodicity signaled in the ISTA Availability Window element	78
Figure 9-788er—Example of mapping of ISTA’s availability bitmap to RSTA’s TSF.....	78
Figure 9-788es—Example of how an RSTA assigns an Availability Window to an ISTA	78
Figure 9-788et—Secure HE-LTF subelement format	78
Figure 9-788eu—Secure HE-LTF Parameters element format	79
Figure 9-788ev—Measurement Result LTF Offset field format.....	80
Figure 9-788ew—Direction Measurement Results element.....	80
Figure 9-788ey—Multiple Best AWV ID element.....	81
Figure 9-788ex—AOA Results field format	81
Figure 9-788ez—Multiple AOD Feedback element.....	82
Figure 9-788fa—AOD Feedback field format.....	82
Figure 9-788fb—PASN Parameters element format.....	83
Figure 9-788fc—PASN Parameters element Control Info field format.....	83
Figure 9-788fd—Comeback Info field format.....	83
Figure 9-788fe—ISTA Passive TB Ranging Measurement Report element format	84
Figure 9-788ff—More & N Timestamp Measurement Reports field format	84
Figure 9-788fg—Timestamp Measurement Report subfield format	85
Figure 9-788fh—Timestamp Error subfield format	85
Figure 9-788fi—RSTA Passive TB Ranging Measurement Report element format	86
Figure 9-788fj—Passive TB Ranging LCI Table Report element format.....	87
Figure 9-788fk—ISTA LCI Report Entry field format	87
Figure 9-788fl—LOS Likelihood element format.....	88
Figure 9-864—Neighbor Report Request frame Action field format.....	89
Figure 9-895—Fine Timing Measurement FTM Request frame Action field format.....	90
Figure 9-896—Fine Timing Measurement (FTM) frame Action field format.....	91
Figure 9-897—Format of the TOD Error field format	91
Figure 9-898—Format of the TOA Error field format	92
Figure 9-898a—Channel Measurement Feedback Type field format	93
Figure 9-909b—Location Measurement Report (LMR) frame Action field format	94
Figure 9-909c—TOD Error field format	95
Figure 9-909d—TOA Error field format	95
Figure 9-909e—ISTA Passive TB Ranging Measurement Report frame format.....	96
Figure 9-909g—Passive TB Ranging LCI Table Countdown Info field format	97
Figure 9-909f—Primary RSTA Broadcast Passive TB Ranging Measurement Report Action field format.....	97
Figure 9-909h—Secondary RSTA Broadcast Passive TB Ranging Measurement Report Action field format.....	98
Figure 11-17—Relationship between state and services between a given pair of nonmesh STAs	104
Figure 11-34—Concurrent FTM sessions	110
Figure 11-34a—Non-TB ranging concurrent FTM sessions.....	111
Figure 11-34b—Example of passive TB ranging used for PSTA location—RSTA1 (AP1) operating passive TB ranging	113
Figure 11-35—Example negotiation and measurement exchange sequence, ASAP = 0, and FTMs Per Burst = 2	129
Figure 11-36—Example negotiation and measurement exchange sequence, ASAP = 1, and FTMs Per Burst = 2.....	130
Figure 11-37—Example negotiation and measurement exchange sequence for a single burst instance, ASAP=1, and FTMs Per Burst = 3	131
Figure 11-37a—TB ranging availability windows each with one instance of a	

polling/sounding/reporting triplet	138
Figure 11-37b—TB ranging availability window with two instances of polling/sounding/reporting triplets within a single TXOP	139
Figure 11-37c—TB ranging availability window with two instances of polling/sounding/reporting triplets in separate TXOPs	139
Figure 11-37d—TB ranging availability window with two ISTAs.....	141
Figure 11-37e—Timing diagram of a measurement sounding phase in TB ranging	143
Figure 11-37f—TB ranging availability window with multiple TF Ranging Sounding frames	144
Figure 11-37g—Measurement sounding phase with I2R time division multiplexing.....	144
Figure 11-37h—TB ranging measurement reporting phase with bidirectional LMR feedback for n ISTAs	145
Figure 11-37i—Non-TB ranging measurement exchange sequence.....	147
Figure 11-37j—Timing diagram of a measurement sounding phase in non-TB ranging.....	149
Figure 11-37k—Non-TB ranging with immediate reporting	150
Figure 11-37l—Non-TB ranging with delayed reporting.....	151
Figure 11-37m—Illustration of min time between measurements and max time between measurements with delayed reporting	151
Figure 11-37n—Non-TB ranging measurement exchange sequence with bidirectional LMR feedback....	151
Figure 11-37o—Overview of the TB ranging measurement exchange with secure HE-LTF	153
Figure 11-37p—Error recovery of TB ranging measurement exchange using secure HE-LTF.....	157
Figure 11-37q—Overview of the non-TB ranging measurement exchange with secure HE-LTF.....	158
Figure 11-37r—Error recovery of non-TB ranging measurement exchange using secure HE-LTF.....	161
Figure 11-37s—ISTA secure HE-LTF octet stream generation.....	163
Figure 11-37t—RSTA secure HE-LTF octet stream generation	164
Figure 11-37u—Timing diagram of a measurement sounding phase in TB ranging based on phase shift of I2R NDP and R2I NDP PPDU's	168
Figure 11-37v—Passive TB ranging polling, measurement sounding, and measurement reporting phases.....	170
Figure 11-37w—Example timing diagram of a Measurement Sounding phase in passive TB ranging	171
Figure 11-37x—Passive TB ranging Measurement Reporting phase	172
Figure 11-37y—Example of passive TB ranging measurement exchanges with PSTOA measurements.....	175
Figure 12-16—Expanded CCMP MPDU	181
Figure 12-26—Expanded GCMP MPDU.....	183
Figure 12-30—Pairwise key hierarchy	187
Figure 12-56—PASN authentication.....	191
Figure 12-57—PASN Wrapped Data format with SAE.....	197
Figure 27-46a—HE Ranging NDP format	207
Figure 27-46b—An example of HE-LTFs in an HE Ranging NDP with NUM_USERS = 1, NUM_STS = 2, and N_LTF_REP = 2.....	209
Figure 27-46c—An example of secure HE-LTFs with NUM_USERS = 2, NUM_STS = [2,1], and LTF_REP = [2,2]	209
Figure 27-46d—HE Ranging NDP format with secure HE-LTFs	210
Figure 27-46e—HE TB Ranging NDP format	210
Figure 27-46f—HE TB Ranging NDP with secure HE-LTFs.....	211
Figure 27-46g—Segment parser distributing pseudorandom octets to the sequences for the lower and upper 80 MHz segments in either the 160 MHz secure HE-LTF, or the 80+80 MHz secure HE-LTF sequence	217
Figure 27-46h—Generation of secure HE-LTF symbols per repetition in a HE Ranging NDP PPDU.....	222
Figure 27-48—DMG secure ranging TRN subfield structure over 2.16 GHz, 4.32 GHz, 6.48 GHz, and 8.64 GHz channel.....	229
Figure AE-1—Attacking secure HE-LTF symbols	245
Figure AE-2—Received SIR after attack	246

IEEE Standard for Information Technology— Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks— Specific Requirements

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 4: Enhancements for Positioning

(This amendment is based on IEEE Std 802.11™-2020 as amended by IEEE Std 802.11ax™-2021, IEEE Std 802.3ay™-2021, IEEE Std 802.3ba™-2021, and IEEE Std 802.11™-2020/Cor 1-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~striking through~~ (to remove old material) and underscoring (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editorial instructions, change markings and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.⁶

⁶ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1. Overview

1.3 Supplementary information on purpose

Insert the following item at the end of the dashed list in 1.3:

- Defines mechanisms for the determination of absolute and relative position with secure measurement and measurement results exchange.

3. Definitions, acronyms, and abbreviations

3.2 Definitions specific to IEEE Std 802.11

Insert new definitions into 3.2 in alphabetical order as follows:

enhanced directional multi-gigabit (EDMG) secure ranging physical layer (PHY) protocol data unit (PPDU): An EDMG single user (SU) PPDU that contains secure training (TRN) subfields in the training (TRN) field to enable secure ranging with physical layer (PHY) level security.

HE-LTF repetitions: Multiple transmissions of HE-LTF symbols in an HE Ranging NDP or HE TB Ranging NDP, where an HE-LTF repetition value of 1 indicates no repetitions, and, for example, a value of 2 or 3 would indicate twice or three times as many HE-LTF symbols, respectively.

LMR frame: A Location Measurement Report frame.

non-TB ranging: A ranging measurement procedure that uses NDP, and is not initiated by a Ranging Trigger frame.

Null-SAC-HE-LTF: An HE-LTF present in an initiating STA (ISTA) to a responding STA (RSTA) null data PPDU (NDP), or RSTA to ISTA NDP in the ranging frame exchange, resulting from a mismatch of sequence authentication code (SAC) subfield in the STA Info field of a Ranging NDP Announcement frame, or the SAC subfield in the Trigger Dependent User Info field in the Ranging Secure Sounding Trigger frame, with either the value of the Validation SAC subfield in the Secure HE-LTF Parameters element in the last transmitted FTM frame, or the last transmitted Location Measurement Report frame to the ISTA, or is equal to 0. The TXVECTOR LTF_KEY and LTF_IV parameter corresponding to this LTF are set to generate any secure HE-LTF or null.

PASN Authentication frame: Authentication frame used in preassociation security negotiation.

Passive Sounding Ranging Trigger frame: A Ranging Trigger frame of Passive Sounding subvariant.

Poll Ranging Trigger frame: A Ranging Trigger frame of Poll subvariant.

preassociation security negotiation (PASN): A mechanism to establish security association and allow management frame protection prior to association.

preassociation security negotiation station (STA): A STA that implements preassociation security negotiation (PASN).

Report Ranging Trigger frame: A Ranging Trigger frame of Report subvariant.

robust Action No Ack frame: An Action No Ack frame with a category value specified in 9.4.1.11 (Table 9-51) of IEEE Std 802.11 with Yes in the Robust column.

Secure Sounding Ranging Trigger frame: A Ranging Trigger frame of Secure Sounding subvariant.

Sounding Ranging Trigger frame: A Ranging Trigger frame of Sounding subvariant.

unassociated range negotiation and measurement management frame protection required (URNM-MFPR): A security policy that specifies whether ranging frames are required to be protected without association.

unassociated range negotiation and measurement management frame protection required exempt 20 MHz (URNM-MFPR-X20): A security policy that specifies whether ranging frames are required to be protected without association if bandwidth greater than 20 MHz is used.

3.4 Acronyms and abbreviations

Insert new abbreviations into 3.4 in alphabetical order as follows:

FPBT	first path beamforming training
FTM	fine timing measurement
FTMR	fine timing measurement request
I2R	ISTA to RSTA
IFTM	initial fine timing measurement
IFTMR	initial fine timing measurement request
ISTA	initiating STA
LMR	Location Measurement Report
PSTA	passive STA
PSTOA	phase shift time of arrival
R2I	RSTA to ISTA
RSTA	responding STA
RSID	ranging session Identifier
SAC	sequence authentication code

Please note that instances of “initial fine timing measurement request” will be changed to “IFTMR” as appropriate in the next revision of IEEE Std 802.11.

Please note that instances of “fine timing measurement” will be changed to “FTM” as appropriate in the next revision of IEEE Std 802.11.

4. General description

4.3 Components of the IEEE 802.11 architecture

4.3.19 Wireless network management

Change the title and content of 4.3.19.19 as follows:

4.3.19.19 Fine timing measurement (FTM)

The Fine timing measurement FTM procedure allows a STA to accurately measure the round trip time (RTT) between it and another STA. If used with multiple peer STAs, the STA can track changes in its relative position with those peer STAs. If the absolute position of the peer STAs is known, then the STA can determine its absolute position. With the regular transfer of Fine Timing Measurement frames it is possible for the recipient STA to track changes in its relative location with other STAs in the environment. A STA executes the FTM procedure with multiple peers and uses the resulting relative location estimates along with the absolute position of each of the peers to determine its position in geospatial coordinates.

The STA can also estimate the direction of the transmission (angle of departure) of frames transmitted to and reception (angle of arrival) of frames received from a peer STA, allowing for estimating position using measurements obtained from frame exchanges with a single peer.

An FTM session is negotiated to determine range estimates by executing one of the measurement exchanges listed below:

- EDCA based exchange of FTM frames where location estimates are based on time of departure and time of arrival of the exchanged FTM frames and their corresponding acknowledgments. When the negotiated FTM session is over EDMG, security parameters can be negotiated to ensure that the measurement exchange is executed with the intended peer.
- Trigger-based (TB) measurement where location estimates are based on the execution of the trigger-based measurement exchange. This mode of operation allows for the execution of the measurement exchange between a responding STA (RSTA) and multiple initiating STAs (ISTAs) at the same time. Optionally, the TB measurement session can be negotiated to enable security parameters enabling mechanisms to ensure that the measurement exchange is executed with the intended peer.
- Non-Trigger-based (non-TB) measurement where location estimates are based on the execution of the non-TB measurement exchange. Optionally, the non-TB measurement session can be negotiated to enable security parameters enabling mechanisms to ensure that the measurement exchange is executed with the intended peer.
- A passive trigger-based (TB) Ranging measurement exchange is defined where a STA can determine its location based on periodic measurement reports from other STAs that execute the passive TB ranging measurement exchange amongst themselves.

The preassociation security negotiation protocol establishes a PTKSA for the exchange of protected frames without association, for example for establishing a secure FTM session and performing the corresponding secure measurement exchange between two unassociated peers.

4.5 Overview of the services

4.5.4 Access control and data confidentiality services

4.5.4.2 Authentication

Change the third and fourth paragraphs of 4.5.4.2 as follows:

IEEE Std 802.11-2020 defines ~~five-six~~ IEEE 802.11 authentication methods: Open System authentication, Shared Key authentication, FT authentication, (IEEE Std 802.11-2020) simultaneous authentication of equals (SAE), ~~and FILS authentication (IEEE Std 802.11-2020) and preassociation security negotiation (PASN) authentication.~~ Open System authentication admits any STA to the DS. Shared Key authentication relies on WEP to demonstrate knowledge of a WEP encryption key. FT authentication relies on keys derived during the initial mobility domain association to authenticate the stations as defined in Clause 13. SAE authentication uses finite field cryptography to prove knowledge of a shared password. FILS authentication allows for faster connection to the network for FILS non-AP STAs by providing authentication, association, and key confirmation information in an efficient number of frame exchanges (see 4.10.3.6). PASN authentication allows for the protection of Management frames without association by establishing a PTKSA using authentication frames. The IEEE 802.11 authentication mechanism also allows definition of new authentication methods, or any combination of these authentication methods.

An RSNA might support one or more of the following authentication methods: SAE authentication, FILS authentication, ~~or both PASN authentication.~~ An RSNA also supports authentication based on IEEE Std 802.1X-2010, or preshared keys (PSKs) after Open System authentication. IEEE 802.1X authentication utilizes the EAP to authenticate STAs and the AS with one another. This standard does not specify an EAP method that is mandatory to implement. See 12.6.5 for a description of the IEEE 802.1X authentication and PSK usage within an IEEE 802.11 IBSS.

Insert a new paragraph at the end of 4.5.4.2 as follows:

PASN authentication is used in an RSN for an infrastructure BSS when it is based on a PMKSA established by another RSN authentication protocol. Otherwise, it does not guarantee mutual authentication, and can be used as a non-RSN protocol in an infrastructure BSS.

6. Layer management

6.3 MLME SAP interface

6.3.5 Authenticate

6.3.5.2 MLME-AUTHENTICATE.request

6.3.5.2.2 Semantics of the service primitive

Change the text of 6.3.5.2.2 as follows:

The primitive parameters are as follows:

```
MLME-AUTHENTICATE.request(  
    PeerSTAAddress,  
    AuthenticationType,  
    AuthenticateFailureTimeout,  
    Content of FT Authentication elements,  
    Content of SAE Authentication frame,  
    Multi-band local,  
    Multi-band peer,  
    Content of FILS Authentication frame,  
    Content of PASN Authentication frame,  
    VendorSpecificInfo  
)
```

Insert a new row above the row for “VendorSpecificInfo” in the unnumbered table in 6.3.5.2.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
Content of PASN Authentication frame	Sequence of elements and fields	As defined in 12.13.3.2, 9.4.2.24, 9.4.2.241, 9.4.2.187, 9.4.2.303, 9.4.2.48	The set of elements and fields to be included in PASN Authentication frames. Present if AuthenticationType indicates PASN authentication and dot11PASNActivated is true, otherwise not present.
...			

6.3.5.3 MLME-AUTHENTICATE.confirm

6.3.5.3.2 Semantics of the service primitive

Change the text of 6.3.5.3.2 as follows:

The primitive parameters are as follows:

```
MLME-AUTHENTICATE.confirm(  
    PeerSTAAddress,  
    AuthenticationType,  
    ResultCode,  
    Content of FT Authentication elements,  
    Content of SAE Authentication frame,  
    Multi-band local,
```

Multi-band peer,
Content of FILS Authentication frame,
Content of PASN Authentication frame.
VendorSpecificInfo
)

Change the row for “AuthenticationType” and insert a new row above the row for “VendorSpecificInfo” in the unnumbered table in 6.3.5.3.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
AuthenticationType	Enumeration	OPEN_SYSTEM, SHARED_KEY FAST_BSS_TRANSITION, SAE, FILS_SHARED KEY_WITHOUT_PFS, FILS_SHARED_KEY_WI TH_PFS, FILS_PUBLIC_KEY_ PASN	Specifies the type of authentication algorithm that was used during the authentication process. This value matches the AuthenticationType parameter specified in the corresponding MLME-AUTHENTICATE.request primitive.
...			
Content of PASN Authentication frame	Sequence of elements and fields	As defined in 12.13.3.2, 9.4.2.241, 9.4.2.24, 9.4.2.187, 9.4.2.303, 9.4.2.48	The set of elements and fields to be included in PASN Authentication frames. Present if AuthenticationType indicates PASN authentication and dot11PASNActivated is true, otherwise not present.
...			

6.3.5.4 MLME-AUTHENTICATE.indication

6.3.5.4.2 Semantics of the service primitive

Change the text of 6.3.5.4.2 as follows:

The primitive parameters are as follows:

MLME-AUTHENTICATE.indication(
PeerSTAAddress,
AuthenticationType,
Content of FT Authentication elements,
Content of SAE Authentication frame,
Multi-band local,
Multi-band peer,
Content of FILS Authentication frame,
Content of PASN Authentication frame.
VendorSpecificInfo
)

Change the row for “AuthenticationType” and insert a new row above the row for “VendorSpecificInfo” in the unnumbered table in 6.3.5.4.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
AuthenticationType	Enumeration	OPEN_SYSTEM, SHARED_KEY, FAST_BSS_TRANSITION, SAE, FILS_SHARED_KEY_WITHOUT_PFS, FILS_SHARED_KEY_WITH_PFS, FILS_PUBLIC_KEY_PASN	Specifies the type of authentication algorithm that was used during the authentication process. <u>This value matches the AuthenticationType parameter specified in the corresponding MLME-AUTHENTICATE.request primitive.</u>
...			
Content of PASN Authentication frame	Sequence of elements and fields	As defined in 12.13.3.2, 9.4.2.24, 9.4.2.241, 9.4.2.187, 9.4.2.303, 9.4.2.48	The set of elements and fields to be included in PASN Authentication frames. Present if AuthenticationType indicates PASN authentication and dot11PASNActivated is true, otherwise not present.
...			

6.3.5.5 MLME-AUTHENTICATE.response

6.3.5.5.2 Semantics of the service primitive

Change the text of 6.3.5.5.2 as follows:

The primitive parameters are as follows:

```
MLME-AUTHENTICATE.response(
    PeerSTAAddress,
    ResultCode,
    Content of FT Authentication elements,
    Content of SAE Authentication frame,
    Multi-band local,
    Multi-band peer,
    Content of FILS Authentication frame,
    Content of PASN Authentication frame,
    VendorSpecificInfo
)
```

Insert a new row above the row for “VendorSpecificInfo” in the unnumbered table in 6.3.5.5.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
Content of PASN Authentication frame	Sequence of elements and fields	As defined in 12.13.3.2, 9.4.2.24, 9.4.2.187, 9.4.2.303, 9.4.2.48	The set of elements and fields to be included in PASN Authentication frames. Present if AuthenticationType indicates PASN authentication and dot11PASNActivated is true, otherwise not present.
...			

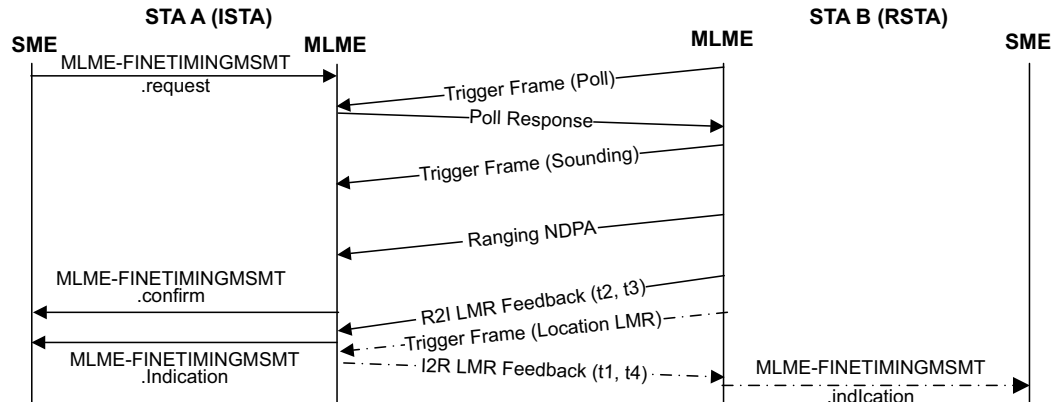


Figure 6-17b—FTM primitives and timestamps reporting capture for TB ranging measurement exchange

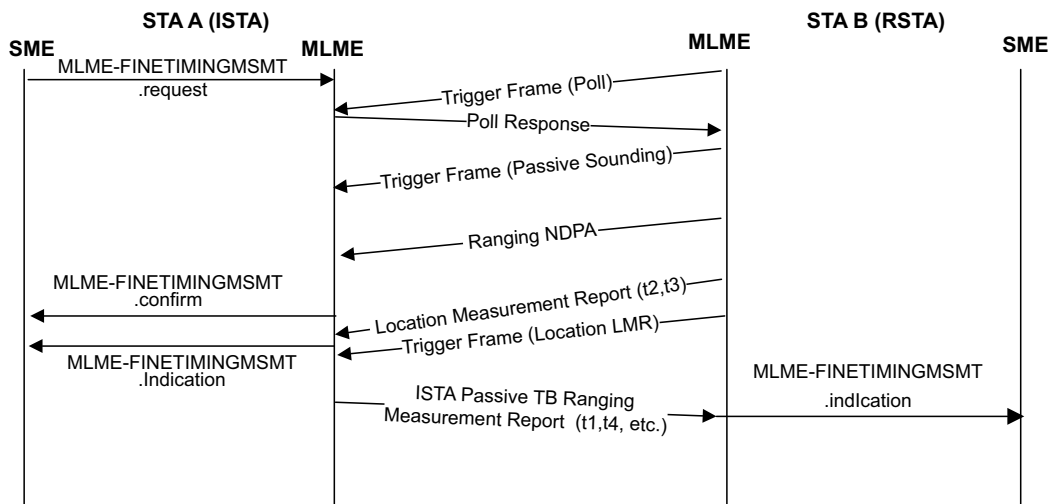


Figure 6-17c—FTM primitives and timestamps reporting capture for passive TB ranging measurement exchange

Change NOTE 1, NOTE 2, and NOTE 3 in 6.3.56.1 as follows:

NOTE 1—In Figure 6-17, t_1 and t_3 correspond to the point in time at which the start of the preamble for the transmitted frame appears at the transmit antenna connector. In Figure 6-17a, Figure 6-17b, and Figure 6-17c, t_1 and t_3 correspond to the point in time at which the transmitted HE TB Ranging NDP and/or HE Ranging NDP appears at the transmit antenna connector. The points where the timestamps are captured are therefore not shown for the non-TB, TB, and passive TB measurement exchanges. An implementation may capture a timestamp during the transmit processing earlier or later than the point at which it actually occurs and offset the value to compensate for the time difference.

NOTE 2—In Figure 6-17, t_2 and t_4 correspond to the point in time at which the start of the preamble for the incoming frame arrives at the receive antenna connector. In Figure 6-17a, Figure 6-17b, and Figure 6-17c, t_2 and t_4 correspond to the point in time at which the incoming HE TB Ranging NDP and/or HE Ranging NDP arrives at the receive antenna connector. The points where the timestamps are captured are therefore not shown for the non-TB and TB measurement exchanges. Because time is needed to detect the frame or the relevant LTF in the preamble and synchronize with its logical structure, an implementation determines when the start of the preamble or the relevant LTF in the preamble for the incoming frame arrived at the receive antenna connector by capturing a timestamp some time after it occurred and compensating for the delay by subtracting an offset from the captured value.

NOTE 3—In the MLME-FINETIMINGMSMT.request primitive corresponding to the message exchange described in Figure 6-17, the t1, Max t1 Error Exponent, t4, and Max t4 Error Exponent parameters are set to the values in the prior MLME-FINETIMINGMSMT.confirm primitive for that Peer MAC Address and with a Dialog Token parameter equal to the Follow Up Dialog Token parameter in the request, or 0 if there was none. In the MLME-FINETIMINGMSMT.confirm primitive the t1, Max t1 Error Exponent, t4, and Max t4 Error Exponent parameters are set to the values determined for the ~~Fine Timing Measurement FTM~~ frame and its acknowledgment. This primitive is not issued if no acknowledgment is received in response to the transmitted FTM frame (see Figure 6-17) or if the corresponding non-TB mode (see Figure 6-17a), or the TB mode (see Figure 6-17b), or the passive TB mode (see Figure 6-17c) Sounding Exchange did not complete. In the MLME-FINETIMINGMSMT.indication primitive the t1, Max t1 Error Exponent, t4, and Max t4 Error Exponent parameters are set to the values in the ~~Fine Timing Measurement FTM~~ frame and the t2, Max t2 Error Exponent, t3, and Max t3 Error Exponent parameters are set to the values determined for the ~~Fine Timing Measurement FTM~~ frame and its acknowledgment, Figure 6-17; or to the values in the corresponding LMR frame; see Figure 6-17a and Figure 6-17b. For passive TB ranging, the reported timestamps and parameters in the MLMEFINETIMINGMSMT.indication primitive are set to the values in the in the corresponding LMR and ISTA Passive TB Ranging Measurement Report frames in Figure 6-17c.

6.3.56.2 MLME-FINETIMINGMSMTRQ.request

6.3.56.2.2 Semantics of the service primitive

Change the text of 6.3.56.2.2 as follows:

The primitive parameters are as follows:

```
MLME-FINETIMINGMSMTRQ.request(
    Peer MAC Address,
    Trigger,
    LCI requested,
    LCI optional subelements,
    Location Civic requested,
    Civic Location Type,
    Location Civic optional subelements,
    Fine Timing Measurement FTM Parameters,
    Ranging Parameters,
    Minimum Required Secure HE-LTF Version,
    Vendor Specific
)
```

Insert two new rows above the row for “VendorSpecific” in the unnumbered table in 6.3.56.2.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
Ranging Parameters	As defined in 9.4.2.298	As defined in 9.4.2.298	Optional element containing the configuration for the requested ranging session.
Minimum Required Secure HE-LTF Version	As defined in 9.4.2.298	As defined in 9.4.2.298	The minimum version number for the secure LTF protocol required by the user or system.
...			

6.3.56.3 MLME-FINETIMINGMSMTRQ.indication

6.3.56.3.2 Semantics of the service primitive

Change the text of 6.3.56.3.2 as follows:

The primitive parameters are as follows:

MLME-FINETIMINGMSMTRQ.indication(
Peer MAC Address,
Trigger,
LCI requested,
LCI optional subelements,
Location Civic requested,
Civic Location Type,
Location Civic optional subelements,
~~Fine Timing Measurement~~ FTM Parameters,
Ranging Parameters,
Minimum Required Secure HE-LTF Version,
Vendor Specific
)

Insert two new rows above the row for “VendorSpecific” in the unnumbered table in 6.3.56.3.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
Ranging Parameters	As defined in 9.4.2.298	As defined in 9.4.2.298	Optional element containing the configuration for the requested ranging session.
Minimum Required Secure HE-LTF Version	As defined in 9.4.2.298	As defined in 9.4.2.298	Optional element. If the negotiation was successful, contains the secure LTF protocol version as assigned for the FTM session. If the negotiation failed, contains the minimum required protocol version allowed by the RSTA (Status Indication equals 2 or 3).
...			

6.3.56.4 MLME-FINETIMINGMSMT.request

6.3.56.4.1 Function

Change 6.3.56.4.1 as follows:

This primitive requests the transmission of a ~~Fine Timing Measurement~~ an FTM frame to a peer entity to initiate any of the following with the specified peer entity: 11.21.6.4.2, 11.21.6.4.3, 11.21.6.4.4, or 11.21.6.4.8.

6.3.56.4.2 Semantics of the service primitive

Change the text of 6.3.56.4.2 as follows:

The primitive parameters are as follows:

```
MLME-FINETIMINGMSMT.request(  
    Peer MAC Address,  
    Dialog Token,  
    Follow Up Dialog Token,  
    t1,  
    Max t1 Error Exponent,  
    t4,  
    Max t4 Error Exponent,  
    LCI Report,  
    Location Civic Report,  
    Fine Timing Measurement FTM Parameters,  
    Ranging Parameters,  
    VendorSpecific  
)
```

Change the unnumbered table in 6.3.56.4.2 as follows:

Name	Type	Valid range	Description	<u>Applies to both non-TB and TB ranging</u>
Peer MAC Address	MAC address	Any valid individual addressed MAC address	The address of the peer MAC entity to which that acknowledges the receipt of the Fine Timing Measurement FTM frame is sent.	<u>Yes</u>
Dialog Token	Integer	0–255	The dialog token to identify the Fine Timing Measurement FTM frame in 11.21.6.4.2. A value of 0 indicates the end of the FTM session.	<u>No</u>
Follow Up Dialog Token	Integer	0–255	The dialog token of a Fine Timing Measurement an FTM frame which that the current frame follows, or 0 if there is no such frame. See 11.21.6 11.21.6.4.2.	<u>No</u>
t1	Integer	$0-(2^{48}-1)$	For EDCA based ranging measurement exchange (11.21.6.4.2), the The value of t1 (see Figure 6-17) for the Fine Timing Measurement FTM frame identified by the Follow Up Dialog Token, in units of picoseconds, or null if the Follow Up Dialog Token is 0.	<u>No</u>
Max t1 Error Exponent	Integer	0–31	The maximum error in the t1 value. This is represented using a function of the Max t1 Error Exponent parameter as defined in Equation (9-4), or is null if the Follow Up Dialog Token is 0.	<u>No</u>
t4	Integer	$0-(2^{48}-1)$	For EDCA based ranging measurement exchange (11.21.6.4.2), the The value of t4 (see Figure 6-17) for the Fine Timing Measurement FTM frame identified by the Follow Up Dialog Token, in units of picoseconds, or null if the Follow Up Dialog Token is 0.	<u>No</u>

Name	Type	Valid range	Description	<u>Applies to both non-TB and TB ranging</u>
Max t4 Error Exponent	Integer	0–31	The maximum error in the t4 value. This is represented using a function of the Max t4 Error Exponent parameter as defined in Equation (9-4), or is null if the Follow Up Dialog Token is 0.	<u>No</u>
LCI Report	As defined in 9.6.7.33	As defined in 9.6.7.33	Optional element to report LCI information of sender.	<u>Yes</u>
Location Civic Report	As defined in 9.6.7.33	As defined in 9.6.7.33	Optional element to report location civic information of sender.	<u>Yes</u>
Fine Timing Measurement FTM Parameters	As defined in 9.4.2.167	As defined in 9.4.2.167	Optional element containing the proposed FTM configuration.	<u>No</u>
<u>Ranging Parameters</u>	<u>As defined in 9.4.2.298</u>	<u>As defined in 9.4.2.298</u>	<u>Optional element containing the configuration for the proposed 802.11az session.</u>	<u>Yes</u>
VendorSpecific	A set of elements	As defined in 9.4.2.25	Zero or more elements.	<u>Yes</u>

6.3.56.4.3 When generated

Change 6.3.56.4.3 as follows:

This primitive is generated by the SME ~~to request that a Fine Timing Measurement frame be sent to a peer entity.~~ in the context of an active FTM Session to initiate a measurement exchange. If the FTM session is:

- EDCA based ranging measurement exchange (11.21.6.4.2): the SME generates this primitive to request that an FTM frame be sent to a peer entity.
- non-TB ranging measurement exchange (11.21.6.4.4): the SME generates this primitive to request that a non-TB ranging measurement exchange be initiated with the specified peer entity.

NOTE—The sounding exchange initiation is constrained to the Min Time Between Measurements and Max Time Between Measurements thresholds that are defined when the corresponding FTM session was established.

- TB Sounding Exchange (11.21.6.4.3): the SME generates this primitive to request that a TB Sounding Exchange be initiated with the specified peer entity.

6.3.56.4.4 Effect of receipt

Change 6.3.56.4.4 as follows:

On receipt of this primitive, if the corresponding measurement exchange is:

- EDCA based ranging measurement exchange (11.21.6.4.2): the MLME constructs a Fine Timing Measurement ~~an FTM~~ frame with the specified parameters. This frame is then scheduled for transmission.
- non-TB ranging measurement exchange (11.21.6.4.4): the MLME generates a Ranging NDP Announcement frame with the specified parameters and transmits it to the specified peer entity.
- TB ranging measurement exchange (11.21.6.4.3): the MLME responds to the next Trigger frame of variant Ranging and subvariant Poll from the specified peer entity.

6.3.56.5 MLME-FINETIMINGMSMT.confirm

6.3.56.5.1 Function

Change 6.3.56.5.1 as follows:

This primitive indicates that a ~~Fine Timing Measurement~~ an FTM frame has been received by the peer STA to which it was sent.

For EDCA based ranging measurement exchange (11.21.6.4.2), this primitive indicates that an FTM frame has been received by the peer STA to which it was sent. For 11.21.6.4.3, or 11.21.6.4.4 this primitive indicates that the corresponding measurement exchange has completed successfully with the specified peer entity.

6.3.56.5.2 Semantics of the service primitive

Change the unnumbered table in 6.3.56.5.2 as follows:

Name	Type	Valid range	Description	<u>Applies to non-TB or TB ranging?</u>
Peer MAC Address	MAC address	Any valid individual addressed MAC address	The address of the peer MAC entity to which that acknowledges the receipt of the Fine Timing Measurement <u>FTM</u> frame.	<u>Yes</u>
Dialog Token	Integer	0–255	The dialog token to identify the Fine Timing Measurement <u>FTM</u> frame in EDCA based ranging measurement exchange (11.21.6.4.2). A value of 0 indicates the end of the FTM session.	<u>No</u>
t1	Integer	0–($2^{48}-1$)	<u>For EDCA based ranging measurement exchange (11.21.6.4.2), the</u> The value of t1 (see Figure 6-17) for the Fine Timing Measurement <u>FTM</u> frame identified by the Dialog Token, in units of picoseconds, or null if the Dialog Token is 0.	<u>No</u>
Max t1 Error Exponent	Integer	0–31	The maximum error in the t1 value. This is represented using a function of the Max t1 Error Exponent parameter as defined in Equation (9-4), or is null if the Dialog Token is 0.	<u>No</u>
t4	Integer	0–($2^{48}-1$)	<u>For EDCA based ranging measurement exchange (11.21.6.4.2), the</u> The value of t4 (see Figure 6-17) for the Fine Timing Measurement <u>FTM</u> frame identified by the Dialog Token, in units of picoseconds, or null if the Dialog Token is 0.	<u>No</u>
Max t4 Error Exponent	Integer	0–31	The maximum error in the t4 value. This is represented using a function of the Max t4 Error Exponent parameter as defined in Equation (9-4), or is null if the Dialog Token is 0.	<u>No</u>

6.3.56.5.3 When generated

Change 6.3.56.5.3 as follows:

This primitive is generated by the MLME when:

- a) ~~an An~~ Ack frame corresponding to the ~~Fine Timing Measurement-FTM~~ frame is received from the ~~peer STA-ISTA~~ when the ranging protocol is EDCA based ranging measurement exchange (11.21.6.4.2), or
- b) When the sounding exchange corresponding to the underlying non-TB (11.21.6.4.4) or TB (11.21.6.4.3) ranging measurement exchange is successfully completed at the ISTA.

6.3.56.6 MLME-FINETIMINGMSMT.indication

6.3.56.6.2 Semantics of the service primitive

Change the text of 6.3.56.4.2 as follows:

The primitive parameters are as follows:

MLME-FINETIMINGMSMT.indication(
Peer MAC Address,
Dialog Token,
Follow Up Dialog Token,
t1,
Max t1 Error Exponent,
t4,
Max t4 Error Exponent,
t2,
Max t2 Error Exponent
t3,
Max t3 Error Exponent,
LCI Report,
Location Civic Report,
~~Fine Timing Measurement-FTM~~ Parameters,
Ranging Parameters,
VendorSpecific
)

Insert a new row above the row for “VendorSpecific” in the unnumbered table in 6.3.56.6.2 as follows (unchanged rows not shown):

Name	Type	Valid range	Description
...			
Ranging Parameters	As defined in 9.4.2.298	As defined in 9.4.2.298	Optional element containing the configuration for the proposed NDP sounding based ranging session.
...			

8. PHY service specification

8.3 Detailed PHY service specifications

8.3.4 Basic service and options

8.3.4.2 PHY SAP inter-(sub)layer service primitives

Insert two new rows at the end of Table 8-2 as follows (unchanged rows not shown):

Table 8-2—PHY SAP inter-(sub)layer service primitives

Primitive	Request	Indication	Confirm
...			
PHY-RXLTFSEQUENCE	X		
PHY-RXTRNSEQUENCE	X		

8.3.4.3 PHY SAP service primitives parameters

Change the row for parameter “RXERROR” and insert a new row for parameter “LTFVECTOR” at the end of Table 8-3 as follows (unchanged rows not shown):

Table 8-3—PHY SAP service primitive parameters

Parameter	Associated primitive	Value
...		
RXERROR	PHY-RXEND.indication	NoError, FormatViolation, CarrierLost, UnsupportedRate, Filtered, <u>IntegrityCheckError</u>
...		
LTFVECTOR	PHY-RXLTFSEQUENCE.request	A set of parameters needed to receive and process the HE Ranging NDP and HE TB Ranging NDP (see Table 27-2a).

8.3.4.4 Vector descriptions

Insert a new row at the end of Table 8-4 as follows (unchanged rows not shown):

Table 8-4—Vector descriptions

Parameter	Associated vector	Value
...		
SECURE_TRN_SEQUENCE	TRNVECTOR	Indicates the Secure TRN bit sequences used in the EDMG secure ranging PPDU. The Secure TRN bit sequences generation is defined in 12.2.11.

8.3.5 PHY SAP detailed service specification

8.3.5.14 PHY-RXEND.indication

8.3.5.14.2 Semantics of the service primitive

Insert a new item at the end of the dashed list in 8.3.5.14.2 as follows:

- *IntegrityCheckError*. This value is used to indicate that during the reception of the HE Ranging NDP or HE TB Ranging NDP, an integrity check was performed and failed.

Insert 8.3.5.18 and 8.3.5.19 (and their subclauses) after 8.3.5.17.4 inserted by IEEE Std 802.11ay-2021, as follows:

8.3.5.18 PHY-RXLTFSEQUENCE.request

8.3.5.18.1 Function

This primitive is a request by the MAC sublayer to the local PHY entity to provide the parameters shown in Table 27-2a for the receipt of an HE Ranging NDP or HE TB Ranging NDP.

8.3.5.18.2 Semantics of the service primitive

The primitive provides the following parameter:

```
PHY-RXLTFSEQUENCE.request(
    LTFVECTOR
)
```

The LTFVECTOR represents a list of parameters needed to receive an HE Ranging NDP or HE TB Ranging NDP, including the HE-LTF configuration and information on how to generate the secure HE-LTF symbols as described in 27.3.18b.6.

8.3.5.18.3 When generated

This primitive is issued by the MAC sublayer to the PHY entity before receiving HE Ranging NDP and HE TB Ranging NDP.

8.3.5.18.4 Effect of receipt

The effect of receipt of this primitive by the PHY entity is to be aware of the number of spatial streams and HE-LTF repetitions that are not signaled in the HE SIG-A of the HE Ranging NDP and HE TB Ranging NDP, as well as to be able to generate the secure LTF symbols based on the parameters in the LTFVECTOR as described in 27.3.18b.6, if secure HE-LTF is used.

8.3.5.19 PHY-RXTRNSEQUENCE.request

8.3.5.19.1 Function

This primitive is a request by the MAC sublayer to the local PHY entity with a TRNVECTOR parameter to provide the Secure TRN bit sequences for the receipt of the EDMG secure ranging PPDU.

8.3.5.19.2 Semantics of the service primitive

The primitive provides the following parameter:

```
PHY-RXTRNSEQUENCE.request(  
    TRNVECTOR  
)
```

The TRNVECTOR represents the Secure TRN bit sequences used in the EDMG secure ranging PPDU.

8.3.5.19.3 When generated

This primitive is issued by the MAC sublayer to the PHY entity before receiving the secure TRN subfield of the EDMG secure ranging PPDU.

8.3.5.19.4 Effect of receipt

The effect of receipt of this primitive by the PHY entity is to make a secure ranging waveform based on the TRNVECTOR.

9. Frame formats

9.3 Format of individual frame types

9.3.1 Control frames

Change the title of 9.3.1.19 (as amended by IEEE Std 802.11ax-2021) as follows:

9.3.1.19 VHT/HE/Ranging NDP Announcement frame format

Change the first paragraph of 9.3.1.19 (as inserted by IEEE Std 802.11ax-2021) as follows:

The VHT/HE/Ranging NDP Announcement frame has ~~two-three~~ variants, the VHT NDP Announcement frame, ~~and the HE NDP Announcement frame, and the Ranging NDP Announcement frame.~~ The ~~two-three~~ formats are distinguished by the setting of the ~~HE subfield NDP Announcement Type subfield~~ in the Sounding Dialog Token field.

Change the now fourth and fifth paragraphs of 9.3.1.19 (as amended by IEEE Std 802.11ax-2021) as follows:

The VHT/HE/Ranging NDP Announcement frame contains at least one STA Info field. If the VHT/HE/Ranging NDP Announcement frame contains only one STA Info field with AID/RSID equal or less than 2007, then in the case of VHT or HE NDP Announcement frames the RA field is set to the address of the STA that can provide feedback (see 10.36.5.2), while in the case of Ranging NDP Announcement frames, the RA address is set to the address of the RSTA or ISTA that is the intended recipient of the frame. If the VHT/HE/Ranging NDP Announcement frame contains more than one STA Info field with AID/RSID equal or less than 2007, then the RA field is set to the broadcast address.

The TA field is set to the address of the STA transmitting the VHT/HE/Ranging NDP Announcement frame or the bandwidth signaling TA of the STA transmitting the VHT/HE/Ranging NDP Announcement frame. In a VHT/HE/Ranging NDP Announcement frame transmitted by a VHT or HE STA in a non-HT or non-HT duplicate format and where the scrambling sequence carries the TXVECTOR parameter CH_BANDWIDTH_IN_NON_HT, the TA field is set to a bandwidth signaling TA. If a Ranging NDP Announcement frame is addressed to a set of ISTAs in which at least two ISTAs have a TB ranging measurement exchange with a different BSSID in the Multiple BSSID set of the RSTA, then the TA field is the transmitted BSSID.

Replace Figure 9-59 with the following figure:

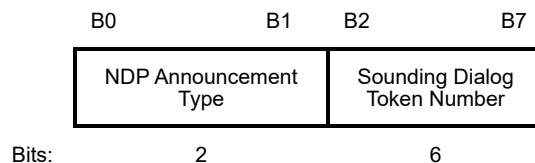


Figure 9-59—Sounding Dialog Token field format

Change the now seventh paragraph of 9.3.1.19 (as inserted by IEEE Std 802.11ax-2021) as follows:

~~The HE subfield in the Sounding Dialog Token field is set to 0 to identify the frame as a VHT NDP Announcement frame and set to 1 to identify the frame as an HE NDP Announcement frame. The NDP~~

Announcement Type field in the Sounding Dialog Token field identifies the NDP Announcement frame variant as defined in Table 9-28f.

Table 9-28f—NDP Announcement frame variant encoding

NDP Announcement Type subfield	NDP Announcement frame variant
0	VHT NDP Announcement frame
1	Ranging NDP Announcement frame
2	HE NDP Announcement frame
3	Reserved

Delete the second paragraph after Figure 9-61a (as inserted by IEEE Std 802.11ax-2021) with content: “The HE subfield in the Sounding Dialog Token field is set to 1 to identify the frame as an HE NDP Announcement frame.”

Insert the following text and Figure 9-61e through Figure 9-61h at the end of 9.3.1.19:

The Ranging NDP Announcement frame uses the same Frame Control subtype as the VHT NDP Announcement frame. The frame format of the Ranging NDP Announcement frame is the same as the HE NDP Announcement frame shown in Figure 9-61a.

The Duration, RA, and TA fields are set as in a VHT NDP Announcement frame.

The Sounding Dialog Token Number subfield in the Sounding Dialog Token (SDT) field contains a value in the range of 0 to 63, which identifies the Measurement Sounding phase of which this transmitted Ranging NDP Announcement frame is part; see 11.21.6.4.3 and 11.21.6.4.4.

The format of the STA Info field in a Ranging NDP Announcement frame, when the AID11 subfield is equal or less than 2007, is defined in Figure 9-61e.

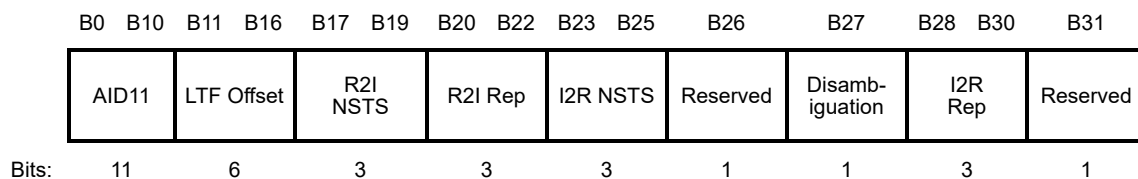


Figure 9-61e—STA Info field format in a Ranging NDP Announcement frame when the AID11 subfield is equal or less than 2007

A Ranging NDP Announcement frame contains one STA Info field with AID11 subfield equal or less than 2007 per STA that is intended to receive this frame.

In the case of the non-TB ranging measurement exchange (see 11.21.6.4.4), there is always only one intended receiver and the RA field is set to the address of that STA.

In the case of the TB ranging measurement exchange (see 11.21.6.4.3), the RA field is set to the broadcast address if more than one STA is intended to receive this frame; otherwise the RA field is set to the address of the STA that is intended to receive this frame.

If the AID11 subfield is equal or less than 2007, it identifies a STA that is intended to receive this frame and assigns the parameters within this STA Info field to this STA. In case of the TB ranging measurement exchange, see 11.21.6.4.3, the AID11 subfield contains the 11 least significant bits of the AID of an associated STA, or the RSID of an unassociated STA, when it is expected to process the following NDP. In case of the non-TB ranging measurement exchange (see 11.21.6.4.4), the intended receiver is identified by the RA field and the AID11 subfield is set to 0.

When used as part of the TB ranging measurement exchange (see 11.21.6.4.3), the LTF Offset, R2I NSTS, and R2I Rep subfields are used to indicate the following R2I NDP's HE-LTF configuration (see 27.3.18a.1); the I2R NSTS and the I2R Rep subfields are reserved.

When used as part of the non-TB ranging measurement exchange, the I2R NSTS and I2R Rep subfields are used to indicate the following I2R NDP's HE-LTF configuration, see 27.3.18a.1, while the R2I NSTS and R2I Rep subfields indicate the HE-LTF configuration of the R2I NDP sent in response by the RSTA, see 11.21.6.4.4.

The LTF Offset subfield is used in the TB ranging measurement exchange protocol with secure LTF see 11.21.6.4.5.2; it indicates the number of HE-LTF to skip when processing the following NDP. The LTF Offset subfield is set to 0 in all other cases.

The R2I NSTS and I2R NSTS subfields indicate the number of spatial streams of the corresponding NDP and is set to the number of spatial streams minus 1.

The R2I Rep and I2R Rep subfields are set to the number of HE-LTF repetitions of the corresponding HE Ranging NDP minus 1; see 27.3.18a.1. If the I2R and R2I Rep subfields have a value equal to 0, then there is no HE-LTF repetition in the I2R and R2I NDP respectively.

The Disambiguation subfield is set to 1 as in the STA Info field in an HE NDP Announcement frame regardless of the value of the AID11 subfield.

The format of the STA Info field in a Ranging NDP Announcement frame when the AID is equal to 2043 is shown in Figure 9-61f. It is used in the non-TB ranging measurement exchange with secure LTF to carry the sequence authentication code (SAC).

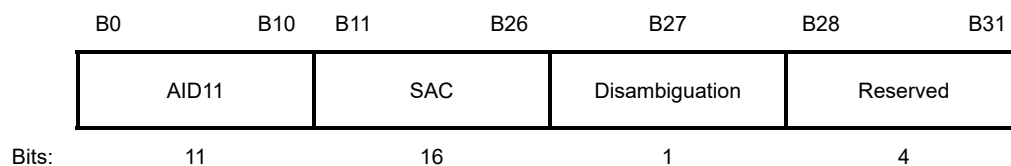


Figure 9-61f—STA Info field format in a Ranging NDP Announcement frame if the AID11 subfield is 2043

The SAC subfield contains the 16-bit SAC used in the non-TB ranging measurement exchange with secure LTF; see 11.21.6.4.5.3.

The format of the STA Info field in a Ranging NDP Announcement frame if the AID11 subfield is equal to 2044 as is shown in Figure 9-61g.

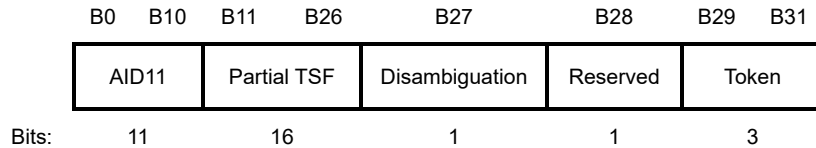


Figure 9-61g—STA Info field format in a Ranging NDP Announcement frame (if the AID11 subfield is equal to 2044)

The STA Info field, with AID11 subfield equal to 2044, is present in Ranging NDP Announcement frames when part of the TB ranging measurement exchange. It is used to carry the Partial TSF subfield. The Partial TSF subfield contains 16 bits of the RSTA's TSF time, TSF[21:6], when the RSTA transmitted the TF Ranging Poll that preceded the Ranging NDP Announcement frame, carrying this STA Info field with AID subfield equal to 2044.

The Token subfield is set to the value of the Token subfield of the TF Ranging Poll whose partial transmission TSF time is carried.

The format of the STA Info field with AID11 subfield equal to 2045 is shown in Figure 9-61h.

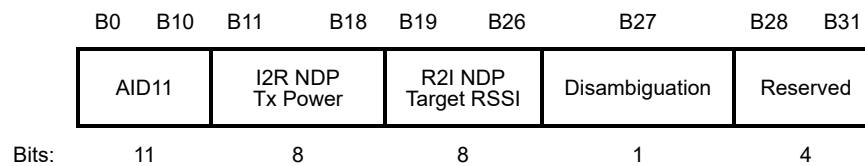


Figure 9-61h—STA Info field format in a Ranging NDP Announcement frame if the AID11 subfield is 2045

The STA Info field with AID11 subfield equal to 2045 is used in the non-TB ranging measurement exchange, see 11.21.6.4.4, to carry the I2R NDP Tx Power and R2I NDP Target RSSI subfields.

The I2R NDP Tx Power subfield indicates the combined average power per 20 MHz bandwidth referenced to the antenna connector, of all antennas used to transmit the following I2R NDP. The transmit power is reported with a resolution of 1 dB, with values in the range 0 to 60 representing –20 dBm to 40 dBm, respectively. Values above 60 are reserved.

The R2I NDP Target RSSI subfield indicates the preferred receive signal power, averaged over the ISTA's antenna connectors, for future R2I NDPs to be transmitted by the RSTA. The preferred receive signal power in units of dBm is $\text{TargetRSSI} = -110 + \text{FVal}$, where FVal is the value of the R2I NDP Target RSSI subfield, except that values above 90 indicate that the ISTA has no receive signal power preference for the R2I NDPs.

9.3.1.22 Trigger frame format

9.3.1.22.1 General

Insert a new row for subfield value “8” above the “Reserved” row and change the “Reserved” row in Table 9-29c (as inserted by IEEE Std 802.11ax-2021) as follows (unchanged rows not shown):

Table 9-29c—Trigger Type subfield encoding

Trigger Type subfield value	Trigger frame variant
...	
8	<u>Ranging</u>
8–15 <u>9–15</u>	Reserved

Change the second paragraph after Table 9-29c (as inserted by IEEE Std 802.11ax-2021) as follows:

The More TF subfield of the Common Info field indicates whether or not a subsequent Trigger frame is scheduled for transmission. The More TF subfield is set as defined in 26.8.2, ~~and 26.8.3.2,~~ and 11.21.6.4.3.

Change the paragraph above Figure 9-64d (as inserted by IEEE Std 802.11ax-2021) as follows:

The User Info field is defined in Figure 9-64d for all Trigger frame variants, except the NFRP Trigger frame, which is defined in 9.3.1.22.9 and the Ranging Trigger frame, which is defined in 9.3.1.22.10.

Change Table 9-29j (as inserted by IEEE Std 802.11ax-2021) as follows:

Table 9-29j—UL Target Receive Power subfield in Trigger frame

UL Target Receive Power subfield	Description
0–90	The expected receive signal power, in units of dBm, is $Target_{pwr} = -110 + F_{val}$, where F_{val} is the subfield value
91–126	Reserved
127	The STA transmits the HE TB PPDU at the STA’s maximum transmit power for the assigned HE-MCS. <u>If the Trigger frame is a Sounding or Passive Sounding Ranging Trigger frame that does not assign an HE-MCS, then the STA’s transmit power is that used for HE-MCS 0. If the Trigger frame is a Secured Sounding Ranging Trigger frame that does not assign an HE-MCS, the assigned HE-MCS is assumed to be HE-MCS 6 in terms of setting the STA’s transmit power.</u> <u>NOTE—</u>The expected receive signal power is then the STA’s maximum transmit power for the assigned HE-MCS minus the path loss.

Insert 9.3.1.22.10 and 9.3.1.22a (and its subclauses) after 9.3.1.22.9 (as inserted by IEEE Std 802.11ax-2021) as follows:

9.3.1.22.10 Ranging Trigger variant

The Ranging Trigger Subtype subfield value in the Trigger Dependent Common Info field of the Ranging Trigger frame, see Table 9-29l, signals the Ranging Trigger frame subvariants, which can be one of five frame types: Poll, Sounding, Secure Sounding, Report, and Passive Sounding Ranging Trigger frame.

The format of the Trigger Dependent Common Info subfield for the Poll, Sounding, Secure Sounding, and Report Ranging Trigger frame, is shown in Figure 9-64m. The Token field in the Trigger Dependent Common Info subfield is used in a Poll Ranging Trigger frame to match it with the partial TSF time in a following Ranging NDP Announcement frame. It is reserved in all other Ranging Trigger subvariants.

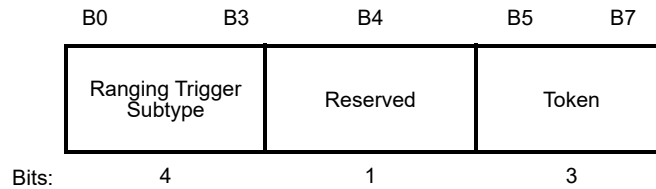


Figure 9-64m—Trigger Dependent Common Info subfield format for the Ranging Trigger subvariants Poll, Sounding, Secure Sounding, and Report

The format of the Trigger Dependent Common Info subfield of Ranging Trigger frame of subvariant Passive Sounding is shown in Figure 9-64n. The Sounding Dialog Token Number subfield contains a value in the range of 0 to 63 which identifies a Measurement Sounding phase (I2RNDP and R2I NDP announced by a Sounding Trigger frame and the Ranging NDP Announcement frame, respectively), and the same value is included in the Sounding Dialog Token field of the Ranging NDP Announcement frame transmitted within the same availability window; see 9.3.1.19.

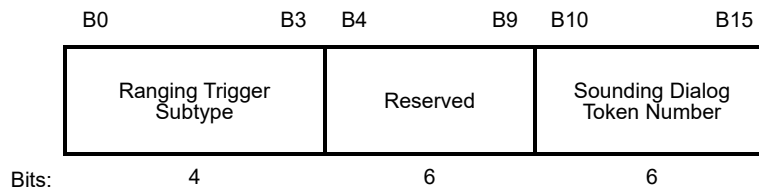


Figure 9-64n—Trigger Dependent Common Info subfield format of the Passive Sounding Ranging Trigger frame

The value of the Ranging Trigger Subtype subfield for the Ranging Trigger frame is defined in Table 9-291.

Table 9-291—Ranging Trigger Subtype subfield encoding

Ranging Trigger Subtype subfield value	Ranging Trigger frame subvariant
0	Poll
1	Sounding
2	Secure Sounding
3	Report
4	Passive Sounding
5–15	Reserved

The RA field, and the CS Required and UL BW subfields in the Common Info field of the Ranging Trigger frame are identical to the Basic Trigger frame described in 26.5.2 and 9.3.1.22, except that the RA field in Ranging Trigger frames with only one User Info field may be either unicast or broadcast.

The More TF subfield of the Common Info field of the Ranging Trigger frame indicates whether a subsequent Poll Ranging Trigger is scheduled for transmission within the availability window as defined in 11.21.6.4.3.

When a Ranging Trigger frame is addressed to ISTAs that all have a TB ranging measurement exchange, see 11.21.6.4.3, with the same RSTA’s BSSID, the TA field of the Ranging Trigger frame is set to the RSTA’s address. The TA field is set to the transmitted BSSID, if the Trigger frame is addressed to a set of ISTAs in which at least two ISTAs have a TB ranging measurement exchange with different BSSIDs in the Multiple BSSID set of the RSTA.

9.3.1.22a Subvariants of Ranging Trigger variant

9.3.1.22a.1 Poll subvariant

The format of the User Info field in the Poll Ranging Trigger frame is defined in Figure 9-64o.

The AID12/RSID12 subfield carries either the 12 LSBs of the AID for an associated ISTA or the 12 LSBs of the RSID for an unassociated ISTA. The RU Allocation, UL FEC Coding Type, UL HE-MCS, UL DCM, SS Allocation/RA-RU Information, UL Target Receive Power subfields are identical to the corresponding subfield in the Basic Trigger frame; see 9.3.1.22.

The Trigger Dependent User Info subfield is not present in the Poll Ranging Trigger frame.

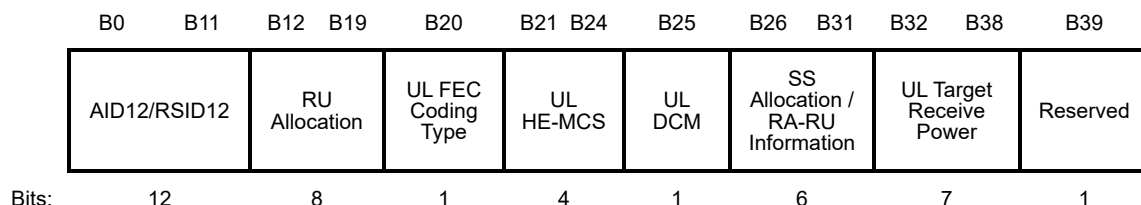


Figure 9-64o—User Info field format for the Poll and Report Ranging Trigger

9.3.1.22a.2 Sounding subvariant

The format of the User Info field in the Sounding Ranging Trigger frame is defined in Figure 9-64p.

The Trigger Dependent User Info subfield is not present in the Sounding Ranging Trigger frame.

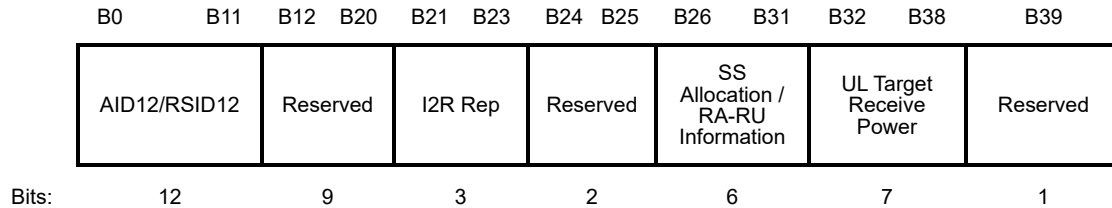


Figure 9-64p—User Info field format for Sounding subvariant

The AID12/RSID12 subfield is identical to the corresponding subfield in the Poll Ranging Trigger frame.

The I2R Rep subfield indicates the number of HE-LTF repetitions in the corresponding HE TB Ranging NDP from the STA indicated in the AID12/RSID12 subfield; the I2R Rep subfield is set to the number of HE-LTF repetitions minus 1. The value of the I2R Rep subfield is the same in all User Info fields in a single Trigger frame. The SS Allocation/RA-RU Information and UL Target Receive Power subfields are identical to the corresponding subfields in the Basic Trigger frame; see 9.3.1.22.

In the Common Info field, the UL STBC, LDPC Extra Symbol Segment, Pre-FEC Padding Factor, and PE Disambiguity subfields are reserved.

The GI And HE-LTF Type subfield in the Common Info field is set to 1 (2x HE-LTF + 1.6 μ s GI). The MU-MIMO HE-LTF Mode subfield in the Common Info field is set to 0 (HE single stream pilot HE-LTF mode).

The Doppler subfield in the Common Info field is set to 0.

NOTE—The UL Length subfield of a Trigger frame is computed using Equation (27-11) (see 26.5.2.2.4), which is based on the TXTIME computed in 27.4.3. In case of Sounding Ranging Trigger frame, the resulting UL Length value is equivalent to $13 + 6 \times N_{LTF-REP} \times N_{HE-LTF}$, where $N_{LTF-REP}$ is the number of HE-LTF repetitions (given by the I2R Rep subfield value plus 1) and N_{HE-LTF} is the number of HE-LTF symbols (given by the Number Of HE-LTF Symbols And Midamble Periodicity subfield).

9.3.1.22a.3 Secure sounding subvariant

The format of the User Info field in the Secure Sounding Ranging Trigger is defined in Figure 9-64q. There is a single User Info field in a Secure Sounding Ranging Trigger frame.

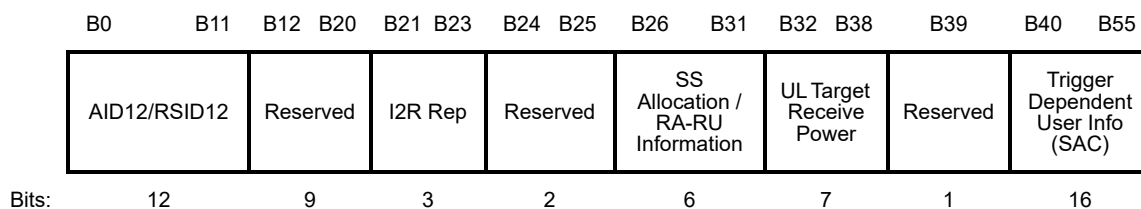


Figure 9-64q—User Info field for Secure Sounding subvariant

The AID12/RSID12 subfield is identical to the corresponding subfield in the Poll Ranging Trigger frame.

The I2R Rep subfield is identical to the corresponding subfield in the Sounding Ranging Trigger frame.

The SS Allocation/RA-RU Information and UL Target Receive Power subfields are identical to the corresponding subfields in the Basic Trigger frame; see 9.3.1.22.

The Trigger Dependent User Info subfield is present in the Secure Sounding Ranging Trigger frame. The Trigger Dependent User Info subfield carries the Sequence Authentication Code (SAC) field. The SAC subfield provides the authentication information for the LTF Sequence Generation information used for the I2R sounding associated with the measurement instance; see 11.21.6.4.5. The length of this subfield is 16 bits.

NOTE—For measurement exchange with secure HE-LTF, the I2R Rep is set to the RSTA Assigned I2R Rep; see 11.21.6.3.4.

In the Common Info field, the MU-MIMO HE-LTF Mode, UL STBC, LDPC Extra Symbol Segment, Pre-FEC Padding Factor, and PE Disambiguity subfields are reserved. The GI And HE-LTF Type and Doppler subfields in the Common Info field are set as in the Sounding Ranging Trigger frame.

9.3.1.22a.4 Report subvariant

The format of the User Info field in the Report Ranging Trigger frame is defined in Figure 9-64o and the subfields of the User Info field are identical to the corresponding subfields in the Poll Ranging Trigger frame.

The Trigger Dependent User Info subfield is not present in the Report Ranging Trigger frame.

9.3.1.22a.5 Passive sounding subvariant

The Passive Sounding Ranging Trigger frame follows the definition of the Sounding Ranging Trigger frame except that the RA field is always set to the broadcast address. There is a single User Info field in a Passive Sounding Ranging Trigger frame.

9.3.3 (PV0) Management frames

9.3.3.2 Beacon frame format

Insert a new row for order 92 in Table 9-32 after the row for order 91 (as inserted by IEEE Std 802.11ba-2021) as follows (unchanged rows not shown):

Table 9-32—Beacon frame body

Order	Information	Notes
...		
92	RSTA Availability Window	The RSTA Availability Window element is optionally present if dot11PassiveTBRangingResponderImplemented is true and a passive TB ranging availability window is present.
...		

9.3.3.11 Authentication frame format

Insert a new row for order 25 in Table 9-40 after the row for order 24 as follows (unchanged rows not shown):

Table 9-40—Authentication frame body

Order	Information	Notes
...		
25	PASN Parameters	A PASN element is present only in certain Authentication frames as defined in Table 9-41.
...		

Insert three new rows at the end of Table 9-41 as follows (unchanged rows not shown):

Table 9-41—Presence of fields and elements in Authentication frames

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements from order 4 onward
...			
PASN Authentication	1	Reserved	RSNE is present. RSNXX is present if any subfield of the Extended RSN Capabilities field in this element, except the Field Length subfield, is nonzero. PASN Parameters element is present. Timeout Interval element may be present. Wrapped Data element is present if wrapped data format in PASN Parameters element is nonzero and not reserved. Fragment element may be present if any of the elements are fragmented.
PASN Authentication	2	Status	RSNE is present and PASN Parameters element is present if Status Code field is 0. RSNXX is present if any subfield of the Extended RSN Capabilities field in this element, except the Field Length subfield, is nonzero. Timeout Interval element may be present. Wrapped data element is present if wrapped data format in PASN Parameters element is nonzero and not reserved and Status Code field is 0. MIC element is present. Fragment element may be present if any of the elements are fragmented and Status Code field is 0.

Table 9-41—Presence of fields and elements in Authentication frames (continued)

Authentication algorithm	Authentication transaction sequence number	Status code	Presence of fields and elements from order 4 onward
PASN Authentication	3	Status	PASN Parameters element is present if Status Code field is 0. Wrapped data element is present if wrapped data format in PASN Parameters element is nonzero and not reserved; and Status Code field is 0. MIC element is present. Fragment element may be present if any of the elements are fragmented and Status Code field is 0.

9.4 Management and Extension frame body components

9.4.1 Fields that are not elements

9.4.1.1 Authentication Algorithm Number field

Change the list of Authentication algorithm numbers after the first paragraph of 9.4.1.1 as follows:

Authentication algorithm number = 0: Open System
 Authentication algorithm number = 1: Shared Key
 Authentication algorithm number = 2: Fast BSS Transition
 Authentication algorithm number = 3: Simultaneous Authentication of Equals (SAE)
 Authentication algorithm number = 4: FILS Shared Key authentication without PFS
 Authentication algorithm number = 5: FILS Shared Key authentication with PFS
 Authentication algorithm number = 6: FILS Public Key authentication
Authentication algorithm number = 7: PASN authentication
 Authentication algorithm number = 65 535: vendor specific use

9.4.1.9 Status Code field

Insert four new rows above the last row and change the last row of Table 9-50 as follows (unchanged rows not shown):

Table 9-50—Status codes

Status code	Name	Meaning
...		
130–135		Reserved.
136	INVALID_PUBLIC_KEY	Public key format is invalid.
137	PASN_BASE_AKMP_FAILED	Failure from Base AKMP processing during PASN.
138	OCI_MISMATCH	OCI does not match received channel information.
130–65 535 <u>139–65 535</u>		Reserved.

9.4.1.11 Action field

Insert two new rows above the row for codes 33–125 (as amended by IEEE Std 802.11ba-2021) and change the row for codes 33–125 in Table 9-50 as follows (unchanged rows not shown):

Table 9-51—Category values

Code	Meaning	See subclause	Robust	Group addressed privacy
...				
33	Reserved			
34	Protected Fine Timing frame	9.6.34	Yes	No
33–125 35–125	Reserved			
...				

9.4.2 Elements

9.4.2.1 General

Change the row for element “Fine Timing Measurement Parameters”, insert 12 new rows above the last reserved row, and change the last reserved row in Table 9-92 as follows (unchanged rows not shown):

Table 9-92—Element IDs

Element	Element ID	Element ID Extension	Extensible	Fragmentable
...				
Fine Timing Measurement <u>FTM</u> Parameters (see 9.4.2.167)	206	N/A	Yes <u>Subelements</u>	No
...				
Secure HE-LTF Parameters	255	94	Yes	No
ISTA Passive TB Ranging Measurement Report (see 9.4.2.304)	255	95	Yes	No
RSTA Passive TB Ranging Measurement Report (see 9.4.2.305)	255	96	Yes	No
Passive TB Ranging LCI Table (see 9.4.2.306)	255	97	Yes	No
ISTA Availability Window	255	98	Yes	No
RSTA Availability Window	255	99	Yes	No
PASN Parameters	255	100	Yes	No
Ranging Parameters	255	101	Yes	No
Direction Measurement Results	255	102	Yes	No
Multiple AOD Feedback	255	103	Yes	No
Multiple Best AWV ID	255	104	Yes	No
LOS Likelihood	255	105	Yes	No
Reserved	255	94–255 <u>106–255</u>		
...				

9.4.2.20 Measurement Request element

9.4.2.20.1 General

Insert a new row above the reserved row and change the reserved row in Table 9-98 as follows (unchanged rows not shown):

Table 9-98—Measurement type definitions for measurement requests

Name	Measurement type
...	
Neighboring DMG APs	17
Reserved	17–254 18–254
...	

9.4.2.21 Measurement Report element

9.4.2.21.10 LCI report (Location configuration information report)

Change Table 9-134 as follows:

Table 9-134—Subelement IDs for LCI Report

Subelement ID	Name	Extensible
0	LCI	No
1	Azimuth Report	Yes
2	Originator Requesting STA MAC Address	No
3	Target MAC Address	No
4	Z	Subelements
5	Relative Location Error	Yes
6	Usage Rules/Policy	Yes
7	Co-Located BSSID List	Yes
8	Relative Compact LCI	No
9	<u>Antenna Placement And Calibration</u>	<u>Yes</u>
8–220 10–220	Reserved	
221	Vendor Specific	Vendor defined
222–255	Reserved	

Insert the following text and Figure 9-256a through Figure 9-256d at the end of 9.4.2.21.10:

The Relative Compact LCI field is formatted as shown in Figure 9-256a.

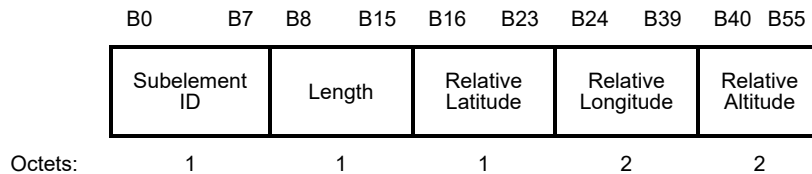


Figure 9-256a—Relative Compact LCI field format

The Relative Compact LCI field is included in the Passive TB Ranging LCI Table Report element; see 9.4.2.306 that contains an RSTA LCI Report field. The reported location of the RSTA serves as the reference location for the reporting of the Relative Latitude, Longitude, and Altitude subfields in the Relative Compact LCI field.

The Relative Latitude subfield contains a signed integer in 2s complement format indicating the latitude offset of the reported location in relation to the reference location, in units of 1.8×10^{-7} deg. (Corresponds to approximately 2 cm at the equator.)

The Relative Longitude subfield contains a signed integer in 2s complement format indicating the longitude offset of the reported location in relation to the reference location, in units of 1.8×10^{-7} deg. (Corresponds to approximately 2 cm at the equator.)

The Relative Altitude subfield contains a signed integer in 2s complement format indicating the elevation offset of the reported location in relation to the reference location, in units of 2 cm.

The Antenna Placement And Calibration subelement contains the antenna placement and the delay information for each of the transmit antennas. The format of the Antenna Placement And Calibration subelement is shown in Figure 9-256b.

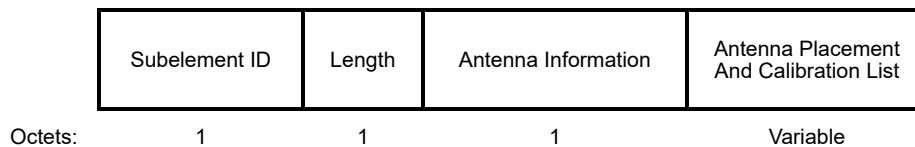


Figure 9-256b—Antenna Placement And Calibration subelement format

The Subelement ID field is equal to the value for Antenna Placement And Calibration in Table 9-134.

The Length field is defined in 9.4.3.

The Antenna Information field is formatted as shown in Figure 9-256c, where the Number of Selected Antennas subfield indicates the total number of the antennas selected for transmission minus one. The total number of the antennas selected for transmission is denoted as N_{Tx_sel} , as shown in Figure 9-256c and Figure 9-256d.

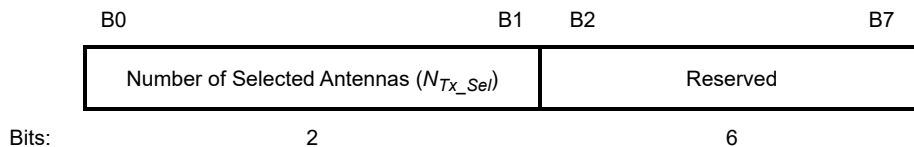
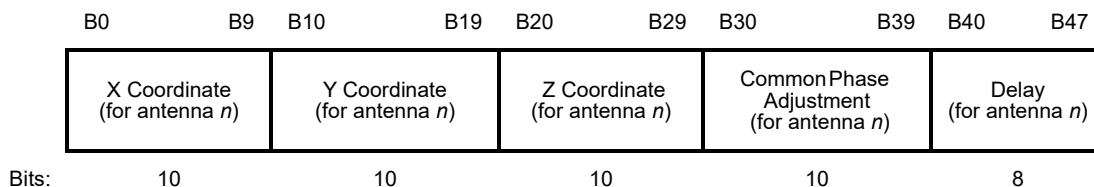


Figure 9-256c—Antenna Information field format

The Antenna Placement And Calibration List field is composed of N_{Tx_Sel} Antenna Placement And Calibration fields, where N_{Tx_Sel} is greater than or equal to one. The format of the Antenna Placement And Calibration field is shown in Figure 9-256d. The order of the antenna ports indicated in the Antenna Placement And Calibration List field matches the order of the rows in the P matrix applied to the space time streams spatial mapped to the antenna ports.



Where “(for antenna n)” represents the n th antenna in the range 1 to N_{Tx_Sel}

Figure 9-256d—Antenna Placement And Calibration field format

The X Coordinate field, Y Coordinate field, and Z Coordinate field contain the placement information of each antenna, where the origin of the axis that is defined by the Latitude, Longitude, and Altitude field values of the LCI subelement, the X axis points to the East, the Y axis points to the North, and the Z axis points up and is perpendicular to the plane formed by the X axis and the Y axis.

The values in the X, Y, and Z Coordinate field range from 0 to 1023, mapping to -512 to 511 , in units of millimeters.

The value in the Common Phase Adjustment subfield indicates the additional phase rotation introduced in the RF chain that is common for all subcarriers. The value for this subfield, denoted k , ranges from 0 to 1023, and indicates a phase adjustment of $k \times \pi / 512$.

9.4.2.24 RSNE

9.4.2.24.3 AKM suites

Insert a new row above the last reserved row and change the last reserved row in Table 9-151 as follows (unchanged rows not shown):

Table 9-151—AKM suite selectors

OUI	Suite type	Meaning			Authentication algorithm numbers (see 9.4.1.1)
		Authentication type	Key management type	Key derivation type	
...					
00-0F-AC	21	PASN-1	PASN	PASN key management defined in 12.13	Defined in 12.13.3
00-0F-AC	21 22 255	Reserved	Reserved	Reserved	Reserved
...					

9.4.2.26 Extended Capabilities element

Insert eight new rows above the last row and change the last row in Table 9-153 as follows (unchanged rows not shown):

Table 9-153—Extended Capabilities field

Bit	Information	Notes
...		
90	Non-TB Ranging Responder	A STA sets the Non-TB Ranging Responder field to 1 if dot11NonTriggeredBasedRangingRespImplemented is true. Otherwise the STA sets the Non-TB Ranging Responder field to 0; see 11.21.6.
91	TB Ranging Responder	A STA sets the TB Ranging Responder field to 1 if dot11TriggerBasedRangingRespImplemented is true. Otherwise the STA sets the TB Ranging Responder field to 0; see 11.21.6.
92	Passive TB Ranging Responder Measurement Support	A STA sets the Passive TB Ranging Responder Measurement Support field to 1 when dot11PassiveTBRangingResponderImplemented is true. Otherwise, the STA sets Passive TB Ranging Responder Measurement Support field to 0. See 11.21.6.4.8.
93	Passive TB Ranging Initiator Measurement Support	A STA sets the Passive TB Ranging Initiator Measurement Support field to 1 when dot11PassiveTBRangingInitiatorImplemented is true. Otherwise, the STA sets the Passive TB Ranging Initiator Measurement Support field to 0. See 11.21.6.4.8.

Table 9-153—Extended Capabilities field (*continued*)

Bit	Information	Notes
94	AOA Measurements Available	A STA sets the AOA Measurement Available field to 1 when dot11AOAMeasurementImplemented is true. Otherwise, the STA sets the AOA Measurement Available field to 0.
95	Phase Shift Feedback Support	A STA sets the Phase Shift Feedback Support field to 1 or 0 as specified in 11.21.6.2. It indicates the LMR transmitted by the STA can carry Phase Shift Feedback.
96	DMG/Location Supporting APs In The Area	An AP STA sets this field to 1 to indicate that APs providing location services using DMG are in the vicinity of the AP STA. Definition of vicinity is implementation dependent. The AP may be capable of providing information about those DMG APs.
97	I2R LMR Feedback Policy	A STA sets the I2R LMR Feedback Policy field to 1 if dot11I2RLMRFeedbackPolicy is true. Otherwise the STA sets the I2R LMR Feedback Policy field to 0. See 11.21.6.3.3. The I2R LMR Feedback Policy field indicates the policy of the STA in the role of an RSTA, and it is reserved for a STA in the role of an ISTA.
90–# 98–n	Reserved	

9.4.2.36 Neighbor Report element

Replace Figure 9-337 (as amended by IEEE Std 802.11ax-2021) with the following figure (containing an additional column for B21):

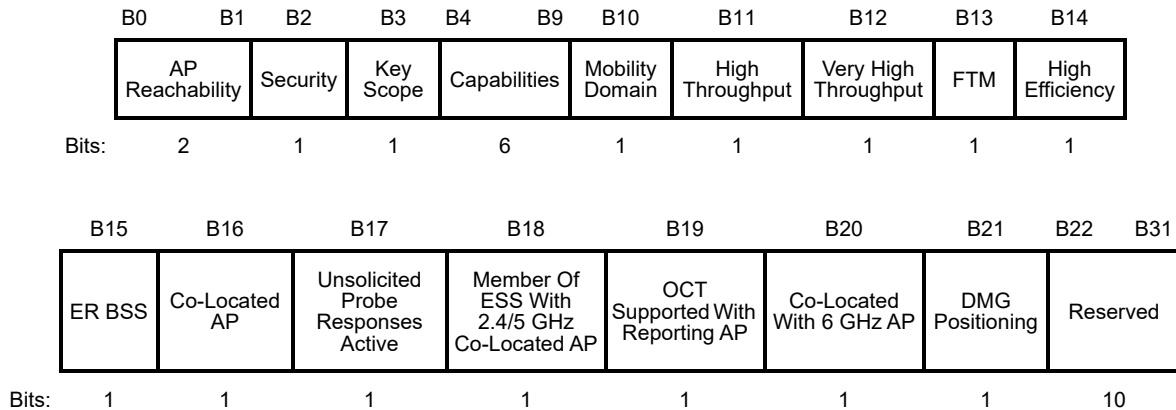


Figure 9-337—BSSID Information field format

Insert a new paragraph after the paragraph beginning “The Co-Located With 6 GHz AP subfield is ...” (inserted by IEEE Std 802.11ax-2021) in 9.4.2.36 as follows:

The DMG Positioning field indicates that the AP supports DMG positioning.

9.4.2.118 MIC element

Change the first paragraph of 9.4.2.118 as follows:

The MIC element provides message integrity to mesh peering Management frames. The format of the MIC element is shown in Figure 9-535.

Replace Figure 9-535 with the following figure (with MIC field length changed to “Variable”):

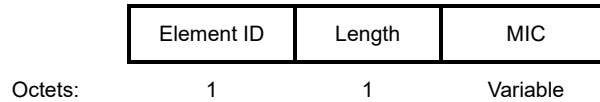


Figure 9-535—MIC element format

Change the third paragraph of 9.4.2.118 as follows:

~~The MIC field contains a message integrity code calculated over~~ In other management frames, the mesh peering Management frame (as specified in 14.5) and the mesh group key handshake frame (as specified in 14.6), the length of the MIC is 16 octets.

In other frames, the MIC field is of variable length, depending on the context that specifies the computation of the MIC.

9.4.2.127 DMG Capabilities element

9.4.2.127.1 General

Replace Figure 9-549 (as modified by IEEE Std 802.11ay-2021) with the following figure (containing two additional columns on the right):

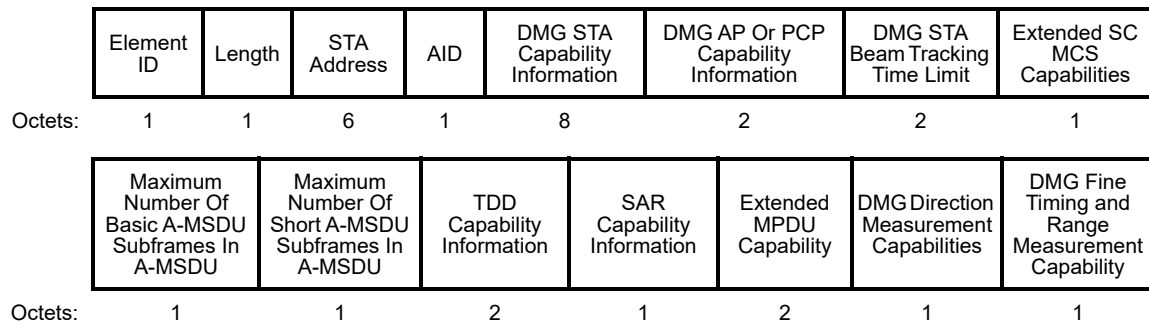


Figure 9-549—DMG Capabilities element format

Insert 9.4.2.127.10 and 9.4.2.127.11 after 9.4.2.127.9 (as inserted by IEEE Std 802.11ay-2021) as follows:

9.4.2.127.10 DMG Direction Measurement Capabilities field

The DMG Direction Measurement Capabilities field, shown in Figure 9-554e, advertises capabilities for performing direction measurement as part of DMG or EDMG exchanges.

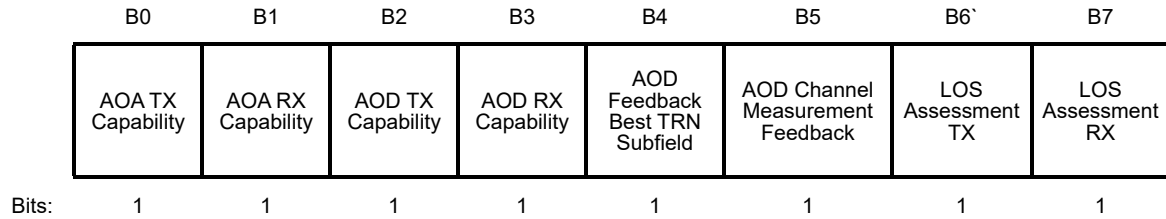


Figure 9-554e—DMG Direction Measurement Capabilities field format

A DMG STA sets the AOA TX Capability subfield to 1 to indicate the ability to attach a TRN field to an FTM/Ack frame for the purpose of allowing the receiver of that frame to perform angle of arrival (AOA) estimation.

A DMG STA sets the AOA RX Capability subfield to 1 to indicate the ability to estimate the AOA based on a TRN field attached to an FTM/Ack frame.

A DMG STA sets the AOD TX Capability subfield to 1 to indicate the ability to attach a TRN field, possibly with different antenna settings to different TRN subfields, to an FTM/Ack frame, for the purpose of allowing the responder to estimate the angle of departure (AOD) of the PPDU.

A DMG STA sets the AOD RX Capability subfield to 1 to indicate the ability to estimate the AOD based on a TRN field attached to an FTM or Ack frame and send a report.

A DMG STA sets the AOD Feedback Best TRN subfield to 1 to indicate the ability to send a best TRN subfield index, based on measurement on a TRN field sent by the receiver STA, for the purpose of AOD estimation.

A DMG STA sets the AOD Channel Measurement Feedback subfield to 1 to indicate the ability to send a Channel Measurement Feedback element based on measurement on a TRN field sent by the peer RSTA, for the purpose of AOD estimation.

The LOS Assessment TX subfield is set to 1 to indicate that the STA can participate as an RSTA in a LOS Assessment exchange by transmitting a LOS Assessment FTM PPDU; see 11.21.6.4.2.5.

The LOS Assessment RX subfield is set to 1 to indicate that the STA can participate in a LOS Assessment exchange as an RSTA by switching polarization at receiving an Ack with TRN field and responding with a Channel Measurement Feedback element in the next FTM frame to the ISTA.

9.4.2.127.11 DMG Fine Timing and Range Measurement Capability Information field

The DMG Fine Timing and Range Measurement Capability Information field, shown in Figure 9-554f, advertises capabilities for performing FTMs and Range Measurements as part of DMG or EDMG exchanges.

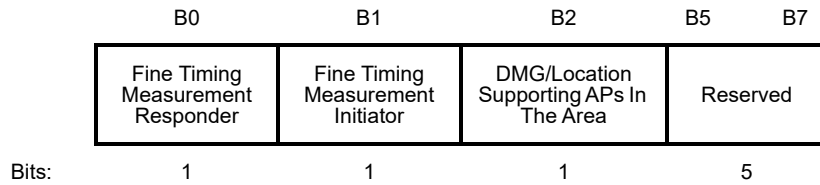


Figure 9-554f—DMG Fine Timing And Range Measurement Capability Information field format

The Fine Timing Measurement Responder field has the same meaning as the Fine Timing Measurement Responder field in the Extended Capabilities element (see 9.4.2.26). The Fine Timing Measurement Initiator field has the same meaning as the Fine Timing Measurement Initiator field in the Extended Capabilities element (see 9.4.2.26).

A DMG STA in the role of AP sets the DMG/Location Supporting APs In The Area field to 1 to indicate that APs providing location services using DMG are in the vicinity of the AP STA and sets it to 0 otherwise. Definition of vicinity is implementation dependent. The AP may be capable of providing information about those DMG APs. The subfield is reserved if the DMG STA is not in the role of an AP.

Change the title of 9.4.2.167 as follows:

9.4.2.167 ~~Fine Timing Measurement~~ FTM Parameters element

Replace Figure 9-624 with the following figure (containing an additional column on the right) and change the figure title:

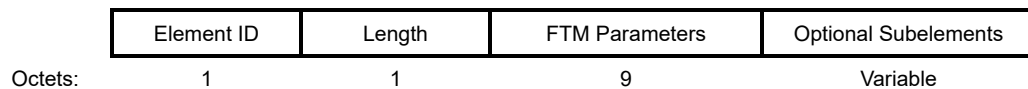


Figure 9-624—~~Fine Timing Measurement~~ FTM Parameters element format

Replace Figure 9-625 with the following figure (with changes to B7, B48, and B49) and change the figure title:

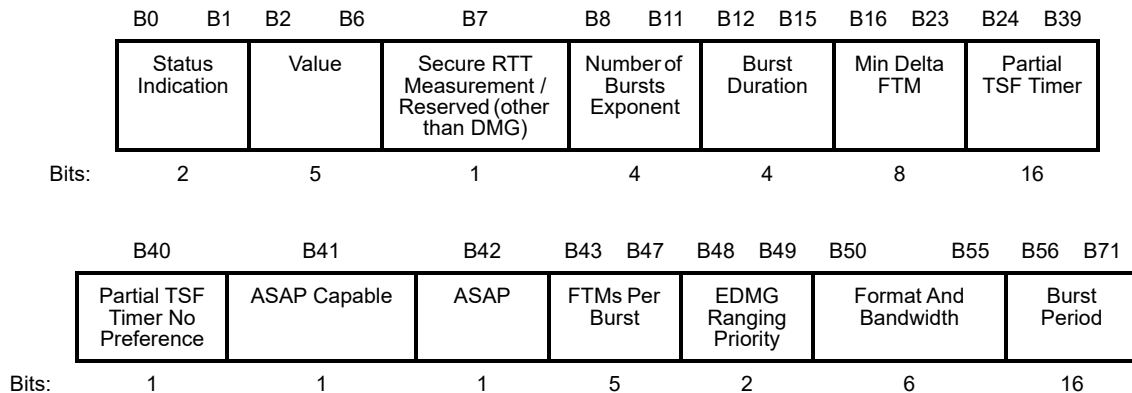


Figure 9-625—~~Fine Timing Measurement~~ FTM Parameters field format

***Insert a new paragraph after the sixth paragraph of 9.4.2.167 (beginning “The Status Indication subfield and Value subfield...”)* as follows:**

For secure EDMG ranging, the Secure RTT Measurement subfield is set to 1 by an ISTA to request a secure RTT measurement exchange between an ISTA and an RSTA; see 11.21.6.3.3. The Secure RTT Measurement subfield is set to 1 by an RSTA to acknowledge a secure RTT measurement exchange, and to 0 otherwise. In cases other than secure EDMG ranging, the Secure RTT Measurement subfield is reserved.

Insert the following text, Table 9-279a, and Table 279b after the now 20th paragraph of 9.4.2.167 (beginning “The FTMs Per Burst subfield indicates...”):

For EDCA based ranging where the value of the Format And Bandwidth field indicates DMG or EDMG format, see Table 9-280, the EDMG Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element in the IFTMR frame contains the ISTA’s ranging priority request, which indicates the time sensitivity of a ranging operation, and it is set according to Table 9-279a. In cases other than EDMG ranging, EDMG Ranging Priority subfield is reserved.

Table 9-279a—Definition of the EDMG Ranging Priority subfield when included in the IFTMR frame

Ranging Priority Request Value	Description	Definition
0	No priority	The ISTA does not know the time sensitivity of the ranging operation.
1	High	The results of the ranging operation is intended for emergency services (e.g., 911 calls)
2	Medium	The completion of the ranging operation impacts user experiences.
3	Low	The completion of the ranging operation is less sensitive to application performance.

For EDCA based ranging where the value of the Format And Bandwidth field indicates DMG or EDMG format, see Table 9-280, the EDMG Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element in the IFTM frame contains the RSTA’s ranging priority response, which indicates whether the RSTA accommodates the ranging priority request of the ISTA, and it is set according to Table 9-279b.

Table 9-279b—Definition of the EDMG Ranging Priority subfield when included in the IFTM frame

Ranging Priority Response Value	Definition
0	The RSTA will be able to prioritize ranging operations to match the ranging priority requested by the ISTA.
1	The RSTA will not be able to prioritize ranging operations to match the ranging priority requested by the ISTA
2	Reserved
3	Reserved

For EDCA based ranging where the value of the Format And Bandwidth field does not indicate DMG or EDMG format, see Table 9-280, the EDMG Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element in the IFTM frame is reserved.

Insert 6 new rows above the row for field value “17–30”, insert 12 new rows above the row for field value “32–63”, and change the rows for field values “17–30” and “32–63” in Table 9-280 as follows (unchanged rows not shown):

Table 9-280—Format And Bandwidth subfield

Field value	Format	Bandwidth (MHz)
...		
17	EDCA based HE	20
18	EDCA based HE	40
19	EDCA based HE	80
20	EDCA based HE	80+80
21	EDCA based HE (two separate RF LOs)	160
22	EDCA based HE (single RF LO)	160
17–30 <u>23–30</u>	Reserved	Reserved
...		
32	EDMG (Single Carrier Mode)	2160
33	EDMG (Single Carrier Mode)	4320
34	EDMG (Single Carrier Mode)	6480
35	EDMG (Single Carrier Mode)	8640
36	EDMG (Single Carrier Mode)	2160+2160
37	EDMG (Single Carrier Mode)	4320+4320
38	EDMG (OFDM)	2160
39	EDMG (OFDM)	4320
40	EDMG (OFDM)	6480
41	EDMG (OFDM)	8640
42	EDMG (OFDM)	2160+2160
43	EDMG (OFDM)	4320+4320
32–63 <u>44–63</u>	Reserved	Reserved

Insert the following text, Table 9-280a, Table 9-280b, and Figure 9-626a through Figure 9-626e at the end of 9.4.2.167:

The Optional Subelements field contains zero or more subelements. The subelement format and ordering of subelements are defined in 9.4.3. The Subelement ID field values for the defined subelements are shown in Table 9-280a.

Table 9-280a—Optional Subelement IDs for FTM Parameters

Subelement ID	Name	Extensible
0	DMG Direction Measurement Parameters	Yes
1	EDMG Specific Parameters	Yes
2–220	Reserved	
221	Vendor Specific	
222–255	Reserved	

The DMG Direction Measurement Parameters subelement has the structure shown in Figure 9-626a.

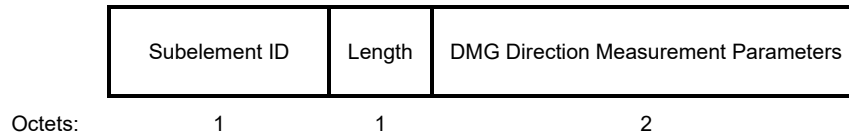


Figure 9-626a—DMG Direction Measurement Parameters subelement format

The format of the DMG Direction Measurement Parameters field is shown in Figure 9-626b.

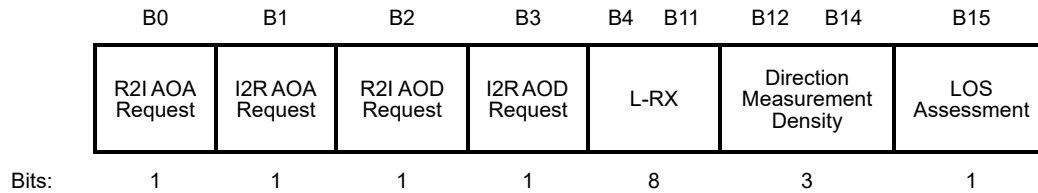


Figure 9-626b—DMG Direction Measurement Parameters field format

A value of 1 in the R2I AOA Request subfield indicates a request for RSTA to ISTA angle of arrival measurement performed at the ISTA.

A value of 1 in the I2R AOA Request subfield indicates a request for ISTA to RSTA angle of arrival measurement performed at the RSTA.

A value of 1 in the R2I AOD Request subfield indicates a request for RSTA to ISTA angle of departure measurement performed at the ISTA.

A value of 1 in the I2R AOD Request subfield indicates a request for ISTA to RSTA angle of departure measurement performed at the RSTA.

The L-RX field indicates how many TRN-units are requested by the sender for angle of arrival measurements. The interpretation of this field is defined in 9.5.4. If either the initiator or the responder is a non-EDMG STA, the value of this field is limited to 32.

The Direction Measurement Density field indicates how often in a burst a direction measurement is performed. The interpretation of the values is defined in Table 9-280b.

Table 9-280b—Direction Measurement Density field interpretation

Value	Interpretation
0	Only the first FTM measurement exchange in a burst is a direction measurement
1	Every measurement in a burst is a direction measurement
2	Every second measurement in a burst is a direction measurement
3	Every 4th measurement in a burst is a direction measurement
4	Every 8th measurement in a burst is a direction measurement
5–7	Reserved

The LOS Assessment field is set to 1 in a DMG Direction Measurement Parameters field sent by an ISTA in an IFTMR frame to indicate a request for LOS assessment in the FTM session (see 11.26.6.4.8).

The EDMG Specific Parameters subelement contains a number of fields that are used to advertise the requested or allocated operation configurations from one EDMG STA to another. The EDMG Specific Parameters subelement is included in the IFTMR frame, as described in 9.6.7.32, and the IFTM frame, as described in 9.6.7.33, if the Secure RTT Measurement subfield of the FTM Parameters field of these frames is equal to 1. The use of the EDMG Specific Parameters subelement is described in 11.21.6.

The format of the EDMG Specific Parameters subelement is shown in Figure 9-626c.

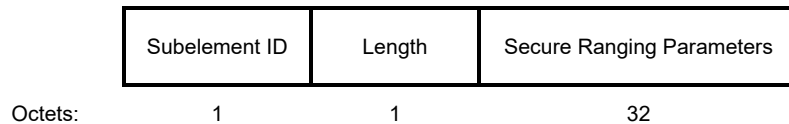


Figure 9-626c—EDMG Specific Parameters subelement format

The format of the Secure Ranging Operation Parameters field is shown in Figure 9-626d.

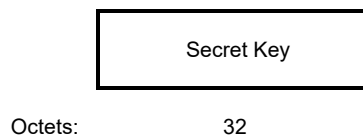


Figure 9-626d—Secure Ranging Operation Parameters field format

The Secret Key subfield is used to carry the secret key, which is used along with Salt value (based on the PMKID), to generate the random sequence(s) as described in 12.2.11.

Change the title and first paragraph of 9.4.2.187 as follows:

9.4.2.187 FILS-Wrapped Data element

The ~~FILS~~-Wrapped Data element is used for the STA and AP to communicate data used by the ~~FILS authentication algorithm~~ RSNA protocols. The format of the ~~FILS~~-Wrapped Data element is defined in Figure 9-664.

Replace Figure 9-664 with the following figure (with “FILS” removed from the figure and the figure title):

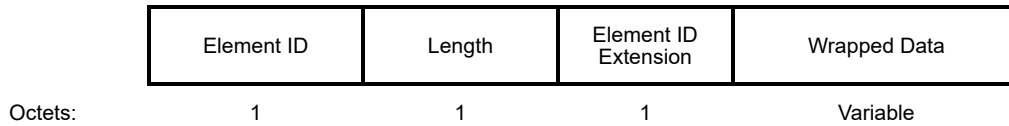


Figure 9-664—Wrapped Data element format

Change the third paragraph of 9.4.2.187 as follows:

The ~~FILS~~-Wrapped Data field is the data used by the FILS authentication algorithm (see 12.11) and PASN authentication algorithm (see 12.13).

Please note that instances of “FILS Wrapped Data” will be changed to “Wrapped Data” as appropriate in the next revision of IEEE Std 802.11.

9.4.2.241 RSN Extension element (RSNXE)

Change Table 9-321 (as amended by IEEE Std 802.11ay-2021, IEEE Std 802.11ba-2021, and IEEE Std 802.11-2020/Cor 1-2022) as follows:

Table 9-321—Extended RSN Capabilities field

Bit	Information	Notes
0–3	Field length	The length of the Extended RSN Capabilities field, in octets, minus 1, i.e., $n - 1$.
4	Protected TWT Operations Support	The STA sets the Protected TWT Operations Support field to 1 when <code>dot11ProtectedTWTOperationsImplemented</code> is true, and sets it to 0 otherwise. See 10.47.1.
5	SAE hash-to-element	The STA supports directly hashing to obtain the PWE instead of looping. See 12.4.4.2.3 and 12.4.4.3.3.
6	Reserved	Used by a specific organization. This reserved bit is used by the Wi-Fi Alliance®: https://www.wi-fi.org . This information is given for the convenience of the user of this standard and does not constitute an endorsement by the IEEE.
7	Protected WUR Frame Support	The STA sets the Protected WUR Frame Support field to 1 when <code>dot11RSNAWURFrameProtectionActivated</code> is true, and sets it to 0 otherwise.
8–10 8	Reserved <u>Secure HE-LTF Support</u>	<u>A STA sets the Secure HE-LTF Support field to 1 when <code>dot11SecureLTFImplemented</code> is true. Otherwise, the STA sets the Secure HE-LTF Support field to 0. See 11.21.6.4.5.</u>
9	<u>Secure RTT Supported</u>	<u>A STA sets the Secure RTT Supported field to 1 if it supports Secure RTT Measurement exchange as defined in 11.21.6.4.2.7. Otherwise, the field is set to 0.</u>
10	<u>URNM-MFPR-X20</u>	<u>A STA sets the URNM-MFPR-X20 field to 1 if <code>dot11RSTARequiresPMFActivated</code> is set to 1. Otherwise, it sets the field to 0. See 11.21.6.3.1 and C.3.</u>

Table 9-321—Extended RSN Capabilities field (*continued*)

Bit	Information	Notes
11	Protected Announce Support	The non-EDMG STA sets the Protected Announce Support field to 1 when dot11ProtectedAnnounceImplemented is true, and sets it to 0 otherwise. See 12.6.20.
<u>12–14</u>	<u>Reserved</u>	
<u>15</u>	<u>URNM-MFPR</u>	<u>A STA sets the URNM-MFPR field to 1 if dot11RSTARequiresPMFActivated is set to 2. Otherwise, it sets the field to 0. See 11.21.6.3.1.</u>
<u>12–(8 × n – 1)</u> <u>16–(8 × n – 1)</u>	Reserved	

9.4.2.265 EDMG Capabilities element

9.4.2.265.2 Beamforming Capability subelement

Replace Figure 9-788aq (as inserted by IEEE Std 802.11ay-2021) with the following figure (with changes to B10 and B20 through B31):

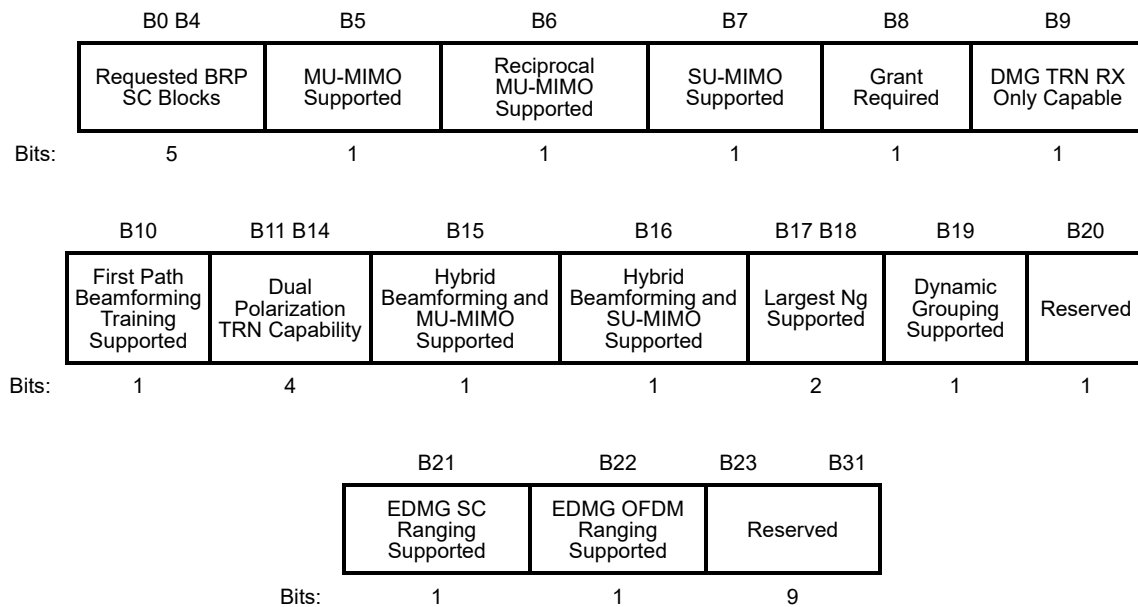


Figure 9-788aq—Data field of the Beamforming Capability subelement format

Insert new text at the end of 9.4.2.265.2 (as inserted by IEEE Std 802.11ay-2021) as follows:

The EDMG SC Ranging Supported subfield is set to 1 to indicate that the EDMG STA is capable of performing range measurement based on FTM using EDMG SC PPDU. This subfield is set to 0 otherwise.

The EDMG OFDM Ranging Supported subfield is set to 1 to indicate that the EDMG STA is capable of performing range measurement based on FTM using EDMG OFDM PPDU. This subfield is set to 0 otherwise. This field is reserved if the Maximum OFDM MCS subfield of the Supported MCS field is set to 0.

Insert 9.4.2.296 through 9.4.2.307 (including Figure 9-788ee through Figure 9-788fl and Table 9-322al through Table 9-322ap) after 9.4.2.295 (as inserted by IEEE Std 802.11ba-2021) as follows:

9.4.2.296 ISTA Availability Window element

The format of the ISTA Availability Window element is shown in Figure 9-788ee.

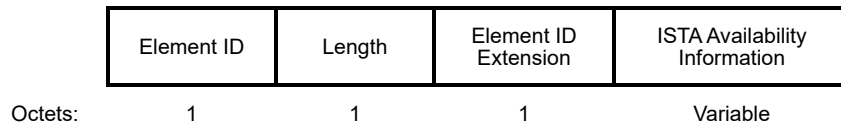


Figure 9-788ee—ISTA Availability Window element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The ISTA Availability Information field format is shown in Figure 9-788ef.

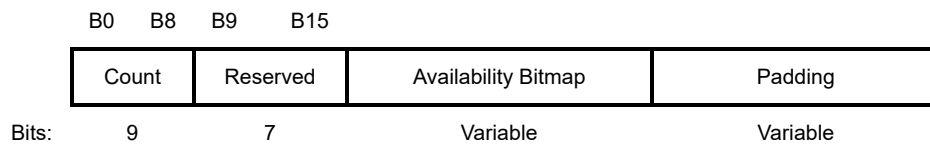


Figure 9-788ef—ISTA Availability Information field format

The Count subfield in the ISTA Availability Information field indicates the size in bits of the Availability Bitmap subfield. The value of this subfield is denoted as “count”.

Each Availability Bit in the Availability Bitmap subfield indicates the ISTA’s availability for TB ranging with the recipient RSTA. The value indicated by each bit in the Availability Bitmap subfield is in units of 10 TUs. Bit B_k (where $0 \leq k \leq \text{count} - 1$) represents the ISTA’s periodic availability for TB ranging with the RSTA in the interval $[t_{start,k}, t_{end,k}]$ repeated every N TUs; see Equation (9-3f):

$$\begin{aligned}
 t_{start,k} &= t_{start,0} + 10k \text{ TU} \\
 t_{end,k} &= t_{start,0} + 10(k+1) \text{ TU} \\
 t_{start,0} &= \text{time 0 per RSTA's TSF} \\
 N &= 10 \times \text{count}
 \end{aligned}
 \tag{9-3f}$$

A value of 1 in an Availability Bit indicates the ISTA’s availability at time $t_{start,k}$ for a duration of 10 TUs, while a value of 0 indicates the ISTA’s unavailability at time $t_{start,k}$ for a duration of 10 TUs.

The Padding subfield may be present in order to render the length of the Availability Bitmap subfield to be a multiple of eight. The value of the bits in the Padding field is reserved.

9.4.2.297 RSTA Availability Window element

The format of the RSTA Availability Window element is shown in Figure 9-788eg.

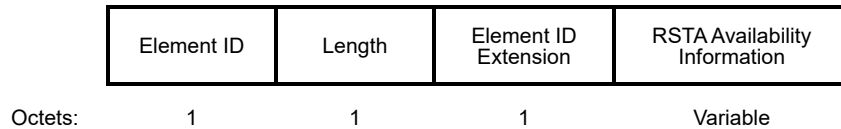


Figure 9-788eg—RSTA Availability Window element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The RSTA Availability Information field format is shown in Figure 9-788eh.

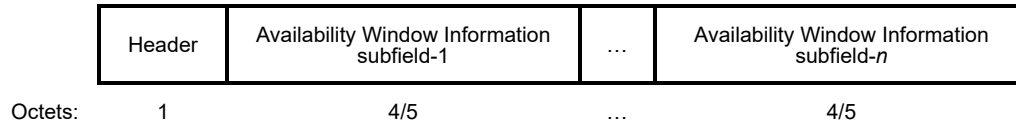


Figure 9-788eh—RSTA Availability Information field format

The Header subfield format in the RSTA Availability Information field is shown in Figure 9-788ei.

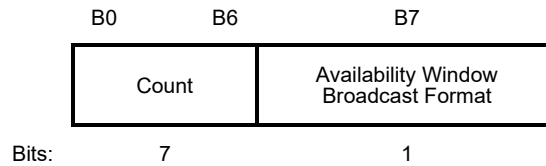


Figure 9-788ei—Header subfield format

The Count subfield in the Header subfield indicates the number of Availability Window Information subfields included in the RSTA Availability information field.

The Availability Window Broadcast Format subfield indicates the format of the Availability Window Information subfields.

When the Availability Window Broadcast Format subfield is set to 0, the format of each Availability Window Information subfield is shown in Figure 9-788ej.

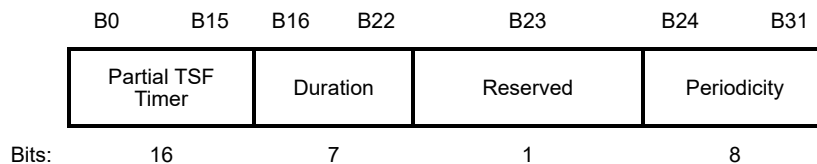


Figure 9-788ej—Availability Window Information field format when the Availability Window Broadcast Format subfield is set to 0

The Partial TSF Timer subfield is derived as described in 9.4.2.167 and indicates the TSF timer of the RSTA at the start of first availability window.

The Duration subfield in the Availability Window Information field indicates the duration of the corresponding availability window in units of 100 μ s (giving it a value from 0 to $\sim$ 12.7 ms). The value 0 is Reserved.

The Periodicity subfield in an Availability Window Information subfield indicates the periodicity of that availability window in units of 100 TU (Giving it a value from 0 to ~26.1 s).

When the Availability Window Broadcast Format subfield is set to one, the format of each Availability Window Information subfield is shown in Figure 9-788ek.



Figure 9-788ek—Availability Window Information field format when the Availability Window Broadcast Format subfield is set to 1

The format of the Partial TSF Timer, Duration, and Periodicity subfields when the Availability Window Broadcast Format subfield is set to 1 are the same as the formats of the subfields when the Availability Window Broadcast Format subfield is set to 0. The Passive TB Ranging Parameters subfield format is shown in Figure 9-788el.

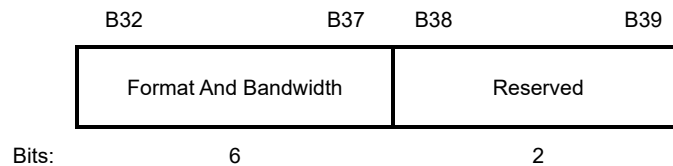


Figure 9-788el—Passive TB Ranging Parameters subfield format

The Format And Bandwidth subfield is defined in Table 9-322a1, and indicates the requested or allocated PPDU format and nominal bandwidth used to transmit the I2R/R2I NDP exchange as part of the passive TB ranging measurement exchange in the passive TB ranging availability window. Depending on the medium availability, the bandwidth used for the exchanged frames is equal to or smaller than the nominal bandwidth.

9.4.2.298 Ranging Parameters element

The Ranging Parameters element is optionally included in the IFTMR frame, as described in 9.6.7.32, and the IFTM frame, as described in 9.6.7.33. The use of the Ranging Parameters element is described in 11.21.6.

The format of the Ranging Parameters element is shown in Figure 9-788em.

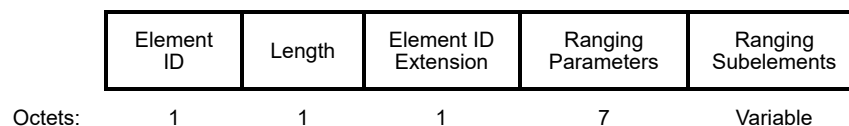


Figure 9-788em—Ranging Parameters element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The format of the Ranging Parameters field is shown in Figure 9-788en.

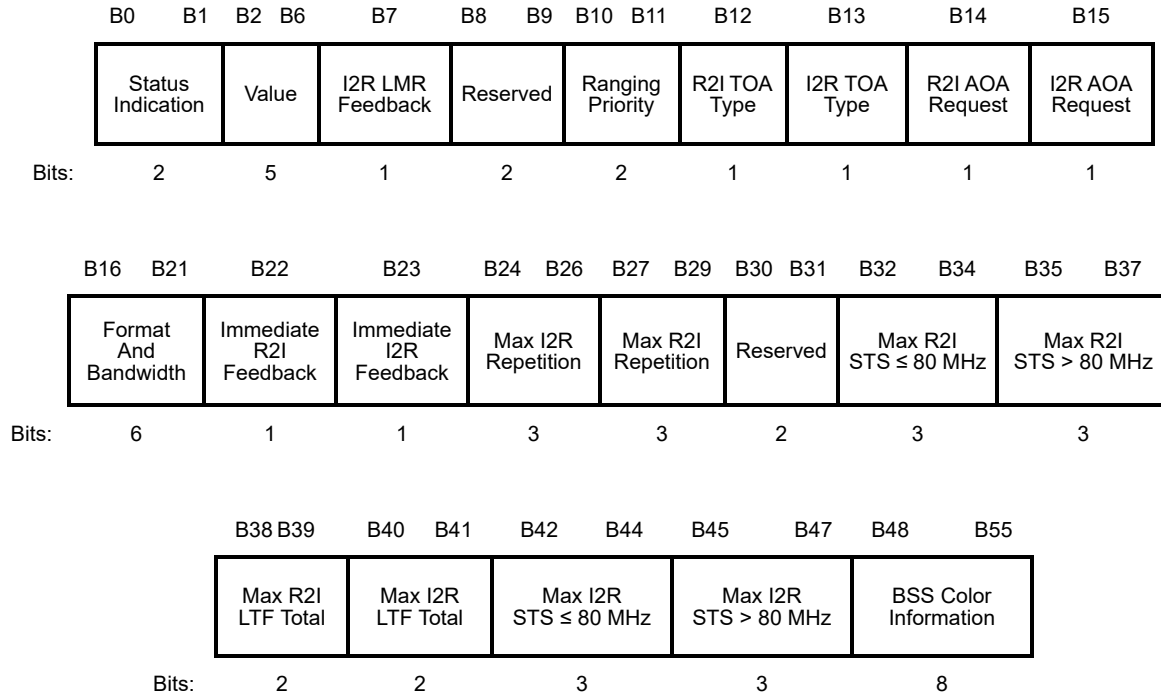


Figure 9-788en—Ranging Parameters field format

The Status Indication subfield indicates the RSTA's response to the Fine Timing Request. The encoding of the Status Indication subfield is shown in Table 9-278.

The Status Indication subfield and Value subfield are reserved in the IFTMR frame. When the Status Indication subfield is set to 3 by the RSTA, the Value subfield contains a duration in units of seconds; otherwise the Value subfield is reserved.

The I2R LMR Feedback subfield in the Ranging Parameters element in the IFTMR frame is set to indicate whether it transmits I2R LMR feedback to the RSTA, if requested by the RSTA; (see 11.21.6.3.3).

The RSTA sets the I2R LMR Feedback subfield in the Ranging Parameters element in the IFTM frame to 1 to indicate it requests the ISTA to transmit I2R LMR feedback to the RSTA and sets it to 0 otherwise (see 11.21.6.3.3, 11.21.6.4.3.4, and 11.21.6.4.4.3).

The R2I TOA Type subfield is set to 1 in the IFTMR frame by the ISTA to indicate that it requests phase shift TOA feedback in the R2I LMR and is set to 0 otherwise. The R2I TOA Type subfield is set to 1 in the IFTM frame by the RSTA to indicate that it reports phase shift TOA feedback in the R2I LMR and is set to 0 otherwise; see 11.21.6.3.3.

The I2R TOA Type subfield is set to 1 in the IFTMR frame by the ISTA to indicate that it supports phase shift TOA feedback in the I2R LMR and is set to 0 otherwise. The I2R TOA type subfield is set to 1 in the IFTM frame by the RSTA to assign phase shift TOA feedback in the I2R LMR and is set to 0 otherwise; see 11.21.6.3.3.

The R2I AOA Requested subfield is set to 1 in the IFTMR frame by the ISTA when it requests the RSTA to include AOA measurements in the AOA Feedback field of the R2I LMR. The R2I AOA Requested subfield is set to 1 in the IFTM frame to indicate that the RSTA includes the AOA measurements in the R2I LMR.

The I2R AOA Requested subfield in the IFTMR frame is set to 1 to indicate that the ISTA supports AOA measurement feedback in the I2R LMR. The I2R AOA Requested subfield in the IFTM frame is set to 1 to request the ISTA to include the AOA measurement feedback in the I2R LMR and is set to 0 otherwise.

The Format And Bandwidth subfield indicates the requested or allocated PPDU format and bandwidth used to transmit the I2R/R2I NDP exchange as part of the non-TB ranging, or TB ranging measurement exchange; see 11.21.6.4.3 and 11.21.6.4.4. The encoding of this subfield is given in Table 9-322a.

Table 9-322a—Format And Bandwidth subfield

Field value	Format	Bandwidth
0	HE	20
1	HE	40
2	HE	80
3	HE	80+80
4	HE (two separate RF LOs)	160
5	HE (single RF LO)	160
6–63	Reserved	Reserved

The values of 3, 4, and 5 indicate STA support for 160 MHz operation as either 80+80, 160 two-LO, or 160 single-LO, respectively in addition to supporting 80 MHz, 40 MHz, and 20 MHz bandwidths (e.g., a field value of 5 does not mean the device supports all 160 MHz options but rather 160 MHz single LO).

The Max I2R Repetition subfield indicates the maximum number of HE-LTF repetitions that the ISTA uses in the preamble of the I2R NDP and is set to the number of HE-LTF repetitions minus 1.

The Max R2I Repetition subfield indicates the maximum number of HE-LTF repetitions that the RSTA uses in the preamble of the R2I NDP and is set to the number of HE-LTF repetitions minus 1.

NOTE—The values of 0 to 7 contained in the Max I2R Repetition and Max R2I Repetition subfields are mapped to the values 1 to 8 for the number of HE-LTF repetitions, respectively.

The Ranging Priority subfield of the Ranging Parameters field of the Ranging Parameters element in the IFTMR frame contains the ISTA's ranging priority request, which indicates the time sensitivity of a ranging operation, and it is set according to Table 9-279a.

For TB ranging, the Ranging Priority subfield of the Ranging Parameters field of the Ranging Parameters element in the IFTM frame contains the RSTA's ranging priority response, which indicates whether the RSTA accommodates the ranging priority request of the ISTA, and it is set according to Table 9-279b in 9.4.2.167.

For non-TB ranging, the Ranging Priority subfield of the Ranging Parameters field of the Ranging Parameters element in the IFTM frame is reserved.

The Immediate R2I Feedback and Immediate I2R Feedback subfields indicate if the measurement results from the current measurement exchange are reported immediately by setting it to 1 (from the current measurement) or delayed by setting it to 0 (from the previous measurement) in the R2I and I2R LMR frame respectively.

The Immediate R2I Feedback subfield is reserved in the IFTMR frame. In the IFTM frame the Immediate R2I Feedback field is set to 1 to indicate that the R2I LMR will be immediate feedback and to 0 to indicate delayed feedback.

The Immediate I2R Feedback field in the IFTMR frame is set to 1 to indicate immediate feedback in the I2R LMR and is set to 0 to indicate delayed feedback. In the IFTM frame the Immediate I2R Feedback field is set to the same value as in the IFTMR frame.

The Max R2I STS $\leq$ 80 MHz subfield indicates for bandwidths less than or equal to 80 MHz the maximum number of space-time streams to be used in the R2I NDP in the session.

The Max R2I STS $>$ 80 MHz subfield indicates for bandwidths greater than 80 MHz the maximum number of space-time streams to be used in the R2I NDP in the session.

The BSS Color Information subfield has the same format as in the BSS Color Information field in the HE Operation element. Each subfield of the BSS Color Information field is set to the same value, as in the HE Operation element transmitted by an RSTA. The BSS Color Information subfield is reserved in the IFTMR frame.

The Max R2I LTF Total and Max I2R LTF Total subfields indicate the maximum number of HE-LTFs to be destined to an ISTA in the R2I NDP and an RSTA in an I2R NDP, respectively, the encoding is given in Table 9-322am. The maximum number of HE-LTFs limits the allowed combinations of number of space-time streams and HE-LTF repetitions.

Table 9-322am—Max R2I/I2R LTF Total subfields

Field value	Max number of HE-LTFs
0	4
1	8
2	16
3	64

The Max I2R STS $\leq$ 80 MHz subfield indicates for bandwidths less than or equal to 80 MHz the maximum number of space-time streams to be used in the I2R NDP in the session.

The Max I2R STS $>$ 80 MHz subfield indicates for bandwidths greater than 80 MHz the maximum number of space-time streams to be used in the I2R NDP in the session.

The Ranging Subelements field contains one or more subelements. The format and ordering of the subelements are defined in 9.4.3. The Subelement ID field values for the defined subelements are shown in Table 9-322an.

Table 9-322an—Ranging Subelement IDs for Ranging Parameters

Subelement ID field	Name	Extensible
0	Non-TB Specific subelement	Yes
1	TB-specific subelement	Yes
2	Secure HE-LTF subelement	Yes
3–220	Reserved	
221	Vendor Specific	
222–255	Reserved	

If the initiator and the responder successfully negotiate an FTM session, and the negotiated ranging protocol is non-TB, then the Non-TB Specific subelement is included in both the IFTMR frame, to describe the requested set of parameters that the initiator proposes to use, and in the IFTM frame.

The format of the Non-TB Specific subelement is as shown in Figure 9-788eo.

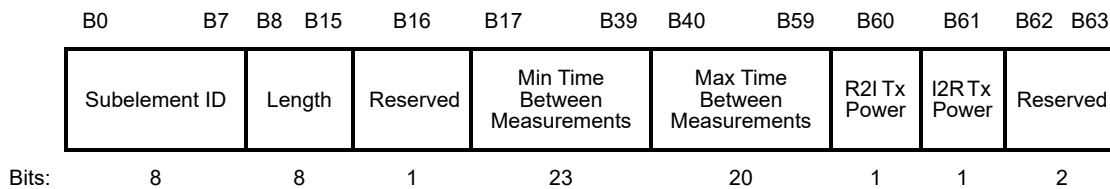


Figure 9-788eo—Non-TB Specific subelement format

The Subelement ID and Length fields are defined in 9.4.3.

The R2I Tx Power field in the IFTMR frame is set to 1 to indicate that the ISTA requests the RSTA to report the TX power of the R2I NDPs in the R2I NPD Tx Power field in the following LMR, and 0 otherwise. In the IFTM frame, the R2I Tx Power field is set to 1 by the RSTA to indicate that it will report the TX power of the R2I NDPs in the R2I NPD Tx Power field of the following LMR, and 0 otherwise.

The I2R Tx Power field in the IFTMR frame is set to 1 to indicate that the ISTA supports announcing the TX power of the I2R NDPs in the STA Info field with the AID11 subfield set to 2045 of the preceding NDP Announcement frame, and 0 otherwise. In the IFTM frame, the I2R Tx Power field is set to 1 by the RSTA to request the ISTA to announce the TX power of the I2R NDPs in the STA Info field with the AID11 subfield set to 2045 of the preceding NDP Announcement frame, and 0 otherwise.

The Min Time Between Measurements field is requested by the ISTA in the IFTMR frame and assigned by RSTA in the IFTM frame, and indicates the minimum time between two consecutive range measurements initiated by an ISTA, in units of 100 μ s.

The Max Time Between Measurements field is requested by the ISTA in the IFTMR frame and assigned by RSTA in the IFTM frame, and indicates the latest time by which the ISTA needs to complete the next round of measurements, in units of 10 ms.

If the initiator and the responder successfully negotiate an FTM session where the negotiated ranging protocol is TB, then the TB Specific subelement is included in both the IFTMR frame, to describe the requested set of parameters that the initiator proposes to use, and in the IFTM frame.

The format of the TB Specific subelement is as shown in Figure 9-788ep.

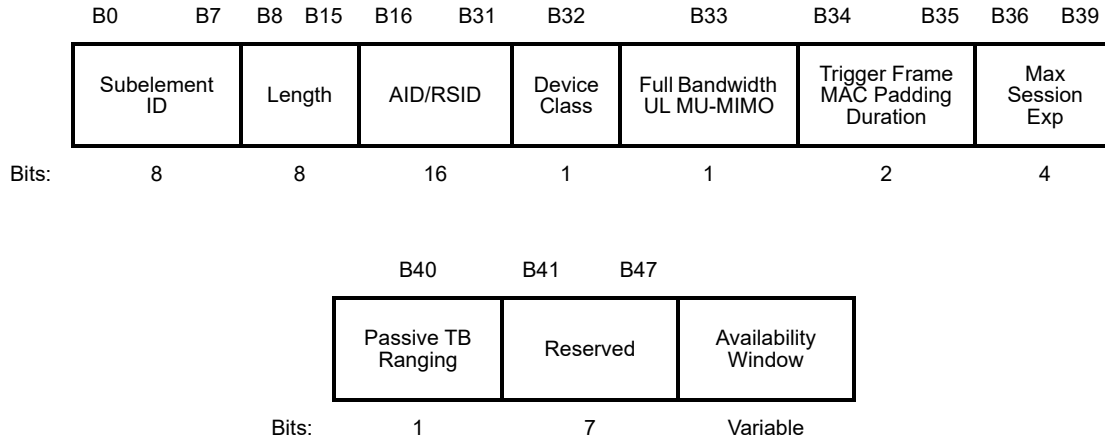


Figure 9-788ep—TB Specific subelement format

The Subelement ID and Length fields are defined in 9.4.3.

The AID/RSID field is reserved in the IFTMR frame. In the IFTM frame it contains an identifier for the ISTA for the duration of the FTM session. If the ISTA is associated with the RSTA, the value is set to the ISTA's AID. If the ISTA is not associated with the RSTA, the AID/RSID field is set to the RSID, which is assigned by the responder to identify the unassociated ISTA, and has the same length as the AID.

The Device Class and Full Bandwidth UL MU-MIMO fields correspond to the Device Class and Full Bandwidth UL MU-MIMO fields defined in Table 9-321b. For associated STAs, the values of the Device Class and Full Bandwidth UL MU-MIMO subfields are equal to the values of the Device Class and Full Bandwidth UL MU-MIMO fields, respectively, that are exchanged during association.

The Trigger Frame MAC Padding Duration field correspond to the Trigger Frame MAC Padding Duration field defined in Table 9-322a. For associated STAs, the value of the Trigger Frame MAC Padding Duration field is equal to the value that was exchanged during association.

The Max Session Exp field indicates the time before which a new measurement exchange between the ISTA and RSTA should be initiated and completed. This value is computed as $2^{(\text{Max Session Exp} + 8)}$ ms. The Max Session Exp field is reserved in an IFTMR frame.

The Passive TB Ranging field is set to 1 by the initiator to request passive TB ranging operation, otherwise it is set to 0. The Passive TB Ranging field is set to 1 by the Responder to grant passive TB ranging operation, otherwise it is set to 0.

The Availability Window field may contain either an ISTA Availability Window element, see Figure 9-788ee, when the containing Ranging Parameters element is in an IFTMR frame; or an RSTA Availability Window element, see Figure 9-788eg, when the containing Ranging Parameters element is in an FTM frame.

Figure 9-788eq, Figure 9-788er, and Figure 9-788es together show an example of how an RSTA assigns an availability window from the received Availability Window element of the ISTA. Figure 9-788eq shows the bitmap in the ISTA Availability Information field of the ISTA Availability Window element included with an IFTMR frame. The bitmap has periodicity of 200 TUs and the RSTA Beacon Interval is 100 TUs. Figure 9-788er shows how the RSTA calculates ISTA's periodic availability from this bitmap relative to

RSTA TSF. Finally, Figure 9-788es shows how the RSTA constructs an availability window with periodicity of 200 TUs requested by the ISTA and with a window duration of 10 TUs. The shaded region indicates the location of the assigned availability windows.

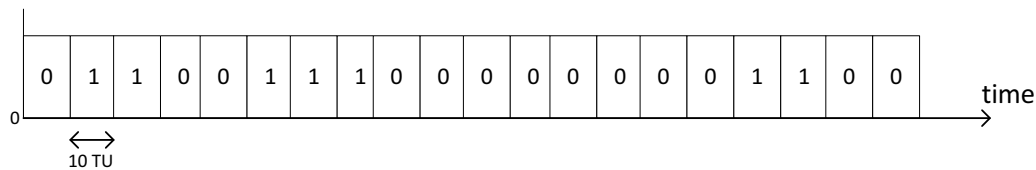


Figure 9-788eq—Example of a bitmap with 200 TU periodicity signaled in the ISTA Availability Window element

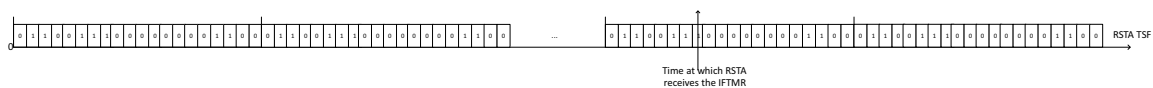


Figure 9-788er—Example of mapping of ISTA's availability bitmap to RSTA's TSF

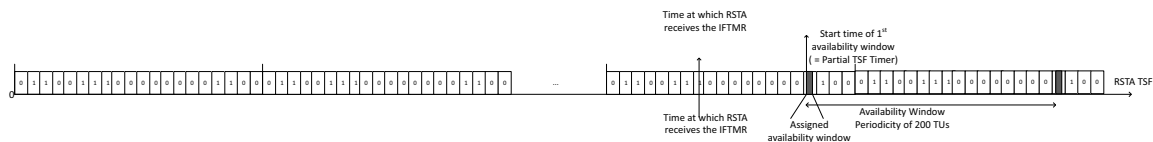


Figure 9-788es—Example of how an RSTA assigns an Availability Window to an ISTA

The Ranging Parameters element in the IFTMR frame includes a Non-TB Specific subelement and/or a TB specific subelement; and the Ranging Parameters element in the corresponding IFTM frame includes either a Non-TB Specific subelement or a TB specific subelement, if the Status Indication subfield in the Ranging Parameters field is set to Success to indicate the range measurement protocol selected by the responder for the negotiated FTM session.

The Secure HE-LTF subelement is included in the IFTMR frame to indicate that the initiator supports use of secure HE-LTF and the associated parameters; it is included in the IFTM, if the initiator and the responder successfully negotiate an FTM session where secure HE-LTF is used.

The format of the Secure HE-LTF subelement is as shown in Figure 9-788et.

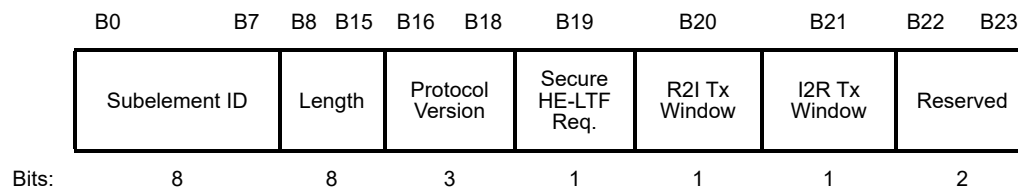


Figure 9-788et—Secure HE-LTF subelement format

The Subelement ID and Length fields are defined in 9.4.3.

The Protocol Version field in the IFTMR frame is set to the value 0 by the ISTA, with values 1 to 7 reserved for future use; see Table 9-322ao. In the IFTM frame the Version field is set to the value 0 by the RSTA, with values 1 to 7 reserved for future use. The interpretation of the version field in the IFTMR frame and IFTM frame, and the possible resulting actions, are described in 11.21.6.3.4.

The Secure HE-LTF Required field is set to 1 by the ISTA to indicate it requires secure HE-LTF to be enabled and is set to 1 by the RSTA to enable a secure HE-LTF measurement exchange between an ISTA and an RSTA. Otherwise the Secure HE-LTF Required field is set to 0.

The R2I Tx Window field in the IFTMR frame is set to 1 to indicate the ISTA requests use of the optional frequency domain Tx Window in the R2I NDPs, and 0 to indicate the default frequency domain Tx window. In the IFTM frame, the R2I Tx Window field is set to 1 to indicate the RSTA will use the optional frequency domain Tx window in the R2I NDPs, and 0 to indicate the default frequency domain Tx window; see Table 9-322ao.

The I2R Tx Window field in the IFTMR frame is set to 1 to indicate that the ISTA supports use of the optional frequency domain Tx window in the I2R NDPs, and 0 to indicate the default frequency domain Tx window. In the IFTM frame, the I2R Tx Window field is set to 1 by the RSTA to request that the ISTA use the optional frequency domain Tx window in the I2R NDPs, and 0 to indicate the default frequency domain Tx window; see Table 9-322ao.

Table 9-322ao—Secure HE-LTF protocol section identifier

Protocol Version	Secure HE-LTF PHY sections
0	27.3.18a
1–7	Reserved

9.4.2.299 Secure HE-LTF Parameters element

The Secure HE-LTF Parameters element is optionally included in the IFTM frame, as described in 9.6.7.33, and the LMR frame, as described in 9.6.7.49 for a secure HE-LTF measurement exchange mode of the non-TB ranging and TB ranging measurement exchange (see 11.21.6.4.5). The use of the Secure HE-LTF Parameters element is described in 11.21.6.

The format of the Secure HE-LTF Parameters element is shown in Figure 9-788eu.

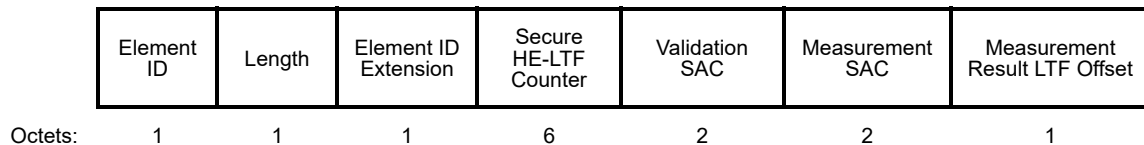


Figure 9-788eu—Secure HE-LTF Parameters element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The Secure HE-LTF Counter field is used to determine the randomized LTF sequence of an I2R NDP and R2I NDP in one of the following secure HE-LTF measurement exchange sequences:

- An I2R NDP and an R2I NDP immediately following a Ranging NDP Announcement frame, in a non-TB ranging measurement exchange (11.21.6.4.4).
- A sequence of an I2R NDP, a Ranging NDP Announcement frame, an R2I NDP, in a TB ranging measurement exchange (11.21.6.4.3).

The Secure HE-LTF Counter field is used in the IFTM frame and the R2I LMR frame when part of a secure HE-LTF measurement exchange; it is reserved otherwise.

The Validation SAC field is a nonzero value associated with the secure HE-LTF counter carried in the same Secure HE-LTF Parameters element and validates the randomized LTF sequence; see 11.21.6.3.4. This field is used in the initial protected FTM frame, the R2I protected LMR frame and is reserved otherwise.

The Measurement SAC field is used to verify that range measurement results contained in the LMR frame are calculated using the same LTF sequence between ISTA and RSTA. The Measurement SAC field is the same value as the SAC subfield in the STA Info field with AID equal to 2043 in the Ranging NDP Announcement frame that solicited the I2R NDP and the R2I NDP in the non-TB ranging measurement exchange or the SAC subfield in the Trigger Dependent User Info of the Secure Sounding Ranging Trigger frame in the TB ranging measurement exchange; see 11.21.6.4.5. This field is reserved in the IFTM frame.

The Measurement Results LTF Offset field is used in TB ranging to verify that the measurement results contained in the LMR frame are calculated using the same LTF Offset for the R2I NDP between RSTA and ISTA. The Measurement Result LTF Offset field has the same value as the LTF Offset subfield in the corresponding STA Info field of the Ranging NDP Announcement frame preceding the R2I NDP, which was used for estimating the measurement results contained in the LMR frame. The Measurement Result LTF Offset field is structured as shown in Figure 9-788ev. This field is not used in the IFTM frame nor in the LMR frames in non-TB ranging.

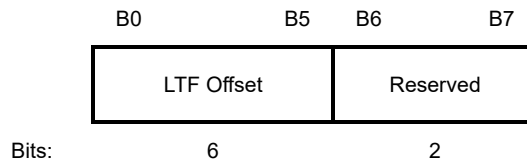


Figure 9-788ev—Measurement Result LTF Offset field format

9.4.2.300 Direction Measurement Results element

The Direction Measurement Results element is used to send angle of arrival (AOA) and angle of departure (AOD) measurement results and is defined in Figure 9-788ew.

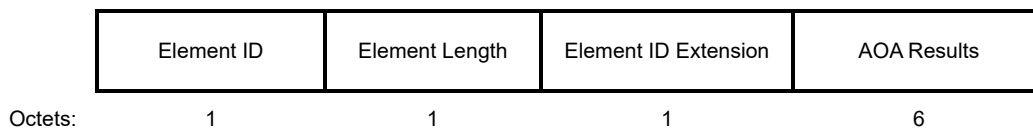


Figure 9-788ew—Direction Measurement Results element

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The AOA Results field is as shown in Figure 9-788ex.

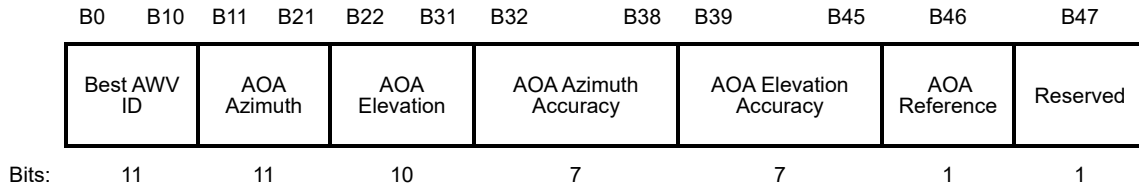


Figure 9-788ex—AOA Results field format

The AOA Azimuth subfield contains the AOA azimuth result in 360°/2048 resolution. This subfield is an unsigned 2s complement number taking values between 0 and 2047. When the AOA Reference subfield is set to 1, the AOA Azimuth subfield is in earth coordinates (i.e., direction 0 is north). When the AOA Reference subfield is set to 0, the AOA Azimuth subfield is in coordinates relative to the orientation of the STA measuring the AOA azimuth.

The AOA Elevation subfield contains the AOA elevation result in 180°/1024 resolution. This subfield is a signed 2s complement number taking values between –512 and 511.

The AOA Azimuth Accuracy subfield contains the AOA azimuth result’s estimated accuracy in 360°/2048 resolution. An accuracy larger than $125 \times 360^\circ/2048$ is represented by the value of 125. A value of 126 indicates no azimuth measurement. A value of 127 indicates no ability to estimate azimuth accuracy.

The AOA Elevation Accuracy subfield contains the AOA elevation result’s estimated accuracy in 360°/2048 resolution. An accuracy larger than $125 \times 360^\circ/2048$ is represented by the value 125. A value of 126 indicates no elevation measurement. A value of 127 indicates no ability to estimate elevation accuracy.

When the AOA Reference subfield is set to 1, the AOA Elevation subfield is in earth coordinates (i.e., elevation 0 is the horizon). When the AOA Reference subfield is set to 0, the AOA elevation is in coordinates relative to the orientation of the STA measuring the AOA elevation.

If the STA sending the frame containing the element is a DMG STA, and the frame containing the Direction Measurement Result element follows an ISTA PPDU that enabled AOD by containing TRN-T subfields, the Best AWV ID subfield contains the index of the TRN-T subfield that was received with the highest SNR. If the ISTA PPDU that enabled AOD contained EDMG TRN-T subfields, then the AWV ID field contains the AWV ID (see 28.9.2.2.5) of the TRN subfields that were received with the highest SNR. If the STA sending the frame containing the element is a non-DMG STA, the Best AWV ID subfield is reserved.

9.4.2.301 Multiple Best AWV ID element

The Multiple Best AWV ID element is used to send multiple TRN subfields indices/AWV IDs from an ISTA to an RSTA and is defined in Figure 9-788ey. The Multiple Best AWV ID element is carried in an FTM frame.

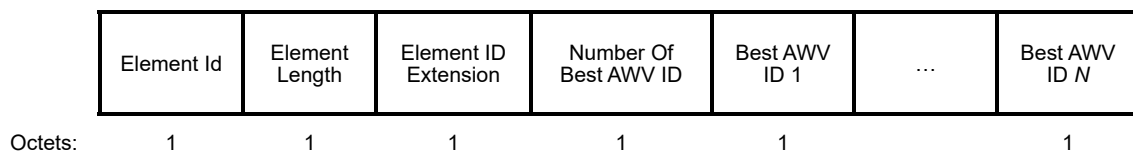


Figure 9-788ey—Multiple Best AWV ID element

The Number Of Best AWV ID field indicates the number of attached Best AWV ID fields. Each Best AWV ID field is 16 bits long. Bits B0–B4 are reserved. It indicates the index of either the TRN subfield index or the AWV ID of the TRN subfield(s) received with highest SNR. The choice of whether TRN subfield index

or AWV ID is used depends on whether the PPDU over which the TRN subfield(s) were selected contained a DMG TRN field or an EDMG TRN field.

9.4.2.302 Multiple AOD Feedback element

The Multiple AOD Feedback element is used to send multiple angle of departure (AOD) results from an RSTA to an ISTA and is defined in Figure 9-788ez. The Multiple AOD Feedback element is carried in an FTM frame.

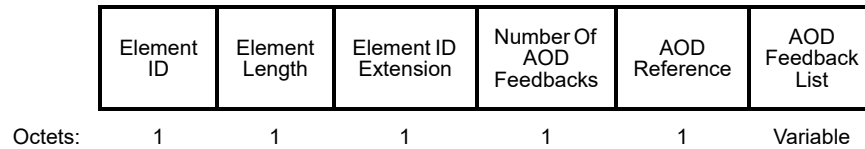


Figure 9-788ez—Multiple AOD Feedback element

The value contained in the Number Of AOD Feedbacks field indicates the number of AOD Feedback fields present in the AOD Feedback List field, and is set to the Number Of Best AWV ID fields present in the last received Multiple Best AWV ID element from the ISTA.

When the AOD Reference field is set to 1, the AOD Elevation subfield is in earth coordinates (i.e., elevation 0 is horizon). When the AOD Reference field is set to 0, the AOD Elevation subfield is in coordinates relative to the orientation of the STA transmitting the measurement signal. When the AOD Reference field is set to 1, the AOD Azimuth subfield is in earth coordinates (i.e., direction 0 is north). When the AOD Reference field is set to 0, the AOD Azimuth subfield is in coordinates relative to the orientation of the STA transmitting the measurement signal.

The AOD Feedback List field is composed of a set of AOD Feedback fields. Each AOD Feedback field has the structure in Figure 9-788fa.

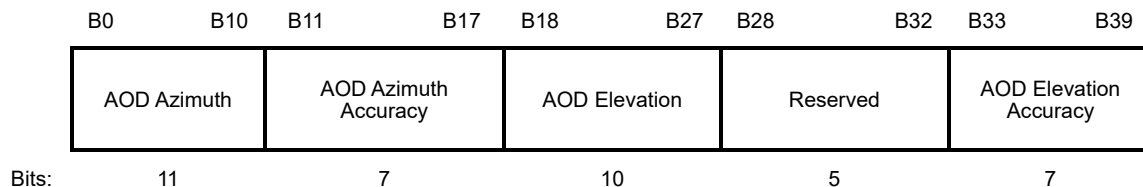


Figure 9-788fa—AOD Feedback field format

The AOD Azimuth subfield contains the angle of departure (AOD) azimuth result in 360°/2048 resolution. This subfield is an unsigned 2s complement number taking values between 0 and 2047.

The AOD Azimuth Accuracy subfield contains the AOD azimuth result's estimated accuracy in 360°/2048 resolution. An accuracy larger than $125 \times 360^\circ/2048$ is represented by the value of 125. A value of 126 indicates no azimuth measurement. A value of 127 indicates no ability to estimate azimuth accuracy.

The AOD Elevation subfield contains the AOD elevation result in 180°/1024 resolution. This subfield is a signed 2s complement number taking values between -512 and 511.

The AOD Elevation Accuracy subfield contains the AOD elevation result's estimated accuracy in 360°/2048 resolution. An accuracy larger than $125 \times 360^\circ/2048$ is represented by the value of 125. A value of 126 indicates no elevation measurement. A value of 127 indicates no ability to estimate elevation accuracy.

9.4.2.303 PASN Parameters element

The Control field, see Figure 9-788fc, indicates the presence or absence of other fields in the PASN Parameters element; see Figure 9-788fb.

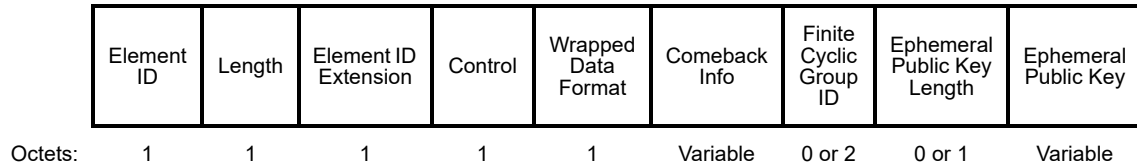


Figure 9-788fb—PASN Parameters element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

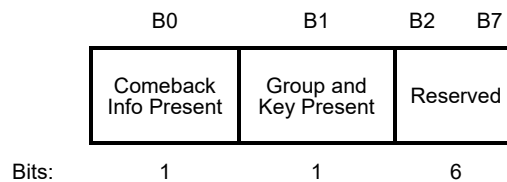


Figure 9-788fc—PASN Parameters element Control Info field format

The Comeback Info Present subfield indicates if the Comeback Info field is included in the PASN Parameters element. The Group and Key Present subfield indicates if the PASN Parameters element includes the Finite Cyclic Group ID, the Ephemeral Public Key Length, and the Ephemeral Public Key fields.

The Wrapped Data Format field indicates the format of data in the Wrapped Data element included along with the PASN Parameters element. The values defined for this format are:

- 0: No wrapped data
- 1: Fast BSS Transition Wrapped Data; see 12.13.6
- 2: FILS Shared Key authentication without PFS Wrapped Data; see 12.13.4
- 3: SAE Wrapped Data; see 12.13.5
- 4–255: Reserved

The Comeback Info field, present only when the corresponding Comeback Info Present subfield in the Control field is set to 1, is of variable length and is formatted as shown in Figure 9-788fd.

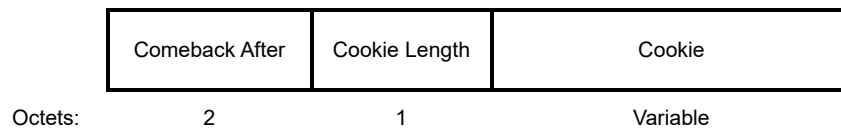


Figure 9-788fd—Comeback Info field format

The Comeback After subfield value is the time in TUs after which the non-AP STA is requested to retry the PASN authentication. The Comeback After subfield is set to 0 to indicate that the operation can be retried with a cookie of nonzero length in the Cookie subfield. The Comeback After subfield is reserved in PASN Authentication frames from a non-AP STA.

The Cookie Length subfield value is the length of the following Cookie subfield. If the Cookie Length subfield is 0, it indicates that there is no Cookie subfield.

The Cookie subfield contains a cookie, which is an opaque sequence of octets generated by an AP STA in an implementation dependent manner; see 12.13.9.

The Finite Cyclic Group ID, Ephemeral Public Key Length, and Ephemeral Public Key fields are present if the Group and Key Present subfield in the Control field is set to 1.

The Finite Cyclic Group ID field indicates the group used in PASN authentication. It has the same semantics as the field Finite Cyclic Group field (9.4.1.42).

The Ephemeral Public Key Length field is set to the length in octets of the Ephemeral Public Key field.

The Ephemeral Public Key field contains the public key encoded using RFC 5480 conventions. Both compressed and uncompressed forms may be used.

9.4.2.304 ISTA Passive TB Ranging Measurement Report element

The ISTA Passive TB Ranging Measurement Report element, defined in Figure 9-788fe, is used to convey measurement results and associated parameters from an ISTA to the RSTA in a passive TB ranging exchange.

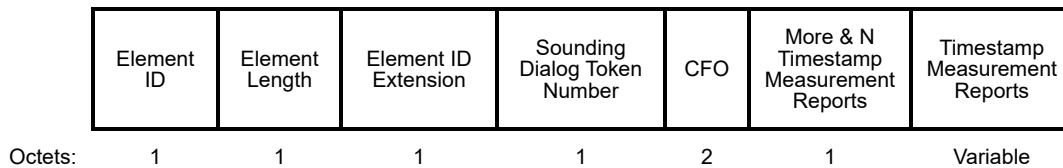


Figure 9-788fe—ISTA Passive TB Ranging Measurement Report element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The value of the Sounding Dialog Token Number field is the value of the Sounding Dialog Token Number subfield in the Ranging NDP Announcement frame of the corresponding measurement sounding phase in which the reported ISTA's timestamps were measured; see 11.21.6.4.3 and 11.21.6.4.8.

The CFO field indicates the reporting ISTA's carrier frequency offset with respect to the RSTA, as a signed integer in 2s complement format in units of 0.01 ppm.

The More & N Timestamp Measurement Reports field is defined as depicted in Figure 9-788ff (More & N Timestamp Measurement Reports field format).

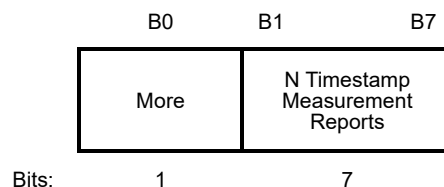


Figure 9-788ff—More & N Timestamp Measurement Reports field format

The More subfield is used to indicate that the ISTA has more timestamps ready to report but is not able to fit them in its allocated resources as given by the RSTA.

The N Timestamp Measurement Reports subfield is an unsigned integer indicating the number of Timestamp Measurement Report subfields present in the Timestamp Measurement Reports field. The value 0 of the N Timestamp Measurement Reports field is reserved.

The Timestamp Measurement Reports field contains one or more Timestamp Measurement Report subfields defined as in Figure 9-788fg.

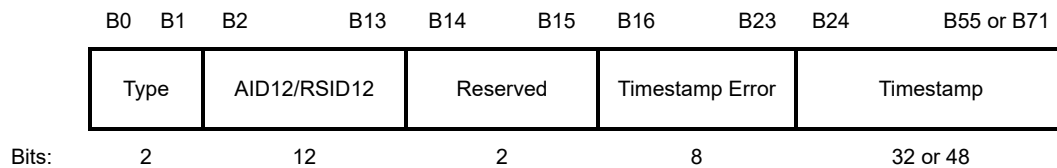


Figure 9-788fg—Timestamp Measurement Report subfield format

The Type subfield is set according to Table 9-322ap.

Table 9-322ap—Type subfield

Field value	Timestamp type	Description
00	TOD	Time of departure timestamp
01	TOA	Time of arrival timestamp
10	PSTOA	Phase shift TOA
11	Reserved	Reserved

The AID12/RSID12 subfield contains the 12 LSBs of the AID, for an associated ISTA, or the 12 LSBs of the RSID, for an unassociated ISTA, of the STA that transmitted the NDP that the timestamp is reported for. When the STA that transmitted the NDP is the RSTA the value zero is reported in the AID12/RSID12 subfield.

The Timestamp Error field is structured as shown in Figure 9-788fh.

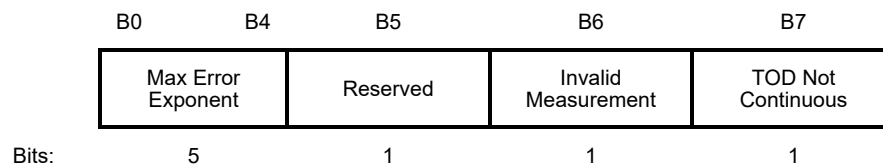


Figure 9-788fh—Timestamp Error subfield format

The maximum errors in the timestamp values are represented using the function defined in Equation (9-3g).

$$E_{max} = \begin{cases} \text{unknown, } F = 0 \\ 2^{F-1}, & 1 \leq F \leq 30 \\ \geq 2^{30}, & F = 31 \end{cases} \quad (9-3g)$$

where

F is the Max Error Exponent

E_{max} is the maximum timestamp error, respectively, in units of picoseconds

The Max Error Exponent subfield contains the Max Error Exponent, F , in Equation (9-3g) for the value specified in the Timestamp subfield.

The Invalid Measurement field is set to 1 to indicate that the timestamp is invalid and set to 0 to indicate that the timestamp is valid.

For a timestamp of type TOD, the TOD Not Continuous subfield indicates that the TOD value is with respect to a different underlying time base than the last transmitted TOD value. It is set to 1 when a discontinuity is present. Otherwise, it is set to 0. For other timestamp types, this field is reserved.

The Timestamp subfield contains a TOD, TOA, or a PSTOA timestamp. The TOD timestamps are represented with 48 bits in units of 1 ps. The TOA and PSTOA timestamps are represented with 32 bits in units of 16 ps.

The TOD timestamp represents the time, with respect to the ISTA's time base, at which the start of the preamble of the NDP in question appeared at the transmit antenna connector.

The TOA timestamp represents the time, with respect to the ISTA's time base, at which the start of preamble of the NDP in question arrived at the receive antenna connector.

The PSTOA timestamp represents the time, with respect to the ISTA's time base, at which the start of preamble of the NDP in question arrived at the receive antenna connector, calculated based on the average linear phase shift between two adjacent tones normalized by the tone spacing. An example of calculation of the phase shift is shown in Annex AD.

9.4.2.305 RSTA Passive TB Ranging Measurement Report element

The RSTA Passive TB Ranging Measurement Report element, defined in Figure 9-788fi, is used to broadcast measurement results and associated parameters from an RSTA to STAs that want to use this information.

Element ID	Length	Element ID Extension	Sounding Dialog Token Number	N Timestamp Measurement Reports	Timestamp Measurement Reports
Octets: 1	1	1	1	1	Variable

Figure 9-788fi—RSTA Passive TB Ranging Measurement Report element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The value of the Sounding Dialog Token Number field is the value of the Sounding Dialog Token Number subfield in the Ranging NDP Announcement frame corresponding to the measurement sounding phase in which the reported RSTA timestamps were measured; see 11.21.6.4.3 and 11.21.6.4.8.

The N Timestamp Measurement Reports field is an unsigned integer indicating the number of Timestamp Measurement Reports. The value 0 of the N Timestamp Measurement Reports field is reserved.

The Timestamp Measurement Reports field contains one or more Timestamp Measurement Report fields defined as in Figure 9-788fg with definitions as detailed in 9.4.2.304.

9.4.2.306 Passive TB Ranging LCI Table element

The Passive TB Ranging LCI Table Report element, defined in Figure 9-788fj, is used by an RSTA to broadcast LCI data for the ISTAs participating in its passive TB ranging exchanges.

	Element ID	Length	Element ID Extension	Number of ISTA LCI Report Entries	ISTA LCI Reports Entries	RSTA LCI Report (optional)	RSTA Location Civic Report (optional)
Octets:	1	1	1	1	Variable	Variable	Variable

Figure 9-788fj—Passive TB Ranging LCI Table Report element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The number of ISTA LCI Reports contained in the ISTA LCI Reports field is indicated by the Number of ISTA LCI Reports ISTA LCI Report field. The format of the ISTA LCI Report field is defined in Figure 9-788fk.

The RSTA LCI Report field is present if the RSTA has never transmitted it before or its content has changed, and it is periodically present otherwise. If present, it contains a Measurement Report element with Measurement Type field equal to LCI, see Table 9-118, which either indicates the LCI of the RSTA and can include the Z subelement, Usage Rules/Policy subelement, and Antenna Placement And Calibration subelement, or indicates an unknown LCI; see 11.21.6.7.

The RSTA Location Civic Report field is optionally present. If present, it contains a Measurement Report element with Measurement Type field equal to Location Civic, see Table 9-118, which either indicates the Civic address of the ISTA or an unknown Civic address; see 11.21.6.7.

ISTA RSID/AID		ISTA MAC ADDRESS		ISTA LCI Report (optional)	ISTA Location Civic Report (optional)
Octets:	2	6	Variable	Variable	

Figure 9-788fk—ISTA LCI Report Entry field format

The ISTA RSID/AID subfield contains the ranging ID of the ISTA reported on.

The ISTA MAC ADDRESS subfield contains the MAC address of the ISTA reported on.

The ISTA LCI Report subfield is present if the RSTA has never transmitted it before or its content has changed and it is periodically present otherwise. If present, it contains a Measurement Report element with Measurement Type field equal to LCI (see Table 9-118), which either indicates the LCI of the ISTA and can include the Z subelement, Usage Rules/Policy subelement, and Antenna Placement And Calibration subelement, or indicates an unknown LCI; see 11.21.6.7.

The ISTA Location Civic Report field is optionally present. If present, it contains a Measurement Report element with Measurement Type field equal to Location Civic (see Table 9-118), which either indicates the Civic address of the ISTA or an unknown Civic address; see 11.21.6.7.

An ISTA LCI Report Entry includes at least one of ISTA LCI Report or ISTA Location Civic Report.

NOTE—To shorten the LCI reporting, the ‘Relative Compact LCI’ Measurement Type can be used in the ISTA’s LCI reporting. In this case the ISTA’s location is reported in relation to the location reported for the RSTA in the Passive TB Ranging LCI Table Report within which the ISTA’s LCI report is contained. See 9.4.2.21.10.

9.4.2.307 LOS Likelihood element

The LOS Likelihood field contains the estimated log likelihood that measurements contained in the Direction Measurement Results element or in the TOD field of the same FTM frame, are line of sight (LOS) measurements. The method to estimate the likelihood is implementation dependent. The format of the LOS Likelihood element is shown in Figure 9-788fl.

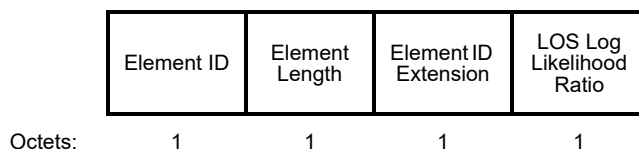


Figure 9-788fl—LOS Likelihood element format

The LOS Log Likelihood Ratio field is a signed 2s complement 8-bit number containing $40 \times \log_{10}$ of the estimated ratio between the probability that measurement is on a LOS path and the probability is on a non-LOS path. This covers the range -31.75 dB to $+31.75$ dB. If this field has a value equal to 255, it indicates that the LOS Log Likelihood Ratio is not available.

9.6 Action frame format details

9.6.6 Radio Measurement action details

9.6.6.6 Neighbor Report Request frame format

Replace Figure 9-864 with the following figure (containing an additional column on the right):

Category	Radio Measurement Action	Dialog Token	SSID (optional)	LCI Measurement Request (optional)	Location Civic Measurement Request (optional)	Neighbor DMG Request (optional)
Octets:	1	1	1	Variable	Variable	Variable

Figure 9-864—Neighbor Report Request frame Action field format

Insert new text at the end of 9.6.6.6 as follows:

The Neighbor DMG Request field is optionally present. If present, it contains a Measurement Request element with Measurement Type field equal to neighboring DMG APs; see Table 9-98. The element indicates a request for Neighbor Report elements containing information about neighboring DMG APs supporting location services. The Enable bit in the Measurement Request Mode field within the Measurement Request element is set to 0.

9.6.7 Public Action details

9.6.7.1 Public Action frames

Insert four new rows above the last row and change the last row of Table 9-364 as follows (unchanged rows not shown):

Table 9-364—Public Action field values

Public Action field value	Description
...	
47	LMR
48	ISTA Passive TB Ranging Measurement Report
49	Primary RSTA Broadcast Passive TB Ranging Measurement Report
50	Secondary RSTA Broadcast Passive TB Ranging Measurement Report
47–255 <u>51–255</u>	Reserved

Change the title of 9.6.7.32 as follows:

9.6.7.32 ~~Fine Timing Measurement~~ **FTM Request frame format**

Replace Figure 9-895 with the following figure (containing three additional columns on the right) and change the figure title:

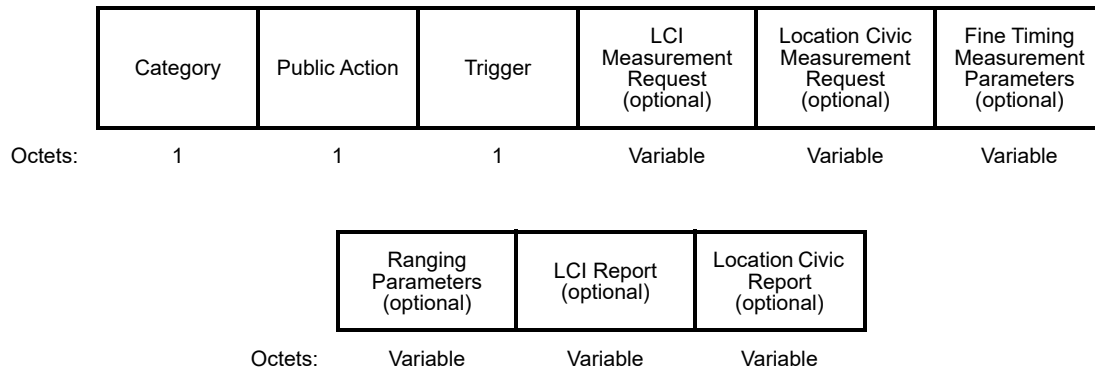


Figure 9-895—~~Fine Timing Measurement~~ FTM Request frame Action field format

Change the fourth (now split into two paragraphs) and seventh paragraphs of 9.6.7.32 as follows:

When the ~~The~~ Trigger field is set to 1, ~~indicates that the initiating STA the ISTA requests that the responding STA RSTA start or continue sending Fine Timing Measurement FTM frames (see 11.21.6).~~
When the ~~The~~ Trigger field set to 0, ~~indicates that the initiating STA the ISTA requests that the responding STA RSTA stop sending Fine Timing Measurement FTM frames.~~

When the Trigger field is set to 2, an EDMG ISTA initiates an FTM measurement exchange using the first path AWV; see 11.21.6.4.2.6. When the Trigger field is set to 3, an EDMG ISTA indicates that the following FTM burst shall contain a LOS assessment measurement. If the FTM burst is performed over the first path AWV and shall contain a LOS assessment measurement, the Trigger field is set to 4. Trigger field values 25–255 are reserved.

~~The Fine Timing Measurement Parameters field is present in the initial Fine Timing Measurement Request frame (see 11.21.6.3) and its retransmissions and is not present in subsequent Fine Timing Measurement Request frames. If present, it contains a Fine Timing Measurement Parameters element as defined in 9.4.2.167. The FTM Parameters field is optionally present. It is present in an IFTMR frame and its retransmissions when negotiating EDCA based ranging; see 11.21.6.3.2.~~

Insert new text at the end of 9.6.7.32 as follows:

The Ranging Parameters field is present in the IFTMR frame, and its retransmissions, when the initiator requires negotiation of parameters with the responder in order to perform non-TB ranging, or TB ranging measurement exchange, see 11.21.6.3. If present, it contains a Ranging Parameters element as defined in 9.4.2.298.

The LCI Report field is optional in an IFTMR frame and is present if the frame contains a TB specific subelement with the Passive TB Ranging field set to 1. The Location Civic Report field is optional in an IFTMR frame and can be present if the frame contains a TB specific subelement with the Passive TB Ranging field set to 1. The LCI Report and Location Civic Report fields are not present in IFTMR frames without a TB specific subelement with the Passive TB Ranging field set to 1. The format of the LCI Report is defined in 9.4.2.21.1 and 9.4.2.21.10. The LCI Report contains the Antenna Placement And Calibration

Replace Figure 9-898 with the following figure (containing an additional column) and change the figure title:

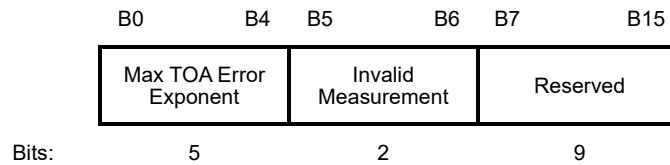


Figure 9-898—~~Format of the TOA Error field~~ format

Change the sixth and seventh paragraphs of 9.6.7.33 as follows:

The Dialog Token field is a nonzero value chosen by the responding STA to identify the ~~Fine Timing Measurement-FTM~~ frame as the first of a pair, with the second or follow-up ~~Fine Timing Measurement-FTM~~ frame to be sent later. The Dialog Token field is set to 0 to indicate the end of the FTM session (see 11.21.6.6 and 11.21.6.4).

The Follow Up Dialog Token field is the nonzero value of the Dialog Token field of the last transmitted ~~Fine Timing Measurement-FTM~~ frame to indicate that it is the follow up ~~Fine Timing Measurement-FTM~~ frame and that the TOD, TOA, Max TOD Error Exponent and Max TOA Error Exponent fields contain the values of the timestamps captured with the first ~~Fine Timing Measurement-FTM~~ frame of the pair. The Follow Up Dialog Token field is 0 to indicate that the ~~Fine Timing Measurement-FTM~~ frame is not a follow up to a last transmitted ~~Fine Timing Measurement-FTM~~ frame. The value 0 in this field also indicates that TOD, TOA, TOD Error, and TOA Error fields are reserved. See 11.21.6.

Change the 10th and 11th paragraphs of 9.6.7.33 as follows:

The TOD field contains a timestamp that represents the time, with respect to a time base, at which the start of the preamble of the last transmitted ~~Fine Timing Measurement-FTM~~ frame appeared at the transmit antenna connector.

The TOA field contains a timestamp that represents the time, with respect to a time base, at which the start of the preamble of the Ack frame to the last transmitted ~~Fine Timing Measurement-FTM~~ frame arrived at the receive antenna connector.

Insert a new paragraph before the last four paragraphs of 9.6.7.33 as follows:

The Invalid Measurement field contains an indication of an invalid TOA field. The Invalid Measurement field is set to 1 to indicate that the TOA value is invalid, and is set to 0 to indicate that the TOA value is valid; values of 2 and 3 are reserved.

Change the last two paragraphs of 9.6.7.33 as follows:

~~The Fine Timing Measurement Parameters field is present in the initial Fine Timing Measurement Frame and is not present in subsequent Fine Timing Measurement frames. If present, it contains a Fine Timing Measurement Parameters element as defined in 9.4.2.167. The FTM Parameters field is optionally present. It is present in an IFTM frame and its retransmissions when negotiating EDCA based ranging; see 11.21.6.3.2.~~

The FTM Synchronization Information field is present in the initial Fine Timing Measurement frame and its retransmissions if any, and in the first ~~Fine Timing Measurement-FTM~~ frame within each burst and its retransmissions if any; otherwise it is not present. If present, it contains an FTM Synchronization Information element with a TSF Sync Info field containing the 4 least significant octets of the TSF at the

responding STA corresponding to the time the responding STA received the last ~~Fine Timing Measurement~~ FTM Request frame with the Trigger field equal to 1.

Insert new text and Figure 9-898a at the end of 9.6.7.33 as follows:

The Ranging Parameters element is optionally present. It is present in an IFTM frame and its retransmissions, when negotiating non-TB ranging; see 11.21.6.4.4 or 11.21.6.4.3. If present, it contains a Ranging Parameters element as defined in 9.4.2.298.

The Secure HE-LTF Parameters element is optionally present. It is present in an IFTM frame, if the IFTMR frame contained a Ranging Parameters element that included a Secure HE-LTF subelement. If present, it contains a Secure HE-LTF Parameters element as defined in 9.4.2.299.

The Channel Measurement Feedback Type field is present in the FTM frame if the Channel Measurement Feedback field is present. It describes the content of the Channel Measurement Feedback element contained in that field. The format of the Channel Measurement Feedback Type field is shown in Figure 9-898a. Each of the subfields is described in Table 9-257.

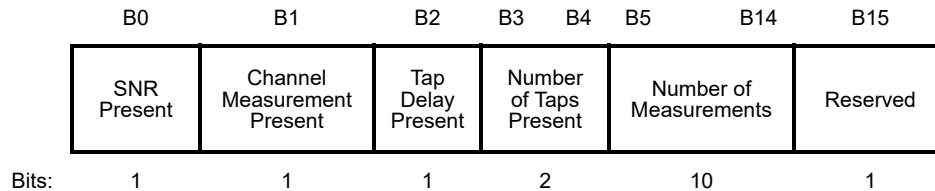


Figure 9-898a—Channel Measurement Feedback Type field format

The Channel Measurement Feedback element is present in the FTM frame if the frame is sent after a LOS Assessment Ack PPDU and optionally in response to an ISTA to RSTA angle of departure TRN field on an Ack frame. This field contains the Channel Measurement Feedback element (see 9.4.2.136).

The Direction Measurement Results field is present in the FTM frame when the frame is sent from an RSTA to an ISTA following an FTM exchange that included ISTA to RSTA AOA measurement. This field contains the Direction Measurement Results element.

The Multiple Best AWV ID field is present in the FTM frame when the frame is sent from an ISTA as part of an AOD feedback exchange after an FTM exchange; see 11.21.6.4.2.4. It contains the Multiple Best AWV ID elements.

The Multiple AOD Feedback element is present in the FTM frame when the frame is sent from an RSTA to an ISTA as part of an AOD feedback exchange after an FTM exchange; see 11.21.6.4.2.4. It contains the Multiple AOD Feedback element.

The LOS Likelihood element may be present in any FTM frame that contain TOA and TOD or Direction Measurement Results on measurements performed over DMG or EDMG PPDU.

Insert 9.6.7.49 through 9.6.7.52 (including Figure 9-909b through Figure 9-909h) after 9.6.7.48 (as inserted by IEEE Std 802.11ay-2021) as follows:

9.6.7.49 Location Measurement Report (LMR) frame format

The LMR frame is an Action No Ack frame of category Public. The LMR frame is used to support the non-TB and TB ranging mechanisms of the FTM procedure described in 11.21.6. The format of the LMR Action field is shown in Figure 9-909b.

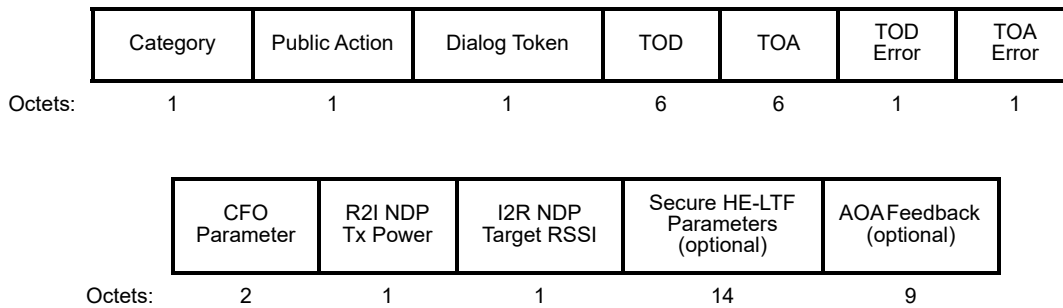


Figure 9-909b—Location Measurement Report (LMR) frame Action field format

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

The Dialog Token field is used to link the LMR to the Measurement Sounding phase that was used for the reported measurement.

The TOD and TOA fields are expressed in units of picoseconds.

The TOD field contains a timestamp that represents the time, with respect to a time base, at which the start of the preamble of the corresponding NDP appeared at the transmit antenna connector. The corresponding NDP in an R2I Location Measurement Report (LMR) frame is an R2I NDP, while in an I2R LMR frame it is an I2R NDP. In both cases the corresponding NDP refers to a measurement exchange that included a Ranging NDP Announcement frame, which carried the matching dialog token that is also included in this LMR.

The TOA field contains a timestamp that represents the time, with respect to a time base, at which the start of the preamble of the corresponding NDP arrived at the receive antenna connector. The corresponding NDP in an R2I LMR frame is an I2R NDP, while in an I2R LMR frame it is an R2I NDP. In both cases the corresponding NDP refers to a measurement exchange that included a Ranging NDP Announcement frame, which carried the matching dialog token that is also included in this LMR.

The TOD Error field format is defined in Figure 9-909c.

The TOA Error field format is defined in Figure 9-909d.

The maximum errors in the TOD and TOA values are represented using the function defined in Equation (9-4a).

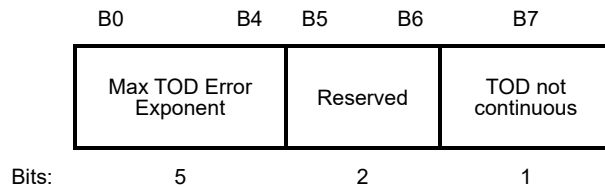


Figure 9-909c—TOD Error field format

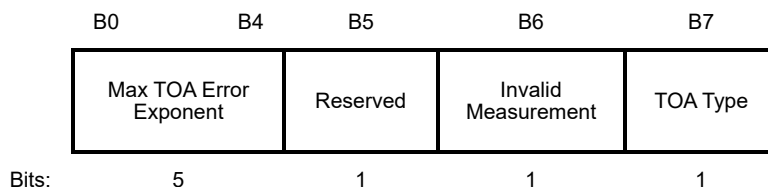


Figure 9-909d—TOA Error field format

$$E_{max} = \begin{cases} \text{unknown, } F = 0 \\ 2^{F-1}, & 1 \leq F \leq 30 \\ \geq 2^{30}, & F = 31 \end{cases} \quad (9-4a)$$

where

F is the Max Error Exponent

E_{max} is the maximum TOD or TOA error, respectively, in units of picoseconds

The Max TOD Error Exponent subfield contains an upper bound for the error exponent in the value specified in the TOD field.

The TOD Not Continuous subfield indicates that the TOD value is with respect to a different underlying time base than the last transmitted TOD value. It is set to 1 when a discontinuity is present. Otherwise, it is set to 0.

The Max TOA Error Exponent subfield contains an upper bound for the error exponent in the value specified in the TOA field.

A value of 0 for the Max TOD Error Exponent or the Max TOA Error Exponent subfield indicates that the upper bound on the error in the corresponding TOD or TOA value is unknown. A value of 31 indicates that the upper bound on the error is greater than or equal to 1.073741824 ms.

The Invalid Measurement subfield indicates whether the TOA field contains a valid value. It is set to 1 to indicate that the TOA value is invalid and the value 0 in this subfield indicates that the TOA value is valid.

The TOA Type subfield indicates if the TOA timestamp was calculated based on the first arrival path of the channel impulse response or the average linear phase across the subcarriers.

In the I2R LMR, the CFO Parameter field indicates the clock rate difference between the ISTA and RSTA; the format is a signed value in units of 0.01 ppm. In the R2I LMR, the value of the CFO Parameter field is reserved.

When negotiated, the R2I NPD Tx Power and I2R NDP Target RSSI fields are used in the in R2I LMR as part of the non-TB ranging measurement exchange, see 11.21.6.4.4, otherwise their values are reserved.

The R2I NDP Tx Power field indicates the combined average power per 20 MHz bandwidth referenced to the antenna connector, of all antennas used to transmit the preceding R2I NDP. The transmit power is reported with a resolution of 1 dB, with values in the range 0 to 60 representing –20 dBm to 40 dBm, respectively. Values above 60 are reserved.

The I2R NDP Target RSSI field indicates the preferred receive signal power, averaged over the RSTA's antenna connectors, for future I2R NDPs transmitted by the ISTA. The preferred receive signal power in units of dBm is $\text{TargetRSSI} = -110 + \text{FVal}$, where FVal is the value of the Target RSSI field, except that values above 90 indicate that the RSTA has no preferred receive signal power for the I2R NDPs.

The Secure HE-LTF Parameters element is present if an ISTA and RSTA have negotiated an FTM session with secure HE-LTF measurement exchange mode; see 11.21.6.3.4. If present, it contains a Secure HE-LTF Parameters element as defined in 9.4.2.299.

The AOA Feedback field is optionally present. If present, it contains a Direction Measurement Results element as defined in 9.4.2.300.

9.6.7.50 ISTA Passive TB Ranging Measurement Report frame format

The ISTA Passive TB Ranging Measurement Report frame is an Action No Ack frame of category Public. The ISTA Passive TB Ranging Measurement Report frame is used to support the passive TB ranging mechanisms of the FTM procedure described in 11.21.6. The format of the ISTA Passive TB Ranging Measurement Report frame is shown in Figure 9-909e.

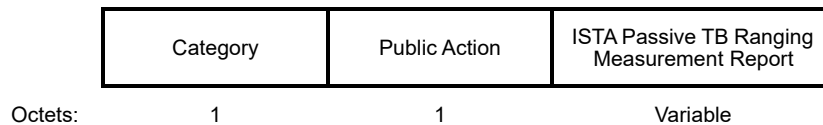


Figure 9-909e—ISTA Passive TB Ranging Measurement Report frame format

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

The ISTA Passive TB Ranging Measurement Report field is defined in 9.4.2.304.

9.6.7.51 Primary RSTA Broadcast Passive TB Ranging Measurement Report frame format

The Primary RSTA Broadcast Passive TB Ranging Measurement Report frame is an Action No Ack frame of category Public. The Primary RSTA Broadcast Passive TB Ranging Measurement Report frame is used to support the passive TB ranging mechanisms of the FTM procedure described in 11.21.6. The format of the Primary RSTA Broadcast Passive TB Ranging Measurement Report Action field is shown in Figure 9-909f.

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

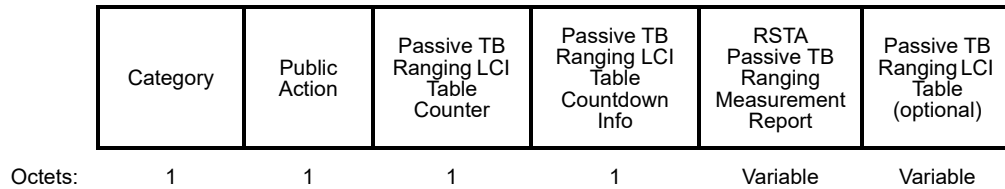


Figure 9-909f—Primary RSTA Broadcast Passive TB Ranging Measurement Report Action field format

The Passive TB Ranging LCI Table Counter field contains a number serving as a reference to the version of the latest Passive TB Ranging LCI Table element transmitted by the RSTA transmitting the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame.

The Passive TB Ranging LCI Table Countdown Info field contains two subfields as shown in Figure 9-909g.

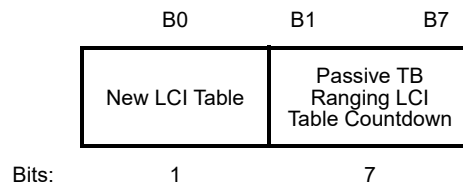


Figure 9-909g—Passive TB Ranging LCI Table Countdown Info field format

The New LCI Table subfield is set to 0 if the current LCI table and LCI table to be transmitted at the end of the countdown are the same, otherwise it is set to 1.

The Passive TB Ranging LCI Table Countdown subfield is an index pointing to the next passive TB ranging availability window where the Passive LCI Table element will be contained in the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame. If the Passive TB Ranging LCI Table element is contained in the current availability window, then the index has the value 0, if the Passive TB Ranging LCI Table element is contained in the next availability window the index has the value 1, if the Passive TB Ranging LCI Table element is contained in the availability window after that the index has value 2, and so on.

The RSTA Passive TB Ranging Measurement Report element is defined in 9.4.2.305.

The Passive TB Ranging LCI Table field is defined in 9.4.2.306.

9.6.7.52 Secondary RSTA Broadcast Passive TB Ranging Measurement Report frame format

The Secondary Broadcast RSTA Passive TB Ranging Measurement Report frame is an Action No Ack frame of category Public. The Secondary RSTA Broadcast Passive TB Ranging Measurement Report frame is used to support the passive TB ranging mechanisms of the FTM procedure described in 11.21.6. The format of the Secondary RSTA Broadcast Passive TB Ranging Measurement Report Action field is shown in Figure 9-909h.

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

Category	Public Action	Number of ISTA Passive TB Ranging Measurement Reports	ISTA Passive TB Ranging Measurement Reports
Octets:	1	1	Variable

**Figure 9-909h—Secondary RSTA Broadcast Passive TB Ranging Measurement Report
Action field format**

The Number of ISTA Passive TB Ranging Measurement Reports field indicates the number of following ISTA Passive TB Ranging Measurement Report fields.

The ISTA Passive TB Ranging Measurement Reports field contains the number of ISTA Passive TB Ranging Measurement Reports in the ISTA Passive TB Ranging Measurement Report elements. The ISTA Passive TB Ranging Measurement Reports element is defined in 9.4.2.304.

Insert 9.6.34 (including Table 9-526m) after 9.6.33 (as inserted by IEEE Std 802.11ba-2021) as follows:

9.6.34 Protected Fine Timing frame details

9.6.34.1 Protected Fine Timing Action field

The Protected Fine Timing Action field, in the one octet immediately after the category field, differentiates between various Protected Fine Timing frames as shown in the Table 9-526m.

Table 9-526m—Protected Fine Timing Action field values

Value	Meaning
0	Reserved.
1	Protected FTMR frame. The format of frame after the Action field is identical to the format of the FTM Request public action frame; see 9.6.7.32. It is carried in a Management Action frame.
2	Protected FTM frame. The format of the frame after the Action field is identical to the format of the FTM public action frame; see 9.6.7.33. It is carried in a Management Action frame.
3	Protected LMR frame. The format of the frame after the Action field is identical to the format of the Location Measurement Report public action frame; see 9.6.7.49. It is carried in a Management Action No Ack frame.
4–255	Reserved.

10. MAC sublayer functional description

10.3 DCF

10.3.2 Procedures common to the DCF and EDCAF

10.3.2.14 Duplicate detection and recovery

10.3.2.14.2 Transmitter requirements

Change the row for “SNS4” and insert a new row for “SNS8” after the row for “SNS7” in Table 10-5 as follows (unchanged rows not shown):

Table 10-5—Transmitter sequence number spaces

Sequence number space identifier	Sequence number space	Applies to	Status	Multiplicity	Transmitter requirements
...					
SNS4	QMF	A QMF STA transmitting a QMF <u>that is neither a Protected Fine Timing frame nor a Public Action LMR</u>	Mandatory	Indexed by <Address 1, AC>	TR2
...					
SNS8	Protected Fine Timing Frame and Public Action LMR	A STA transmitting Protected Fine Timing frames or Public Action LMR	Mandatory	Single Instance	
...					

10.3.2.14.3 Receiver requirements

Change the row for “RC6” and insert a new row for “RC13” after the row for “RC12” in Table 10-6 as follows (unchanged rows not shown):

Table 10-6—Receiver caches

Receiver cache identifier	Cache name	Applies to	Status	Multiplicity/Cache size	Receiver requirements
...					
RC6	QMFs	A STA receiving an individually addressed QMF <u>that is neither a Protected Fine Timing frame nor a Public Action LMR</u>	Mandatory	Indexed by: <Address 2, AC, sequence number, fragment number> The most recent cache entry per <Address 2, AC, sequence-number, fragment-number>.	RR2 RR3 RR5
...					
RC13	Protected Fine Timing frame and Public Action LMR	A STA receiving Protected Fine Timing frames or Public Action LMR	Mandatory	Indexed by: <Address 2, sequence number, fragment number>. At least the most recent cache entry per <Address 2>.	RR1 RR2 RR5
...					

10.12 A-MPDU operation

10.12.3 Minimum MPDU start spacing rules

Insert a new paragraph at the end of 10.12.3 as follows:

For FTM procedure measurement exchange with an unassociated STA, the transmitting STA shall use one of the values specified in Table 9-185, for the t_{MMSS} .

10.23 HCF

10.23.2 HCF contention based channel access (EDCA)

10.23.2.8 Multiple frame transmission in an EDCA TXOP

Insert a new item after the item beginning “A BFRP Trigger frame...” (inserted by IEEE Std 802.11ax-2021) at the end of the dashed list after the first paragraph of 10.23.2.8 as follows:

- A Ranging NDP Announcement frame followed after SIFS by an HE Ranging NDP followed after SIFS by an HE Ranging NDP followed after SIFS by an LMR frame.

10.42 DMG beamforming

10.42.10 EDMG beamforming

10.42.10.6 First path beamforming training

Replace the text in 10.42.10.6 (inserted by IEEE Std 802.11ay-2021) with the following text:

First path beamforming training (FPBT) is a procedure used to determine the best AWP corresponding to the first path. The first path is defined to be the propagation path between TX and RX that is estimated to have the shortest time of flight. In line of sight (LOS) conditions, the first path corresponds to the LOS path. If several AWP have the same estimated shortest time of flight, the beamforming training shall select among them the first path as the one with best quality. The method a STA uses to determine the first path and the first path with best quality is implementation dependent and beyond the scope of this standard.

Prior to performing the FPBT, the two EDMG STAs are assumed to know their best transmit AWP and receive AWP for data communication. A FPBT shall be completed within the CBAP or SP in which it was initiated. An EDMG STA that has the First Path Beamforming Training Supported subfield in the STA's RSNXE set to 1 is FPBT capable.

An EDMG STA shall not initiate FPBT with a peer EDMG STA that is not FPBT capable.

The FPBT procedure is the same as BRPTXSS procedure (see 10.42.10.5) except for the following differences:

- FPBT shall be performed in SISO configuration defined in 10.42.10.5.2.2 with any valid combination of TXVECTOR parameter, CH_BANDWIDTH, and CH_AGGREGATION. FPBT shall not be performed in MIMO configuration. All the phases of the BRP-TXSS procedure shall be supported in FPBT. Selection of which phases are used is dependent on the DMG antenna reciprocity and antenna pattern reciprocity properties of the involved EDMG STAs as defined in their DMG STA Capability Information field.
- In the setup phase, the EDMG STA shall transmit a BRP frame with the First Path Beamforming Training subfield set to 1 to request FPBT. An EDMG STA that is FPBT capable, and receives a BRP frame with the First Path Beamforming Training subfield equal to 1, shall set the First Path Beamforming Training subfield to 1 in the BRP frame that it sends in response to the reception of the BRP frame.
- In a BRP transaction (Initiator BRP TXSS, receive training phase of the responder, Responder BRP TXSS, and receive training phase of the initiator) that is part of a first path beamforming training, all transmitted BRP frames shall have the First Path Beamforming Training subfield set to 1 and shall have the FIRST_PATH_TRAINING parameter in the TXVECTOR set to 1. In such a transaction, all TX and RX beamforming training using the TRN field is used to find the AWP of the first path and not the best path. The AWP selected by the FPBT shall not be used for data transmission. It shall be used only in the FTM procedure defined in 11.21.6.4.2.6.

The TRN field of the EDMG BRP-RX PPDU used for the receive training phase of the responder:

- Shall be transmitted with the best first path AWP identified in the preceding Initiator BRP TXSS phase.
- Shall be received with the DMG antenna corresponding to the best first path AWP configuration identified in the preceding Initiator BRP TXSS phase.
- When the receive training phase of the initiator is performed, the TRN field of the EDMG BRP-RX PPDU used in this phase shall be transmitted with the best first path AWP identified in the preceding Responder BRP TXSS phase or in the receive training of the responder, as defined in 10.42.10.5.2.2.

- Shall be received with the DMG antenna corresponding to the best first path AWV configuration identified in the Responder BRP TXSS or in the Initiator BRP TXSS, or with multiple DMG antennas, as defined in 10.42.10.5.2.

11. MLME

11.3 STA authentication and association

11.3.1 State variables

Insert a new item and note after the first item in the dashed list in 11.3.1 as follows:

- *State 1a:* Authenticated for non-DMG STAs and for DMG STAs that have performed PASN authentication (12.13). Unassociated. Association is not possible from this state without non-PASN IEEE 802.11 authentication.

NOTE—This state is used by STAs that wish to exchange robust management frames for example, perform secure ranging, but do not wish to exchange data frames.

11.3.2 State transition diagram for nonmesh STAs

Replace Figure 11-17 with the figure shown on the next page:

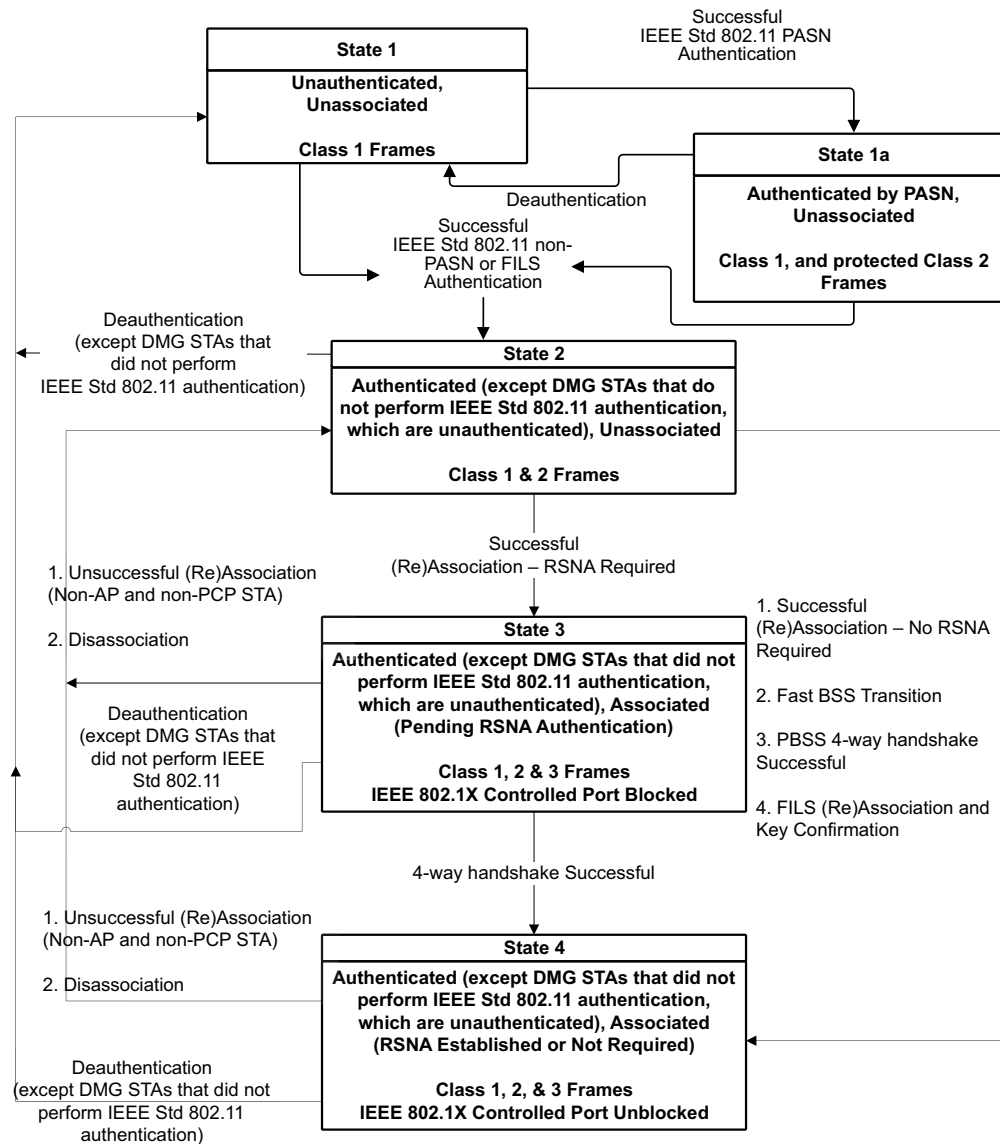


Figure 11-17—Relationship between state and services between a given pair of nonmesh STAs

11.3.3 Frame filtering based on STA state

Insert new list item b) (including its sub-items 1) and 2)) for Class 1a frames after list item a) for Class 1 frames and renumber items for Class 2 and Class 3 frames to be c) and d), respectively, in the tiered list after the sixth paragraph of 11.3.3 as follows:

- b) Class 1a frames
 - In an infrastructure BSS when PTKSA from PASN authentication exists
 - 1) Protected Fine Timing frames (9.6.34)
 - 2) Unicast SA Query (11.13)

Change the eighth paragraph of 11.3.3 as follows:

A STA shall not transmit Class 2 frames unless in State 2 or State 3 or State 4. In State 1a, a STA shall not transmit frames other than Class 1 and Class 1a frames.

11.3.4 Authentication and deauthentication

11.3.4.1 General

Change the second and fourth paragraphs of 11.3.4.1 as follows:

Successful authentication other than PASN authentication sets the state for a STA's state to State 2, if it was in State 1 or State 1a. Successful PASN authentication sets the STA's state to State 1a, if it was in State 1. PASN authentication is disallowed in states other than State 1. Unsuccessful authentication leaves the state for the STA's state unchanged.

If STA A in an infrastructure BSS receives an unprotected Class 2 frame or a Class 3 frame from STA B that is not authenticated with STA A (i.e., the state for STA B is State 1) when the state for STA B is State 1a, STA A shall discard the frame. If the frame has an individual address in the Address 1 field, the MLME of STA A shall send a Deauthentication frame to STA B.

Insert a new paragraph after the fourth paragraph of 11.3.4.1 as follows:

If STA A in an infrastructure BSS receives any frame other than Class 1 and Class 1a frames (see 11.3.3) from STA B when in State 1a, STA A shall discard the frame.

11.3.4.2 Authentication—originating STA

Insert a new list item after item b) 4) in the tiered list in 11.3.4.2 as follows:

- 5) For PASN authentication, the authentication mechanism described in 12.13.

Change items c) and d) in the tiered list in 11.3.4.2 as follows:

- c) For any authentication protocol that is not PASN, if the authentication was successful within the AuthenticateFailureTimeout, the state for the indicated STA shall be set to State 2 if it was in State 1 or State 1a; the state shall remain unchanged if it was other than State 1 or State 1a.
- d) For PASN authentication, if the authentication was successful within the AuthenticateFailureTimeout, the state for the indicated STA shall be set to State 1a if it was State 1; PASN authentication is disallowed in states other than State 1.
- e) ~~d)~~ The MLME shall issue an MLME-AUTHENTICATE.confirm primitive to inform the SME of the result of the authentication.

11.3.4.3 Authentication—destination STA

Insert new item e) after item d) (re-numbering the remaining items) in the lettered list in 11.3.4.3 and change items f) and g) (now items g) and h)) as follows:

- e) If PASN authentication is being used, the MLME shall issue an MLME-AUTHENTICATE.indication primitive to inform the SME of the authentication request, and the SME shall execute the procedure described in 12.13.

- f) ~~e)~~ If the STA is in an IBSS and management frame protection was not negotiated when the PTKSA(s) were created, the SME shall delete any PTKSA, GTKSA, IGTKSA and temporal keys held for communication with the originating STA by using the MLME-DELETEKEYS.request primitive (see 12.6.18).
- g) ~~f)~~ Upon receipt of an MLME-AUTHENTICATE.response primitive, if the ResultCode is not SUCCESS, the MLME shall transmit an Authentication frame with the corresponding status code, as defined in 9.4.1.9, and the state for the originating STA shall be left unchanged. The Authentication frame is constructed using the appropriate procedure in 12.3.3.2, 12.3.3.3, 13.5, ~~or 13.6, or 12.13.~~
- h) ~~g)~~ Upon receipt of an MLME-AUTHENTICATE.response primitive, if the ResultCode is SUCCESS, the MLME shall transmit an Authentication frame that is constructed using the appropriate procedure in 12.3.3.2, 12.3.3.3, 13.5, ~~or 13.6, or 12.13,~~ with a status code of SUCCESS; and the The state for the originating STA shall be set to State 2 if it was in State 1 or State 1a when PASN authentication procedure was not used; the state shall remain unchanged if it was other than State 1. The state for the originating STA shall be set to State 1a if it was in State 1 and PASN authentication procedure was used. PASN authentication shall be disallowed in states other than State 1.

11.3.4.4 Deauthentication—originating STA

Change item b) in the lettered list in 11.3.4.4 as follows:

- b) On receipt of the MLME-DEAUTHENTICATE.request primitive, if the state for the indicated STA is State 1a, State 2, State 3, or State 4, the MLME shall generate a Deauthentication frame to be transmitted to the indicated STA.

11.3.4.5 Deauthentication—destination STA

Change the second paragraph of 11.3.4.5 as follows:

Otherwise, upon receipt of a Deauthentication frame from a STA for which the state is State 1a, State 2, State 3, or State 4, the destination STA shall deauthenticate with the originating STA using the following procedure:

11.3.5 Association, reassociation, and disassociation

11.3.5.1 General

Change paragraphs three to seven of 11.3.5.1 as follows:

Successful association enables a STA to exchange Class 3 frames. Successful association sets the state for a non-FILS STA to State 3 or State 4. Successful association sets the state for FILS STAs to State 4. Association and reassociation are not allowed in State 1 and State 1a. In order to associate or reassociate, a STA in State 1 or State 1a shall perform an IEEE 802.11 non-PASN authentication or FILS authentication and transition to State 2.

Successful reassociation enables a STA to exchange Class 3 frames. Unsuccessful reassociation ~~when not in State 1~~ leaves the state for a STA state unchanged [with respect to the AP or PCP that was sent the Reassociation Request (which may be the current STA)]. Successful reassociation sets the state for a non-FILS STA to State 3 or State 4 (with respect to the AP or PCP that was sent the Reassociation Request frame). Successful reassociation ~~when not in State 1~~ sets the state for a STA to State 2 (with respect to the current AP or PCP, if this is not the AP or PCP that was sent the Reassociation Request frame). Successful reassociation sets the state for a FILS STA to State 4 (with respect to the AP or PCP that was sent the Reassociation Request

frame) and enables it to exchange Class 3 frames. Reassociation shall be performed only if the originating STA is already associated in the same ESS.

Disassociation notification when not in State 1 or State 1a sets the state for a non-FILS STA to State 2. Disassociation notification when not in State 1 or State 1a sets the state for a FILS STA to State 1. The STA shall become associated again prior to sending Class 3 frames. A STA may disassociate a peer STA at any time, for any reason. The state shall remain unchanged upon disassociation notification in State 1a.

If non-DMG STA A in an infrastructure BSS receives a Class 3 frame from STA B that is authenticated but not associated with STA A (i.e., the state for STA B is State 2 or State 1a), STA A shall discard the frame. If the frame has an individual address in the Address 1 field, the MLME of STA A shall send a Disassociation frame to STA B.

If DMG STA A in an infrastructure BSS receives a Class 3 frame from STA B that is not associated with STA A (i.e., the state for STA B is State 2 or State 1a), STA A shall discard the frame. If the frame has an individual address in the Address 1 field, the MLME of STA A shall send a Disassociation frame to STA B.

Change the last paragraph of 11.3.5.1 as follows:

Association is not applicable in an IBSS. In an infrastructure BSS, association is required to exchange Class 3 frames. In a PBSS, association is optional. APs and PCPs do not initiate association.

11.3.5.2 Non-AP and non-PCP STA association initiation procedures

Change item a) in the lettered list in 11.3.5.2 as follows:

- a) If the state for the AP or PCP is State 1 or State 1a, the MLME shall inform the SME of the failure of the association by issuing an MLME-ASSOCIATE.confirm primitive, and this procedure ends.

11.3.5.3 AP or PCP association receipt procedures

Change item b) in the lettered list in 11.3.5.3 as follows:

- b) If the state for the STA is State 1 and the STA is a non-DMG STA, the SME shall refuse the association request by issuing an MLME-ASSOCIATE.response primitive with ResultCode NOT_AUTHENTICATED. If the state for the STA is State 1a and the STA is a non-DMG STA, the SME shall refuse the association request by issuing an MLME-ASSOCIATE.response primitive with ResultCode PASN_NOT_AUTHORIZED.

11.3.5.4 Non-AP and non-PCP STA reassociation initiation procedures

Change item a) in the lettered list in 11.3.5.4 as follows:

- a) If the STA is not associated in the same ESS or the state for the new AP or PCP is State 1 or State 1a, the MLME shall inform the SME of the failure of the reassociation by issuing an MLME-REASSOCIATE.confirm primitive, and this procedure ends.

11.3.5.5 AP or PCP reassociation receipt procedures

Change item b) in the lettered list in 11.3.5.5 as follows:

- b) If the state for the STA is State 1 or State 1a and the STA is a non-DMG STA, the SME shall refuse the reassociation request by issuing an MLME REASSOCIATE.response primitive with ResultCode NOT_AUTHENTICATED. If the state for the STA is State 1a and the STA is a non-DMG STA, the SME shall refuse the reassociation request by issuing an MLME REASSOCIATE.response primitive with ResultCode PASN_NOT_AUTHORIZED.

11.3.5.6 Non-AP and non-PCP STA disassociation initiation procedures

Change item b) in the lettered list in 11.3.5.6 as follows:

- b) The state for the AP or PCP shall be set to State 2 if it was not State 1 or State 1a. In the case of an MM-SME coordinated STA, the MLME shall perform this for each STA whose address was included in the MMS parameter of the MLME-ASSOCIATE.request or MLME-REASSOCIATE.request primitive that established the association.

11.3.5.8 AP or PCP disassociation initiation procedure

Change item b) in the lettered list in 11.3.5.8 as follows:

- b) The state for the STA shall be set to State 2, if it was not State 1 or State 1a. The MM-SME shall perform this process for each STA whose address was included in the MMS parameter of the MLME-ASSOCIATE.request or MLME-REASSOCIATE.request primitive that established the association.

11.10 Radio measurement procedures

11.10.10 Usage of the neighbor report

11.10.10.2 Requesting a neighbor report

Insert a new paragraph at the end 11.10.10.2 as follows:

To request a list of neighboring APs that support DMG location services, the STA shall transmit a Neighbor Report Request frame that includes a Neighbor DMG request field with a Measurement Request element with the value of its Measurement Type field equal to Neighboring DMG APs. A STA shall not send this type of request to an AP unless it received an Extended Capabilities element from the AP with the DMG Location Supporting APs Information field equal to 1. The Neighbor Report Request frame shall include an SSID element with the SSID set to wildcard SSID.

11.10.10.3 Responding to a neighbor report request

Insert a new paragraph at the end 11.10.10.3 as follows:

An AP that has set the DMG/Location Supporting APs In The Area field to 1 in the Extended Capabilities element that receives a Neighbor Report Request frame that includes a Measurement Request element with the value of its Measurement Type field equal to Neighboring DMG APs, shall respond with a Neighbor Report Measurement frame with a list of Neighbor Report elements of neighboring DMG APs supporting location services. Per each DMG AP, the Neighbor Report element shall include a Measurement Report subelement

with the Measurement Type field equal to LCI. If available to the sending AP, the Neighbor Report element shall include a TSF subelement. The definition of which APs are neighboring APs is implementation dependent. If the AP sending is collocated with a DMG AP that is location capable, the collocated DMG AP should be included in the list.

11.13 SA Query procedures

Change the first item in the dashed list following the third paragraph of 11.13 as follows:

- The STA is not currently associated to the STA that sent the SA Query Request frame and there is no PTKSA established by PASN authentication.

11.21 Wireless network management procedures

11.21.6 Fine timing measurement (FTM) procedure

Change the title of 11.21.6.1 as follows:

11.21.6.1 ~~Overview~~General

Replace the content of 11.21.6.1 (including Figure 11-34) with the following text and subclauses 11.21.6.1.1 through 11.21.6.1.3 (including Figure 11-34, Figure 11-34a, and Figure 11-34b) as follows:

The FTM procedure allows a STA to determine its range, relative range, and its direction to or from another STA using time of flight (TOF), time difference of arrival (TDOA), and phase measurement. In order for a STA to obtain its location, the STA may perform this procedure with multiple STAs whose locations are known.

An FTM session is an instance of an FTM procedure between an ISTA and an RSTA along with the associated scheduling and operational parameters (see 9.4.2.167 and 9.4.2.298). An FTM session is composed of a negotiation, measurement exchange, and termination.

An RSTA might be required to establish overlapping FTM sessions with a large number of ISTAs (e.g., an AP providing measurements to STAs at a mall or a store). On the other hand, an ISTA might have multiple ongoing FTM sessions on the same or different channels with different RSTAs, while being associated with an AP for the exchange of data or signaling.

Since some of the ISTA's activities may be nondeterministic and might have higher precedence than the FTM session (e.g., data transfer interaction with an associated AP), a conflict might prevent the ISTA from being available at the scheduled time window(s) for executing the ranging measurement exchange(s). The FTM procedure provides mechanisms as described in 11.21.6.1.1 and 11.21.6.1.2 to ensure that the ISTA is available to execute the ranging measurement exchange as scheduled.

11.21.6.1.1 EDCA based Ranging and TB ranging overview

To support the constraints of both the initiating and responding STAs, during negotiation the ISTA initially requests a preferred periodic time window allocation. The RSTA subsequently responds by accepting or overriding the allocation request based on its resource availability and capability.

The ISTA in Figure 11-34 establishes sessions with RSTA 1 and RSTA 2 on different channels. Since the ISTAs may not be able to be on channel (for instance, the ISTA is in the middle of some other activity) at the start of the negotiated time window (i.e., burst instance in the case of EDCA based ranging measurement

exchange or availability window in the case of TB ranging measurement exchange (11.21.6.4.3) to execute the measurement exchange, within each instance of the negotiated time window, the ISTA indicates whether it is available to start the measurement exchange.

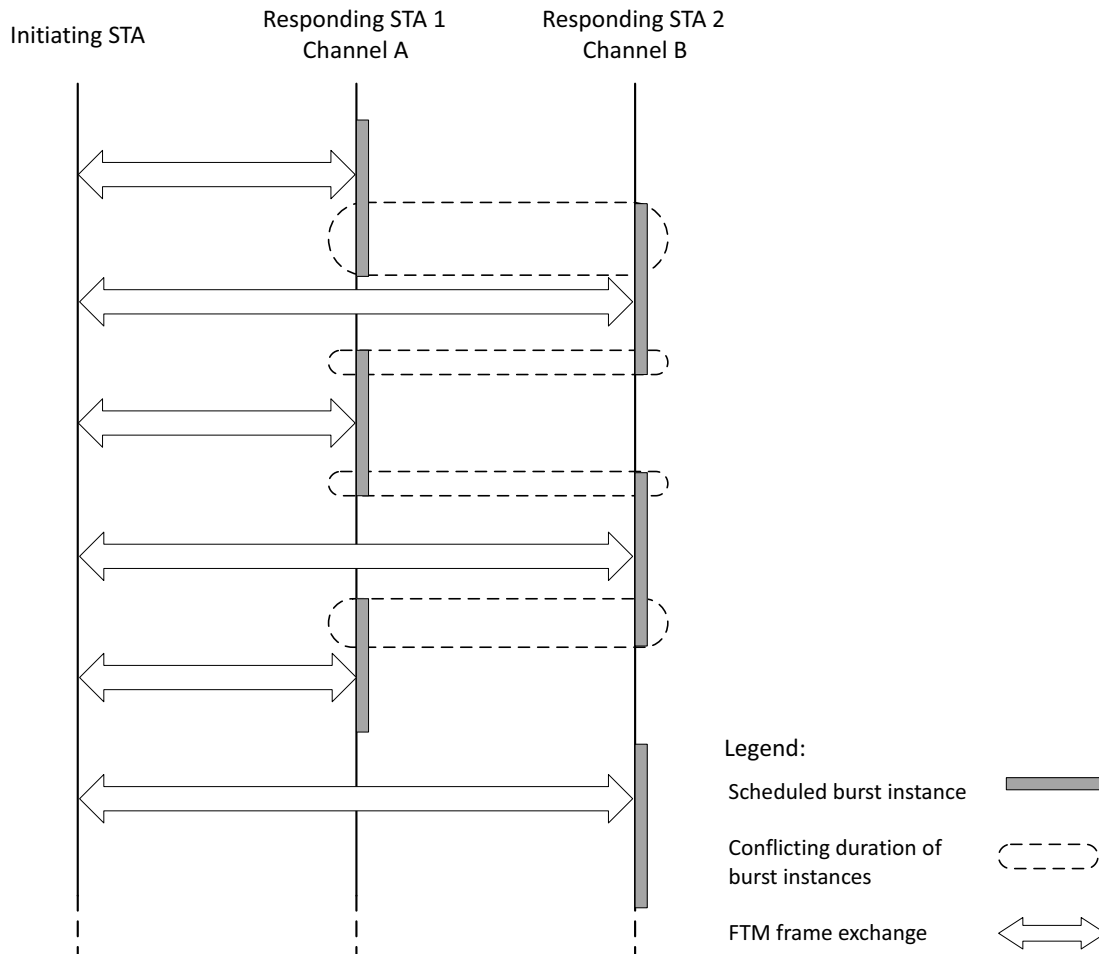


Figure 11-34—Concurrent FTM sessions

- In EDCA based ranging measurement exchange the ISTA indicates its availability to start the measurement exchange by sending an FTM Request frame with the Trigger field set to 1 during the corresponding burst instance.
- In TB ranging measurement exchange (11.21.6.4.3), the ISTA indicates its availability to start the measurement exchange by responding to the Poll Ranging Trigger from the RSTA.

11.21.6.1.2 Non-TB ranging overview

Non-TB ranging measurement exchange is used by the ISTA to determine the measurement timing, based on its scheduling conflicts with other activities and the availability window parameters to identify the availability window instance, which is a time window referenced to the end of the previous measurement instance. During this measurement time window, the ISTA may come to the channel at any time and use contention based access to initiate a new measurement exchange. Because of conflict arising due to other activities, the ISTA does not start measurement at start of availability window while the RSTA waits for the start of measurement exchange. Figure 11-34a shows an example of concurrent non-TB ranging sessions. The dotted region

indicates that the non-TB ranging measurement exchange does not always start at the beginning of the time window since the ISTA may have been active on another channel.

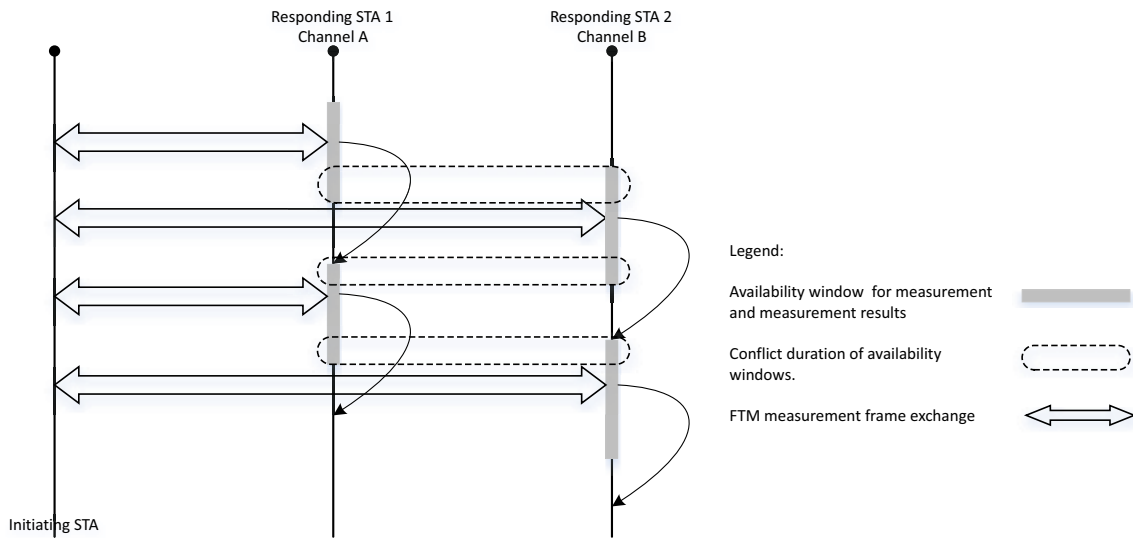


Figure 11-34a—Non-TB ranging concurrent FTM sessions

The ISTA in Figure 11-34a, establishes sessions with RSTA 1 and RSTA 2 on different channels. Scheduling of each availability window instance is determined by the timing of the previous measurement instance with the corresponding RSTA. The constraints for an ISTA to initiate the non-TB ranging measurement exchange with each RSTA may be different. The RSTAs shall remain available within the entire availability window instance for the ISTA to initiate the non-TB ranging measurement exchange.

11.21.6.1.3 Passive TB ranging overview

Passive TB ranging is a variant of the TB ranging mode referred to in 11.21.6. In all aspects, except where explicitly stated differently, the passive TB ranging mode, its protocols, procedures, components, and definitions follow the rules for TB ranging.

The rules and procedures related to the use of protected management frames and secure HE-LTF in the TB ranging protocol does not apply to passive TB ranging.

NOTE 1—For example, the text in the following subclauses, and their subclauses, apply also to passive TB ranging:

- Subclause 11.21.6.1.1
- Subclause 11.21.6.3.3
- Subclause 11.21.6.4.3
- Subclause 11.21.6.5
- Subclause 11.21.6.5.1
- Subclause 11.21.6.6.2

NOTE 2—Below is a list of example exceptions for passive TB ranging where it does not follow the rules for TB ranging:

- The rules and procedures specific for the secure version of TB ranging does not apply to passive TB ranging.
- The RSTA uses the Ranging Trigger frame of subvariant Passive Sounding for its sounding trigger frames.
- The ISTAs use HE Ranging NDPs for their I2R NDPs.

- The ISTAs do not use the LMR frame for reporting of I2R LMR but instead use the ISTA Passive TB Ranging Measurement Report frame.
- The RSTA sends the Primary and Secondary RSTA Broadcast Passive TB Ranging Measurement Report frames at the end of the measurement exchange.
- The number of spatial streams (NSTS) for passive TB ranging is limited to 4.
- When phase shift feedback is negotiated for passive TB ranging, both the RSTA and the ISTA measure and report PSTOAs, in addition to measuring and reporting TOAs.

The passive TB ranging mode consists of ranging exchanges between an RSTA and a set of ISTAs. These ranging exchanges and associated measurement reporting are set up such that an RSTA or ISTA can listen in to them and use the ranging exchanges and reported ranging measurements to estimate its differential distance to pairs formed by RSTAs and/or ISTAs. The listening STA, a passive STA (PSTA), is not itself an active transmitting participant in the ranging exchanges. That is, the PSTA can passively estimate its differential distances to the pairs of RSTAs and/or ISTAs. For a detailed description of how this differential distance can be calculated; see 11.21.6.4.8.3. It can then use these differential distances together with knowledge of the RSTA and ISTA locations to estimate its own location.

The RSTA centric scheduling for passive TB ranging operation operates as the RSTA centric scheduling for TB ranging operation referred to in 11.21.6.1.1. The availability window is here referred to as a passive TB ranging availability window. The passive TB ranging is scheduled by the RSTA in an availability window used for passive location.

In order to announce the scheduling and parameters of the availability window for passive TB ranging, the RSTA includes an RSTA Availability Window element (see 9.4.2.297) in its Beacon frame (see 9.3.3.2). Here, the Availability Window Broadcast Format subfield in the Header subfield in the RSTA Availability Information field in the RSTA Availability Window element is set to 1 to indicate that this is an RSTA Availability Window element that is used to signal availability windows for passive TB ranging.

The purpose of the announcement of the availability window for the passive TB ranging is to enable PSTAs to listen to the passive TB ranging exchanges that are occurring during the availability windows.

Consider passive TB ranging between a set of RSTAs and a set of ISTAs as depicted in Figure 11-34b. Here, three access points, AP1, AP2, and AP3, act as three passive TB ranging responder stations, RSTA1, RSTA2, and RSTA3. RSTA1 operates ranging exchanges in an availability window for passive TB ranging and is performing passive TB ranging exchanges with ISTA1, ISTA2, ISTA3, ISTA4, ISTA5, and ISTA6.

The PSTA is depicted, with the dashed arrows, as listening in to the passive TB ranging exchange between RSTA1 and ISTA4. Generalizing, the PSTA has the opportunity to receive the transmissions of all passive TB ranging exchanges occurring. The reception of each of these ranging exchange transmissions enables the PSTA to estimate its differential distance with respect to the RSTA–ISTA pair and use this information towards its location estimation.

In addition to the ranging exchanges between the ISTAs and RSTA1, the passive TB ranging protocol also allows the same ISTAs to measure time of arrivals of each other's ranging NDPs. An example of one such occurrence is depicted in Figure 11-34b in the form of the dotted double arrow between ISTA1 and ISTA2.

Furthermore, if one of the other APs in Figure 11-34b temporarily takes on the role of being an ISTA, it may also participate in RSTA1's passive TB ranging operation and perform passive TB ranging exchanges with RSTA1.

At other time intervals, RSTA2 and RSTA3 may operate their own passive TB ranging exchanges in their own availability windows. In each case, the ISTAs may perform passive TB ranging exchanges with RSTA2 or RSTA3.

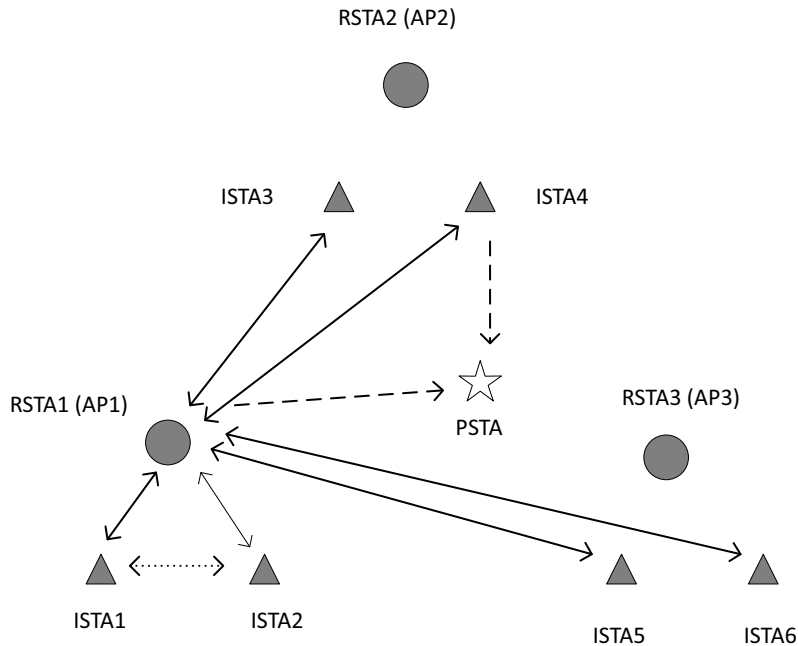


Figure 11-34b—Example of passive TB ranging used for PSTA location—RSTA1 (AP1) operating passive TB ranging

Each of the access points operating as RSTA1, RSTA2, and RSTA3, announces the timing and bandwidth of its ranging availability window for passive TB ranging in its beacon in an RSTA Availability Window element. By listening to the AP's beacons, the PSTA acquires this information for the different RSTA availability windows for passive TB ranging.

The PSTA listens to all of these ranging exchanges. Considering all ranging exchanges between all RSTAs and all ISTAs, the PSTA has the opportunity to overhear a large set of ranging exchanges between RSTAs and ISTAs in different locations, enabling the use of all of them towards its location estimation and also mitigating non-LOS issues between the ISTA(s)/RSTA involved in the ranging measurements.

The locations of the RSTA–ISTA or ISTA–ISTA pairs together with their differential distance with respect to the PSTA, generates a set of hyperbolic curves to which the PSTAs location is constrained. The intersection of these hyperbolic curves, or the best approximation to their intersection, is then an estimate of the PSTA's location.

11.21.6.2 FTM capabilities

Change the last two paragraphs of 11.21.6.2 as follows:

A STA in which `dot11FineTimingMsmtRespActivated` is false shall set the Fine Timing Measurement Responder field of the Extended Capabilities element for non-DMG STA or the Fine Timing Measurement Responder subfield of the DMG Fine Timing and Range Measurement Capability Information field of the DMG Capabilities element for DMG STA to 0.

A STA in which `dot11FineTimingMsmtInitActivated` is false shall set the Fine Timing Measurement Initiator field of the Extended Capabilities element for non-DMG STA or the Fine Timing Measurement Initiator subfield of the DMG Capabilities element for DMG STA to 0.

Insert new text at the end of 11.21.6.2 as follows:

A STA in which dot11NonTriggerBasedRangingRespImplemented is true shall set the Non-TB Ranging Responder field of the Extended Capabilities element to 1. Otherwise, it shall set the Non-TB Ranging Responder field of the Extended Capabilities element to 0.

A STA in which dot11TriggerBasedRangingRespImplemented is true shall set the TB Ranging Responder field of the Extended Capabilities element to 1. Otherwise, it shall set the TB Ranging Responder field of the Extended Capabilities element to 0.

A STA in which dot11SecureLTFImplemented is true shall set the Secure HE-LTF Support field of the RSNXE to 1. Otherwise, it shall set the Secure HE-LTF Support field to 0.

A STA in which dot11RSTARequiresPMFActivated has the value Required (2) shall set the URNM-MFPR field of the RSNXE to 1. Otherwise, it shall set the URNM-MFPR field to 0.

A STA in which dot11RSTARequiresPMFActivated has the value Required-X20M (1) shall set the URNM-MFPR-X20 field of the RSNXE to 1. Otherwise, it shall set the URNM-MFPR-X20 field to 0.

A STA in which dot11DMGOptionImplemented is true and dot11FineTimingMsmtRespActivated is true shall set the Fine Timing Measurement Responder subfield of the DMG Fine Timing and Range Measurement Capability Information field to 1.

A STA in which dot11DMGOptionImplemented is true and dot11FineTimingMsmtInitActivated is true shall set the Fine Timing Measurement Initiator subfield of the DMG Fine Timing and Range Measurement Capability Information field to 1.

A STA in which dot11PassiveTBRangingResponderImplemented is true shall set the Passive TB Ranging Responder Measurement Support field of the Extended Capabilities element to 1. Otherwise, it shall set the Passive TB Ranging Responder Measurement Support field of the Extended Capabilities element to 0.

A STA in which dot11PassiveTBRangingInitiatorImplemented is true shall set the Passive TB Ranging Initiator Measurement Support field of the Extended Capabilities element to 1. Otherwise, it shall set the Passive TB Ranging Initiator Measurement Support field of the Extended Capabilities element to 0.

A STA shall set the Phase Shift Feedback Support field in the Extended Capabilities element to 1 if dot11PhaseShiftFeedbackImplemented is true and one of the following is true:

- dot11TriggerBasedRangingRespImplemented
- dot11NonTriggerBasedRangingRespImplemented
- dot11PassiveTBRangingInitiatorImplemented
- dot11PassiveTBRangingResponderImplemented

Otherwise it shall set the Phase Shift Feedback Support field in the Extended Capabilities element to 0.

A STA in which either dot11NonTriggerBasedRangingRespImplemented is true or dot11TriggerBasedRangingRespImplemented is true shall set the I2R LMR Feedback Policy field of the Extended Capabilities element to 1 if dot11I2RLMRFeedbackPolicy is true. The STA shall set the I2R LMR Feedback Policy field of the Extended Capabilities element to 0 if dot11I2RLMRFeedbackPolicy is false.

An RSTA in which dot11PhaseShiftFeedbackImplemented is true shall set the Phase Shift TOA Feedback Support field in the Extended Capabilities element to 1 to indicate the RSTA's capability to support phase shift TOA feedback in the R2I LMR.

Change the title of 11.21.6.3 as follows:

11.21.6.3 ~~Fine timing measurement~~ FTM procedure negotiation

Replace the content of 11.21.6.3 with subclauses 11.21.6.3.1 through 11.21.6.3.9 (including Table 11-11a) as follows:

11.21.6.3.1 General

In order to initiate an FTM procedure, a STA that supports the FTM procedure as an initiator (referred to as an ISTA) shall transmit an FTM Request frame. This frame is called the IFTMR frame. After transmission of this frame, the ISTA shall be ready to receive an FTM frame.

A STA that supports the FTM procedure as a responder (referred to as an RSTA) shall not transmit FTM frames addressed to a peer STA unless the RSTA has received an IFTMR frame from the peer STA.

The first FTM frame in the FTM session is called the initial FTM (IFTM) frame. The RSTA should transmit an IFTM frame within 10 ms in response to the IFTMR frame. This IFTM frame shall include either an FTM Parameters element or a Ranging Parameters element. The value of the Status Indication field indicates the outcome of the request.

NOTE 1—In an IFTM frame, the RSTA might indicate that the request for an FTM session is successful, even if the IFTM frame includes at least one of the following:

- A Measurement Report element that indicates an unknown LCI.
- A Measurement Report element that indicates an unknown civic location.

NOTE 2—If the ISTA and the RSTA are associated, the channel bandwidth used in the frame exchanges between the ISTA and the RSTA during the ranging measurement exchange is constrained by the rules listed in 10.6.5.7 and in 10.6.6.6.

If the ISTA and the RSTA are associated, a PTKSA is established as defined in 12.6.1.1.6, 12.7.6, 12.11.2, and 13.5. RSN capabilities MFPC and MFPR determine if the PTKSA between RSTA and ISTA is established or not.

If the ISTA and the RSTA are not associated the PTKSA is established using the preassociation security negotiation mechanism as defined in 12.13, and conditions to establish the context are defined below.

A secure FTM session is established when an ISTA and an RSTA establish a PTKSA and use it to exchange a Protected FTM Request Action frame, serving as the IFTMR frame, and the corresponding Protected FTM Action frame, serving as the IFTM frame, and the negotiation completes successfully; see 9.6.34. This applies to the following cases:

- A TB ranging measurement exchange.
- A non-TB ranging measurement exchange.
- An EDCA based ranging measurement exchange with a Format And Bandwidth field indicating DMG or EDMG format; see Table 9-280.

An ISTA shall not initiate FTM negotiation with a Protected FTM Request Action frame, that is, setup a secure FTM session, unless attempting to negotiate one of the cases listed above. A DMG or EDMG STA may use Protected Fine Timing frames in a DMG or EDMG EDCA based ranging session, respectively.

In the cases listed previously:

- If an RSTA has set the URNM-MFPR field in the RSNXE to 1, regardless of the setting in the URNM-MFPR-X20 field, an ISTA shall establish a PTKSA with that RSTA prior to initiating an FTM procedure negotiation with that RSTA.
- If the RSTA has set the URNM-MFPR-X20 field in the RSNXE to 1, an ISTA shall establish a PTKSA with that RSTA prior to initiating an FTM procedure negotiation with that RSTA unless the Format And Bandwidth subfield of the Ranging Parameters field (see 9.4.2.298) in the IFTMR indicates a 20 MHz Bandwidth.
- An RSTA shall reject an FTM request, if a PTKSA was required, and the ISTA has not successfully set up a PTKSA to allow protection of the FTMR frame, FTM and LMR frames exchanged between the RSTA and the ISTA.

The RSTA may accept the request in the cases not listed above.

11.21.6.3.2 Negotiation for EDCA based ranging measurement exchange

The IFTMR frame shall have:

- The Trigger field set to 1
- A set of scheduling parameters that describe the ISTA's availability for measurement exchange

in an FTM Parameters element using an FTM Parameters field present in an IFTMR frame (see 11.21.6.3) and its retransmission, and shall not be present in any subsequent FTM Request frames. If present, it contains the FTM Parameters element as defined in 9.4.2.168.

In the case when an ISTA requests for a format equal to VHT and/or EDCA based HE and bandwidth equal to 160 MHz (80+80, 160 single RF LO, or 160 two separate RF LOs), the RSTA shall assign the same format and bandwidth as requested by the ISTA for 160 MHz bandwidth, otherwise it shall assign format and bandwidth less than or equal to 80 MHz. An ISTA performing an FTM procedure with an RSTA that is an AP shall support non-ASAP operation. An ISTA performing an FTM procedure with an RSTA that is not an AP shall support ASAP operation.

The ISTA shall indicate, in the Format And Bandwidth field, a format and bandwidth that it supports, and this may be different from the BSS bandwidth.

The ISTA shall not indicate No preference in the Format And Bandwidth field unless it supports all formats and bandwidths that are applicable to the FTM session being negotiated.

NOTE 1—The use of phase relations, per subclauses 19.3.11.11.4 and 21.3.7.5, between the 20 MHz parts in non-HT duplicate PPDU that contain Ack frames transmitted by the ISTA is required to enable the RSTA to accurately determine the TOA of these frames, and is thus required for the ISTA to support FTM using Ack frames transmitted in non-HT duplicate PPDU. Alternatively, an ISTA that supports transmission of Ack frames in VHT or HT PPDU in response to FTM frames sent by the RSTA in VHT or HT PPDU, respectively, is not required to support transmission of Ack frames in non-HT duplicate PPDU in response to such FTM frames.

The ISTA shall indicate an EDCA based HE format in the Format And Bandwidth field sent to an RSTA if the STAs are operating in the 6 GHz band, at least one of the STAs does not support TB, or does not support non-TB ranging; otherwise the STA shall not indicate an EDCA based HE format in the Format And Bandwidth field.

A STA that supports TB or non-TB ranging is not required to support a Format And Bandwidth value of EDCA based HE.

For EDCA based ranging where the value of the corresponding Format And Bandwidth field value that indicates DMG or EDMG format, see Table 9-282, the ISTA shall indicate, in the Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element in the IFTMR frame, its ranging priority as described in Table 9-279a. Otherwise, the Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element is reserved.

When the request was successful:

- The RSTA shall indicate, in the Format And Bandwidth field, a format and bandwidth that it supports. The RSTA should indicate the same format and bandwidth in the Format And Bandwidth field as that requested by the ISTA, if the RSTA supports this. The RSTA shall not assign a value that indicates:
 - A bandwidth wider than requested
 - An EDCA based HE format if VHT, DMG, HT mixed, or non-HT format was requested
 - A VHT format if DMG, HT-mixed, or non-HT format was requested
 - An HT format if DMG or non-HT format was requested
 - A DMG format if EDCA based HE, VHT, HT-mixed, or non-HT format was requested
 - An EDCA based HE format only if EDCA based HE was requested, see 26.17.2, and the STA is operating in the 6 GHz band; otherwise the STA shall not indicate EDCA based HE format
 - A VHT (160 with two separate RF LOs) or a VHT (80+80) if a VHT (160 with single RF LO) was requested
 - A VHT (160 with single LO) or a VHT (80+80) if a VHT (160 with two separate RF LOs) was requested
 - A VHT (160 with single LO) or a VHT (160 with two separate RF LOs) if a VHT (80+80) was requested
 - An EDCA based HE (160 with two separate RF LOs) or an EDCA based HE (80+80) if an EDCA based HE (160 with single RF LO) was requested
 - An EDCA based HE (160 with single LO) or an EDCA based HE (80+80) if an EDCA based HE (160 with two separate RF LOs) was requested
 - An EDCA based HE (160 with single LO) or an EDCA based HE (160 with two separate RF LOs) if an EDCA based HE (80+80) was requested
 - An EDMG format if a DMG format was requested
 - An EDMG OFDM format if an EDMG SC format was requested
- An RSTA that is an AP shall support and select non-ASAP operation when so requested by an ISTA.
- An RSTA that is not an AP shall support and select ASAP operation when so requested by an ISTA.
- If the RSTA is ASAP capable, the RSTA's selection of ASAP should be the same as that requested by the ISTA.
- The RSTA's selection of the Min Delta FTM field value shall be greater than or equal to the value requested by the ISTA.
- The RSTA's selection of the Number of Bursts Exponent field value shall be 0 if the ISTA requested it to be 0.
- The RSTA's selection of the Burst Duration field value should be less than or equal to the one requested by the ISTA if the requested FTMs Per Burst field value is set to a value indicating no preference, subject the recommendations below and the RSTA's policy on the maximum and minimum Burst Duration field values.
- If the Number of Bursts Exponent field is set to 0 and the ASAP field is set to 1, the Burst Duration field value should be set to indicate the value BD_1 , defined as follows:

$$BD_1 = ((N_{FTMPB} \times (K + 1)) - 1) \times T_{MDFTM} + T_{FTM} + aSIFSTime + T_{Ack}$$

where

N_{FTMPB}	is the value of the FTMs Per Burst field
K	is the maximum number of FTM frame retransmissions the RSTA might attempt
T_{MDFTM}	is the duration indicated by the Min Delta FTM field of the FTM Parameters field of the IFTM frame (FTM_1)
T_{FTM}	is the duration of the IFTM frame if the FTMs Per Burst field of the FTM Parameters field of FTM_1 is set to 1, and the duration of the non IFTM frame otherwise
T_{Ack}	is the duration of the Ack frame expected as a response

- Otherwise, the Burst Duration field value should be set to indicate a value greater than or equal to the following value:

$$BD_1 + T_{FTMR} + aSIFSTime + T_{Ack} + T_{ACCESS_FTM}$$

where

T_{FTMR}	is the duration of an FTM Request frame without a Measurement Request element and without an FTM Parameters element
T_{ACCESS_FTM}	is the estimated medium access time for the first FTM frame in a burst

- The RSTA's selection of the value of the FTMs Per Burst field should be the same as the one requested by the ISTA if the requested value of the Burst Duration field is set to a value indicating no preference, see Table 9-279, subject to the RSTA's policy on the maximum value of the FTMs Per Burst field.
- The RSTA's selection of Burst Period shall be greater than or equal the RSTA's selection of Burst Duration.
- When the Format And Bandwidth subfield indicates DMG or EDMG format, the RSTA's setting of the Ranging Priority is as defined in Table 9-279b.

NOTE 2—Apart from the Status Indication, Value, ASAP, Number of Bursts Exponent, Min Delta FTM, and Burst Period fields, the other fields in the FTM Parameters element in the IFTM frame have no constraints.

When the RSTA cannot support the initiator's Min Delta FTM or Number of Bursts Exponent constraints, the RSTA shall set the Status Indication field to Request incapable and the FTM session ends. When the RSTA is unable to fulfill the request by the ISTA, the RSTA shall set the Status Indication field to Request failed and the FTM session ends.

11.21.6.3.3 Negotiation for TB and non-TB ranging measurement exchange

For TB and non-TB ranging Measurement exchange the IFTMR frame shall have:

- The Trigger field set to 1.
- A set of scheduling parameters in a Ranging Parameters element that describe the ISTA's availability for measurement exchange.

For TB and non-TB ranging measurement exchange, the IFTM frame shall include a Ranging Parameters element containing either the non-TB specific subelement, or the TB specific subelement.

The ISTA shall include one ISTA Availability Window element in the TB specific subelement in the IFTMR frame indicating its availability for TB ranging as well as the requested periodicity. The periodicity of the availability windows requested by the ISTA is expressed in units of 10 TUs in the Count subfield in the ISTA Availability Information field of the ISTA Availability Window element. The value of the Count subfield in

the ISTA Availability Information field of the ISTA Availability Window element shall be a multiple of the Beacon Interval of the RSTA in units of 10 TUs.

When the RSTA sets the Status Indication field in an IFTM frame to 1 (Successful), it shall not include a Non-TB Specific subelement or TB-specific subelement, if the same subelement was not also included in the IFTMR frame by the ISTA.

When the RSTA includes a TB-specific subelement in an IFTM frame and the Status Indication field in the IFTM frame is equal to 1, then the RSTA shall include an RSTA Availability Window element in the IFTM frame. The RSTA Availability Information field in the RSTA Availability Window element shall contain exactly one Availability Window Information field. The Availability Window Information field represents the availability window assigned by the RSTA to the ISTA. The Availability Window Broadcast Format subfield in the Header subfield in the RSTA Availability Information field in this RSTA Availability Window element is set to 0.

When the RSTA includes a TB-specific subelement in an IFTM frame and the Status Indication field in the IFTM frame is equal to 1, the subelement contains an AID/RSID field assignment to the ISTA. The RSID and the AID are derived from the same ID number space and are non-conflicting.

When the RSTA includes a TB-specific subelement in an IFTM frame and the Status Indication field in the IFTM frame is equal to 1, the RSTA shall assign the value of the Max Session Exp field in the TB Ranging Specific subelement in the Ranging Parameters element in the initial FTM frame. The value of this field shall be larger than the assigned periodicity signaled in the Periodicity subfield in the Availability Window Information field in the TB ranging Specific subelement.

When a Ranging Parameters element is included in the IFTMR frame, the ISTA shall indicate the following parameters in the Ranging Parameters field:

- Maximum supported bandwidth in the Format And Bandwidth subfield.
- Maximum number of LTF repetitions it is capable of receiving in the preamble of the R2I NDP, in the Max R2I Repetition subfield.
- Maximum number of LTF repetitions it is capable of transmitting in the preamble of the I2R NDP, in the Max I2R Repetition subfield.
- Maximum number of space-time streams it is capable of receiving in the R2I NDP for bandwidths less than or equal to 80 MHz, in the Max R2I STS $\leq$ 80 MHz subfield.
- Maximum number of space-time streams it is capable of receiving in the R2I NDP for bandwidths greater than 80 MHz, in the Max R2I STS $>$ 80 MHz subfield.
- Maximum number of space-time streams it is capable of transmitting in the I2R NDP for bandwidths less than or equal to 80 MHz, in the Max I2R STS $\leq$ 80 MHz subfield.
- Maximum number of space-time streams it is capable of transmitting in the I2R NDP for bandwidths greater than 80 MHz, in the Max I2R STS $>$ 80 MHz subfield.
- Maximum number of LTFs in total it is capable of receiving, including all repetitions, in the R2I NDP, in the Max R2I LTF Total subfield.
- Maximum number of LTFs in total it is capable of transmitting, including all repetitions, in the I2R NDP, in the Max I2R LTF Total subfield.
- Immediate or delayed feedback capability in the I2R LMR by setting the Immediate I2R Feedback subfield to 1 or 0, respectively.

When based on the policy at the ISTA, the ISTA does not share measurement results with the RSTA, the ISTA shall set the I2R LMR Feedback subfield in the Ranging Parameters field, in the IFTMR frame, to 0. Otherwise the ISTA shall set the I2R LMR Feedback subfield to 1.

When the I2R LMR Feedback subfield in the IFTMR frame is equal to 0, and the I2R LMR Feedback Policy field in the Extended Capabilities element is set to 1 by the RSTA, then the RSTA shall not reject the request because the I2R LMR Feedback subfield in the Ranging Parameters field was set to 0, and shall set the I2R LMR Feedback in the IFTM frame to 0.

When the I2R LMR Feedback subfield in the IFTMR frame is equal to 0, and the I2R LMR Feedback Policy field in the Extended Capabilities element is set to 0 by the RSTA, then the RSTA shall set the I2R LMR Feedback subfield to 1 to indicate it requests the ISTA to transmit the I2R LMR, or to 0 otherwise. In the former case, the ISTA may either proceed with measurement exchange; see 11.21.6.4.3 and 11.21.6.4.4, or terminate the FTM session (see 11.21.6.6).

When the I2R LMR Feedback subfield in the IFTMR frame is equal to 1, then the RSTA shall set the I2R LMR Feedback subfield to 1 to indicate it requests the ISTA to transmit the I2R LMR, or to 0 otherwise.

NOTE 1—The setting of the I2R LMR Feedback subfield to 1 in the Ranging Parameters field in the Ranging Parameters element contained in the IFTMR frame and IFTM frame is based on higher layer agreements.

NOTE 2—Because the FTM procedure executes at the PHY/MAC layer, an RSTA accepting a ranging request despite the ISTA having set the I2R LMR Feedback subfield in the Ranging Parameters field in the IFTMR frame to 0 enables use cases where the ISTA might share its location information at a higher layer.

When the ISTA sets the I2R AOA Requested subfield to 1 in the IFTMR frame, the RSTA may set the I2R AOA Requested subfield in the corresponding IFTM frame to 1 to confirm that the ISTA shall deliver the I2R AOA measurement result in any transmitted I2R LMR, and set to 0 otherwise.

When the ISTA sets the R2I AOA Requested subfield to 1 in the IFTMR frame, the RSTA may set R2I AOA Requested subfield in the corresponding IFTM frame to 1 to confirm that the RSTA shall deliver R2I AOA measurement results in the R2I LMR, and set to 0 otherwise.

When the ISTA indicated support to announce the TX power of its I2R NDPs by setting the I2R Tx Power field in the Non-TB specific subelement of the Ranging Parameters element in the IFTMR frame to 1, the RSTA may set the corresponding field in the IFTM frame to 1 to request the ISTA to announce the TX power of its I2R NDs, and 0 otherwise.

When the ISTA requested the RSTA to announce the TX power of its R2I NDPs by setting the R2I Tx Power field in the Non-TB Specific subelement to 1, the RSTA may set the corresponding field in the IFTM frame to 1. Otherwise, the RSTA shall set the R2I Tx Power field to 0.

For TB ranging and non-TB ranging, the ISTA shall indicate, in the Ranging Priority subfield of the Ranging Parameters field of the Ranging Parameters element in the IFTMR frame, its Ranging Priority according to Table 9-279a.

For TB ranging, the RSTA shall indicate, in the Ranging Priority subfield of the Ranging Parameters field of the Ranging Parameters element in the IFTM frame, whether it accommodates the ranging priority request transmitted by the ISTA according to Table 9-279b.

When the negotiation is successful for TB ranging and non-TB ranging, the corresponding IFTM frame from the RSTA shall include a Ranging Parameters element with the parameters that define the negotiated range measurement session. The RSTA shall indicate the following parameters in the Ranging Parameters field:

- In the Format And Bandwidth subfield, it assigns the maximum allowed bandwidth used during measurement exchanges (referred to as RSTA Assigned Max Bandwidth). This value shall not be greater than the value in the corresponding IFTMR frame.

- In the Max R2I Repetition subfield, it assigns the maximum number of HE-LTF repetitions in the preamble of the R2I NDP for this session (referred to as RSTA Assigned R2I Rep). This value shall not be greater than the value in the corresponding IFTMR frame.
- In the Max I2R Repetition subfield, it assigns the maximum number of HE-LTF repetitions in the preamble of the I2R NDP for this session (referred to as RSTA Assigned I2R Rep). This value shall not be greater than the value in the corresponding IFTMR frame.
- In the Max R2I STS ≤ 80 MHz subfield, either the maximum number of space-time streams it is capable of transmitting in the R2I NDP for bandwidths less than or equal to 80 MHz, or the value in the corresponding IFTMR frame, whichever is smaller (referred to as RSTA Assigned R2I STS ≤ 80 MHz).
- In the Max R2I STS > 80 MHz subfield, either the maximum number of space-time streams it is capable of transmitting in the R2I NDP for bandwidths greater than 80 MHz, or the value in the corresponding IFTMR frame (referred to as RSTA Assigned R2I STS > 80 MHz).
- In the Max I2R STS ≤ 80 MHz subfield, either the maximum number of space-time streams it is capable of receiving in the I2R NDP for bandwidths less than or equal to 80 MHz, or the value in the corresponding IFTMR frame, whichever is smaller (referred to as RSTA Assigned I2R STS ≤ 80 MHz).
- In the Max I2R STS > 80 MHz subfield, either the maximum number of space-time streams it is capable of receiving in the I2R NDP for bandwidths greater than 80 MHz, or the value in the corresponding IFTMR frame, whichever is smaller (referred to as RSTA Assigned I2R STS > 80 MHz).
- In the Max R2I LTF Total subfield, either the maximum number of LTFs in total it is capable of transmitting to this ISTA, including HE-LTF repetitions, in the R2I NDP, or the value in the corresponding IFTMR frame, whichever is smaller (referred to as RSTA Assigned R2I LTF Total).
- In the Max I2R LTF Total subfield, either the maximum number of LTFs in total it is capable of receiving, including HE-LTF repetitions, in the I2R NDP, or the value in the corresponding IFTMR frame, whichever is smaller (referred to as RSTA Assigned I2R LTF Total).
- Immediate or delayed feedback in the R2I LMR by setting the Immediate R2I Feedback subfield to 1 or 0, respectively.

An ISTA and an RSTA may negotiate a phase shift feedback mode of the non-TB ranging and TB ranging measurement exchange (11.21.6.4.3), for either the R2I LMR and/or I2R LMR. In this case, instead of the TOA t_2 of the I2R NDP, the R2I LMR carries the phase shift tp_2 of I2R NDP and instead of the TOA t_4 of the R2I NDP, the I2R LMR carries phase shift tp_4 of R2I NDP. The ISTA and RSTA can use Equation (11-6c) and Equation (11-6d) to derive the RTT.

NOTE 3— tp_2 and tp_4 are reported in the same units of time (not angle) as the TOA; see 9.6.7.48.

If an RSTA has set the Phase Shift TOA Feedback Support field to 1 in the Extended Capabilities element, then to request phase shift TOA feedback in the R2I LMR, an ISTA shall set the R2I TOA Type subfield in the Ranging Parameter field in an IFTMR frame to 1. To assign phase shift TOA feedback in the R2I LMR the RSTA shall set the R2I TOA subfield in the Ranging Parameter field in an IFTM frame to 1, otherwise it shall set it to 0. If the RSTA sets the R2I TOA Type subfield in the Ranging Parameter field in an IFTM frame to 1, the RSTA shall report the phase shift TOA feedback (tp_2 of the I2R NDP) in the R2I LMR.

An ISTA that has set the I2R LMR feedback subfield in the Ranging Parameters field in an IFTMR frame to 1, shall set the I2R TOA Type subfield to 1 to indicate that it supports phase shift TOA feedback in the I2R LMR, and set it to 0 otherwise. When the ISTA indicates that it does not support phase shift TOA feedback in the I2R LMR, then the RSTA shall set the I2R Type subfield to 0 in the IFTM frame. When the ISTA indicates support for phase shift TOA feedback in the I2R LMR, the RSTA shall set the I2R TOA Type subfield in the Ranging Parameters field of an IFTM frame to 1 to assign phase shift TOA feedback in the I2R LMR, otherwise it shall

set it to 0. If the RSTA sets the I2R TOA Type subfield in the Ranging Parameters field of an IFTM frame to 1, the ISTA shall report phase shift TOA feedback (tp4 of the R2I NDP) in the I2R LMR.

An RSTA shall reject a request for TB ranging from an ISTA if the RSTA cannot assign the ISTA to an availability window that overlaps with a 10 TU interval in which the ISTA is available (as signaled by the ISTA Availability Window element in the IFTMR frame).

When an RSTA sets the Status Indication field in an IFTM frame to 1 (Successful), it shall include an FTM Synchronization Information element in this IFTM frame.

When the RSTA includes a TB-specific subelement in an IFTM frame and the Status Indication field in the IFTM frame is equal to 2 (Request incapable) or 3 (Request failed), then the RSTA may include an RSTA Availability Window element in the IFTM frame. The RSTA Availability Information field in the RSTA Availability Window element shall contain one or more Availability Window Information fields. Each Availability Window Information field represents an availability window that the RSTA can assign to that ISTA if requested by the ISTA in future. The passive TB ranging availability window bit in this Availability Window Information subfield is set to 0.

When Management Frame Protection is negotiated for TB and non-TB ranging, a STA shall:

- Use Protected Fine Timing Action frames for FTMR frames, FTM frames, and LMR frames for ranging negotiation and measurement
- Discard unprotected variants of these frames upon reception

An RSTA shall reject a request, unless the request is for passive TB ranging, if it has set the URNM-MFPR field of the RSNXE to 1, and the ISTA has not successfully set up a PTKSA to protect IFTMR, IFTM, and LMR frames exchanged between the RSTA and the ISTA.

Upon reception of an IFTMR frame with the Format And Bandwidth subfield set to a value of 3, 4, or 5 representing the ISTA's support for one of the 160 MHz BW options, the RSTA shall respond with the same requested value in the Format And Bandwidth subfield in the IFTM frame, if it supports the requested 160 MHz BW option, otherwise respond with a value less than 3.

NOTE 4—The PTKSA can either be established as a result of a successful association between the RSTA and ISTA; or as a result of the ISTA successfully completing PASN as described in 12.13.

11.21.6.3.4 Negotiation for secure HE-LTF in the TB and non-TB ranging measurement exchange

An ISTA and an RSTA may activate a secure HE-LTF measurement exchange for non-TB ranging or TB ranging that uses randomized LTF sequences in the I2R NDPs and R2I NDPs, refer to 11.21.6.4.5.

An RSTA in which dot11SecureLTFImplemented is true shall set the Secure HE-LTF Support field in the RSNXE to 1. An ISTA in which dot11SecureLTFImplemented is true shall include the Secure HE-LTF subelement in the Ranging Parameters element in an IFTMR frame.

When an RSTA has set the Secure HE-LTF Support field to 1 in the RSNXE, then to request a secure HE-LTF measurement exchange mode with the RSTA, an ISTA with dot11SecureLTFImplemented equal to true shall include the Secure HE-LTF subelement in the Ranging Parameters element in the IFTMR frame and set the value of the Secure HE-LTF Required field in the Secure HE-LTF subelement to 1.

When an ISTA has included the Secure HE-LTF subelement in the Ranging Parameters element in its IFTMR frame and set the value of the Secure HE-LTF Required field to 1, then the RSTA that has set the value of the Status Indication field to 1 (i.e., Successful) shall include the Secure HE-LTF subelement in the Ranging

Parameter element in its IFTM frame and set the value of the Secure HE-LTF Required field to 1 to assign a secure HE-LTF measurement exchange mode with the ISTA.

When an ISTA has included the Secure HE-LTF subelement in the Ranging Parameters element in an IFTMR frame, then to assign a secure HE-LTF measurement exchange mode with the ISTA, an RSTA with `dot11SecureLTFImplemented` equal to true shall include a Secure HE-LTF subelement in the Ranging Parameters element in an IFTM and set its Secure HE-LTF Required field to 1.

When an ISTA has included the Secure HE-LTF subelement in the Ranging Parameters element in its IFTMR frame and set the value of the Secure HE-LTF Required field to 1, the ISTA shall set the Max R2I Repetition and Max I2R Repetition subfields to a value greater than 0.

When an RSTA assigns a secure HE-LTF measurement exchange by including a Secure HE-LTF subelement in the Ranging Parameters element in its IFTM frame and setting the Secure HE-LTF Required subfield to 1, the RSTA shall set the *RSTA Assigned R2I Rep* to the Max R2I Repetition subfield value in the IFTMR frame which shall be greater than 0, and the RSTA shall set *RSTA Assigned I2R Rep* to a value greater than 0 and less than or equal to the Max I2R Repetition subfield value in the IFTMR frame, where the *RSTA Assigned R2I Rep* and *RSTA Assigned I2R Rep* specify the number of HE-LTF repetitions in the preamble of the R2I and I2R NDP for this session, respectively.

When an ISTA has included the Secure HE-LTF subelement in the Ranging Parameters element in the IFTMR frame, it shall indicate to the RSTA a request to use the optional frequency domain Tx window in the R2I NDPs by setting the R2I Tx Window field to 1, otherwise shall set it to 0; and indicate its support of the optional frequency domain Tx window in the I2R NDPs by setting the I2R Tx Window field to 1, otherwise shall set it to 0.

When an RSTA has assigned a secure HE-LTF measurement exchange mode, by including a Secure HE-LTF subelement in the Ranging Parameters element in the IFTM frame, and setting its Secure HE-LTF Required field to 1, the RSTA shall also include a Secure HE-LTF Parameters element in the IFTM frame that contains the values of the Validation SAC and Secure HE-LTF Counter subfields.

The secure HE-LTF negotiation supports negotiation of the secure HE-LTF protocol version; see Table 9-322ao, which allows an RSTA and ISTA to negotiate the highest mutually supported secure HE-LTF protocol version. The secure HE-LTF protocol version is indicated in the Protocol Version field within the Secure HE-LTF subelement. If an RSTA includes a Secure HE-LTF element in the IFTM frame, it shall set the Protocol Version subfield to either the same or a lower value than the ISTA's Secure Protocol Version sent in the IFTMR frame with the Status Indication subfield equal to 1 (i.e., Success), or a higher value than the ISTA's Secure Protocol Version with the Status Indication subfield equal to 2 (i.e., 'Request Incapable. Do not send same request again'). Upon receiving an IFTM frame with Status Indication subfield set to 1 (i.e., successful) that includes a Secure HE-LTF subelement, the ISTA shall either proceed with the RSTA's Protocol Version assignment, or if it does not meet its required security level, shall terminate by transmitting an FTMR frame with trigger field set to 0.

NOTE—The secure HE-LTF protocol version negotiation provides user, or system, feedback about the protocol version used, as reported by the MLME. If the negotiation results in termination, user feedback can also be given about the mismatch of versions, helping them understand why it has failed, and what they need to do to address the problem.

When an RSTA has initiated a secure measurement exchange, by including a Secure HE-LTF subelement in the Ranging Parameters element in the IFTM frame, and set the Secure HE-LTF Required field to 1, then it shall indicate to the ISTA whether the I2R and R2I NDPs will use the optional frequency domain Tx window by setting the I2R Tx Window and R2I Tx Window fields in the Secure HE-LTF subelement respectively; see Figure 9-788et.

- If the ISTA had set the I2R Tx Window field to value 1 in the IFTMR frame and the RSTA sets the corresponding I2R Tx Window field in the IFTM frame to value 1, the ISTA shall use the optional frequency domain Tx Window in the I2R NDPs.
- If the ISTA had set the R2I Tx Window field to value 1 in the IFTMR frame and the RSTA sets the corresponding R2I Tx Window field to value 1 in the IFTM frame, the RSTA shall use the optional frequency domain Tx window in the R2I NDPs.

The Measurement SAC field in the Secure HE-LTF Parameters element is reserved in any IFTM frame.

When `dot11SecureLTFImplemented` is true, prior to generating the new secure HE-LTF counter for a given PTKSA, the RSTA initializes a monotonically increasing 48-bit counter Secure-LTF-Counter to 0. The RSTA also derives a Secure-LTF-Key-Seed as follows:

$$\text{Secure-LTF-Key-Seed} = \text{HMAC-Hash}(\text{KDK}, \text{"Secure HE-LTF key seed"})$$

Where the KDK is derived as part of PTKSA establishment, Hash is the hash determined by the AKM and used to derive the PTK. If the Extended Key ID for Individually Addressed Frames subfield of the RSN Capabilities field is 1 for both the Authenticator and the Supplicant, there may be multiple KDKs each identified by a different Key ID (0 or 1). In such a case, the KDK chosen to derive the Secure-LTF-Key-Seed shall be the most recent.

Similarly, when `dot11SecureLTFImplemented` is true, an ISTA also derives the same Secure-LTF-Key-Seed prior to initiating a secure FTM negotiation.

For each secure FTM negotiation, prior to transmission of IFTM, and for each measurement within an FTM session, an RSTA shall increment the Secure-LTF-Counter by 1. The Secure-LTF-Counter is included in the Secure HE-LTF Parameters element conveyed to the ISTA. An ISTA determines the Secure-LTF-Counter to use for a measurement using the Secure HE-LTF Counter and Validation SAC received from the RSTA in the most recent protected IFTM or LMR frame. The Secure-LTF-Counter shall be reset when a new KDK is derived as part of a new PTK derivation and it shall continue, and not be reset, for each secure FTM negotiation using a given KDK.

11.21.6.3.5 Capability negotiation for EDCA based ranging with the Format And Bandwidth field set to value in range 31–43

A STA that supports first Path Beamforming Training shall set the First Path Beamforming Training Supported field of the Beamforming Capability subelement in the EDMG Capabilities element to 1. Otherwise it shall set the First Path Beamforming Training Supported field to 0.

A STA that supports secure RTT measurement as described in 11.21.6.4.2.7 shall set the Secure RTT Supported field in the RSNXE to 1. Otherwise, it shall set the Secure RTT Supported field to 0. A STA shall not set the Secure RTT Supported field in the RSNXE to 1 if it has not also set the First Path Beamforming Training Supported field of the Beamforming Capability subelement in the EDMG Capabilities element to 1.

A STA that supports EDMG SC Ranging shall set the EDMG SC Ranging Supported field of the Beamforming Capability subelement in the EDMG Capabilities element to 1. Otherwise it shall set the EDMG SC Ranging Supported field to 0.

A STA that supports EDMG OFDM Ranging shall set the EDMG OFDM Ranging Supported field of the Beamforming Capability subelement in the EDMG Capabilities element to 1. Otherwise, it shall set the EDMG OFDM Ranging Supported field to 0; see 11.21.6.3.6.

11.21.6.3.6 Negotiation for secure EDMG TRN in EDCA based ranging measurement

An ISTA may request a Secure RTT measurement by setting the Secure RTT Measurement subfield in the Measurement Parameters field in the initial Protected FTM Request frame. An ISTA shall not set the Secure RTT Measurement subfield to 1 in a request to an RSTA if the RSTA has not set the Secure RTT Supported field in the RXSNE. An ISTA that sets the Secure RTT Measurement subfield to 1 shall generate a 32-octet random Secret Key and include it in the Secure Ranging Operation Parameters field; see Figure 9-626d, in the initial Protected FTM Request frame. An RSTA that supports Secure RTT measurement shall acknowledge a request for Secure RTT measurement by setting the Secure RTT Measurement subfield in the Measurement Parameters field in the initial Protected FTM frame.

11.21.6.3.7 Direction measurement negotiation for DMG STAs

A STA that supports inclusion of a TRN field in the transmitted FTM frame and the corresponding acknowledgment to facilitate angle of arrival measurements at the recipient shall set the AOA TX Capability subfield in the DMG Direction Measurement Capabilities field in the DMG Capabilities element to 1. Otherwise it shall set the AOA TX Capability subfield to 0.

A STA that supports angle of arrival estimation using the TRN field included in the received FTM frame or the corresponding acknowledgment shall set the AOA RX Capability subfield in the DMG Direction Measurement Capabilities field in the DMG Capabilities element to 1. Otherwise, it shall set the AOA RX Capability subfield to 0.

A STA that supports inclusion of direction setting specific TRN field(s) in the transmitted FTM frame and the corresponding acknowledgment to facilitate angle of departure estimation(s) at the recipient shall set the AOD TX Capability subfield in the DMG Direction Measurement Capabilities field in the DMG Capabilities element to 1. Otherwise, it shall set the AOD TX Capability subfield to 0.

A STA that supports angle of departure estimation using the TRN field(s) included in the received FTM frame or the corresponding acknowledgment shall set the AOD RX Capability subfield in the DMG Direction Measurement Capabilities field in the DMG Capabilities element to 1. Otherwise, it shall set the AOD RX Capability subfield to 0.

A pair of ISTA and RSTA are defined to be mutually direction measurement capable if both the ISTA and the RSTA indicated support for DMG direction measurement by including the DMG Direction Measurement Capabilities field in the DMG Capabilities element, and by the ISTA setting at least one of the first four subfields (AOA TX Capability, AOA RX Capability, AOD TX Capability, AOD RX Capability) of this field to 1 and the RSTA setting one the corresponding subfields (AOA RX Capability, AOA TX Capability, AOD RX Capability, AOD TX Capability) of this field to 1. A DMG ISTA may request that FTM exchanges will include direction measurement if the ISTA and RSTA are mutually direction measurement capable.

A DMG ISTA may request initiator AOA measurement, responder AOA measurement, initiator AOD measurement, and responder AOD measurement, by including a DMG Direction Measurement Parameters subelement in the FTM Parameters element transmitted in the FTMR frame. Valid combinations of AOA and AOD requests and the corresponding required capabilities are shown in Table 11-11a. The L-RX field in the DMG Direction Measurement Specific Parameters subelement shall be set to the number of TRN units the ISTA needs for AOA estimation in the case where R2I AOA was requested, otherwise it shall be set to 0.

The RSTA should respond within 10 ms from the IFTMR frame with an IFTM frame with an FTM Parameters element with a DMG Direction Measurement Parameters subelement. The requested AOA/AOD I2R/R2I parameters in the IFTM frame shall be set to the same value as those requested in the IFTMR frame. The L-RX field shall be set to the number of TRN units the RSTA needs for AOA estimation in the case where I2R AOA was requested. The RSTA sets the Direction Measurement Density field.

Table 11-11a—Valid combinations of direction measurements requests and the required capabilities at the initiator and responder

Requested parameters				Required initiator capabilities				Required responder capabilities			
R2I AOA	I2R AOD	I2R AOA	R2I AOD	AOA RX	AOA TX	AOD RX	AOD TX	AOA RX	AOA TX	AOD RX	AOD TX
Y				Y					Y		
	Y						Y			Y	
		Y			Y			Y			
			Y			Y					Y
Y	Y			Y			Y		Y	Y	
		Y	Y		Y	Y		Y			Y
Y		Y		Y	Y			Y	Y		
	Y		Y			Y	Y			Y	Y

The AOA and AOD parameters requested in the IFTMR frame shall be compatible with the corresponding AOA/AOD TX/RX capabilities as shown in Table 11-11a.

11.21.6.3.8 LOS assessment negotiation for EDMG STAs

An EDMG STA that supports LOS assessment by transmitting a LOS Assessment FTM PPDU shall set the dot11LOSAssessmentTXImplemented to true and shall set it to false otherwise.

An EDMG STA that in the role of RSTA supports LOS assessment by switching polarization while receiving a LOS Assessment Ack PPDU and responds with Channel Measurement feedback shall set the dot11LOSAssessmentRXImplemented to true and shall set it to false otherwise.

An EDMG STA for which the dot11LOSAssessmentTXImplemented is true shall set to 1 the LOS Assessment TX Capability subfield in the DMG Direction Measurement Capabilities field.

An EDMG STA for which the dot11LOSAssessmentRXImplemented is true shall set to 1 the LOS Assessment RX capability and the LOS Assessment TX Capability subfields in the DMG Direction Measurement Capabilities field.

An EDMG STA in ISTA role for which the dot11LOSAssessmentTXImplemented is true may establish an FTM session that contains LOS assessment exchanges with another EDMG STA that will take the RSTA role if that STA's LOS Assessment RX capability subfields in the DMG Direction Measurement Capabilities field is set to 1.

The ISTA requests the FTM session by setting the LOS Assessment field to 1 in a DMG Direction Measurement Parameters subelement in the IFTMR frame. The responding RSTA shall set to 1 the LOS Assessment field in the DMG Direction Measurement Parameters element in the IFTM frame.

11.21.6.3.9 Passive TB ranging measurement negotiation

The passive TB ranging measurement negotiation follows the rules and procedures for the TB ranging measurement negotiation detailed in 11.21.6.3.3, unless explicitly stated otherwise.

A STA in which `dot11PassiveTBRangingResponderImplemented` is true shall set the Passive TB Ranging Responder Measurement Support field in the Extended Capabilities element to 1.

When an RSTA has set the Passive TB Ranging Responder Measurement Support field to 1 in the Extended Capabilities element it transmits, an ISTA with `dot11PassiveTBRangingInitiatorImplemented` equal to true may set the Passive TB Ranging field in the TB specific subelement in an IFTMR frame to 1 to request a passive TB ranging measurement session between the ISTA and the RSTA.

When requesting, or responding to a request for, passive TB ranging, the I2R LMR Feedback subfield in the Ranging Parameters field of the Ranging Parameters element in the IFTMR and IFTM frames, respectively, is reserved.

NOTE—This is because in passive TB ranging, the transmission of the ISTA Passive TB Ranging Measurement Report frame is mandatory; see 11.21.6.4.8.4.

To assign an ISTA passive TB ranging session, the RSTA shall respond with the Passive TB Ranging subfield in the Ranging Parameters field set to 1 in the corresponding IFTM frame.

When an ISTA sets the Passive TB Ranging field in the TB specific subelement in an IFTMR frame to 1, the ISTA shall not include the Secure HE-LTF subelement in the Ranging Parameters element in an IFTMR frame.

The negotiation of phase shift feedback in passive TB ranging follows the same behavior as for TB ranging.

If the ISTA sets the Passive TB Ranging field to 1, it shall include an LCI report in the FTM Request frame.

11.21.6.4 Measurement exchange

Insert 11.21.6.4.1 at the beginning of 11.21.6.4 (before the current content of 11.21.6.4) as follows:

11.21.6.4.1 FTM measurement exchange overview

FTM measurement has three basic ranging mechanisms:

- EDCA based ranging described in 11.21.6.4.2
- TB ranging described in 11.21.6.4.3 and 11.21.6.4.8
- Non-TB ranging described in 11.21.6.4.4

The frequency of the clock for the FTM timestamps shall be derived from the same reference oscillator as the transmit center frequency and the symbol clock frequency.

NOTE—The transmit center frequency and symbol clock frequency are derived from the same reference oscillator, as per the specifications for the different PHYs.

Insert new subclause headings for 11.21.6.4.2 and 11.21.6.4.2.1 after 11.21.6.4.1 (thereby making what was previously the content of 11.21.6.4, now the content of 11.21.6.4.2.1) as follows:

11.21.6.4.2 EDCA based ranging measurement exchange

11.21.6.4.2.1 Common behavior for EDCA based ranging measurement exchange

Change the first four paragraphs of former 11.21.6.4 (now 11.21.6.4.2.1) as follows:

The time windows during which TOD measurement for the transmission of the ~~Fine Timing Measurement~~ FTM frame and the TOA measurement on the receipt of the acknowledgment corresponding to the transmitted ~~Fine Timing Measurement~~ FTM frame are performed are known as burst instances. ~~Fine Timing Measurement~~

FTM frames over which TOD and TOA measurements are performed shall not be sent outside burst instances. The timing of the burst instances is defined by the following parameters:

- Partial TSF Timer: The partial TSF timer, as defined in 9.4.2.167, at the beginning of the first burst instance.
- Burst Duration: The duration of each burst instance starting at the boundary of a burst period.
- Burst Period: The interval from the beginning of one burst instance to the beginning of the following burst instance.

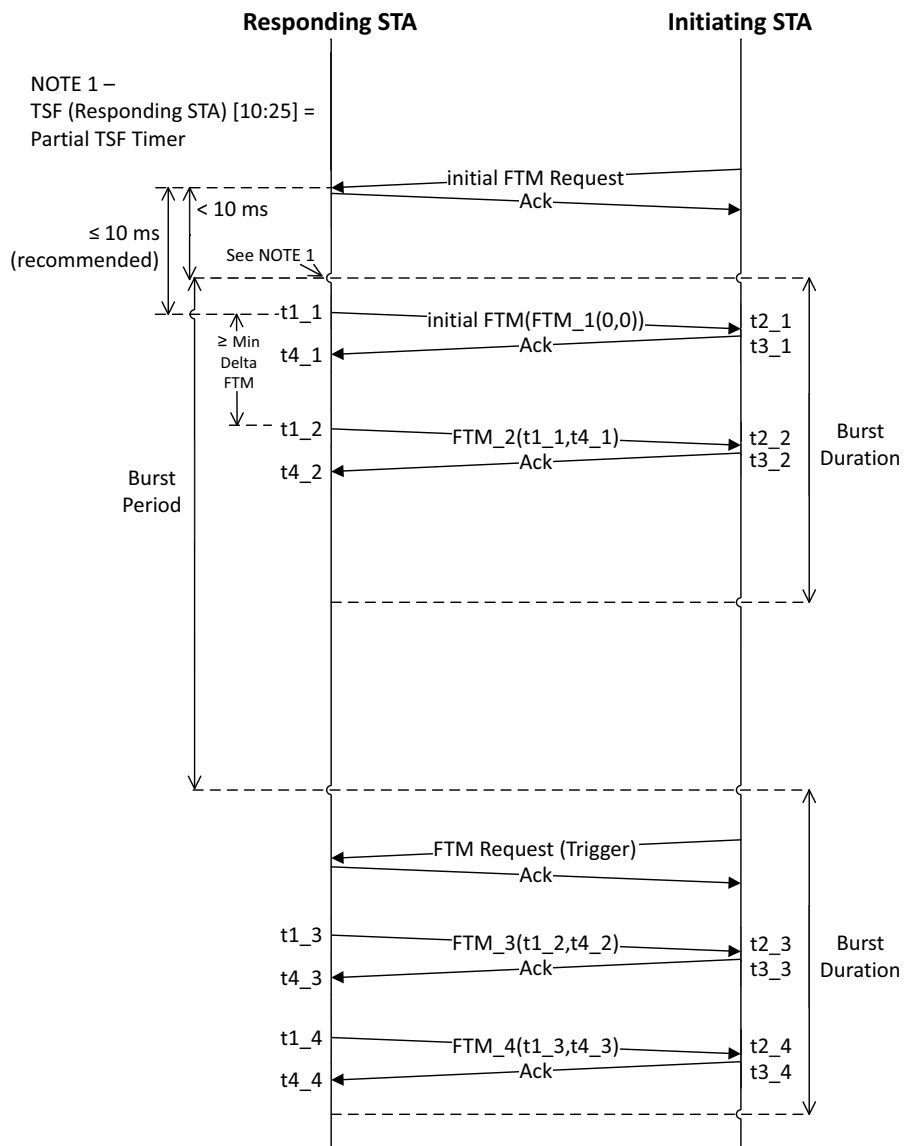
The ~~initiating STA, ISTA~~ shall transmit a ~~Fine Timing Measurement Request~~ an FTMR frame without a Measurement Request element, without a ~~Fine Timing Measurement~~ an FTM Parameters element and with the Trigger field set to 1 as soon as it is available on channel during a burst instance. This indicates to the ~~responding STA, RSTA~~ its availability for the remainder of the burst instance. In response, the ~~responding STA, RSTA~~ shall transmit an Ack frame and should transmit FTMs Per Burst ~~Fine Timing Measurement~~ FTM frames before the end of the burst instance. In addition, the ~~initiating STA, ISTA~~ shall be ready to receive a ~~Fine Timing Measurement~~ an FTM frame. The first ~~Fine Timing Measurement~~ FTM frame and its retransmissions in the burst instance from the ~~responding STA, RSTA~~ shall include an FTM Synchronization Information element in the FTM Synchronization Information field. The TSF Sync Info field in the FTM Synchronization Information element contained in the ~~Fine Timing Measurement~~ FTM frame might be used by the ~~initiating STA, ISTA~~ to synchronize its TSF with the ~~responding STA's, RSTA's~~ in order to determine the start of subsequent burst instances. Subsequent ~~Fine Timing Measurement~~ FTM frames within the burst instance shall not include a ~~Fine Timing Measurement~~ an FTM Parameters element and shall not include an FTM Synchronization Information field (for additional constraints on the ~~Fine Timing Measurement~~ FTM frame see 11.21.6.7). Consecutive ~~Fine Timing Measurement~~ FTM frames to a given peer STA shall be spaced at least Min Delta FTM apart. Within a burst instance the ~~initiating STA, ISTA~~ shall perform FTM on each ~~Fine Timing Measurement~~ FTM frame addressed to it, except the last ~~Fine Timing Measurement~~ FTM frame. The ~~initiating STA, ISTA~~ may perform FTM on the last ~~Fine Timing Measurement~~ FTM frame in a burst instance.

NOTE 1—If the ~~initiating STA, ISTA~~ successfully transmits a ~~non-initial Fine Timing Measurement Request~~ non-IFTMR frame late in a burst instance, fewer than FTMs per burst might be successfully transmitted by the ~~responding STA, RSTA~~ in the burst instance. If a ~~Fine Timing Measurement~~ an FTM frame, except for the ~~initial Fine Timing Measurement~~ IFTM frame in the ASAP=0 case, is sent outside a burst instance, it might not be acknowledged.

The first burst instance shall start at the value indicated by the Partial TSF Timer subfield in the ~~initial Fine Timing Measurement~~ IFTM frame, regardless of the ASAP field's value. When ASAP is set to 1 by the ~~responding STA, RSTA~~, the Partial TSF Timer subfield value shall be set to a value less than 10 ms from the reception of the most recent ~~initial Fine Timing Measurement Request~~ IFTMR frame.

The scheduling parameters of an FTM session are illustrated in Figure 11-35.

Replace Figure 11-35 with the following figure:



**Figure 11-35—Example negotiation and measurement exchange sequence, ASAP = 0,
and FTMs Per Burst = 2**

Change the fifth paragraph of former 11.21.6.4 (now 11.21.6.4.2.1) as follows:

The ~~initiating STA, ISTA~~ can request the ~~responding STA, RSTA~~ to start the burst instances “as soon as possible” by setting the ASAP subfield to 1. The scheduling in this case is illustrated in Figure 11-36.

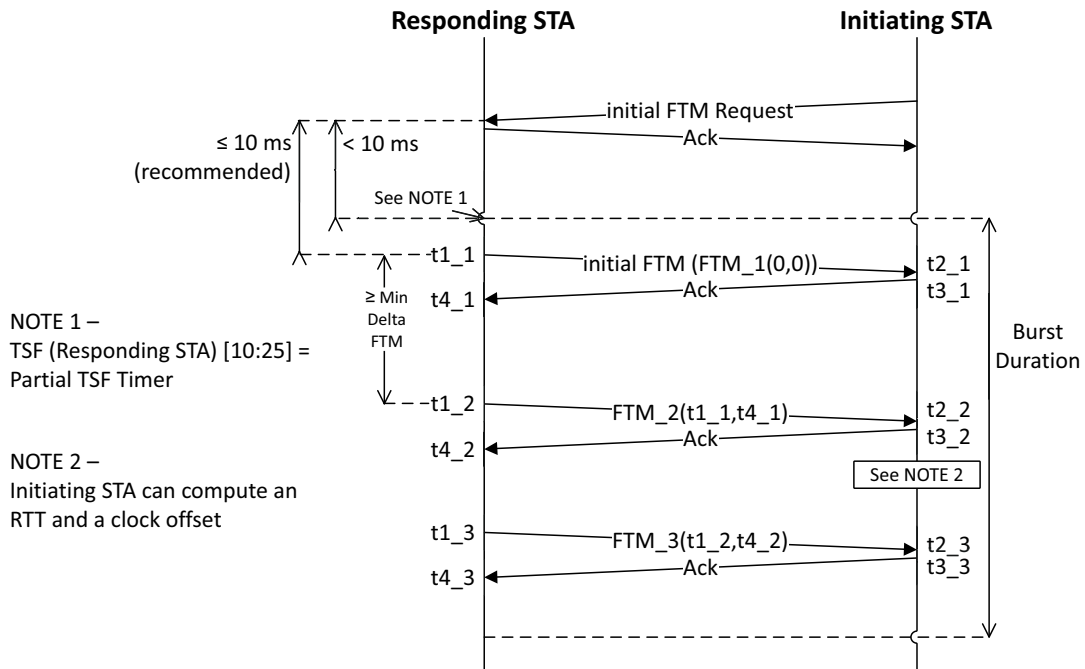


Figure 11-37—Example negotiation and measurement exchange sequence for a single burst instance, ASAP=1, and FTMs Per Burst = 3

For non-DMG STAs, both the ~~Fine Timing Measurement~~ FTM frame and the corresponding Ack frame shall be transmitted using a single transmit chain.

The ~~responding STA RSTA~~ shall not transmit ~~Fine Timing Measurement~~ FTM frames using Clause 15 or Clause 16 formats, MCS 32 format, or HT-greenfield format.

The ~~responding STA RSTA~~ should transmit ~~Fine Timing Measurement~~ FTM frames with the format and bandwidth it indicated in the ~~initial Fine Timing Measurement~~ IFTM frame.

For the ~~Fine Timing Measurement~~ FTM frames transmitted during the FTM session:

- The ~~responding STA RSTA~~ shall not use a bandwidth wider than it indicated in the ~~initial Fine Timing Measurement~~ IFTM frame.
- The ~~responding STA RSTA~~ shall not use an HE format if it indicated VHT, HT-mixed, or non-HT format in the ~~initial Fine Timing Measurement~~ IFTM frame.
- The ~~responding STA RSTA~~ shall not use a VHT format if it indicated HT-mixed or non-HT format in the ~~initial Fine Timing Measurement~~ IFTM frame.
- The ~~responding STA RSTA~~ shall not use an HT format if it indicated non-HT format in the ~~initial Fine Timing Measurement~~ IFTM frame.

The ASAP Capable subfield shall be set to 1 in the ~~initial Fine Timing Measurement~~ IFTM frame if a ~~responding STA RSTA~~ is ASAP-capable; otherwise it shall be set to 0.

A ~~responding STA RSTA~~ transmits ~~Fine Timing Measurement~~ FTM frames in overlapping pairs of consecutive frames. For example, in Figure 11-36, FTM_1 and FTM_2, FTM_2 and FTM_3, and FTM_3 and FTM_4 are overlapping pairs of consecutive frames. The first ~~Fine Timing Measurement~~ FTM frame of a pair of consecutive ~~Fine Timing Measurement~~ FTM frames contains a nonzero value in the Dialog Token field. The

follow up ~~Fine Timing Measurement-FTM~~ frame contains a Follow Up Dialog Token field set to the value of the Dialog Token field in the first frame of the consecutive pair. Dialog Token field values of consecutive ~~Fine Timing Measurement-FTM~~ frames shall be consecutive, except when the value wraps around to 1 or in the last ~~Fine Timing Measurement-FTM~~ frame in an FTM session. With the first ~~Fine Timing Measurement-FTM~~ frame, both STAs capture timestamps. The ~~responding STA RSTA~~ captures the time at which the ~~Fine Timing Measurement-FTM~~ frame is transmitted ($t1$). The ~~initiating STA ISTA~~ captures the time at which the ~~Fine Timing Measurement-FTM~~ frame arrives ($t2$) and the time at which the Ack response is transmitted ($t3$). The ~~responding STA RSTA~~ captures the time at which the Ack frame arrives ($t4$). See Figure 6-17. In the follow up ~~Fine Timing Measurement-FTM~~ frame, in the same or the subsequent burst, the ~~responding STA RSTA~~ transfers the timestamp values it captured ($t1$ and $t4$) to the ~~initiating STA ISTA~~. In this follow up ~~Fine Timing Measurement-FTM~~ frame, the timestamp values ($t1$ and $t4$) shall be the measurement according to the ~~responding STA's RSTA's~~ clock (i.e., without applying any frequency offset correction to the time bases).

NOTE 2—A ~~Fine Timing Measurement-An FTM~~ frame can contain nonzero values in both the Dialog Token and Follow Up Dialog Token fields, meaning that the Action frame contains follow up information from a previous measurement, and new Timestamp values are captured to be sent in a future follow up ~~Fine Timing Measurement-FTM~~ frame.

When the ASAP subfield is set to 0 by a ~~responding STA RSTA~~, the Follow Up Dialog Token, TOD, TOA, TOD Error, and TOA Error fields in the ~~Fine Timing Measurement-FTM~~ frame following the ~~initial Fine Timing Measurement-FTM~~ frame shall be reserved.

When the ASAP subfield is set to 1 by a ~~responding STA RSTA~~, the timestamps for the ~~initial Fine Timing Measurement-FTM~~ frame shall be captured and sent in the following ~~Fine Timing Measurement-FTM~~ frame. In the ~~Fine Timing Measurement-FTM~~ frame following the ~~initial Fine Timing Measurement-FTM~~ frame, the TOD and TOA fields shall be set to the timestamps associated with the ~~initial Fine Timing Measurement-FTM~~ frame.

The round trip time (RTT) is defined by Equation (11-5).

$$RTT = [(t4' - t1') - (t3 - t2)] \quad (11-5)$$

where $t1'$ and $t4'$ are the time at which the ~~Fine Timing Measurement-FTM~~ frame was transmitted and the time at which the Ack was received, respectively, as determined by the ~~initiating STA ISTA~~.

At the ~~initiating STA ISTA~~, the mechanism by which $t1'$ and $t4'$ are derived from the TOD and TOA fields is implementation dependent.

The FTM procedure can also be used to synchronize a local clock between STAs. One higher-layer protocol for synchronizing a local clock time between STAs is specified in IEEE Std 802.1AS.

The SME at the ~~initiating STA ISTA~~ may estimate the offset of the local clock relative to that at the ~~responding STA RSTA~~ using clock offset as defined by Equation (11-6).

$$clock\ offset = [(t2 - t1') - (t4' - t3)]/2 \quad (11-6)$$

NOTE 3—The ~~initiating STA ISTA~~ might track this clock offset over time to derive an estimate of the difference between the ~~initiating STA's ISTA's~~ time base and the ~~responding STA's RSTA's~~ time base, and thereby improve the accuracy of its derivation of $t1'$ and $t4'$ from the TOD and TOA fields.

If the Ack frame for a transmitted ~~Fine Timing Measurement-FTM~~ frame is not received, the ~~responding STA RSTA~~ shall not retry the frame. In order to send the frame again, the ~~responding STA RSTA~~ shall send a ~~Fine Timing Measurement-an FTM~~ frame with the same Action frame body as the ~~Fine Timing Measurement-FTM~~ frame for which the Ack was not received, except for updating the Dialog Token if it was nonzero. The Sequence Number in the MAC header is also updated. This is called an *FTM retransmission*.

~~A responding STA RSTA that transmits a Fine Timing Measurement an FTM frame with the ASAP subfield set to 0 shall set the Partial TSF Timer subfield to an offset value D_{TSF} from the partial value of the responding STA's RSTA's TSF timer at the time of the transmission of the Ack to the last Fine Timing Measurement Request FTMR frame from the initiating STA ISTA, subject to~~

$$D_{TSF} \geq K_I \times T_{MDFTM} + T_{FTM_I} + aSIFSTime + T_{Ack} + T_{ACCESS_FTM}$$

where

K_I	is the maximum number of initial Fine Timing Measurement IFTM frame (FTM_1) retransmissions the responding STA RSTA might attempt
T_{MDFTM}	is the value of the Min Delta FTM subfield of the Fine Timing Measurement FTM Parameters field of FTM_1
T_{FTM_I}	is the duration of the FTM_1
T_{Ack}	is the duration of the Ack frame expected as a response
T_{ACCESS_FTM}	is the estimated medium access time for FTM_1

NOTE 4—This Partial TSF Timer subfield ought to result in the ~~Fine Timing Measurement Request FTMR~~ frame not being transmitted before a successful transmission of FTM_1.

When neither an Ack to the ~~initial Fine Timing Measurement IFTM~~ frame nor a ~~Fine Timing Measurement Request an FTMR~~ frame has been received by the ~~responding STA RSTA~~, the ~~responding STA RSTA~~ shall not terminate the FTM session before the time indicated by the Partial TSF timer plus the Burst Duration.

The ~~responding STA RSTA~~ shall set the Follow Up Dialog Token field to 0 in the ~~initial Fine Timing Measurement IFTM~~ frame and its FTM retransmissions.

The ~~responding STA RSTA~~ shall set the Dialog Token field to 0 in the last ~~Fine Timing Measurement FTM~~ frame and its FTM retransmissions.

Insert a new paragraph at the end of former 11.21.6.4 (now 11.21.6.4.2.1) as follows:

When a Secure FTM Session is established as described in 11.21.6.3.1, the FTM frames transmitted during the execution of Measurement Exchange shall be Protected Fine Timing Action frames; see 9.6.34.

Insert 11.21.6.4.2.2 through 11.21.6.4.2.7 (including Table 11-11b) after 11.21.6.4.2.1 as follows:

11.21.6.4.2.2 DMG general measurement exchange

An EDMG ISTA and RSTA that have both indicated support for first path beam forming by setting to 1 the First Path Training Supported field in the Beamforming Capability subelement of the EDMG Capability element shall perform first path beamforming training as defined in 10.42.10.6 prior to any FTM exchange.

A DMG/EDMG ISTA/RSTA performs an FTM exchange that does not require AOA or AOD measurements as defined in 11.21.6.4.2.1. To perform an FTM exchange that does require AOD or AOD measurements, it follows the procedure in 11.21.6.4.2.3. In both these cases, when the first path AWV setting is not used in the exchange, the Trigger field shall be set to 1 in the FTM Request frame that initiates the exchange. In both cases the same AWV used for data transfer between the STAs shall be used for transmission and reception of the preamble and data portion of the PPDU.

An EDMG ISTA initiates an FTM measurement exchange using the first path AWV setting by setting the value of the Trigger field of the FTM Request frame that initiates the exchange to 2. An EDMG ISTA may send an FTM Request frame with the trigger set to 2 only if the RSTA has set the First Path Beamforming Training Supported subfield to 1 in the Beamforming field of the EDMG Capabilities element and the ISTA

and RSTA have performed beamforming training for first path as defined in 10.42.10.6. When the trigger is equal to 2 in the FTM request frame that initiates the burst, the first path AWV settings shall be used in transmission and reception of FTM and Ack frames sent by the ISTA and RSTA during the FTM burst.

An EDMG ISTA sets a value for the Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element in the IFTMR frame from the values in Table 9-279a. The RSTA responds to this request by setting a value for the Ranging Priority subfield of the FTM Parameters field of the FTM Parameters element in the IFTM frame from Table 9-279b in response. How an RSTA supports ranging priority is implementation dependent.

11.21.6.4.2.3 DMG/EDMG AOA/AOD measurement exchange

A DMG ISTA/RSTA pair that has agreed on performing direction measurement by agreeing on either R2I AOA, I2R AOD, I2R AOA, or R2I AOD shall add TRN fields to FTM exchanges in the burst according to the Direction Measurement Density sent by the RSTA in the IFTM frame.

A DMG ISTA/RSTA pair that has agreed on either R2I AOA, I2R AOD, I2R AOA, or R2I AOD using the procedure described in 11.21.6.3.7 shall be denoted as Direction Measurement FTM pair.

An FTM exchange in which the Trigger field in the FTM Request that initiated the exchange is set to 2 is denoted as a first path AWV FTM exchange.

In a Direction Measurement FTM pair, the first measurement in the burst shall include either R2I AOA, I2R AOD, I2R AOA, or R2I AOD for all values of the Direction Measurement Density.

In a Direction Measurement FTM pair that agreed on R2I AOA, the RSTA shall add a TRN field to the FTM frames in the exchanges specified by the Direction Measurement Density field by setting the TRN_LEN/EDMG_TRN_LEN to the value of the L_RX field sent by the ISTA and PPDU_TYPE/EDMG_PPDU_TYPE to 0. In a first path AWV FTM exchange the RSTA shall also set the TXVECTOR parameter FIRST_PATH_AWV_TRN to FIRST_PATH_AWV_ON_TRN in the FTM frames it sends to the ISTA. The ISTA may receive the TRN field using implementation dependent AWV settings.

In a Direction Measurement FTM pair that agreed on I2R AOA, the ISTA shall add a TRN field to the Ack frames in the exchanges specified by the Direction Measurement Density field by setting the TRN_LEN/EDMG_TRN_LEN to the value of the L_RX field of the DMG Direction Measurement Parameters received from the RSTA and PPDU_TYPE/EDMG_PPDU_TYPE to 0. In a first path AWV FTM exchange the ISTA shall also set the TXVECTOR parameter FIRST_PATH_AWV_TRN to FIRST_PATH_AWV_ON_TRN in the Ack frames it sends to the RSTA. The RSTA may receive the TRN field using implementation dependent AWV setting. The RSTA shall provide the AOA measurement results in the Direction Measurement Result element included in the next FTM frame sent to the ISTA.

In a Direction Measurement FTM pair that agreed on R2I AOD, the RSTA shall add a TRN field to the FTM frames in the exchanges specified by the Direction Measurement Density field by setting the TRN_LEN/EDMG_TRN_LEN to a nonzero value and PPDU_TYPE/EDMG_PPDU_TYPE to 1. In a first path AWV FTM exchange the RSTA shall also set the TXVECTOR parameter FIRST_PATH_AWV_TRN to FIRST_PATH_AWV_ON_TRN in the FTM frames it sends to the ISTA. The ISTA shall receive the TRN field using the first path AWV setting. The RSTA may use implementation dependent AWV (such as sectors) in the TRN field.

In a Direction Measurement FTM pair that agreed on I2R AOD, the ISTA shall add a TRN field to the Ack frames in the exchanges specified by the Direction Measurement Density by setting the TRN_LEN/EDMG_TRN_LEN to a nonzero value and PPDU_TYPE/EDMG_PPDU_TYPE to 1. In a first path AWV FTM exchange the ISTA shall also set the TXVECTOR parameter FIRST_PATH_AWV_TRN to FIRST_PATH_AWV_ON_TRN in the Ack frames it sends to the RSTA. The RSTA shall receive the TRN

field using the first path AWV setting. The ISTA may use implementation dependent AWV (such as sectors) in the TRN field. The RSTA shall set the Best AWV ID field in the FTM frames sent to the ISTA following these Ack frames to the AWV ID or the Best Sector Index of the TRN field (if the Ack was an EDMG/DMG PPDU, respectively). If the RSTA has set the AOD Channel Measurement Feedback subfield to 1 in the DMG Direction Measurement Capabilities field, it shall also include a Channel Measurement Feedback Type field and a Channel Measurement Feedback field in the FTM frames sent to the ISTA following the reception of the Ack frames that its RXVECTOR parameter PPDU_TYPE is equal to TRN-T or EDMG_PPDU_TYPE equal to EDMG-TRN-T, and TRN-LEN greater than 0 or EDMG-TRN-LEN greater than 0.

If the Trigger field in the FTMR frame that initiated an FTM exchange in a Direction Measurement FTM pair is equal to 2, all the preamble and PSDU parts of the PDUs sent during the exchange shall be transmitted and received using the first path AWV setting. The TRN fields of these PDUs shall also be transmitted and received using the first path AWV setting, except for AOD TRN fields, which may be transmitted with changing AWV setting, and AOA TRN fields, which may be received with changing AWV setting.

11.21.6.4.2.4 DMG AOD feedback exchange after an FTM exchange

When an ISTA and RSTA agreed on performing an R2I AOD measurement FTM exchange as described in 11.21.6.4.2.3 the ISTA needs to send AOD measurement results to the RSTA, so that the RSTA may use these to generate the AOD estimates and send the results back to the ISTA.

To enable this, the ISTA collects the best sector results throughout the burst.

After the burst the ISTA shall send an FTM frame containing a Multiple AWV ID element with the Trigger field set to 0, to the RSTA. The Number Of Best AWV ID field in the Multiple AWV ID element shall be set to the number of FTM frames in the burst that contained a TRN field enabling AOD estimation. When the RSTA is ready with the AOD results, it should send an FTM frame containing a Multiple AOD Feedback element containing the AOD results. The Number Of AOD Feedbacks field in the Multiple AOD Feedback element shall be set to the value of the Number Of Best AWV ID field in the Multiple AWV ID element in the FTM frame from the ISTA that elicited the Multiple AOD feedback.

11.21.6.4.2.5 DMG LOS assessment for EDCA based ranging measurement exchange

A LOS assessment FTM exchange may provide an ISTA with information about whether the link with the ISTA is over a non-LOS path. A LOS assessment FTM exchange is a DMG FTM burst as defined in 11.21.6.4.2.2 or 11.21.6.4.2.3 in which one FTM frame from the RSTA to the ISTA is a LOS assessment FTM PPDU. A LOS assessment FTM burst over the non-first path AWV is identified by the RSTA setting the FTM trigger to 3 at the FTMR. A LOS assessment FTM burst over the first path AWV is identified by setting the FTM trigger to 4. An ISTA shall not initiate an FTM burst by an FTMR with the trigger set to 3 or 4 unless the RSTA has indicated it may participate in such exchange by setting to 1 the LOS Assessment TX subfield in the DMG Direction Measurement Capabilities field of the DMG Capabilities element.

A LOS assessment FTM PPDU is defined as a PPDU that contains an FTM frame that is sent with the TXVECTOR parameters set as follows: DUAL_POLARIZATION_TRNS set to 1, EDMG_TRN_LEN set to a value greater than 0, and EDMG_PPDU_TYPE set to EDMG-TRN-R.

The RSTA and the ISTA shall switch polarization while transmitting and receiving the TRN field of the LOS assessment FTM PPDU as described in 28.9.2.2.5.

If the RSTA has set the LOS Assessment RX subfield to 1 in the DMG Direction Measurement Capabilities field, the ISTA shall respond to a LOS assessment FTM PPDU with a LOS Assessment Ack PPDU.

A LOS Assessment Ack PPDU is defined to be a PPDU that contains an Ack frame and has the DUAL_POLARIZATION_TRNS field set to 1, the EDMG_TRN_LEN set to a value greater than 0 and the

EDMG_PPDU_TYPE set to EDMG-TRN-R. The ISTA and the RSTA shall switch polarization while transmitting and receiving the TRN field of the LOS assessment FTM PPDU and the LOS assessment Ack PPDU as described in 28.9.2.2.5. The RSTA shall send a channel measurement element based on the measurements it performed on the Ack PPDU TRN field in the next FTM frame sent to the ISTA.

11.21.6.4.2.6 DMG first path AWV for EDCA based ranging measurement exchange

A first path AWV FTM burst is started when an ISTA sends the RSTA an FTM Request frame with Trigger field set to 2. An EDMG ISTA may send an FTM Request frame with the trigger set to 2 only if the RSTA has set the First Path Training Supported subfield to 1 in the Beamforming field of the EDMG Capabilities and the ISTA and RSTA have performed beamforming training for first path as defined in 10.42.10.6. All the FTM frames sent from the RSTA to the ISTA during the FTM burst and all the Ack frames sent from the RSTA to the ISTA in response to the FTM frame shall have the setting described in Table 11-11b, except when used for AOD and AOA estimation.

Table 11-11b—TXVECTOR parameter setting for first path AWV FTM exchange

TXVECTOR Parameter	Value
EDMG_PPDU_TYPE	EDMG-TRN-T
EDMG_TRN_LEN	1
RX_TRN_PER_TX_TRN	1
EDMG_TRN_P	0
EDMG_TRN_M	3
EDMG_TRN_N	3
TRN_SEQ_LENGTH	Normal
FIRST_PATH_AWV_TRN	FIRST_PATH_AWV_ON_TRN

The setting of the TXVECTOR parameters when the frames are used for AOA and AOD estimation is described in 11.21.6.4.2.3.

When an FTM frame is sent with the TXVECTOR parameter FIRST_PATH_AWV_TRN set to FIRST_PATH_AWV_ON_TRN, the Ack frame sent in response to this frame shall have the TXVECTOR parameter FIRST_PATH_AWV_TRN set to FIRST_PATH_AWV_ON_TRN. In both these frames the TRN field shall be received by the ISTA and RSTA using the first path AWV receive antenna setting.

11.21.6.4.2.7 DMG secure measurement exchange for EDMG STAs

Setup of the Secure EDMG Measurement exchange is described in 11.21.6.3.6. The Secure EDMG Measurement exchange protocol follows the procedure as described in 11.21.6.4.2 with the following changes:

- The Protected FTM Request frame shall be used by the ISTA to initiate the exchange.
- The ISTA shall use the Protected FTM frame during the exchange.
- The PPDUs carrying the Protected FTM frames transmitted by the RSTA and the Acks transmitted by the ISTA shall be based on the format as described in 28.9.3. In these PPDUs, the TXVECTOR parameter SECURED_TRN shall be set to SECURED_TRN.
- The PPDU carrying the protected FTM frames transmitted by the responder to initiator shall use the first path AWVs obtained during first path beamforming training as described in 10.42.10.6.

- The PPDU carrying the Ack frame transmitted by the initiator to the responder shall use the first path AWWs obtained during first path beamforming training as described in 10.42.10.6.
- If the Ack frame for an FTM frame is not received, the RSTA may retransmit the FTM frame. In this case, the RSTA shall send an FTM frame with the same Action frame body as the FTM frame for which the Ack was not received, except for updating the Dialog Token field if it was nonzero, and a new Secure TRN Sequence shall be used. The Sequence Number field in the MAC header is also updated.

The following rules apply to the FTM retransmission of the secure EDMG measurement exchange:

- a) The t_2 measured by the RSTA and the t_4 measured by the ISTA shall be based on the secure TRN subfields. For verification of security, the RSTA and ISTA should estimate the time separately on each secure TRN subfield and verify that values estimated on different TRN subfields are equal up to a small estimation error.
- b) If FTM retransmission is successful, the ISTA may derive the RTT using the received TOA (t_4) and TOD (t_1) values included in the retransmitted FTM frame, and t_2 and t_3 values estimated locally in the prior (or preceding) FTM.
- c) During the FTM retransmission, the t_2 value estimated by the ISTA shall not be used for RTT calculation in the subsequent FTM. This is because the t_2 was estimated by correlating two unmatched secure TRNs as a result of FTM retransmission.
- d) During the FTM retransmission, the t_4 value estimated by the RSTA shall not be included in the subsequent FTM frame. This is because the t_4 was estimated by correlating two unmatched secure TRNs as a result of FTM retransmission.
- e) Due to rules c) and d), the FTM immediately after the FTM retransmission shall not be used by an ISTA for RTT calculation.
- f) A successful FTM retransmission shall only be counted towards to the total number of FTMs Per Burst value.

If an RSTA does not perform FTM retransmission, so that the maximum number of FTM frame retransmissions the RSTA might attempt is zero, the RSTA shall send a new FTM frame (with TOA=0, TOD=0, updated Dialog Token) with new Secure TRN (associated with the updated dialog token) appended when an FTM transmission in a secure EDMG FTM exchange fails.

A STA shall maintain the Dialog Token Counter, and update the Dialog Token Counter to (Dialog Token field value – 1) mod 254 + 1, where the Dialog Token field value corresponds to the value of the Dialog Token field either in the lastly transmitted FTM frame if the STA is an RSTA or in the lastly received FTM frame if the STA is an ISTA.

A STA transmitting an EDMG secure ranging PPDU to a peer STA shall set the TXVECTOR parameters as follows:

- The SECURED TRN parameter is set to SECURED_TRN.
- The SECURE_TRN_SEQUENCE parameter contains the Secure TRN bit sequences associated with the Dialog Token Counter.

After transmitting an EDMG secure ranging PPDU, a STA's MAC sublayer issues a PHY-RXTRNSEQUENCE.request with a TRNVECTOR parameter that provides the TRN sequence associated with the Dialog Token Counter.

When an RSTA transmits an EDMG secure ranging PPDU, it shall include the Protected FTM frame in which the Dialog Token field is set to the Dialog Token Counter.

An ISTA shall transmit an Ack frame to the received Protected FTM frame as an EDMG secure ranging PPDU. An ISTA shall discard the timestamps measured from an EDMG secure ranging PPDU if the Dialog Token field in the Protected FTM frame in the EDMG secure ranging PPDU does not match its Dialog Token Counter.

NOTE—It is recommended that an ISTA or RSTA discards ranging measurements when it detects that the transmit center frequency offset between the ISTA and the RSTA exceeds the allowed tolerance from the values specified in 20.3.3.2.1.

Insert 11.21.6.4.3 through 11.21.6.4.9 (including Figure 11-37a through Figure 11-37y) after 11.21.6.4.2.7 as follows:

11.21.6.4.3 TB ranging measurement exchange

11.21.6.4.3.1 General

TB ranging is the dynamic trigger-based variant of the FTM procedure. The TB ranging measurement exchange consists of one or more scheduled availability windows. The TB ranging measurement exchange is dynamic, as the actual number of ISTAs participating in the measurement exchange can vary across availability windows. Each ISTA that is assigned to the scheduled availability window may participate in or skip the corresponding measurement exchange. The availability windows are scheduled periodic time windows assigned to ISTAs during negotiation (see 11.21.6.3.3).

Within each availability window the RSTA and ISTAs shall not transmit or trigger transmission of any Data frames; they shall only perform ranging activities related to Polling, Measurement Sounding, and Measurement Reporting phases, as well as signaling of modification of availability window parameters; see 11.21.6.5.1, and TB ranging session termination, see 11.21.6.6.2. Each availability window by default consists of a single TXOP and may be extended to multiple TXOPs by announcement, if a single TXOP is insufficient to accommodate all ISTAs that responded to the poll; see 11.21.6.4.3.2, and 11.21.6.4.3.3. Each availability window of the TB ranging measurement exchange consists of one or more triplets of sequential phases: Polling phase, Measurement Sounding phase, and Measurement Reporting phase. Figure 11-37a shows an example of two availability windows, each composed of a single triplet of Polling, Measurement Sounding, and Measurement Reporting phases. An RSTA and ISTA participating in TB ranging shall perform any Polling, Measurement Sounding, and Measurement Reporting phases only within the availability windows.

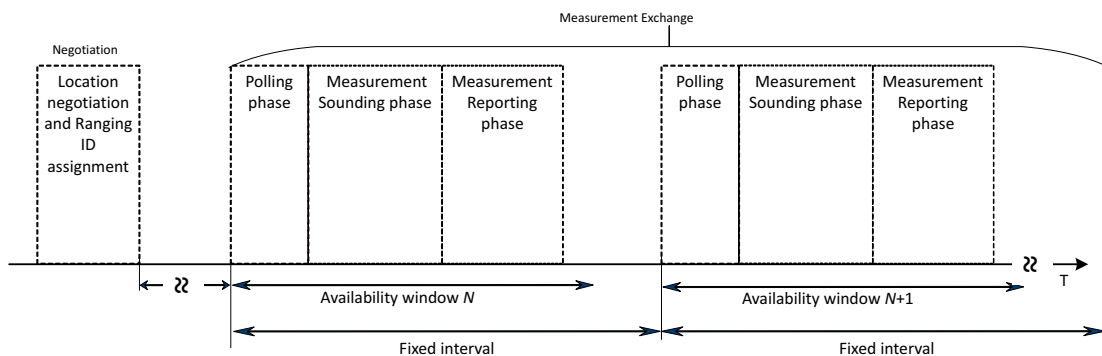


Figure 11-37a—TB ranging availability windows each with one instance of a polling/sounding/reporting triplet

Each availability window typically contains a single poll, where the RSTA should poll all the ISTAs assigned to that availability window. If the available bandwidth is insufficient to allow for the polling of all ISTAs

assigned to the availability window with one poll, the RSTA shall indicate that one or more extra polling/sounding/reporting triplets can be expected within the availability window (see example in Figure 11-37b, and Figure 11-37c. All instances of polling/sounding/reporting triplets shall be completed before the end of the availability window.

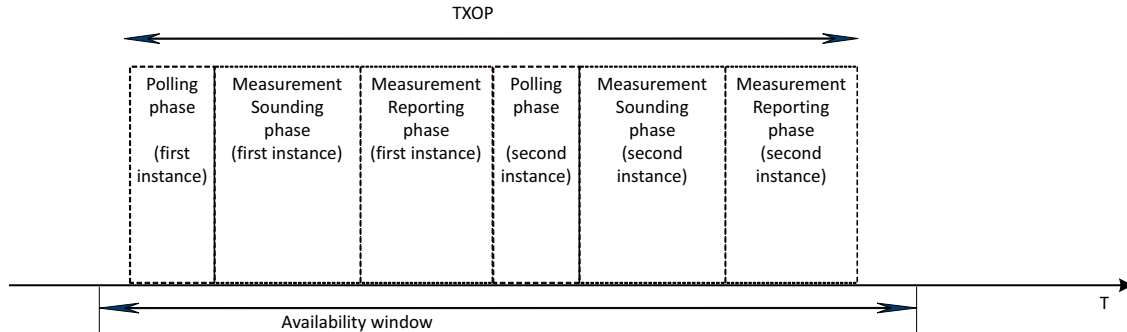


Figure 11-37b—TB ranging availability window with two instances of polling/sounding/reporting triplets within a single TXOP

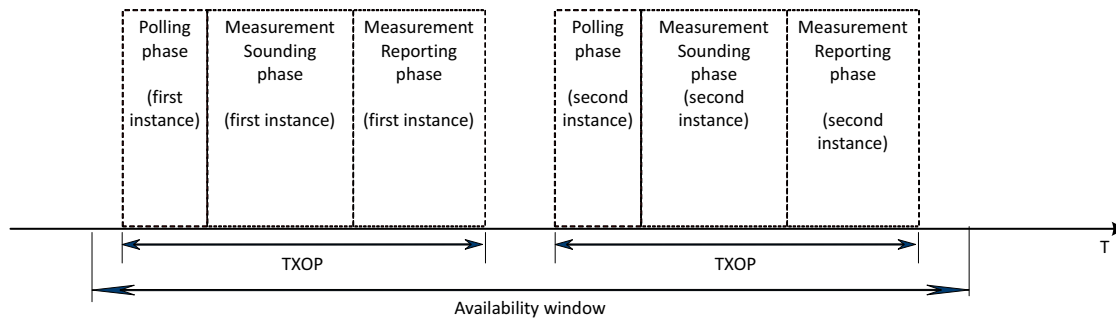


Figure 11-37c—TB ranging availability window with two instances of polling/sounding/reporting triplets in separate TXOPs

During the availability window, measurement resources and results shall be made available to each ISTA whose poll response was received at the RSTA; see 11.21.6.4.3.3 and 11.21.6.4.3.4. This may also lead to extra instances of polling/sounding/reporting triplets, even if all ISTAs assigned to this availability window were polled in the first Polling phase instance (e.g., if the RSTA is not able to accommodate all ISTAs that responded in a single measurement sounding phase instance; see 11.21.6.4.3.3).

The timestamps reported within each measurement sounding phase shall be derived from a clock that runs continuously during the measurement sounding phase.

If there is a discontinuity in the clock for the FTM timestamping between two reported TOD timestamps, then the TOD Not Continuous subfield in the TOD Error field of the LMR frame shall be set to 1. Otherwise it shall be set to 0.

NOTE—In some use cases it is of interest to have a clock that runs continuously across subsequent ranging availability windows. For this reason it is desirable for the clock for the FTM timestamping to run continuously.

Within each availability window, an RSTA shall use an AID or Ranging Session ID (RSID) to identify an associated or unassociated ISTA, respectively. The AID and RSID assignment shall be nonconflicting and shall have the same size and valid address space (as defined in 9.4.1.8 and 26.17.4). The RSID usage shall follow the same rules as that of AIDs for HE operations. The RSIDs are assigned to unassociated ISTAs during the FTM negotiation; see 11.21.6.3.

An RSTA shall follow the rules defined in 26.5.2 when transmitting any Trigger frames of variant Location for TB ranging with the following rules:

- A Ranging Trigger frame shall be carried in an S-MPDU if the Ranging Trigger frame is carried in a VHT PPDU or HE PPDU.
- An RSTA shall not transmit a Ranging Trigger frame in a VHT MU PPDU or HE MU PPDU.

An RSTA shall not transmit a Sounding Ranging Trigger frame soliciting an HE TB Ranging NDP that uses UL MU-MIMO, that is, where the same RU is allocated to multiple ISTAs, to any ISTA from which it has not received a TB specific subelement in the Ranging Parameters element with the Full Bandwidth UL MU-MIMO field equal to 1.

If required, an ISTA shall transmit any FTM Request frames outside of availability windows allocated to itself. Inside availability windows allocated to itself, an ISTA shall not transmit any frame except when assigned UL resources by a TF transmitted by the RSTA.

An RSTA, in which `dot11MultiBSSIDImplemented` is true, and that transmits a Ranging Trigger frame or a Ranging NDP Announcement frame to a set of ISTAs, in which at least two ISTAs have a TB ranging measurement exchange with different BSSIDs in the Multiple BSSID set of the RSTA shall set the TA field of these frames to the transmitted BSSID. Any ISTA that negotiates a TB ranging measurement exchange session with a nontransmitted BSSID of an RSTA shall support the reception of a Control frame with TA equal to the transmitted BSSID.

An ISTA shall follow the rules defined in 26.5.2 when transmitting any HE TB PPDU for TB ranging with the exceptions defined in 11.21.6.4.3.2, 11.21.6.4.3.3, and 11.21.6.4.3.4.

11.21.6.4.3.2 Polling phase of TB ranging

The Polling phase is the first phase of each polling/sounding/reporting triplet.

In each Polling phase instance, the RSTA shall transmit a Poll Ranging Trigger frame, see 9.3.1.22.10. Once a response is received, the RSTA shall proceed to the Measurement Sounding phase; otherwise it may transmit additional Poll Ranging Trigger frames, until it receives a response from at least one ISTA. The RSTA shall allocate each RU in the TF Ranging poll to only one ISTA. Any ISTA addressed by a User Info field in a TF Ranging Poll frame that intends to participate in the measurement sequence within this availability window shall send a CTS-to-self in an S-MPDU within an HE TB PPDU in its designated RU allocation as identified in the TF Ranging Poll frame, otherwise shall not send a CTS-to-self to avoid resource allocation in this measurement sequence; see Figure 11-37c.

If the available bandwidth does not allow for the polling of all ISTAs assigned to this availability window using a single TF Ranging Poll frame, the RSTA shall attempt to schedule one or more extra polling/sounding/reporting triplets within the availability window. The RSTA shall indicate the extra polling/sounding/reporting triplets by setting the More TF subfield in the Common Info field to 1 and the RA field to the broadcast address in the TF Ranging Poll frame, and in TFs in subsequent Polling, Measurement Sounding, and Measurement Reporting phases in the same availability window. If the RSTA had set the More TF subfield to 1 in the preceding Ranging Trigger frame, and if there are no additional polling/sounding/reporting triplets in the same availability window, the RSTA shall set the More TF subfield in the Common Info field to 0 and the RA field to the broadcast address in the next Ranging Trigger frame within

that availability window. On receipt of such a frame, an ISTA that has not been addressed by a User Info field in the TF, may enter doze state, if no other condition requires this STA to remain awake. Any extra polling/sounding/reporting triplets can either be transmitted in the same TXOP (see example in Figure 11-37b) or a new TXOP (see example in Figure 11-37c), depending on the maximum allowed TXOP duration and the predicted length of the extra instances of polling/sounding/reporting triplets.

11.21.6.4.3.3 Measurement Sounding phase of TB ranging

The measurement-sounding phase commences SIFS time after the Polling phase and is the second phase of each polling/sounding/reporting triplet; see Figure 11-37d. The measurement sounding phase consists of one or more Sounding Ranging Trigger frames (see 9.3.1.22.10) allocating uplink resources to one or more ISTAs; see Figure 11-37a and Figure 11-37c. The Ranging variant Trigger frame of Sounding subvariant is called the TF Ranging Sounding frame. Each TF Ranging Sounding frame shall allocate uplink resources for one or more ISTAs' I2R NDP multiplexed in the spatial stream domain covering the full bandwidth. The format of the I2R NDP is an HE TB Ranging NDP; see 27.3.18a.2. SIFS time after receiving the last I2R NDP, the RSTA shall transmit an NDP Announcement frame followed by an R2I NDP; the Ranging NDP Announcement frame is described in 9.3.1.19 and the R2I NDP is an HE Ranging NDP, as described in 27.3.18a.1. Figure 11-37d shows an availability window with an RSTA and two ISTAs (ISTA 1 and ISTA 4) responding to the poll. The Ranging NDP Announcement frame's STA Info fields specify all the ISTAs that will use the R2I NDP, which shall include the ISTAs that were allocated uplink resources in the measurement sounding phase.

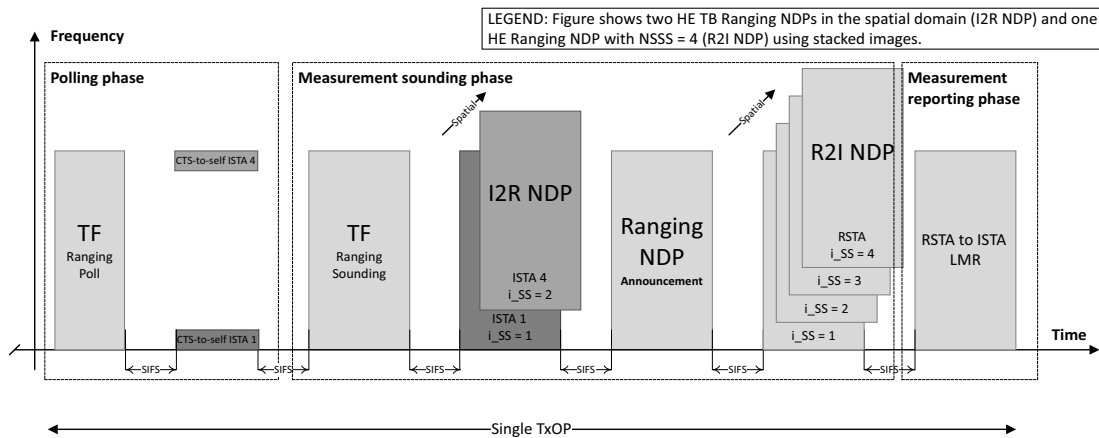


Figure 11-37d—TB ranging availability window with two ISTAs

The RSTA shall select a bandwidth value for the measurement sounding phase that is less than or equal to the RSTA Assigned Max Bandwidth of each of the ISTAs that are being allocated resources during this measurement instance. It may be different from the bandwidth used in the Polling phase, but shall adhere to the rules of multiple frame transmission in an EDCA TXOP; see 10.23.2.8.

The RSTA shall set the TXVECTOR parameter CH_BANDWIDTH of the Trigger frame Ranging Sounding to that same bandwidth and use the same value for the UL BW subfield of the Common Info field of said Trigger frame. When transmitting the Ranging NDP Announcement frame and R2I NDP, the RSTA shall set the TXVECTOR parameter CH_BANDWIDTH to that same bandwidth.

In the TF Ranging Sounding, the RSTA shall set the SS Allocation subfield and the I2R Rep subfield of the User Info fields corresponding to each of the ISTAs triggered by the Trigger frame in the following way:

- The Number of Spatial Streams in each SS Allocation subfield shall not exceed the *RSTA Assigned I2R STS* ≤ 80 MHz for the corresponding ISTA, if the UL BW subfield in the Common Info field

indicated a bandwidth less than or equal to 80 MHz, and not exceed the *RSTA Assigned I2R STS > 80 MHz* for the corresponding ISTA otherwise.

- All the I2R Rep subfields in the User Info fields of the TF Ranging Sounding shall be set to the same value. This value indicates the number of LTF repetitions in the I2R NDP preamble and shall not exceed any of the *RSTA Assigned I2R Rep* corresponding to the ISTA triggered by this Trigger frame.
- The product of the number of LTF repetitions, indicated in each of the I2R Rep subfields of the User Info fields, and the number of HE-LTF symbols, indicated in the Number Of HE-LTF Symbols And Midamble Periodicity subfield in the Common Info field, shall not exceed the *RSTA Assigned I2R LTF Total* for any of the ISTA triggered by this Trigger frame.

After transmission of the TF Ranging Sounding, the RSTA's MAC sublayer shall issue a PHY-RXLTFSEQUENCE.request primitive with an LTFVECTOR containing the following parameters:

- The SECURE_LTF_FLAG parameter set to 0.
- The LTF_NSTS and LTF_REP parameter vectors set to the same values as indicated, respectively, by the SS Allocation and I2R Rep subfields of all the User Info fields.

Similarly, in the Ranging NDP Announcement frame, the RSTA shall set the R2I NSTS subfield and R2I Rep subfield of the STA Info fields corresponding to each of the ISTAs addressed by that frame in the following way:

- The R2I NSTS subfield value shall not exceed the *RSTA Assigned R2I STS ≤ 80 MHz* for the corresponding ISTA, if the TXVECTOR parameter CH_BANDWIDTH for this Ranging NDP Announcement frame is less than or equal to 80 MHz, and not exceed *RSTA Assigned R2I STS > 80 MHz* for the corresponding ISTA otherwise.
- The number of LTF repetitions in the R2I Rep subfield shall be set to a value not to exceed the *RSTA Assigned R2I Rep* for the corresponding ISTA.
- The combination of the values of the R2I NSTS and the R2I Rep shall not lead to a total number of LTF that exceeds the *RSTA Assigned R2I LTF Total* for each corresponding ISTA.

After reception of the Ranging NDP Announcement frame, the ISTAs' MAC sublayer shall issue a PHY-RXLTFSEQUENCE.request primitive with an LTFVECTOR containing the following parameters:

- The SECURE_LTF_FLAG parameter set to 0.
- The LTF_NSTS and LTF_REP parameters set to the same values as indicated, respectively, by the R2I NSTS and R2I Rep subfields of the STA Info field addressed to it.

The RSTA shall set a different value than the previous transmission in the Sounding Dialog Token Number field in its transmitted Ranging NDP Announcement frame as part of each Measurement Sounding phase. Measurement instances are associated with the Sounding Dialog Token Number field value.

The RSTA may schedule some ISTAs that replied during the Polling phase to the first measurement sounding phase instance and other ISTAs to one of possibly multiple extra measurement sounding phase instances; see Figure 11-37b and Figure 11-37c. The RSTA shall only schedule measurement sounding resources to an ISTA in a measurement sounding instance if a valid poll response was received from that ISTA in the corresponding Polling phase instance. This may require an RSTA to poll an ISTA multiple times. This is necessary, for example, if different ISTAs have varying, incompatible, RSTA Assigned Max Bandwidth values or if the RSTA wants to limit the time duration of each range measurement sounding instance.

Both RSTA and ISTA perform RTT measurements by capturing the timestamps of the NDP. The ISTA shall record the time at which the I2R NDP is transmitted (t1). The RSTA shall then capture the time at which the I2R NDP arrives (t2) and shall record the time at which the R2I NDP is transmitted (t3). The ISTA shall finally capture the time at which the R2I NDP arrives (t4); see Figure 11-37e. The timestamp values t2 and t3 shall be

measured according to the RSTA's clock (i.e., without applying any frequency offset correction to the time basis).

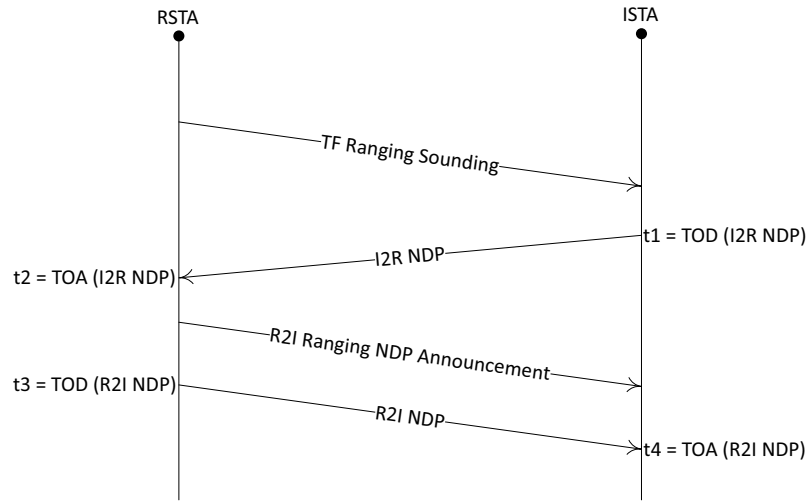


Figure 11-37e—Timing diagram of a measurement sounding phase in TB ranging

The Round Trip Time (RTT_{ISTA}) observed by ISTA is defined as:

$$\text{RTT}_{\text{ISTA}} = [(t4 - t1) - (t3' - t2')]$$

where $t3'$ and $t2'$ are the time at which the R2I NDP was transmitted and the time at which the I2R NDP was received, respectively, as converted by the ISTA from the RSTA's time basis to its own time basis.

The mechanism by which the ISTA derives $t3'$ and $t2'$ from the TOD and TOA fields of the relevant R2I LMR (see 11.21.6.4.3.4) is implementation dependent.

When the I2R LMR Feedback is negotiated, the RSTA can compute the RTT as:

$$\text{RTT}_{\text{RSTA}} = [(t4' - t1') - (t3 - t2)]$$

where $t1'$ and $t4'$ are the time at which the I2R NDP was transmitted and the time at which the R2I NDP was received, respectively, as converted by the RSTA from the ISTA's time basis to its own time basis. The mechanism by which the RSTA derives $t4'$ and $t1'$ from the TOD and TOA fields of the relevant I2R LMR is implementation dependent; see 11.21.6.4.3.4.

NOTE 1—When using CFO in the conversion from the ISTA's time basis to the RSTA's, the RSTA uses the CFO reported in the CFO Parameter field of the I2R LMR.

NOTE 2—Refer to 27.3.15.3 for carrier frequency offset (CFO) correction requirements for HE TB PPDU transmission.

In an LMR, the TOA field contains a timestamp that represents the time, with respect to a time base, at which the start of the preamble of the corresponding NDP arrived at the receive antenna connector. The TOD field contains a timestamp that represents the time, with respect to the same time base, at which the start of the preamble of the corresponding NDP appeared at the transmit antenna connector.

If the measurement sounding phase includes more than a single TF Ranging Sounding frame (see Figure 11-37f) each ISTA and RSTA pair shall refer to the t1 and t2 of the I2R NDP transmitted by that ISTA, while t3 and t4 shall be based on the single R2I NDP received by all ISTAs; see Figure 11-37g.

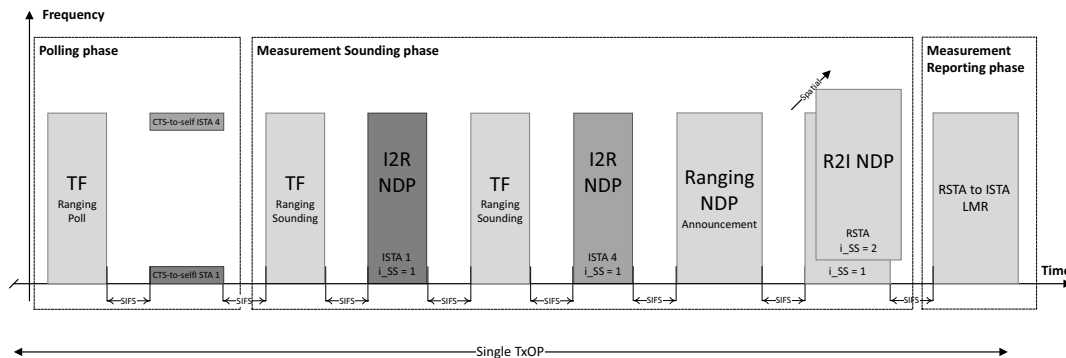


Figure 11-37f—TB ranging availability window with multiple TF Ranging Sounding frames

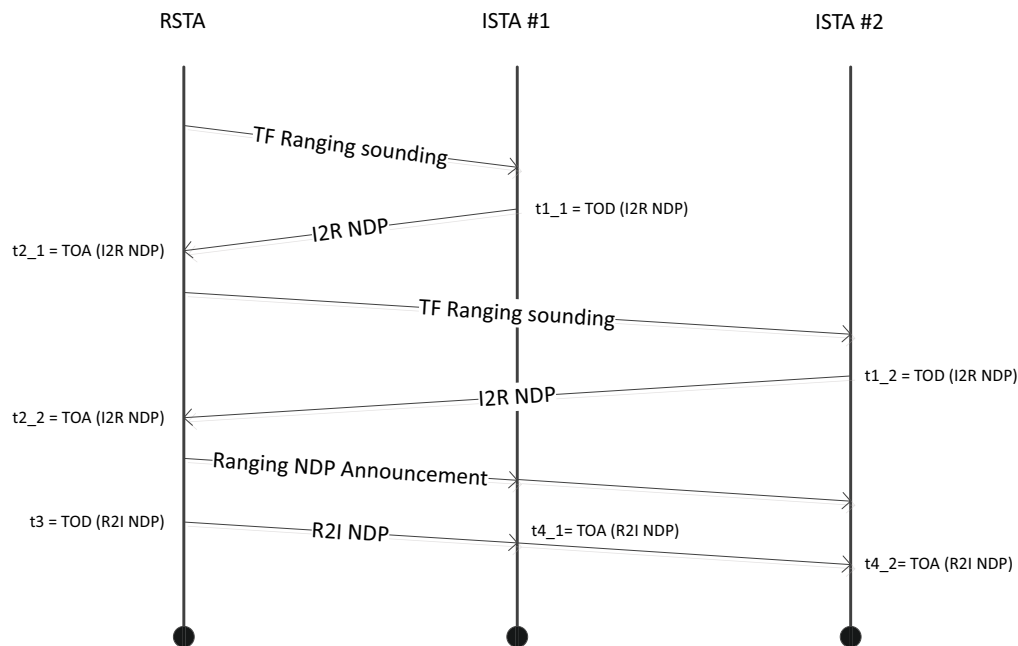


Figure 11-37g—Measurement sounding phase with I2R time division multiplexing

If due to retransmission an ISTA receives multiple TF Ranging Sounding frames and transmits multiple I2R NDPs, the ISTA and RSTA shall refer to the t1 and t2 of the I2R NDP last transmitted.

The uplink power control, timing, and frequency synchronization requirements of unassociated STAs performing TB ranging shall follow the same rules as those of associated HE STAs.

To aid in synchronizing the TSF time at the ISTAs, the RSTA maintains a trigger poll counter. Before transmitting a TF Ranging Poll frame, the RSTA shall increase the trigger poll counter by one (modulo 8) and set the Token subfield of the Trigger Dependent Common Info subfield to the value of the trigger poll counter.

When transmitting a Ranging NDP Announcement frame as part of the TB ranging measurement exchange, an RSTA shall include a value in the Partial TSF subfield in the STA Info field with the AID11 subfield equal to 2044, that is equal to the RSTA's TSF[21:6] at the time of transmission of the preceding TF Ranging Poll frame. Specifically, the time that the first data symbol of the PSDU of said frame was transmitted to the PHY plus the RSTA's delays through its local PHY from the MAC-PHY interface to its interface with the WM.

Additionally, the RSTA shall set the Token subfield in the STA Info field with the AID11 subfield equal to 2044 to the same trigger poll counter value as the Token subfield in the TF Ranging Poll frame whose partial TSF time is carried in the Ranging NDP Announcement frame.

NOTE 3—An ISTA that tries to synchronize to the RSTA's TSF time will need to keep track of the difference between its local TSF[63:22] and the RSTA's TSF[63:22] when updating the TSF[21:6]. When receiving a partial TSF value in a received Ranging NDP Announcement frame from the RSTA, to synchronize its TSF time with the RSTA's TSF time in order to determine the start of a subsequent TB ranging availability window:

- The ISTA should check if its TSF[21:6] at the reception of a TF Ranging Poll frame was larger than the received Partial TSF and the absolute difference is more than 2^{15} , then the ISTA should increase the RSTA's tracked TSF [63:22] value by 1.
- If the ISTA's TSF [21:6] at the reception of the TF Ranging Poll frame is less than the Partial TSF and the absolute difference is more than 2^{15} , the ISTA should decrease the RSTA's tracked TSF [63:22] value by 1.

11.21.6.4.3.4 Reporting phase of TB ranging measurement

The last phase of each polling/sounding/reporting triplet is the measurement reporting phase, which is transmitted a SIFS time after the measurement sounding phase; see Figure 11-37c. The measurement results shall be carried in LMR frames; see 9.6.7.49. LMR frames shall carry measurement results from the RSTA to the ISTA, and if negotiated also from the ISTA to the RSTA; see Figure 11-37h. If the Range Reporting is performed in the context of a Secure FTM Session, see 11.21.6.3, the corresponding LMR and FTM (see 11.21.6.5.1) frames shall be transmitted using Protected Fine Timing Action frames, and see 9.6.34.

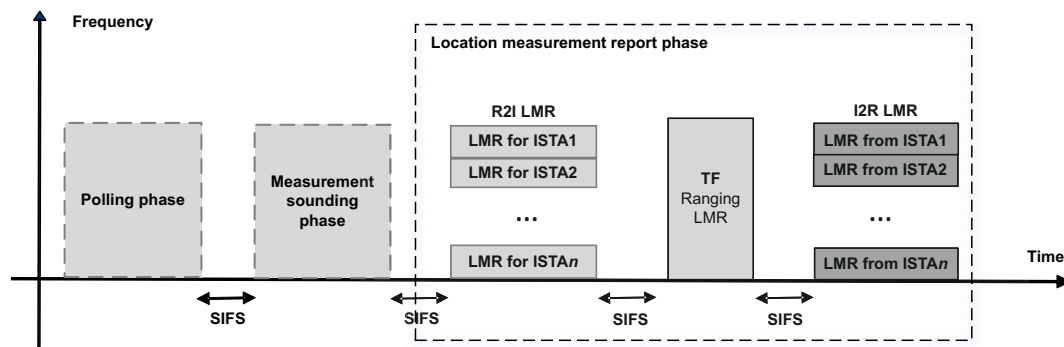


Figure 11-37h—TB ranging measurement reporting phase with bidirectional LMR feedback for n ISTAs

The feedback type of the I2R and R2I LMRs may be either immediate (i.e., from the current availability window) or delayed (i.e., from the last availability window in which the ISTA responded to the TF Ranging Poll frame and the RSTA allocated resources to that ISTA during the measurement sounding phase). The LMR feedback (immediate/delayed) is indicated by the RSTA during the negotiation (see 11.21.6.3.3).

The Dialog Token field in the LMR frame shall be identical to the Sounding Dialog Token field in the corresponding Ranging NDP Announcement frame in the Measurement Sounding phase from which the reported TOA and TOD values were measured; see 11.21.6.4.3.3.

NOTE 1—LMR feedback is carried in Action No-Ack frames and is therefore neither acknowledged nor retransmitted; see 9.6.7.49.

The RSTA shall transmit an R2I LMR to all ISTAs that were allocated resources in the preceding measurement sounding phase. All the R2I LMR frames shall be carried in one HE MU PPDU, where each RU contains only one user; if there is only one R2I LMR it may be carried in an HE SU PPDU.

If I2R LMR was negotiated, the RSTA shall assign uplink resources to the ISTAs using a Report Ranging Trigger frame; see 9.3.1.22.10. The Ranging variant Trigger frame of report subvariant is called the TF Ranging LMR.

A TB ranging measurement reporting phase including the optional I2R LMR is illustrated in Figure 11-37h. If the I2R LMR was negotiated by one or more ISTAs, then SIFS time after transmitting out the R2I LMR, the RSTA transmits a TF Ranging LMR to solicit the I2R LMR frame(s). This TF shall allocate uplink resources to ISTAs that negotiated I2R LMR and were allocated resources in the preceding measurement sounding phase. The RSTA shall allocate each RU in the TF Ranging LMR to only one ISTA. In response to the TF Ranging LMR, each addressed ISTA shall respond by transmitting an I2R LMR frame.

If an ISTA negotiated delayed I2R LMR reporting, and if the TOA measurement for the previous availability window is not ready, then the ISTA shall not respond to the TF Ranging Poll frame in the Polling phase of any availability window until the I2R LMR is ready.

For delayed reporting, the first instance of the R2I LMR and the optional I2R LMR do not have valid TOA/TOD timestamps to include, in this case the RSTA and the ISTA shall set the Invalid Measurement subfield in the TOA Error field of the respective LMR to 1.

In TB ranging, the PHY shall issue the PHY-RXEND.indication primitive with error condition IntegrityCheckError if the PHY detects the integrity check error in the reception of the corresponding HE Ranging NDP or HE TB Ranging NDP. If the PHY of an RSTA issues a PHY-RXEND.indication primitive with error condition IntegrityCheckError, the RSTA shall set the Invalid Measurement field in the R2I LMR frame carrying the TOA measured from the I2R NDP to 1. Correspondingly, if I2R LMR was negotiated between the ISTA and RSTA and the PHY of the ISTA issues a PHY-RXEND.indication primitive with error condition IntegrityCheckError, the ISTA shall set the Invalid Measurement field in the I2R LMR carrying the TOA measured from the R2I NDP to 1.

NOTE 2—When a STA detects that the transmit center frequency offset (CFO) between the ISTA and the RSTA exceeds the allowed tolerance from the values specified in 27.3.19.3 and 27.3.15.3, this can be an indication of a security attack.

If I2R LMR reporting was negotiated, then the ISTA shall include a CFO parameter in the I2R LMR; see 9.6.7.49. The ISTA shall estimate the CFO parameter based on the PPDU carrying the TF Ranging Sounding frame that solicits the I2R NDP from the ISTA. The RSTA may account for clock rate differences between ISTA and RSTA based on the CFO parameter included in the received I2R LMR. The mechanism by which t_4 and t_1 are adjusted by the RSTA is implementation specific. The CFO parameter refers to the t_1 and t_4 indicated in the same I2R LMR.

If the Invalid Measurement field in an R2I LMR or I2R LMR is equal to 1, the RSTA or ISTA receiving the LMR should discard the TOA carried in the LMR.

In TB ranging measurement reporting phase, if R2I LMR reporting or I2R LMR reporting carries phase shift feedback, then the R2I LMR reporting or the I2R LMR reporting shall be immediate feedback.

In TB ranging measurement reporting phase, if R2I AOA feedback was negotiated, the RSTA shall include the optional AOA feedback subfield in the R2I LMR frame and if I2R LMR reporting was negotiated in addition

to I2R AOA Feedback reporting then the ISTA shall include the optional AOA feedback subfield in the I2R LMR frame. The AOA field contains the Direction Measurement Results element described in 9.4.2.300.

11.21.6.4.4 Non-TB ranging measurement exchange

11.21.6.4.4.1 General

In Non-TB ranging, the protocol operates in an ISTA centric scheduling FTM mode; whenever the medium is available, an ISTA may initiate the measurement. The RSTA can only limit the frequency with which the ISTA can initiate measurements, by setting a minimum time interval between subsequent range measurements.

11.21.6.4.4.2 Measurement sounding phase of non-TB ranging

An ISTA shall initiate a non-TB ranging measurement instance by transmitting a Ranging NDP Announcement frame addressed to the RSTA, followed by an I2R NDP SIFS after. In response to the correctly received Ranging NDP Announcement frame addressed to itself, the RSTA shall transmit an R2I NDP; see Figure 11-37i. I2R NDP and R2I NDP refer to HE Ranging NDPs. The measurement-reporting phase consists of an LMR frame, which is a Location Measurement Report as defined in 9.6.7.49.

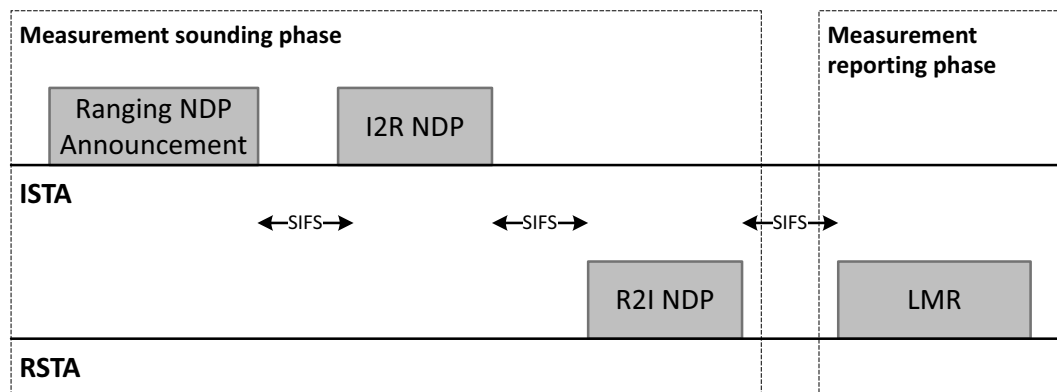


Figure 11-37i—Non-TB ranging measurement exchange sequence

For immediate feedback the LMR carries measurement results of the current measurement exchange, while for delayed feedback the LMR carries measurement results of the previous measurement exchange; see 11.21.6.4.4.3.

An ISTA shall set the Min Time Between Measurements field and the Max Time Between Measurements field in the Non-TB Specific subelement in the Ranging Parameters element in the IFTMR frame, where Max Time Between Measurements shall be larger than Min Time Between Measurements and takes into account the measurement exchange duration. An RSTA shall assign the value of the Min Time Between Measurements and Max Time Between Measurements in the Non-TB Specific subelement in the Ranging Parameters element in the initial FTM frame, where Max Time Between Measurements shall be larger than Min Time Between Measurements and takes into account of the measurement exchange duration.

An ISTA shall not initiate a new measurement exchange sequence until the assigned minimum time interval between subsequent range measurements has elapsed; see Figure 11-37k. An ISTA should complete the measurement sequence before the assigned maximum time interval between subsequent range measurements has elapsed.

An ISTA may use any AC to transmit the Ranging NDP Announcement frame.

The ISTA maintains a sounding dialog token counter modulo 64 for each FTM session. When transmitting a Ranging NDP announcement frame to an RSTA, the Sounding Dialog Token Number subfield in the Sounding Dialog field shall be set to the value of the corresponding counter; after which the counter shall be incremented by 1.

After transmitting the Ranging NDP Announcement frame and I2R NDP, the ISTA shall wait for a time interval of $aSIFSTime + aSlotTime + aRxPHYStartDelay$. This interval begins when the MAC receives a PHY-TXEND.confirm primitive for I2R NDP. If a PHY-RXSTART.indication primitive does not occur during the time interval, the ISTA shall conclude that the transmission of the Ranging NDP Announcement frame + I2R NDP has failed and abort the current measurement exchange. If a PHY-RXSTART.indication primitive occurred during the time interval, the ISTA tries to receive the R2I NDP and the LMR frame from the RSTA addressed by the Ranging NDP Announcement frame. If the LMR is received from the RSTA, the frame exchange initiated by the Ranging NDP Announcement frame is complete, otherwise the ISTA shall conclude that the current measurement exchange has failed.

The ISTA may invoke the backoff procedure by an EDCAF when the last frame of the measurement exchange initiated by the Ranging NDP Announcement frame has completed and the TXNAV timer has expired.

In the Non-TB ranging measurement exchange sequence, the ISTA shall transmit the Ranging NDP Announcement frame with the same bandwidth as the I2R NDP to reserve the medium. The Ranging NDP Announcement frame shall be unicast with the RA field set to the address of the RSTA, and contain one STA Info field with the AID11 subfield set to 0. If negotiated, the NPD Announcement frame shall contain another STA Info field with AID11 subfield set to 2045, and the I2R Tx Power subfield shall be set to indicate the TX power of the following I2R NDP. If the STA Info field with AID11 subfield set to 2045 is included, the ISTA shall set the R2I NDP Target RSSI subfield to either its preferred receive signal power or a reserved value.

The RSTA shall transmit the R2I NDP with the same bandwidth as the Ranging NDP Announcement frame, while the LMR may be transmitted at a different bandwidth, according to the rules of multiple frame transmission in an EDCA TXOP, see 10.23.2.8. The allowed bandwidths for the Ranging NDP Announcements, I2R NDP, and R2I NDP, shall be less than or equal to the RSTA Assigned Max Bandwidth.

Accordingly:

- An ISTA transmitting an I2R NDP shall set the TXVECTOR parameter CH_BANDWIDTH to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging NDP Announcement frame.
- An RSTA transmitting an R2I NDP shall set the TXVECTOR parameter CH_BANDWIDTH to the bandwidth of the Ranging NDP Announcement frame and/or the I2R NDP; which are obtained from the RXVECTOR parameter CH_BANDWIDTH of the Ranging NDP Announcement frame or I2R NDP, respectively. For the Ranging NDP Announcement frame, when not received in an HE/VHT/HT PPDU: from the RXVECTOR parameter CH_BANDWIDTH_IN_NON_HT when the Ranging NDP Announcement frame is received in a non-HT duplicate PPDU and is 20 MHz when the Ranging NDP Announcement frame is received in a non-HT PPDU.

If the bandwidth is less than or equal to 80 MHz, the ISTA shall set the I2R NSTS subfield and the R2I NSTS subfield in the STA Info field of the Ranging NDP Announcement frame each to a value not to exceed the *RSTA Assigned I2R STS ≤ 80 MHz* and *RSTA Assigned R2I STS ≤ 80 MHz*, respectively. If the bandwidth is greater than 80 MHz, the ISTA shall set these same subfields to values not to exceed the *RSTA Assigned I2R STS > 80 MHz* and *RSTA Assigned R2I STS > 80 MHz*, respectively.

The ISTA shall set the I2R Rep subfield and R2I Rep subfield in the STA Info field of the Ranging NDP Announcement frame each to a value not to exceed the *RSTA Assigned I2R rep* and *RSTA Assigned R2I rep*, respectively. Furthermore, the total number of LTF in the I2R NDP and R2I NDP, based on the number of

spatial streams and repetitions, shall not exceed the *RSTA Assigned I2R LTF Total* and *RSTA Assigned R2I LTF Total*, respectively.

After transmission of the Ranging NDP Announcement frame, the ISTA's MAC sublayer shall issue a PHY-RXLTFSEQUENCE.request primitive with a LTFVECTOR containing the following parameters:

- The SECURE_LTF_FLAG parameter set to 0.
- The LTF_NSTS and LTF_REP parameters set to the same values as indicated, respectively, by the R2I NSTS and R2I Rep subfields in the STA Info field with the AID11 subfield less than or equal to 2007.

After reception of the Ranging NDP Announcement frame, the RSTA's MAC sublayer shall issue a PHY-RXLTFSEQUENCE.request primitive with a LTFVECTOR containing the following parameters:

- The SECURE_LTF_FLAG parameter set to 0.
- The LTF_NSTS and LTF_REP parameters set to the same values as indicated, respectively, by the I2R NSTS and I2R Rep subfields in the STA Info field with the AID11 subfield less than or equal to 2007.

Both RSTA and ISTA perform RTT measurements by capturing the timestamps of the NDP. The ISTA shall record the time at which the I2R NDP is transmitted (t_1). The RSTA shall then capture the time at which the I2R NDP arrives (t_2) and shall record the time at which the R2I NDP is transmitted (t_3). The ISTA shall finally capture the time at which the R2I NDP arrives (t_4); see Figure 11-37j. The timestamp values t_2 and t_3 shall be measured according to the RSTA's clock (i.e., without applying any frequency offset correction to the time basis).

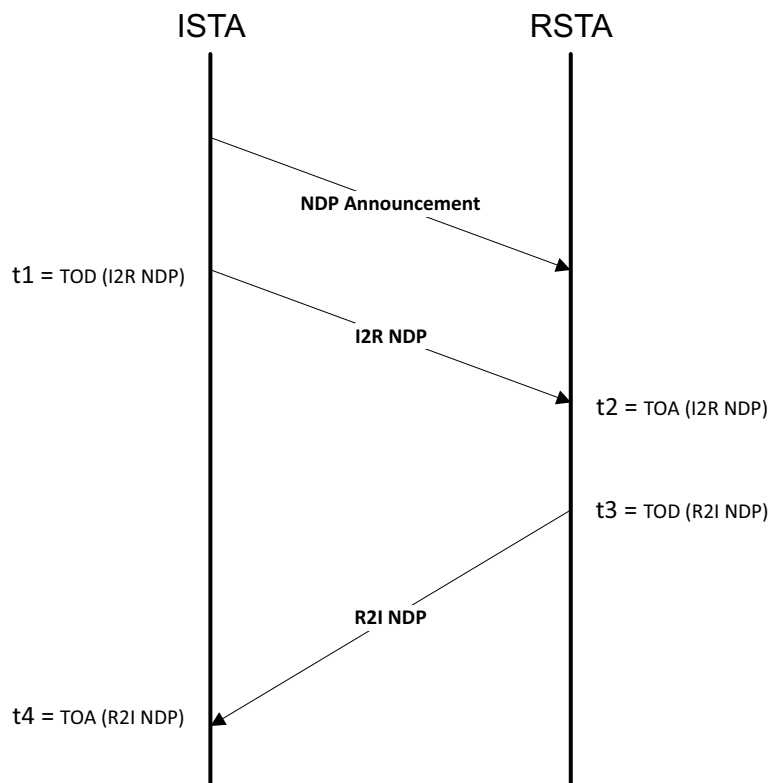


Figure 11-37j—Timing diagram of a measurement sounding phase in non-TB ranging

The Round Trip Time (RTT_{ISTA}) based on first path reporting is defined as:

$$RTT_{ISTA} = [(t_4 - t_1) - (t_3' - t_2')]$$

where t_3' and t_2' are the time at which the R2I NDP was transmitted and the time at which the I2R NDP was received, respectively, as converted by the ISTA from the RSTA's time basis to its own time basis.

The mechanism by which the ISTA derives t_3' and t_2' from the TOD and TOA fields of the relevant LMR is implementation dependent.

The Round Trip Time (RTT_{RSTA}) based on first path reporting is defined as:

$$RTT_{RSTA} = [(t_4' - t_1') - (t_3 - t_2)]$$

where t_1' and t_4' are the time at which the I2R NDP was transmitted and the time at which the R2I NDP was received, respectively, as converted by the RSTA from the ISTA's time basis to its own time basis.

The mechanism by which the RSTA derives t_1' and t_4' from the TOD and TOA fields of the relevant LMR is implementation dependent.

In an LMR, the TOA field contains a timestamp that represents the time, with respect to a time base, at which the start of the preamble of the corresponding NDP arrived at the receive antenna connector. The TOD field contains a timestamp that represents the time, with respect to the same time base, at which the start of the preamble of the corresponding NDP appeared at the transmit antenna connector.

11.21.6.4.4.3 Non-TB ranging measurement reporting phase

In non-TB ranging, the ranging protocol supports both immediate and delayed reporting.

An RSTA or ISTA indicates immediate reporting by setting the immediate R2I Or I2R Feedback subfield in the Ranging Parameters field of the FTM session frames that enable the measurement exchange, to 1. In immediate reporting, the TOA feedback corresponding to the current measurement exchange sequence shall be reported in the current measurement exchange; see Figure 11-37k.

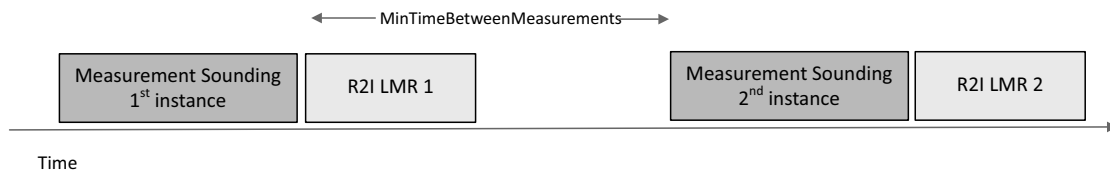


Figure 11-37k—Non-TB ranging with immediate reporting

An RSTA or ISTA indicates delayed reporting by setting the Immediate R2I Or I2R Feedback subfield in the Ranging Parameters field to 0. In delayed reporting, the TOA and TOD values in the current LMR shall carry the measurement results of the previous round. In this case, the first instance of the R2I LMR and the optional I2R LMR do not have valid TOA/TOD timestamps to include, and the RSTA and the ISTA shall set the Invalid Measurement subfield in the TOA Error field of the corresponding LMR to 1; see Figure 11-37l.

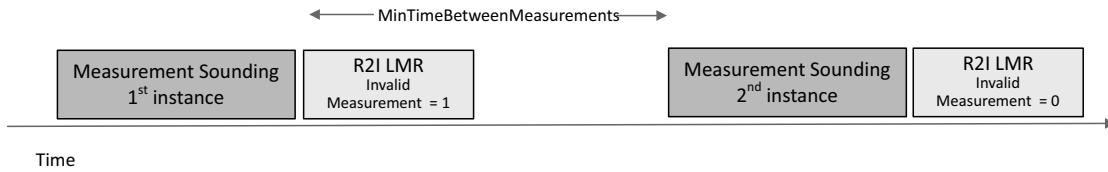


Figure 11-37l—Non-TB ranging with delayed reporting

An RSTA that indicated delayed reporting shall provide TOA feedback to the ISTA when the ISTA initiates another measurement sequence after Min Time Between Measurements, but before Max Time Between Measurements; see Figure 11-37m. This TOA feedback is carried in the LMR frame of this new measurement sequence and can be either valid or invalid as indicated by the Invalid Measurement field in the LMR frame.

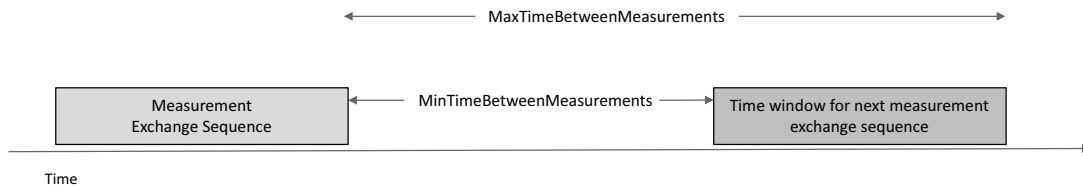


Figure 11-37m—Illustration of min time between measurements and max time between measurements with delayed reporting

The Dialog Token field of the LMR frame shall be copied from the Sounding Dialog Token Number subfield in the Ranging NDP Announcement frame that preceded the NDP which is used for the reported measurement.

If negotiated, the RSTA shall set the R2I TX Power field in the R2I LMR to report the TX power of the preceding R2I NDP; otherwise the R2I Tx Power field shall be set to a reserved value. The RSTA shall set the I2R NDP Target RSSI field to its preferred receive signal power or a reserved value.

If I2R LMR feedback is negotiated, the non-TB ranging measurement exchange (11.21.6.4.4) sequence shall follow the sequence shown in Figure 11-37n. After SIFS time of receiving the R2I LMR frame, the ISTA shall transmit the I2R LMR frame to the RSTA. The feedback type of I2R LMR could be either immediate or delayed.

NOTE 1—LMR feedback is carried in an Action No-Ack frame and is therefore neither acknowledged nor retransmitted; see 9.6.7.49.

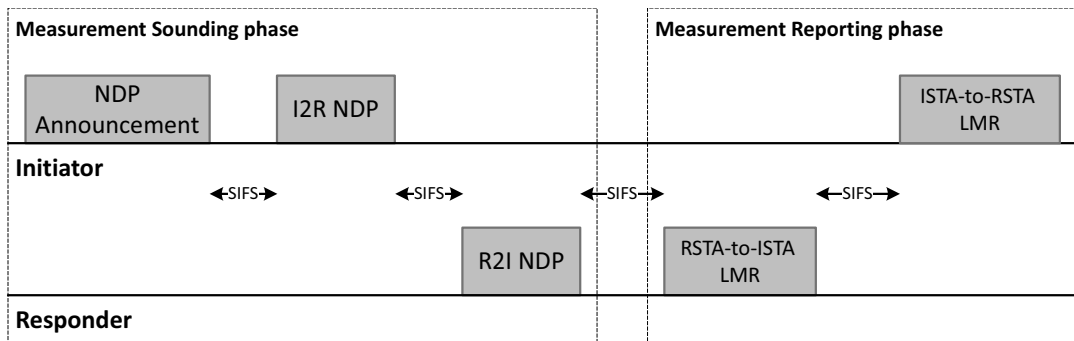


Figure 11-37n—Non-TB ranging measurement exchange sequence with bidirectional LMR feedback

The data rate or MCS used for transmitting the R2I and I2R LMR frames is solely decided by the transmitter of each of the frames. The bandwidth used to transmit the R2I LMR frame shall be no greater than that of the soliciting NDP Announcement frame, and the transmit bandwidth of the I2R LMR frame shall be no greater than the bandwidth of the preceding R2I LMR frame.

In non-TB ranging, the PHY shall issue the PHY-RXEND.indication primitive with error condition IntegrityCheckError if the PHY detects the integrity check error in the reception of the corresponding HE Ranging NDP. If the PHY of an RSTA issues a PHY-RXEND.indication primitive with error condition IntegrityCheckError, the RSTA shall set the Invalid Measurement field in the R2I LMR frame carrying the TOA measured from the I2R NDP to 1. Similarly, if I2R LMR was negotiated between the ISTA and RSTA and the PHY of an ISTA issues a PHY-RXEND. Indication primitive with error condition IntegrityCheckError, the ISTA shall set the Invalid Measurement field in the I2R LMR carrying the TOA measured from the R2I NDP to 1.

NOTE 2—When a STA detects that the transmit center frequency offset (CFO) between the ISTA and the RSTA exceeds the allowed tolerance from the values specified in 27.3.19.3 and 27.3.15.3, this can be an indication of a security attack.

In non-TB ranging, both RSTA and ISTA should measure the CFO value based on reception of I2R NDP and R2I NDP, respectively. Therefore, the CFO parameter field in the I2R LMR, if negotiated, and R2I LMR are reserved. The RSTA and ISTA may account for clock rate differences between ISTA and RSTA, respectively, based on their own measured CFO value. The mechanism by which t_4 and t_1 are adjusted by the RSTA, and t_2 and t_3 are adjusted by the ISTA is implementation specific.

If the Invalid Measurement field in R2I LMR or I2R LMR is equal to 1, the RSTA or ISTA receiving the LMR should discard the TOA carried in the LMR.

In the non-TB ranging measurement reporting phase, if R2I LMR reporting or I2R LMR reporting carries phase shift feedback, then the R2I LMR reporting or the I2R LMR reporting shall be immediate feedback.

In the non-TB ranging measurement reporting phase, if R2I AOA feedback was negotiated the RSTA shall include the optional AOA feedback subfield in the R2I LMR frame and if R2I LMR reporting was negotiated in addition to I2R AOA Feedback reporting then the ISTA shall include the optional AOA feedback subfield (see 9.4.2.300) in the I2R LMR frame. The AOA field contains the Direction Measurement Results element described in 9.4.2.300.

11.21.6.4.5 Secure HE-LTF in the TB and non-TB ranging measurement exchange protocol

11.21.6.4.5.1 General

Both the TB and the non-TB ranging measurement exchanges allow for the use of secure HE-LTF for PHY security if the ISTA and RSTA have established a secure HE-LTF measurement setup as defined in 11.21.6.3.4. The frame exchange sequences stay nominally the same as described in 11.21.6.4.3 and 11.21.6.4.4, except that the HE Ranging NDP and HE TB Ranging NDP use secure HE-LTF as described in 27.3.18a.1 and 27.3.18a.2, respectively. To use the secure HE-LTF, the ISTA and RSTA need to share and communicate pseudorandom bit sequences that are used to generate and demodulate the secure HE-LTF, the details and management thereof is described in 11.21.6.4.5.2 to 11.21.6.4.5.6.

11.21.6.4.5.2 TB ranging measurement exchange with secure HE-LTF

An example of the negotiation and two TB Ranging Measurement Exchanges with secure HE-LTF is shown in Figure 11-37o, where the LTF_VALID_SAC and SEC_LTF_CTR refer to the values of the Validation SAC and Secure HE-LTF Counter subfields, respectively. The first value of the Validation SAC and its associated Secure HE-LTF Counter subfields shall be included in a protected IFTM frame, and thereafter any subsequent

value of the Validation SAC and its corresponding Secure HE-LTF Counter subfield shall be included in a protected R2I LMR frame. The value of the Validation SAC subfield shall also be the same value of the SAC subfield in the Trigger Dependent User Info field of the Secure Sounding Ranging Trigger frame. The description of how these fields are set in the TB ranging measurement exchange is given next.

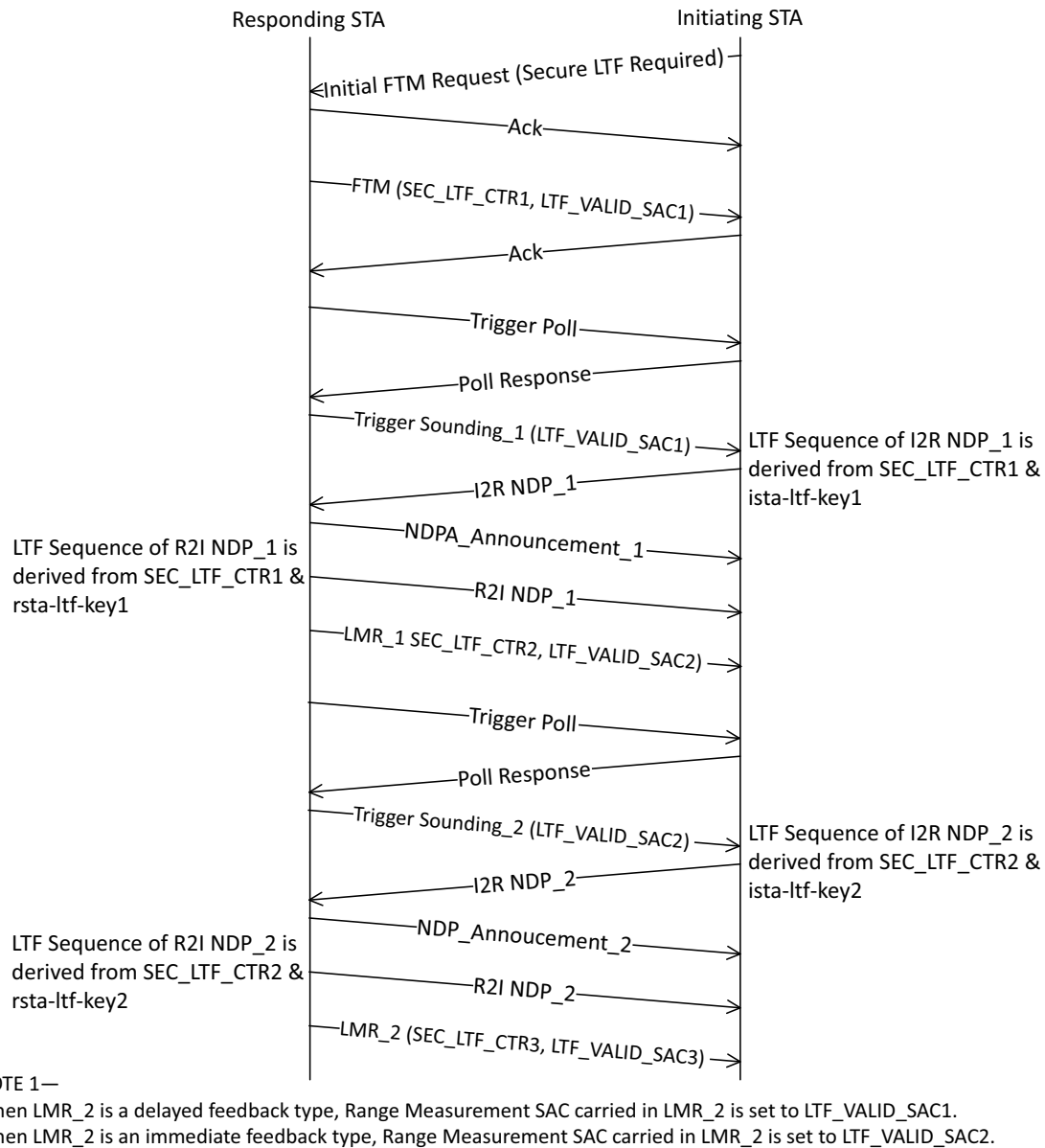


Figure 11-37o—Overview of the TB ranging measurement exchange with secure HE-LTF

In a TB ranging measurement exchange with secure HE-LTF where there are multiple ISTAs involved in the measurement sequence, the RSTA shall transmit a Secure Sounding Ranging Trigger frame that includes a single User Info field to trigger a single ISTA at a time.

When an RSTA has established a secure HE-LTF measurement exchange with an ISTA as specified in 11.21.6.3.4, the RSTA that sends a Secure Sounding Ranging Trigger frame to the STA shall set:

- The SAC subfield in the Trigger Dependent User Info field corresponding to the ISTA in the Secure Sounding Ranging Trigger frame to the same value as in the Validation SAC field in the Secure HE-LTF Parameters element in the last transmitted protected IFTM, or last transmitted protected LMR frame, to the ISTA.

The RSTA shall set the I2R Rep subfield of the User Info field corresponding to the ISTA in the Secure Sounding Ranging Trigger frame equal to the value of the *RSTA Assigned I2R Rep* corresponding to the ISTA, where the value of the *RSTA Assigned I2R Rep* shall be greater than 0.

After transmission of the Secure Sounding Ranging Trigger frame to the ISTA, the RSTA's MAC sublayer shall issue a PHY-RXLTFSSEQUENCE.request primitive with a LTFVECTOR containing the following parameters:

- The SECURE_LTF_FLAG parameter set to 1.
- The LTF_NSTS and LTF_REP parameters set to the same values, respectively, as indicated by the SS Allocation and I2R Rep subfields of the User Info field.
- The LTF_KEY and LTF_IV parameters that are set to *ista-ltf-key* and *ltf-iv*, respectively, for generating the secure HE-LTF based on the value of the Secure HE-LTF Counter subfield in the Secure HE-LTF Parameters element in the last transmitted protected IFTM frame or last transmitted protected LMR frame to the ISTA; see 11.21.6.4.5.4.
- The TX_WINDOW_FLAG set to 1 if the RSTA and ISTA have negotiated to use the optional frequency domain Tx window for I2R NDP; it is set to 0 otherwise.
- The LTF_OFFSET set to 0.

When the RSTA receives the HE TB Ranging NDP from the ISTA, the RSTA shall:

- a) Send a Ranging NDP Announcement frame.
- b) Send an HE Ranging NDP with the TXVECTOR parameters LTF_KEY and LTF_IV set to the *rsta-ltf-key* and *ltf-iv*, respectively, for generating secure HE-LTF based on the value of the Secure HE-LTF Counter subfield in the Secure HE-LTF Parameters element in the last transmitted protected IFTM frame or last transmitted protected LMR frame to the ISTA; see 11.21.6.4.5.4.
- c) Send a protected LMR frame that includes the Secure HE-LTF Parameters element to the ISTA.

Otherwise, the RSTA shall follow the rules in 10.22.2.2 as the frame exchange is not successful.

When an RSTA sends a Ranging NDP Announcement frame, it shall set the LTF Offset subfield in the STA Info fields to values that satisfy Equation (11-6a) and Equation (11-6b).

$$\text{Offset}_i \leq \sum_{k \in \text{MaxOffset}_i} \text{N_LTF}_k \times \text{Rep}_k \quad (11-6a)$$

$$\begin{aligned} \text{Offset}_i &\geq \text{Offset}_j + \text{N_LTF}_j \times \text{Rep}_j, \quad \forall j \in \text{MinOffset}_i \\ \text{If } \text{Offset}_i &= 0 \text{ then } \text{N_STS}_i \geq \text{N_STS}_j, \quad \forall j \end{aligned} \quad (11-6b)$$

where

- Offset_n represents the LTF Offset subfield value of the n th STA Info field in the Ranging NDP Announcement frame.
- N_STS_n represents the R2I NSTS subfield value plus 1 of the n th STA Info field in the Ranging NDP Announcement frame.

- N_LTF_n represents the number of HE-LTF symbols based on N_STS_n , see Table 21-13, for the n th STA Info field in the Ranging NDP Announcement frame.
- Rep_n represents the R2I Rep subfield value plus 1 of the n th STA Info field in the Ranging NDP Announcement frame.
- MinOffset represents the set of indexes of the STA Info fields of which the LTF Offset subfield values are less than the LTF Offset subfield value of the i th STA Info field in the Ranging NDP Announcement frame.
- MaxOffset represents the set of indexes of all STA Info fields excluding the i th STA Info field.

The RSTA shall set the R2I Rep subfield in each of the STA Info fields in the Ranging NDP Announcement frame equal to the *RSTA Assigned R2I Rep* for each of the corresponding ISTAs, where all of the *RSTA Assigned R2I Rep* shall be greater than 0.

When an ISTA receives a Secure Sounding Ranging Trigger frame from an RSTA in which the value of the SAC subfield in the Trigger Dependent User Info field is equal to the value of the Validation SAC subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame, or last protected LMR frame, received from the RSTA, the ISTA shall send an HE TB Ranging NDP with the TXVECTOR parameters LTF_KEY and LTF_IV that are set to *ista-ltf-key* and *ltf-iv*, respectively, for generating the secure HE-LTF based on the value of the Secure HE-LTF Counter subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame, or last protected LMR frame, received from the RSTA; see 11.21.6.4.5.4.

When an ISTA receives a Secure Sounding Ranging Trigger frame from an RSTA in which the value of the SAC subfield in the Trigger Dependent User Info field is not equal to the value of the Validation SAC subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame or last protected LMR frame received from the RSTA, the ISTA shall send an HE TB Ranging NDP with the TXVECTOR parameters LTF_KEY and LTF_IV that are set to the *ista-ltf-key* and *ltf-iv*, respectively, for generating any secure HE-LTF.

When an ISTA receives a Ranging NDP Announcement frame from an RSTA in which the AID11 subfield in the STA Info field contains the 11 least significant bits of the AID or RSID of the ISTA, the ISTA shall issue a PHY-RXLTFSEQUENCE.request primitive with the following LTFVECTOR parameters:

- The SECURE_LTF_FLAG parameter set to 1.
- The LTF_NSTS, LTF_REP, and LTF_OFFSET parameters set to the same values, respectively, as indicated by the R2I NSTS, R2I Rep, and LTF Offset subfields of the STA Info field addressed to it.
- The LTF_KEY and LTF_IV parameters that are set to the *rsta-ltf-key* and *ltf-iv* for generating the secure HE-LTF based on the value of the Secure LTF Counter subfield in the Secure LTF Parameters element in the last protected IFTM frame, or last protected LMR frame received from the RSTA; see 11.21.6.4.5.4.
- The TX_WINDOW_FLAG set to 1 if the ISTA and RSTA have negotiated to use the optional frequency domain Tx window for R2I NDP; it is set to 0 otherwise.

When an LMR frame contains range measurement results measured from an I2R NDP or an R2I NDP, an RSTA that transmits an R2I LMR frame, or when negotiated, an ISTA that transmits an I2R LMR frame, shall include the Secure HE-LTF Parameters element in the protected LMR frame:

- The Measurement SAC subfield in the Secure HE-LTF Parameters element in the protected LMR frame shall be set to the same value as in the SAC subfield in the Trigger Dependent User Info field in the Secure Sounding Ranging Trigger destined to the ISTA receiving or transmitting this protected LMR frame.
- The Measurement Result LTF Offset subfield in the Secure HE-LTF Parameters element in the protected LMR frame shall be set to the same value as in the LTF Offset subfield of the STA Info field in the Ranging NDP Announcement frame that preceded the R2I NDP destined to the ISTA for this measurement result.

When an ISTA or RSTA receives the R2I or I2R protected LMR frame, the ISTA or RSTA shall compare the value of the Measurement Result LTF Offset subfield with the value of the LTF Offset subfield in the corresponding STA Info field of the Ranging NDP Announcement frame, and if these two values don't match, the ISTA or RSTA shall discard the measurement results carried in the protected LMR frame.

When a frame exchange that consists of receiving an HE TB Ranging NDP and transmitting HE Ranging NDP results into a Null-SAC-HE-LTF, the RSTA shall not use the TOA and TOD value of the HE TB Ranging NDP and HE Ranging NDP, respectively, and shall set the Invalid Measurement Indication subfield to 1 in the TOA error field in the protected LMR carrying the TOA value of the I2R HE TB Ranging NDP.

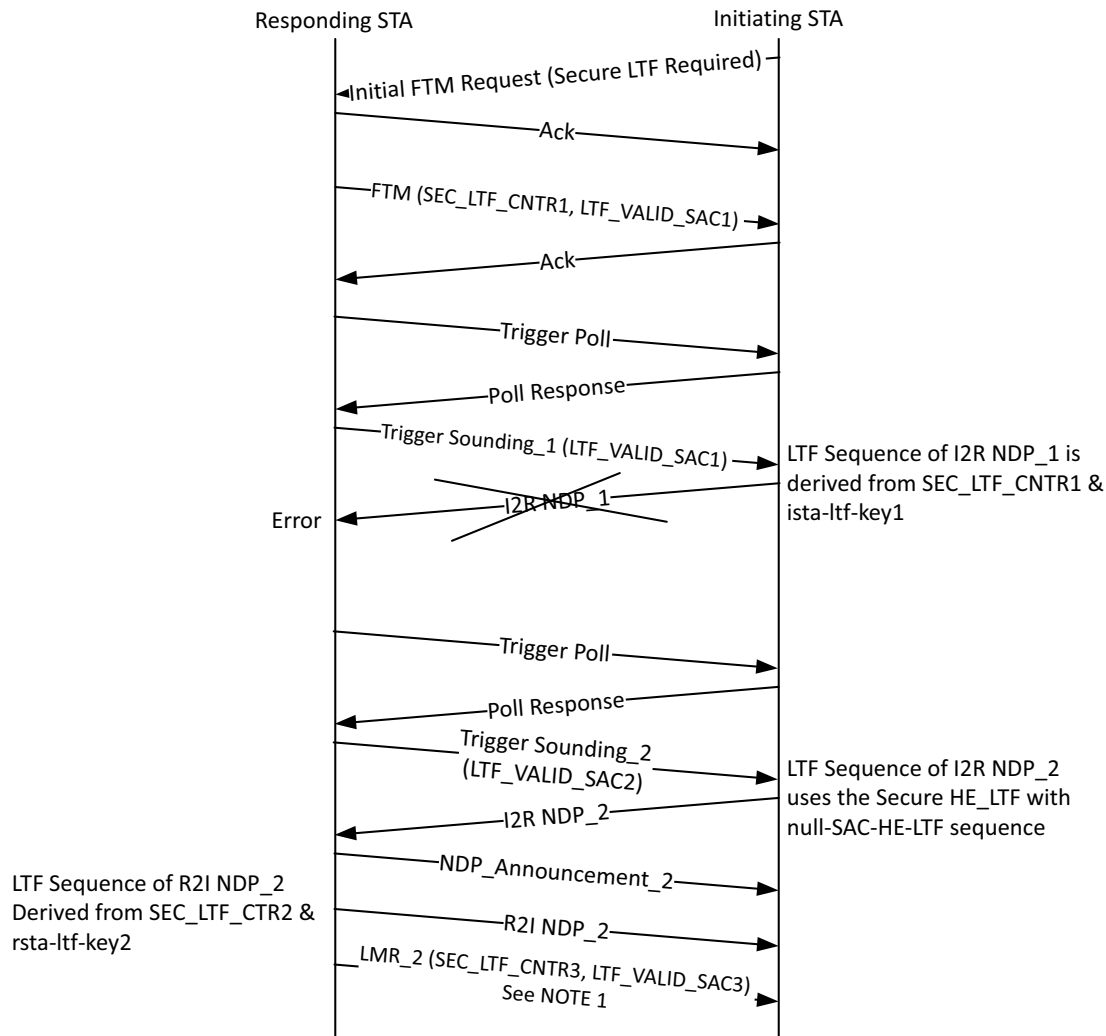
When an ISTA receives a Secure Sounding Ranging Trigger frame from an RSTA in which the value of the SAC subfield in the Trigger Dependent User Info field is not equal to the value of the Validation SAC subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame or last protected LMR frame received from the RSTA, the ISTA shall:

- Not use the TOD value of the I2R HE TB Ranging NDP.
- Not use the TOA value of the R2I HE Ranging NDP, and shall set the Invalid Measurement Indication subfield to 1 in the TOA Error field in the protected LMR carrying the TOA value of the HE Ranging NDP if the I2R LMR transmission from the ISTA was negotiated.

The STA shall discard the SAC value used in the frame exchange and shall not use the same SAC value in the current measurement exchange.

NOTE—In TB ranging measurement exchange with secure HE-LTF, the RSTA never sends the SAC subfield with value 0 in the Trigger Dependent User Info field in the Secure Sounding Ranging Trigger frame.

When there is a transmission failure within a secure measurement exchange sequence, the recovery procedure of the LTF_VALID_SAC and its associated SEC_LTF_CTR parameters is illustrated in Figure 11-37p.

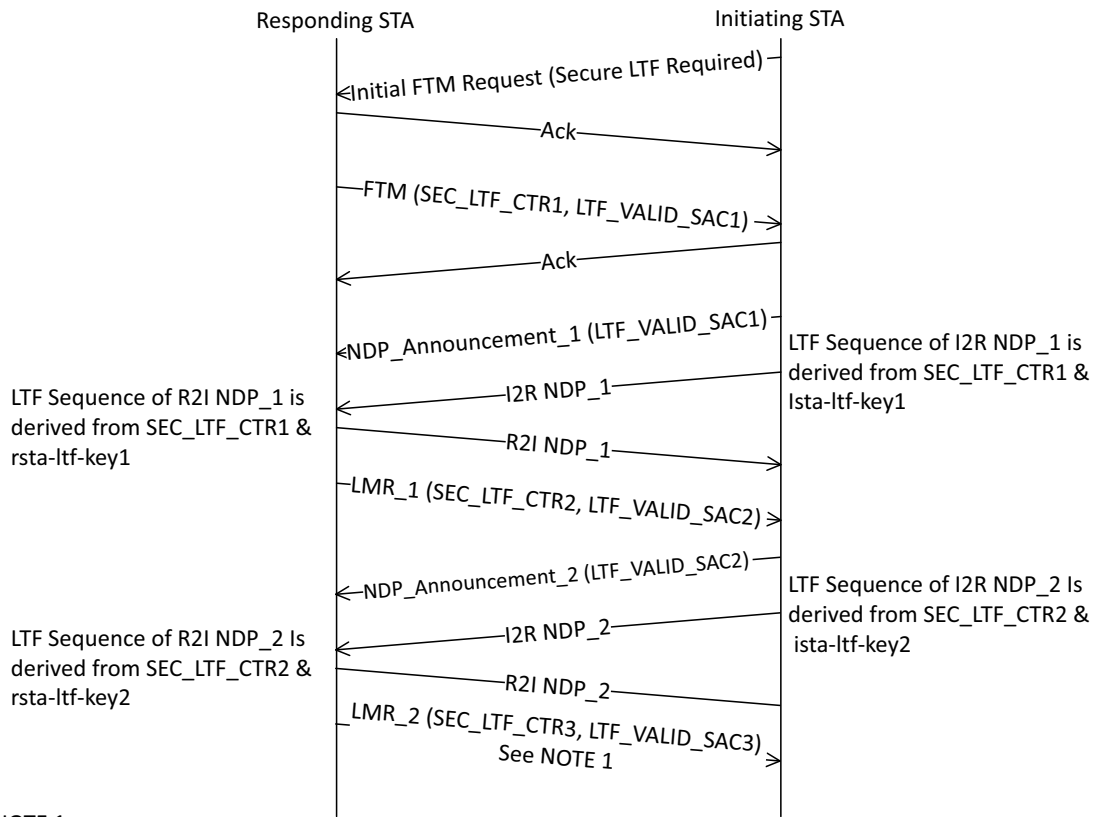


NOTE 1—
Range measurement result carried in LMR_2 is invalid and Range Measurement SAC field carried in LMR_2 is reserved, regardless of whether LMR_2 is a delayed feedback type or an immediate feedback type.

Figure 11-37p—Error recovery of TB ranging measurement exchange using secure HE-LTF

11.21.6.4.5.3 Non-TB ranging measurement exchange with secure HE-LTF

An overview of the negotiation and two non-TB ranging measurement exchanges are illustrated in Figure 11-37q. The LTF_VALID_SAC and its associated SEC_LTF_CNTR parameter shall be included in a protected IFTM frame and a protected R2I LMR frame. The LTF_VALID_SAC shall also be included in the SAC subfield in the Ranging NDP Announcement frame. The detailed description of how the fields in these frames are set follows.



NOTE 1—

When LMR is a delayed feedback type, Range Measurement SAC carried in LMR_2 is set to LTF_VALID_SAC1.
When LMR is an immediate feedback type, range measurement SAC carried in LMR_2 is set to LTF_GEN_SAC2

Figure 11-37q—Overview of the non-TB ranging measurement exchange with secure HE-LTF

When an ISTA has established a secure HE-LTF measurement exchange with an RSTA as specified in 11.21.6.3.4, this ISTA shall set the following subfields in any Ranging NDP Announcement frame addressed to that RSTA as follows:

- The SAC subfield in the STA Info field with AID11 equal to 2043 in the Ranging NDP Announcement frame is set to the same value contained in the Validation SAC subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame, or last protected R2I LMR frame, received from the RSTA.
- Otherwise the SAC subfield in the STA Info field with AID11 equal to 2043 in the Ranging NDP Announcement frame is set to a value of 0 to indicate that a new SEC_LTF_CTRL and the corresponding LTF_VALID_SAC parameter are needed.

The ISTA shall set the I2R Rep subfield and R2I Rep subfield of the STA Info field in the Ranging NDP Announcement frame to the values of *RSTA Assigned I2R Rep* and the *RSTA Assigned R2I Rep*, respectively, corresponding to the RSTA. Both values of the *RSTA Assigned R2I Rep* and *RSTA Assigned I2R Rep* shall be greater than 0.

An ISTA that sends an I2R NDP a SIFS after transmission of the Ranging NDP Announcement frame shall set the TXVECTOR parameters LTF_KEY and LTF_IV as follows:

- Either to the value of Null-SAC-HE-LTF if the SAC subfield in the STA Info field with AID11 equal to 2043 in the Ranging NDP Announcement frame, is equal to 0.
- Or the *ista-ltf-key* and *ltf-iv* for generating secure HE-LTF based on the value of the Secure HE-LTF Counter subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame or last protected LMR frame, received from the RSTA; see 11.21.6.4.5.4.

After transmission of the Ranging NDP Announcement frame to the RSTA, the ISTA's MAC sublayer shall issue a PHY-RXLTFSEQUENCE.request primitive with an LTFVECTOR containing the following parameters:

- The SECURE_LTF_FLAG parameter set to 1.
- The LTF_NSTS and LTF_REP parameters set to the same values as indicated, respectively, by the R2I NSTS and R2I Rep subfields in the STA Info field with the AID11 subfield less than or equal to 2007.
- The LTF_KEY and LTF_IV parameters that are set to either the value of Null-SAC-HE-LTF, if the SAC subfield in the STA Info field with AID11 equal to 2043 in the Ranging NDP Announcement frame is equal to 0; or the *rsta-ltf-key* and *ltf-iv* for generating a secure HE-LTF based on the values of the Secure HE-LTF Counter subfield in the Secure HE-LTF Parameters element in the last protected IFTM frame, or last protected LMR frame received, from the RSTA; see 11.21.6.4.5.4.
- The TX_WINDOW_FLAG set to 1 if the ISTA and RSTA have negotiated to use the optional frequency domain Tx window for R2I NDP; it is set to 0 otherwise.
- The LTF_OFFSET set to 0.

When an RSTA receives a Ranging NDP Announcement frame from an ISTA in which the SAC subfield in the STA Info field with AID11 equal to 2043 is not equal to the value of the Validation SAC subfield in the Secure HE-LTF Parameters element in the last transmitted protected IFTM frame or last transmitted protected LMR frame to the ISTA, the RSTA shall:

- Send an HE Ranging NDP to the ISTA with the TXVECTOR parameters *rsta-ltf-key* and *ltf-iv* for generating any secure HE-LTF to the ISTA, only if the RSTA receives an HE Ranging NDP from the ISTA a SIFS after the ranging NDP Announcement frame.
- Send a protected LMR frame with a Secure HE-LTF Parameters element containing the SEC_LTF_CTR and the corresponding LTF_VALID_SAC parameters to the ISTA, only if the RSTA receives an HE Ranging NDP from the ISTA a SIFS after the ranging NDP Announcement frame.

When an RSTA receives a Ranging NDP Announcement frame from an ISTA in which the value of the SAC subfield in the STA Info field with AID11 equal to 2043 is equal to the value of the Validation SAC subfield in the Secure HE-LTF Parameters element in the last transmitted protected IFTM frame or last transmitted protected LMR frame to the ISTA, the RSTA shall:

- Send an HE Ranging NDP with the TXVECTOR parameters *rsta-ltf-key* and *ltf-iv* for generating a secure HE-LTF based on the values of the Secure HE-LTF Counter in the Secure HE-LTF Parameters element in the last transmitted protected IFTM frame, or last transmitted protected LMR frame to the ISTA, only if the RSTA receives an HE Ranging NDP from the ISTA a SIFS after the ranging NDP Announcement frame; see 11.21.6.4.5.4.
- Send a protected LMR frame that includes the Secure HE-LTF Parameters element to the ISTA, only if the RSTA receives an HE Ranging NDP from the ISTA a SIFS after the ranging NDP Announcement frame.

When an RSTA receives a Ranging NDP Announcement frame from an ISTA, the RSTA shall also issue a PHY-RXLTFSEQUENCE.request primitive with an LTFVECTOR with the following parameters:

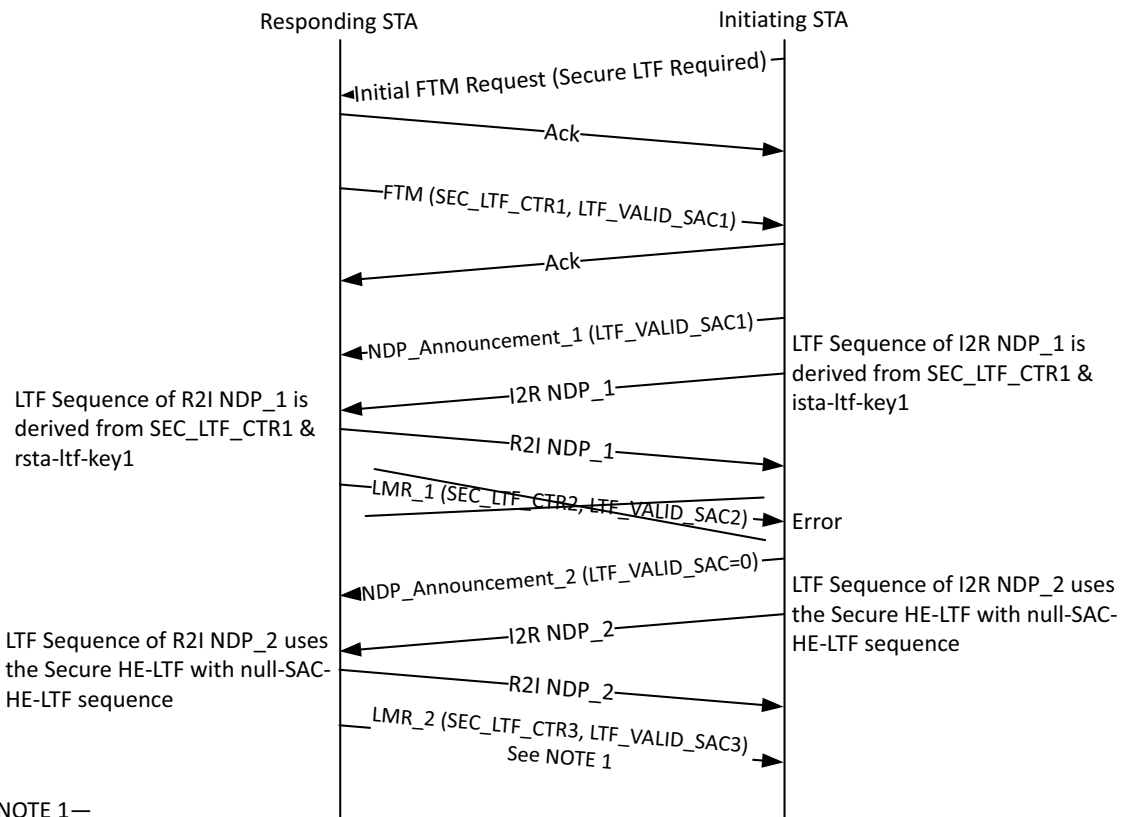
- The SECURE_LTF_FLAG parameter set to 1.
- The LTF_NSTS and LTF_REP parameters set to the same values as indicated, respectively, by the I2R NSTS and I2R Rep subfields in the STA Info field with the AID11 subfield less than or equal to 2007.
- The LTF_KEY and LTF_IV parameters that are set to the *ista-ltf-key* and *ltf-iv* for receiving a secure HE-LTF based on the values of the Secure LTF Counter and corresponding Validation SAC subfields in the Secure LTF Parameters element in the last transmitted protected IFTM frame, or last transmitted protected LMR frame to the ISTA; see 11.21.6.4.5.4.
- The TX_WINDOW_FLAG set to 1 if the ISTA and RSTA have negotiated to use the optional frequency domain Tx window for I2R NDP; it is set to 0 otherwise.
- The LTF_OFFSET set to 0.

An RSTA transmitting an R2I LMR frame, or an ISTA when negotiated to transmit LMR frame, containing range measurement results measured from an I2R NDP and an R2I NDP, shall include the Secure HE-LTF Parameters element in the protected LMR frame and set the Measurement SAC subfield in the Secure HE-LTF Parameters element in the protected LMR frame to the same value as in the SAC subfield in the STA Info field with AID11 equal to 2043 in the Ranging NDP Announcement frame that solicited the I2R NDP and the R2I NDP.

When a STA sends a Null-SAC-HE-LTF in a HE Ranging NDP, the STA shall not use the TOD value of the HE Ranging NDP for the secure range measurement exchange.

When a STA receives a Null-SAC-HE-LTF in a HE Ranging NDP, the STA shall not use the TOA value of the HE Ranging NDP, and shall set the Invalid Measurement Indication subfield to 1 in the TOA Error field in the protected LMR carrying the TOA value of the HE Ranging NDP.

When there is a transmission failure within a secure measurement exchange, the recovery procedure of the LTF_VALID_SAC is illustrated in Figure 11-37r.



NOTE 1—

When LMR is a delayed feedback type, Range Measurement SAC carried in LMR_2 is set to LTF_VALID_SAC1. When LMR is an immediate feedback type, range measurement result carried in LMR_2 is invalid and Range Measurement SAC field carried in LMR_2 is reserved.

Figure 11-37r—Error recovery of non-TB ranging measurement exchange using secure HE-LTF

A STA shall discard the LTF_VALID_SAC parameter used in the frame exchange and shall not use the same LTF_VALID_SAC parameter in the current measurement exchange.

11.21.6.4.5.4 Overview of secure HE-LTF octet stream generation

This subclause describes mechanisms for generating the SAC, LTF protection keys, counters, and the pseudorandom octet stream used to randomize the input to the modulation and per stream phase rotation for constructing secure HE-LTFs. The mechanism is illustrated in Figure 11-37s and Figure 11-37t.

For each secure measurement (e.g., NDP exchanges), a SAC and two secret keys *ista-ltf-key* and *rsta-ltf-key* shall be derived by the ISTA and the RSTA independently as follows:

$$\text{SAC-and-LTF-Keys} = \text{KDF-Hash-Length}(\text{Secure-LTF-Key-Seed}, \text{"Secure HE-LTF Expansion"}, \text{Secure-LTF-Counter})$$

where

KDF and Hash are the key derivation function and hash function determined by the AKM used to derive the PTKSA from which the Secure-LTF-Key-Seed was derived

Length	is equal to 272 (bits)
SAC	is equal to L(SAC-and-LTF-Keys, 0, 16)
<i>ista-ltf-key</i>	is equal to L(SAC-and-LTF-Keys, 16, 128)
<i>rsta-ltf-key</i>	is equal to L(SAC-and-LTF-Keys, 144, 128)

The *ista-ltf-key* shall be used to generate the pseudorandom octet stream to protect all of the LTFs in PPDU_s transmitted by the ISTA. The *rsta-ltf-key* shall be used to generate the pseudorandom octet stream to protect all of the LTFs in PPDU_s transmitted by the RSTA. The ISTA and RSTA shall use the same derivation and derive identical keys.

With the SAC constructed as above, an attacker not knowing the Secure-LTF-Key-Seed would not be able to predict the SAC that would be used for a given measurement.

Integer to octet string conversion (MSB first) specified in 12.4.7.2.2 shall be used to encode the value of the Secure HE-LTF Counter subfield in the KDF as well as in the transmitted LTF sequence information. The counter shall be padded with leading (MSB) 0s to be exactly 6 octets.

The 16-octet IV input *ltf-iv* to the secure HE-LTF bit generator shall be constructed as follows:

- First 6 octets shall be the transmitter MAC address (A2).
- Next 6 octets shall encode the value of the Secure HE-LTF Counter subfield with the encoding convention described above.
- Final 4 octets shall be used as a 32-bit block counter. The block counter shall be initialized to 0 before the NDP with secure HE-LTFs is transmitted or received. The counter shall be incremented by 1 each time an AES block is output by the generator.

Each time pseudorandom bits are required to protect the LTFs in an NDP, the secure HE-LTF bit generator generates the required number of AES blocks using the AES algorithm (see FIPS 197) with the corresponding 128-bit key and 128-bit IV inputs. The output of the AES-128 counter encryption is a 128-bit integer. It shall be converted using the conventions specified in 12.4.7.2.2 to obtain the octet stream used for secure HE-LTF generation.

NOTE 1—A 6-octet parameter representing the value of the Secure HE-LTF Counter subfield is sufficient because a unicast protected management frame that uses a 6-octet PN is used to convey the LTF sequence information that carries the counter.

NOTE 2—The pseudorandom bit generator is based on AES-128 Counter mode approved by NIST (see NIST SP 800-90A R1 [B57a]).

The number of pseudorandom bits that can be generated without violating security guarantees of the scheme is 2^{39} without updating the key or reseeding the generator with additional entropy.

Secure HE-LTF measurement requires a variable number of octets from the pseudorandom octet stream depending on the TXVECTOR or RXVECTOR parameters for the LTFs as described in section 27.3.18b.2. The initial block counter value used to construct the *ltf-iv* for setting the TXVECTOR and RXVECTOR parameters shall be 0.

11.21.6.4.5.5 Secure HE-LTF octet stream generation on an ISTA

Figure 11-37s illustrates the scheme for generating the SAC, LTF protection keys, and counter values used to generate a pseudorandom octet stream on an ISTA.



For each secure measurement on an ISTA, the following parameters shall be used for generating the pseudorandom octet stream used to construct the LTFs for NDPs:

- The *ista-ltf-key* and *rsta-ltf-key* for transmitted NDP (TXVECTOR parameter LTF_KEY) and received NDP (RXVECTOR parameter LTF_KEY), respectively.
- The *ltf_iv* for transmitted NDP (TXVECTOR parameter LTF-IV) and received NDP (RXVECTOR parameter LTF-IV) with corresponding ISTA MAC address and RSTA MAC address, respectively, together with the value of the Secure HE-LTF Counter subfield and the block counter.

11.21.6.4.5.6 Secure HE-LTF input octet stream generation on an RSTA

Figure 11-37t illustrates the scheme for generating the SAC, LTF protection keys, and counter values used to generate a pseudorandom octet stream on an RSTA.

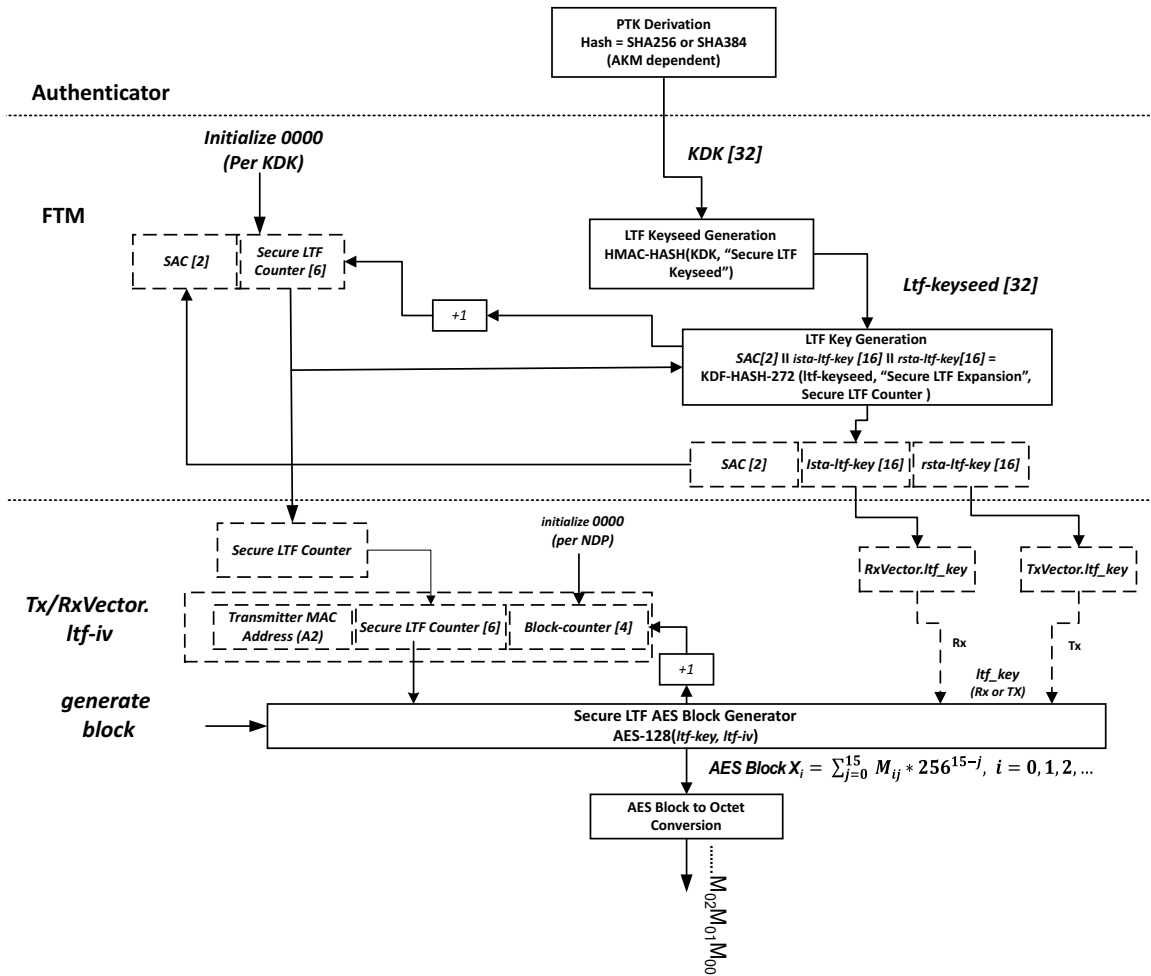


Figure 11-37t—RSTA secure HE-LTF octet stream generation

For each secure measurement on an RSTA, the following parameters are provided for generating the pseudorandom octet stream to construct the LTFs for NDPs:

- The *ista-ltf-key* and *rsta-ltf-key* for received NDP (RXVECTOR parameter LTF_KEY) and transmitted NDP (TXVECTOR parameter LTF_KEY), respectively.
- The *ltf_iv* for received NDP (RXVECTOR parameter LTF-IV) and for transmitted NDP (TXVECTOR parameter LTF-IV) with corresponding ISTA MAC address and RSTA MAC address, respectively, together with the values of the Secure HE-LTF Counter subfield and the block counter.

NOTE—In an R2I NDP used for range measurement, LTFs assigned to each of the recipient STAs would use their corresponding pseudorandom octet stream derived from the key material from the corresponding PTKSA for the recipient.

The values of the Secure HE-LTF Counter subfield shall be maintained for the lifetime of a PTKSA used for secure HE-LTF measurements. It shall not be reset between measurements and shall not be reset for multiple FTM negotiations using the same PTKSA.

The values of the Secure HE-LTF Counter subfield used for each measurement protected bits is derived from a given Secure-LTF-Key-Seed (and its KDK and the PTKSA) and shall be unique. When the derived SAC is equal to 0, the RSTA shall increment the secure HE-LTF counter parameter by 1 and derive the SAC until a nonzero SAC value is obtained. An RSTA shall also increment the secure HE-LTF counter parameter by 1 each time an *ista-ltf-key* and *rsta-ltf-key* are derived.

11.21.6.4.6 Transmission of a ranging NDP

An RSTA transmitting an HE Ranging NDP to one or more peer ISTAs shall set the TXVECTOR parameters as follows:

- The FORMAT parameter is set to HE_SU.
- The RANGING_FLAG is present.
- The UPLINK_FLAG parameter is set to 0.
- The APEP_LENGTH parameter is set to 0.
- The SECURE_LTF_FLAG is set as follows:
 - Is set to 0 in the TB ranging measurement exchange (11.21.6.4.3) and non-TB ranging measurement exchange (11.21.6.4.4).
 - Is set to 1 in the TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.2) and the non-TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.3).
- The TX_WINDOW_FLAG is set to 1 if the SECURE_LTF_FLAG is set to 1 and the RSTA and ISTA have negotiated to use the optional frequency domain Tx window for R2I NDPs; it is set to 0 otherwise.
- In the TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.2), the NUM_USERS parameter is set to the number of ISTAs to which the HE Ranging NDP is transmitted.
- In the Non-TB ranging measurement exchange (11.21.6.4.4), the TXPWR_LEVEL_INDEX parameter is set to a value that matches the Tx Power value indicated in the R2I NDP Tx Power field in the following LMR frame, except if the value in the R2I NDP Tx Power field was set to a reserved value.
- The DOPPLER parameter is set to 0.
- The NUM_STS parameter is set as follows:
 - In the TB ranging measurement exchange (11.21.6.4.3), set to the same value as the R2I NSTS subfield in the STA Info field in the preceding Ranging NDP Announcement frame plus 1.
 - In the TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.2):
 - When the HE Ranging NDP is transmitted to more than one ISTA, the NUM_STS[*p*] parameter is set to the same value as the R2I NSTS subfield in the STA Info field addressed to the corresponding STA *p* in the preceding Ranging NDP Announcement frame plus 1.
 - When the HE Ranging NDP is transmitted to one ISTA, the NUM_STS parameter is set to the same value as the R2I NSTS subfield in the STA Info field with AID11 subfield equal or less than 2007 in the preceding Ranging NDP Announcement frame plus 1.
 - In the Non-TB ranging measurement exchange (11.21.6.4.4) and the non-TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.3), set to the same value as the R2I NSTS subfield in the STA Info field with AID11 subfield equal or less than 2007 in the preceding Ranging NDP Announcement frame plus 1.
- The LTF_REP parameter is set as follows:
 - In the TB ranging measurement exchange (11.21.6.4.3), set to the same value as the R2I Rep subfield in the STA Info field in the preceding Ranging NDP Announcement frame plus 1.

- In the TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.2):
 - When the HE Ranging NDP is transmitted to more than one ISTA, the LTF_REP[*p*] parameter is set to the same value as the R2I Rep subfield in the STA Info field addressed to the corresponding STA *p* in the preceding Ranging NDP Announcement frame plus 1.
 - When the HE Ranging NDP is transmitted to one ISTA, the LTF_REP parameter is set to the same value as the R2I Rep subfield in the STA Info field with AID11 subfield equal or less than 2007 in the preceding Ranging NDP Announcement frame plus 1.
- In the non-TB ranging measurement exchange (11.21.6.4.4) and the non-TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.3), set to the same value as the R2I Rep subfield in the STA Info field with AID11 subfield less than or equal to 2007 in the preceding Ranging NDP Announcement frame plus 1.
- The CH_BANDWIDTH parameter is set as follows:
 - In the TB ranging measurement exchange (11.21.6.4.3), and TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.2), set to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging Sounding Trigger frame.
 - In the non-TB ranging measurement exchange (11.21.6.4.4) and non-TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.3), set to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging NDP Announcement frame.
- In the TB and non-TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5), the LTF_KEY parameter is set as defined in 11.21.6.4.5.2 and 11.21.6.4.5.3. Otherwise, the LTF_KEY parameter is not present.
- In the TB ranging measurement exchange with secure HE-LTF, the LTF_OFFSET parameter is set as defined in 11.21.6.4.5.2. Otherwise, the LTF_OFFSET parameter is not present.
- The HE_LTF_TYPE parameter is set to 2xHE-LTF.
- The GI_TYPE parameter is set to 1u6s_GI.
- The SPATIAL_REUSE parameter is set to SRP_AND_NON-SRG_OBSS-PD_PROHIBITED.
- The BSS_COLOR parameter is set to the value indicated in the BSS Color subfield of the HE Operation element transmitted by the RSTA.
- The TXOP_DURATION parameter is set to either 127 or a value defined in Equation (26-3), replacing $D_{\text{HE_NDPA}}$ by $D_{\text{Ranging_NDPA}}$, which is the value of the Duration/ID field in the MAC header of the preceding Ranging NDP Announcement frame.

An ISTA transmitting an HE Ranging NDP shall set the TXVECTOR parameters as follows:

- The FORMAT parameter is set to HE_SU.
- The RANGING_FLAG is present.
- The UPLINK_FLAG parameter is set to 1.
- The APEP_LENGTH parameter is set to 0.
- The SECURE_LTF_FLAG is set as follows:
 - Is set to 0 in the non-TB ranging measurement exchange (11.21.6.4.4).
 - Is set to 1 in the non-TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.3).
- The TX_WINDOW_FLAG is set to 1 if the SECURE_LTF_FLAG is set to 1 and the RSTA and ISTA have negotiated to use the optional frequency domain Tx window for I2R NDPs; it is set to 0 otherwise.
- The DOPPLER parameter is set to 0.
- The NUM_STS parameter is set to the same value as the I2R NSTS subfield in the STA Info field with AID11 subfield equal or less than 2007 in the preceding Ranging NDP Announcement frame plus 1.

- The LTF_REP parameter is set to the same value as the I2R Rep subfield with AID11 subfield equal or less than 2007 in the STA Info field in the preceding Ranging NDP Announcement frame plus 1.
- The TXPWR_LEVEL_INDEX parameter is set to a value that matches the Tx Power value indicated in the I2R NDP Tx Power subfield in the STA Info field with the AID11 subfield set to 2045 in the preceding Ranging NDP Announcement frame, except if the value in the I2R NDP Tx Power subfield was set to a reserved value.
- The CH_BANDWIDTH is set to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging NDP Announcement frame.
- In the non-TB ranging measurement exchange with secure HE-LTF, the LTF_KEY parameter is set as defined in 11.21.6.4.5.3. Otherwise, the LTF_KEY parameter is not present.
- The HE_LTF_TYPE parameter is set to 2xHE-LTF.
- The GI_TYPE parameter is set to 1u6s_GI.
- The SPATIAL_REUSE parameter is set to SRP_AND_NON-SRG_OBSS-PD_PROHIBITED.
- The BSS_COLOR parameter is set to the value indicated in the BSS Color subfield of the HE Operation element received from the RSTA.
- The TXOP_DURATION parameter is set to either 127 or a value defined in Equation (26-3), replacing $D_{\text{HE_NDPA}}$ by $D_{\text{Ranging_NDPA}}$, which is the value of the Duration/ID field in the MAC header of the preceding Ranging NDP Announcement frame.

An ISTA transmitting an HE TB Ranging NDP to an RSTA shall set the TXVECTOR parameters as follows:

- The FORMAT parameter is set to HE_TB.
- The RANGING_FLAG is present.
- The APEP_LENGTH parameter is set to 0.
- The SECURE_LTF_FLAG is set as follows:
 - Is set to 0 in the TB ranging measurement exchange (11.21.6.4.3).
 - Is set to 1 in the TB ranging measurement exchange with secure HE-LTF (11.21.6.4.5.2).
- The TX_WINDOW_FLAG is set to 1 if the SECURE_LTF_FLAG is set to 1 and the RSTA and ISTA have negotiated to use the optional frequency domain Tx window for I2R NDPs; it is set to 0 otherwise.
- The DOPPLER parameter is set to 0.
- The NUM_STS parameter is set to the same value as the Number Of Spatial Streams subfield in the SS Allocation field in the User Info field in the preceding Ranging Sounding Trigger frame.
- The LTF_REP parameter is set to the same value as the I2R Rep subfield in the User Info field in the preceding Ranging Sounding Trigger frame plus 1.
- The CH_BANDWIDTH parameter is set to the same value as the TXVECTOR parameter CH_BANDWIDTH in the preceding Ranging Sounding Trigger frame.
- In the TB ranging measurement exchange with secure HE-LTF, the LTF_KEY parameter is set as defined in 11.21.6.4.5.2. Otherwise, the LTF_KEY parameter is not present.
- The HE_LTF_TYPE parameter is set to 2xHE-LTF.
- The GI_TYPE parameter is set to 1u6s_GI.
- The SPATIAL_REUSE parameter is set to SRP_AND_NON-SRG_OBSS_PD_PROHIBITED.
- The BSS_COLOR parameter is set to the value indicated in the BSS Color subfield of the HE Operation element received from the RSTA.

The TXOP_DURATION parameter is set as defined in 26.11.5.

11.21.6.4.7 Time of arrival estimation using phase shift feedback

Based on Figure 11-37u, and Equation (11-6c), to enable the ISTA to derive the RTT, the RSTA needs to compute TOA t_2 and feed t_2 and t_3 back to the ISTA using R2I LMR. Instead of utilizing TOA t_2 for RTT computation, a phase shift feedback can be prepared by the RSTA and fed back to the ISTA for deriving RTT.

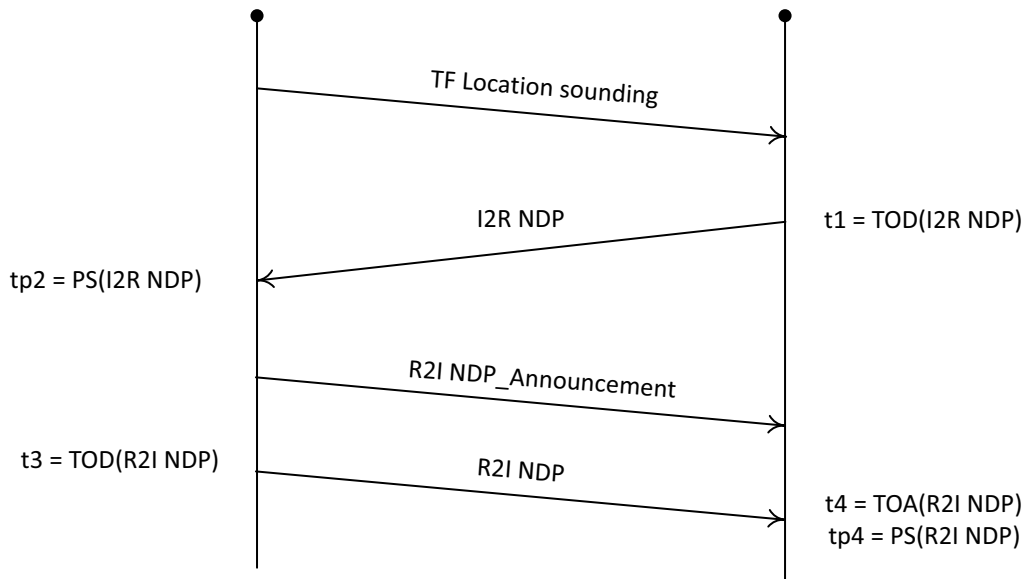


Figure 11-37u—Timing diagram of a measurement sounding phase in TB ranging based on phase shift of I2R NDP and R2I NDP PPDU

As shown in Figure 11-37u, in the phase shift feedback method, the ISTA transmits an I2R NDP at TOD t_1 , and the RSTA determines the phase shift tp_2 of the I2R NDP. The RSTA transmits an R2I NDP at TOD t_3 , and the ISTA determines the phase shift tp_4 and TOA t_4 of the R2I NDP. tp_2 and tp_4 are determined from the phase slope of the frequency domain channel estimation of the corresponding NDP. An example of calculation of the phase shift is shown in Annex AD.

The RSTA sends the R2I NDP at TOD t_3 , and after receiving the R2I NDP, the ISTA calculates the phase shift tp_4 and TOA t_4 of R2I NDP. The value of tp_2 and tp_4 are calculated utilizing the frequency domain channel estimation of I2R NDP and R2I NDP.

The phase shift is defined as the average linear phase shift between two adjacent tones normalized by the tone spacing. To enable the ISTA to calculate the RTT, the RSTA should feed phase shift tp_2 and TOD t_3 back to the ISTA using R2I LMR, and the ISTA can calculate the RTT according to Equation (11-6c).

$$RTT_{ISTA} = (t_4 - t_1) - (t_3 - t_2''), \text{ with } t_2'' = tp_2 - (tp_4 - t_4) \quad (11-6c)$$

When the I2R LMR with phase shift feedback is negotiated between ISTA and RSTA, I2R LMR carries phase shift tp_4 and TOD t_1 , and then the RSTA can calculate the RTT according to Equation (11-6d).

$$RTT_{RSTA} = (t_4'' - t_1) - (t_3 - t_2), \text{ with } t_4'' = tp_4 - (tp_2 - t_2) \quad (11-6d)$$

11.21.6.4.8 Measurement exchange in passive TB ranging mode

11.21.6.4.8.1 General

As stated in 11.21.6.1.3, the passive TB ranging mode is a variant of the TB ranging mode. In all aspects, except where explicitly stated differently, the passive TB ranging mode, its protocols, procedures, components, and definitions follow the rules for TB ranging mode.

In particular, the measurement exchanges for passive TB ranging follow the rules and procedures described in 11.21.6.4.3, with subclauses, unless explicitly stated otherwise.

In passive TB ranging, the RSTA shall transmit the Passive Sounding Ranging Trigger frame instead of the Sounding Ranging Trigger frame. Upon receiving of the Passive Sounding Ranging Trigger frame, the ISTA shall respond with an HE Ranging NDP instead of an HE TB Ranging NDP; see 11.21.6.4.8.3 for further details.

Furthermore, the RSTA shall broadcast two frames, the Primary and Secondary RSTA Broadcast Passive TB Ranging Measurement Report frames containing measurement data and related information; see 11.21.6.4.8.4 for further details.

The passive TB ranging exchanges shall only occur in an availability window assigned for Passive TB ranging.

For passive TB ranging, the timestamps reported within each availability window shall be derived from a clock that runs continuously during the availability window.

If there is a discontinuity in the clock for the FTM timestamping between two reported TOD timestamps, then the TOD Not Continuous subfield in the Timestamp Error subfield of the Timestamp Measurement Report subfield in the ISTA Passive TB Ranging Measurement Report element shall be set to 1. Otherwise it shall be set to 0.

11.21.6.4.8.2 Polling phase of passive TB ranging

The Polling phase of passive TB ranging follows the same rules and procedures for the Polling phase of TB ranging described in 11.21.6.4.3.2.

11.21.6.4.8.3 Passive TB ranging measurement sounding phase

The passive TB ranging measurement sounding follows the same rules and procedures for the measurement sounding for TB ranging described in 11.21.6.4.3.3, unless explicitly stated otherwise.

The second phase of the passive TB ranging measurement sequence, after the passive TB ranging polling phase, is called the passive TB ranging measurement sounding phase. The passive TB ranging measurement sounding phase may include one or more Passive Sounding Ranging Trigger frames and HE Ranging NDP exchanges, a Ranging NDP Announcement frame, and an HE Ranging NDP transmission; see Figure 11-37v.

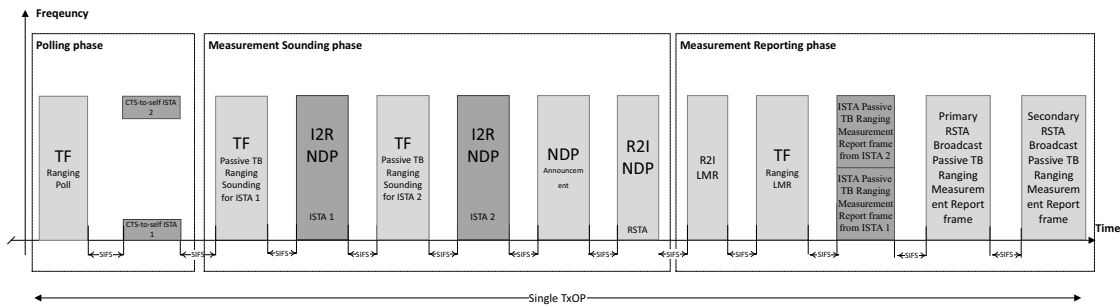


Figure 11-37v—Passive TB ranging polling, measurement sounding, and measurement reporting phases

In passive TB ranging, for each ISTA, the RSTA shall transmit a Passive Sounding Ranging Trigger frame, which includes a single User Info field.

An RSTA shall transmit one or more Passive Sounding Ranging Trigger frames, each of which is addressed to a single ISTA, the first one coming a SIFS time after the TB Polling phase.

An ISTA addressed by the AID/RSID in the Passive Sounding Ranging Trigger frame shall transmit an HE Ranging NDP a SIFS time after the reception of the Passive TB Ranging Trigger frame.

An RSTA transmitting a Passive Sounding Ranging Trigger frame shall not use a bandwidth wider than that indicated in the IFTM frame sent to the ISTA, and the RSTA shall set the TXVECTOR parameter CH_BANDWIDTH to be the same value as the UL BW subfield of the Common Info field in the Passive Sounding Ranging Trigger frame.

NOTE—Generally a PSTA benefits from consistent ranging measurement performance when an RSTA initiates a passive TB ranging sequence with the nominal advertised bandwidth in every TXOP.

An RSTA transmitting a Ranging NDP Announcement frame and an HE Ranging NDP after receiving an HE Ranging NDP as a response to a Passive Sounding Ranging Trigger frame shall set the TXVECTOR parameter CH_BANDWIDTH to be the same value as the BW subfield of the Common Info field in the Passive Sounding Ranging Trigger frame.

An ISTA transmitting an HE Ranging NDP as a response to a Passive Sounding Ranging Trigger frame shall set the TXVECTOR parameter CH_BANDWIDTH to be the same value as the UL BW subfield of the Common Info field in the Passive Sounding Ranging Trigger frame.

As in TB ranging, an ISTA participating in a passive TB ranging exchange shall measure the TOD of its own HE Ranging NDP and either the TOAs, or both the TOAs and the phase shift feedback TOAs (PSTOAs), when it receives the RSTA's HE Ranging NDP. In addition, optionally the ISTA may also measure and report either the TOAs, or both the TOAs and the PSTOAs, when it receives the HE Ranging NDPs transmitted by the other ISTAs participating in the passive TB ranging exchange. By reporting the timestamps for when it received the other ISTAs NDP transmissions, the quality of the location estimate for a PSTA listening in to the passive TB ranging exchanges can be improved.

The value of NSTS used in the passive TB ranging exchanges shall be less than or equal to 4.

When phase shift feedback is negotiated between an ISTA and an RSTA in passive TB ranging, the protocol for the Measurement Sounding phase differs from passive TB ranging with TOA feedback on the following points:

- The RSTA shall measure phase shift feedback TOA (PSTOA), in addition to measuring the TOA, on the I2R NPD it receives from the ISTA.
- The ISTA shall measure:
 - The phase shift TOA (PSTOA), in addition to measuring the TOA, for the R2I NDP it receives from the RSTA,
 - And may also measure phase shift TOA(s) (PSTOAs), in addition to measuring the TOA(s), for the I2R NDP(s) it receives from other ISTA(s).

See Figure 11-37w for an example of timestamps measured by the RSTA, ISTA, and a PSTA in a passive TB ranging measurement exchange. The timestamp values t_1 , t_2 , t_3 , and t_4 are analogous to the corresponding labeled timestamps in 11.21.6.4.3.3. The timestamps t_5 and t_6 are the times at which the I2R NDP and R2I NDPs arrive at the PSTA, respectively.

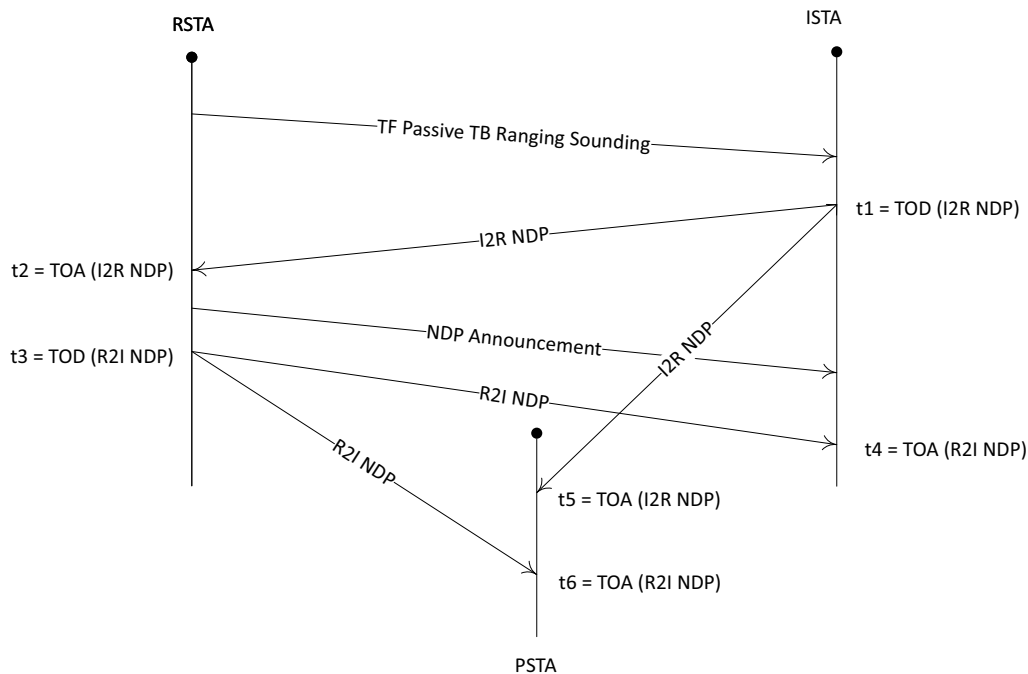


Figure 11-37w—Example timing diagram of a Measurement Sounding phase in passive TB ranging

The PSTA may use the ISTA's and RSTA's timestamps, together with its own measured TOAs of the ranging NDPs, t_5 and t_6 , to calculate its differential time of flight to the RSTA and the ISTA.

The differential time-of-flight (DToF) from PSTA to RSTA and ISTA (DToF_PRI) is defined by Equation (11-6e).

$$\text{DToF_PRI} = \text{ToF_PR} - \text{ToF_PI} \quad (11-6e)$$

where

ToF_PR is the time of flight between the PSTA and the RSTA
ToF_PI is the time of flight between the PSTA and the ISTA

The differential time of flight $DTof_PRI$ can be computed as per Equation (11-6f):

$$DTof_PRI = t_6 - t_5 - 0.5 \times t_3' + 0.5 \times t_2' - 0.5 \times t_4' + 0.5 \times t_1' \quad (11-6f)$$

where

- t_1' and t_4' are the time at which the I2R NDP was transmitted from the ISTA and the time at which the R2I NDP was received by the ISTA, respectively, converted by the PSTA from the ISTA's time basis to the PSTA's time basis.
- t_2' and t_3' are the time at which the I2R NDP was received by the RSTA and the time at which the R2I NDP was transmitted by the RSTA, respectively, converted by the PSTA from the RSTA's time basis to the PSTA's time basis.

At the PSTA, the mechanism by which t_1' and t_4' is derived from t_1 , t_4 , the ISTA's reported CFO, and the PSTA's CFO measured with respect to the RSTA, is implementation dependent.

At the PSTA, the mechanism by which t_2' and t_3' is derived from t_2 , t_3 , and the PSTA's CFO measured with respect to the RSTA, is implementation dependent.

By multiplying the differential time of flight, $DTof_PRI$, with the speed of light, the differential distance from PSTA to RSTA and ISTA can be computed.

See 11.21.6.4.9 for how the PSTA's differential distance to the RSTA and the ISTA can be computed using PSTOAs measured by the RSTA and the ISTA.

11.21.6.4.8.4 Passive TB ranging Measurement Reporting phase

The passive TB ranging measurement reporting follows the same rules and procedures for the measurement reporting for TB ranging described in 11.21.6.4.3.4, unless explicitly stated otherwise.

The last phase of the passive TB ranging measurement sequence is the passive TB ranging Measurement Reporting phase and occurs a SIFS time after the passive TB ranging Measurement Sounding phase; see Figure 11-37x for a depiction of the passive TB ranging Measurement Reporting phase.

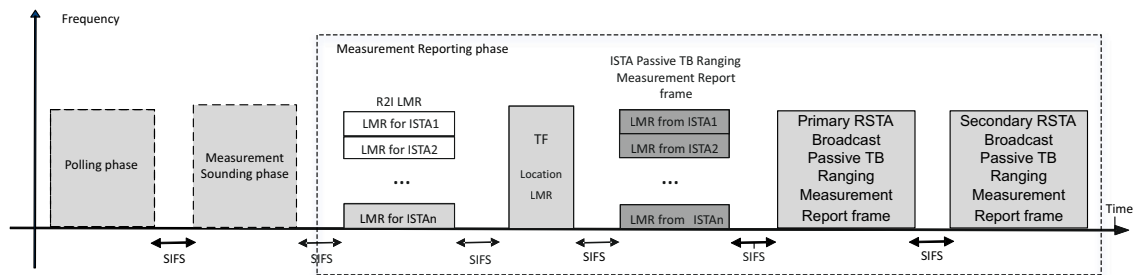


Figure 11-37x—Passive TB ranging Measurement Reporting phase

In the passive TB ranging Measurement Reporting phase, an RSTA shall send a Passive TB Ranging Measurement Report frame and the Report Ranging Trigger frame to one or more ISTAs that sent an HE Ranging NDP in the preceding passive TB ranging Measurement Sounding phase. An ISTA addressed by the Report Ranging Trigger frame shall transmit an ISTA Passive TB Ranging Measurement Report frame a SIFS time after the Report Ranging Trigger frame transmission to report its I2R LMR.

In order to facilitate broadcasting of the ISTA's timestamps by the RSTA, the ISTA Passive TB Ranging Measurement Report frame shall be transmitted as a public Action frame.

The ISTA Passive TB Ranging Measurement Report element, see 9.4.2.304, in ISTA Passive TB Ranging Measurement Report frames shall contain:

- A Sounding Dialog Token Number identifying the measurement sounding phase in which the reported ISTA's timestamps were measured.
- The CFO of the ISTA with respect to the RSTA.
- The TOD timestamp for the I2R NDP that the ISTA transmitted—labeled with the AID12/RSID12 of the ISTA.
- The TOA timestamp for the R2I NDP that the ISTA received from the RSTA.
- Optionally, the TOA timestamps for the I2R NDPs received from other ISTAs participating in the passive TB ranging (i.e., polling, sounding, and reporting triplet) identified by the Sounding Dialog Token Number—labeled with their respective AID12/RSID12s.

If phase shift TOA reporting has been negotiated, the ISTA Passive TB Ranging Measurement Report element shall also include:

- The PS-TOA timestamp of the R2I NDP that the ISTA received from the RSTA.
- Optionally, the PS-TOAs for the I2R NDPs received from other ISTAs participating in the passive TB ranging (i.e., polling, sounding, and reporting triplet) identified by the Sounding Dialog Token Number—labeled with their respective AID12/RSID12s.

The ISTA Passive TB Ranging Measurement Report frame shall include an entry for the ISTA's I2R NDP TOD.

The ISTA shall set the More subfield in the More & N Timestamp Measurements Report field in the ISTA Passive TB Ranging Measurement Report element contained in the ISTA Passive TB Ranging Measurement Report frame to 1 if it has more timestamps ready to report but does not have space in its allocated resources by the RSTA for ISTA Passive TB Ranging Measurement Report frame. Else the ISTA shall set the More subfield to 0.

The RSTA shall send the Primary and Secondary RSTA Broadcast Passive TB Ranging Measurement Report frames, the Primary a SIFS time after receiving the ISTA Passive TB Ranging Measurement Report frames from the ISTA and the Secondary a SIFS following the Primary; see Figure 11-37x.

The Primary RSTA Broadcast Passive TB Ranging Measurement Report frame shall contain the following:

- Passive TB Ranging LCI Table Counter.
- Passive TB Ranging LCI Table Countdown Info.
- RSTA Passive TB Ranging Measurement Report.
- Passive TB Ranging LCI Table (optionally present).

Each time an RSTA transmits a Primary RSTA Broadcast Passive TB Ranging Measurement Report frame, it shall set the Passive TB Ranging LCI Table Counter value such as to refer to the latest version of the Passive TB Ranging LCI Table element transmitted by the RSTA.

If a Passive TB Ranging LCI Table element is included in the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame, and this element has different content as compared to the last transmitted Passive TB Ranging LCI Table element, then the Passive TB Ranging LCI Table Counter shall be incremented by 1 (modulo 256) from the value associated with the previous Passive TB Ranging LCI Table element content. (The first time the RSTA transmits a Passive TB Ranging LCI Table element the value shall be set to 0). This

new value of the Passive TB Ranging LCI Table Counter is now associated with this new version of the Passive TB Ranging LCI Table element.

If a Passive TB Ranging LCI Table element is included in the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame, and this element has the same content as the last transmitted Passive TB Ranging LCI Table element, then the Passive TB Ranging LCI Table Counter value shall be the value associated with this last version of the Passive TB Ranging LCI Table element transmitted by the RSTA.

If a Passive TB Ranging LCI Table element is not included in the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame, then the Passive TB Ranging LCI Table Counter value shall be the value associated with the last version of the Passive TB Ranging LCI Table element transmitted by the RSTA.

When the Passive TB Ranging LCI Table is present in the Primary Broadcast Passive TB Ranging Measurement Report frame, the RSTA LCI Report field of the Passive TB Ranging LCI Table Report element shall contain the Antenna Placement And Calibration subelement if the RSTA has `dot11PassiveTBRangingAODImplemented` equal to true, and shall not contain the Antenna Placement And Calibration subelement if the RSTA has `dot11PassiveTBRangingAODImplemented` equal to false.

When the Passive TB Ranging LCI Table is present in the Primary Broadcast Passive TB Ranging Measurement Report frame, the corresponding entry of the ISTA LCI Reports Entries field of the Passive TB Ranging LCI Table Report element shall contain the Antenna Placement And Calibration subelement if the ISTA has `dot11PassiveTBRangingAoDImplemented` equal to true, and shall not contain the Antenna Placement And Calibration subelement if the ISTA has `dot11PassiveTBRangingAODImplemented` equal to false.

The Secondary RSTA Broadcast Passive TB Ranging Measurement Report frame shall contain the following:

- ISTA Passive TB Ranging Measurement Reports.

See 9.6.7.52.

When phase shift feedback is negotiated between an ISTA and an RSTA in Passive TB ranging, the protocol for the measurement reporting phase differs from Passive TB ranging with TOA feedback on the following points:

- The RSTA shall report its measured PSTOA in the R2I LMR frame.
- The ISTA shall report its measured PSTOA(s), in addition to its measured TOA(s), in the ISTA Passive TB Ranging Measurement Report frame.
 - The PSTOAs are indicated as phase shift TOA timestamps by setting the Measurement Report field of the ISTA Passive TB Ranging Measurement Report element, see 9.4.2.304, to the value 10 (PSTOA).
- In the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame, the RSTA shall send a broadcast frame containing its measured PSTOA, in addition to its measured TOA, for the I2R NDPs it has received from the ISTA.
- In the Secondary RSTA Broadcast Passive TB Ranging Measurement Report frame the RSTA shall rebroadcast the timestamps the ISTA has reported to the RSTA. As the ISTA has negotiated phase shift feedback, these would contain PSTOAs in addition to TOAs.

When phase shift feedback is negotiated in Passive TB ranging, the reporting by both the RSTA and the ISTA of phase shift TOAs, the TOD, and CFO shall be immediate feedback. The reported TOAs may be immediate or delayed feedback. When the TOA feedback is delayed, the dialog token shall refer to the previous measurement instance for the RSTA–ISTA pair.

The TODs and PSTOAs measured by the RSTA shall be broadcast in the Primary RSTA Broadcast Passive TB Ranging Measurement Report frame in the reporting phase following the measurement exchange in which they were measured.

The CFO, TOD, and PSTOAs reported by the ISTA shall be rebroadcasted in the Secondary RSTA Broadcast Passive TB Ranging Measurement Report frame in the reporting phase following the measurement exchange in which they were measured.

11.21.6.4.9 Passive TB ranging differential time-of-flight calculations using phase shift TOA timestamps

In Figure 11-37y, passive TB ranging measurement exchanges and their reception by a passive station, a PSTA, are depicted for the case when the RSTA and the ISTA are measuring and reporting phase shift TOAs, PSTOAs; see Annex AD for how the phase shift TOAs are calculated.

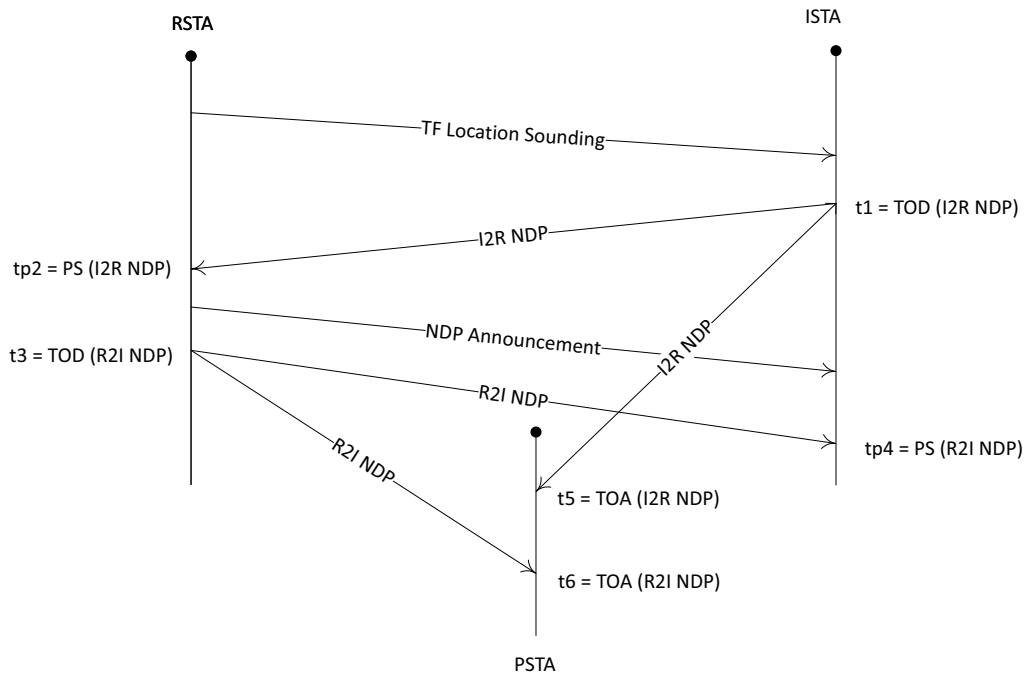


Figure 11-37y—Example of passive TB ranging measurement exchanges with PSTOA measurements

The RSTA measures the PSTOA tp2 and the ISTA measures the PSTOA tp4, in addition to measuring their TODs t3 and t1, respectively. The ISTA also measures its CFO with respect to the RSTA. The ISTA reports its CFO and timestamps to the RSTA and the RSTA broadcasts these as well as its own timestamps to the PSTA.

The ISTA also measures and reports the TOA timestamps for the ranging NDPs it receives and the RSTA also measures and broadcasts the TOAs of the ranging NDPs it receives, as well as the ISTAs reported TOAs, though these timestamps are not used in the calculations described here.

The differential time of flight from the PSTA to the RSTA and the ISTA (DTof_PRI) is defined by Equation (11-6g):

$$\text{DTof_PRI} = \text{ToF_PR} - \text{ToF_PI} \quad (11-6g)$$

where

ToF_PR is the time of flight between the PSTA and the RSTA
ToF_PI is the time of flight between the PSTA and the ISTA

The differential time of flight (DTof_PRI) can then be computed as per Equation (11-6h):

$$\text{DTof_PRI} = t_6 - t_5 - 0.5 \times t_3' + 0.5 \times tp_2' - 0.5 \times tp_4' + 0.5 \times t_1' \quad (11-6h)$$

where

timestamp t1' is the time at which the I2R NDP was transmitted from the ISTA.
timestamp tp4' is the PSTOA measurement for the time at which the R2I NDP was received by the ISTA, converted from the ISTA's time basis to the PSTA's time basis.
timestamp tp2' is the PSTOA measurement for the time at which the I2R NDP was received by the RSTA.
timestamp t3' is the time at which the R2I NDP was transmitted from the RSTA, converted from the RSTA's time basis to the PSTA's time basis.

At the PSTA, the mechanism by which t1' and tp4' is derived from t1, tp4, the ISTA's reported CFO, and the PSTA's CFO measured with respect to the RSTA is implementation dependent.

At the PSTA, the mechanism by which t2' and tp3' is derived from t2, tp3, and the PSTA's CFO measured with respect to the RSTA is implementation dependent.

Change the title and content of 11.21.6.5 as follows:

11.21.6.5 ~~Fine timing measurement~~ FTM parameter modification

During an FTM session, an ~~initiating STA-ISTA~~ may terminate the current session and request a new session with modified session parameters by transmitting a ~~Fine Timing Measurement~~ an FTM Request frame with Trigger field set to 1 ~~and including a new Fine Timing Measurement Parameters element.~~

This FTM Request shall include:

- A new FTM Parameters element if the corresponding FTM session is using an EDCA based measurement exchange; see 11.21.6.4.2 or,
- A Ranging Parameters element if the corresponding FTM session is of type non-TB ranging; see 11.21.6.4.4, or TB ranging; see 11.21.6.4.3.

NOTE—This allows up to one ranging session between a given ISTA and RSTA at any time.

The existing FTM session is terminated upon reception of such a ~~Fine Timing Measurement~~ an FTM Request frame. This ~~Fine Timing Measurement~~ FTM Request frame is an ~~initial Fine Timing Measurement Request~~ IFTMR frame for the new FTM session, which follows the behavior described in 11.21.6.3.

Insert 11.21.6.5.1 at the end of 11.21.6.5 as follows:

11.21.6.5.1 Availability window parameter modification

In an FTM session based on TB ranging, the RSTA may modify the availability window parameters corresponding to the FTM session by transmitting an A-MPDU containing an FTM frame and an R2I LMR whenever the RSTA is permitted to transmit such an LMR to that ISTA. The FTM frame is of type Action No Ack and shall contain a Ranging Parameters field containing a TB-Specific subelement. The Availability Window field in the TB-Specific subelement indicates the parameters of the new availability window assigned

to the corresponding FTM session. In the FTM frame the Follow Up Dialog Token field is set as 0. The existing ranging session continues based on current availability window parameters until the start of the new availability window signaled in the FTM frame.

Change the title of 11.21.6.6 as follows:

11.21.6.6 ~~Fine timing measurement~~ FTM session termination

Insert a new subclause heading for 11.21.6.6.1 at the beginning of 11.21.6.6 (thereby making what was previously the content of 11.21.6.6, now the content of 11.21.6.6.1) as follows:

11.21.6.6.1 EDCA based ranging session termination

Change the content of former 11.21.6.6 (now 11.21.6.6.1) as follows:

An FTM session terminates after the last burst instance, as indicated in the Number of Bursts Exponent, Burst Duration, FTMs Per Burst and Burst Period fields in the ~~initial Fine Timing Measurement~~ FTM frame.

An FTM session may be terminated before then through one of the following:

- At any time during the FTM session when the ~~responding STA-RSTA~~ is permitted to transmit a ~~Fine Timing Measurement~~ an FTM frame (see 11.21.6.4), the ~~responding STA-RSTA~~ sends a ~~Fine Timing Measurement~~ an FTM frame with the Dialog Token field set to 0.
- At any time during the FTM session when the initiating STA is permitted to transmit a ~~Fine Timing Measurement~~ an FTM Request frame (see 11.21.6.4), the initiating STA sends a ~~Fine Timing Measurement~~ an FTM Request frame with the Trigger field set to 0. This frame shall not include the following:
 - A Measurement Request element
 - ~~A Fine Timing Measurement~~ An FTM Parameters element
- At any time during the FTM session when the initiating STA is permitted to transmit a ~~Fine Timing Measurement~~ an FTM Request frame (see 11.21.6.4), the initiating STA terminates the current session and requests a new session with modified ~~Fine Timing Measurement~~ FTM parameters (see 11.21.6.5).
- After the number of burst instances indicated in the Number of Bursts Exponent field in the initial ~~Fine Timing Measurement~~ FTM frame has been reached.

When the FTM session is a Secure FTM Session, the FTM frames transmitted shall be the Protected Fine Timing Action frames; see 9.6.34.

Insert 11.21.6.6.2 after 11.21.6.6.1 as follows:

11.21.6.6.2 TB ranging and non-TB ranging session termination

A non-TB ranging FTM session may be terminated if an ISTA does not initiate a non-TB ranging measurement exchange (11.21.6.4.4) with an RSTA before the duration signaled in the Max Time Between Measurements field assigned to that ISTA by the RSTA has elapsed from either the end of the successful FTM session negotiation or the beginning of the last successful measurement exchange, whichever is later.

A TB ranging FTM session may be terminated if the ISTA fails to respond to a TF Ranging Poll frame and receive one TF Ranging Sounding frame or TF Ranging Secured Sounding frame containing its AID/RSID at least once within the Max Session Expiry interval. The Max Session Expiry interval starts from either the end of the successful FTM session negotiation or the beginning of the last successful TB ranging measurement exchange (11.21.6.4.3), whichever is later. The length of this interval is equal to the duration signaled in the

Max Session Exp field, present in the TB Ranging Specific subelement in the Ranging Parameters field in the initial FTM frame.

A TB ranging or a non-TB ranging FTM session may be terminated through one of the following:

- At any time during the session when the RSTA is permitted to transmit an R2I LMR frame, the RSTA transmits an A-MPDU containing an LMR frame and an FTM frame with the Dialog Token field set to 0 and of type Action No ACK. In the FTM frame the Follow Up Dialog Token field is set as 0. The FTM frame shall not include any Ranging Parameters field.
- At any time during the session the ISTA sends an FTM Request frame with the Trigger field set to 0. This frame shall not include the following:
 - Ranging Parameters element.
 - Measurement Request element.
- At any time during the session the ISTA terminates the current session and requests a new session with modified ranging parameters; see 11.21.6.5.

When the ranging session is a Secure FTM Session, the corresponding FTM frames transmitted shall be Protected Fine Timing Action frames; see 9.6.34.

11.21.6.7 LCI and Location Civic retrieval using FTM procedure

Change the first four paragraphs of 11.21.6.7 as follows:

Within the FTM procedure, a STA, to request the LCI of ~~a responding STA~~ an RSTA that transmits an Extended Capabilities element with the Fine Timing Measurement Responder field set to 1, shall include a Measurement Request element with Measurement Type field equal to LCI within the ~~Fine Timing Measurement FTM Request frame~~.

Within the FTM procedure, a STA, to request the Location Civic of ~~a responding STA~~ an RSTA that transmits an Extended Capabilities element with the Fine Timing Measurement Responder field set to 1, shall include a Measurement Request element with Measurement Type field equal to Location Civic within the ~~Fine Timing Measurement FTM Request frame~~.

A Measurement Request element with Measurement Type field equal to LCI shall not be included in a ~~Fine Timing Measurement FTM Request frame~~ unless the ~~Fine Timing Measurement FTM Request frame~~ also includes ~~a Fine Timing Measurement FTM Parameters element or a Ranging Parameters element~~. A Measurement Request element with Measurement Type field equal to Location Civic shall not be included in a ~~Fine Timing Measurement FTM Request frame~~ unless the ~~Fine Timing Measurement FTM Request frame~~ also includes ~~a Fine Timing Measurement FTM Parameters element or a Ranging Parameters element~~.

When ~~a responding STA~~ an RSTA that has both dot11FineTimingMsmtRespActivated and dot11RMLCIConfigured equal to true receives a Measurement Request element with Measurement Type field equal to LCI within an ~~initial Fine Timing Measurement Request FTM frame~~, the ~~responding STA~~ RSTA shall include a Measurement Report element with Measurement Type field equal to LCI in the ~~initial Fine Timing Measurement FTM frame~~. If the maximum horizontal or vertical location error of the ~~responding STA~~ RSTA relative to a reference STA is known and this relative error is smaller than the absolute error indicated in the LCI subelement, then the ~~responding STA~~ RSTA may include a Relative Location Error subfield in the Measurement Report field. The ~~responding STA~~ RSTA shall not include a Measurement Report element with Measurement Type field equal to LCI in the ~~Fine Timing Measurement FTM frame~~ in the current FTM session unless the ~~responding STA~~ RSTA's LCI has changed since the last time it was reported to the initiating STA.

11.24 Quality-of-service Management frame (QMF)

11.24.1 General

11.24.1.2 Default QMF policy

Insert a new row for “Public Action - LMR” above the row for “Radio measurement” and a new row for “Protected Fine Timing Frame” above the row for “Vendor-specific Protected” in Table 11-16 as follows (unchanged rows not shown):

Table 11-16—Default QMF policy

Description	Management Frame Subtype value from Table 9-1	Category value from Table 9-51	Action field	QMF access category
...				
Public Action—LMR	1101	4	47	AC_VO
...				
Protected Fine Timing frame	1101	34	Any	AC_VO for Protected FTM Request and Protected FTM. N/A for Protected LMR.
...				

12. Security

12.2 Framework

12.2.4 RSNA establishment

Insert new item g) after item f) in the lettered list after the first paragraph of 12.4.2 as follows:

- g) If an RSNA uses PASN authentication, an RSNA capable the STA establishes an RSNA as described in 12.13.

Change the last paragraph of 12.2.4 as follows:

An RSNA STA shall not associate if Shared Key authentication was invoked prior to RSN association. PASN authentication may be used within an RSNA. However, an RSNA STA shall allow association only if authentication other than PASN was invoked prior to RSN association.

12.2.7 Requirements for management frame protection

Change the first two paragraphs of 12.2.7 as follows:

The robust Management frames are Disassociation, Deauthentication, ~~and robust Action~~, and robust Action No Ack frames.

Action frames and Action No Ack frames specified with “No” in the “Robust” column of Table 9-51 are not robust Management frames and shall not be protected.

Insert 12.2.11 after 12.2.10 as follows:

12.2.11 EDMG secure ranging sequences

EDMG secure ranging uses Secure TRN subfields, see 28.9.3, as part of the TRN fields of EDMG PPDU. Those TRN subfields are based on bit sequences henceforth denoted as secure TRN sequences. These secure TRN bit sequences are generated as follows:

The Secret Key is 32 octets randomly generated by the ISTA and sent in the Secure Ranging Operation Parameters field, see Figure 9-626d, of the EDMG Specific Parameters subelement of the Fine Timing Parameters element sent by the ISTA to the RSTA in the initial Protected FTM request; see 11.21.6.3.6. The Secret Key is used as Input Key Material (IKM) to generate pseudorandom secure TRN sequences that are used to construct secure ranging waveforms at the ISTA and RSTA. The Secret Key and Salt shall be discarded after the FTM session is terminated. Each FTM session shall have a different Secret Key.

Pseudorandom secure TRN sequences are generated using HKDF (Hashed Message Authentication Code (HMAC)-based Key Derivation Function) specified in IETF RFC 5869:

The hash function to be employed in HKDF is SHA-256.

The Secret Key is to be used as the IKM in HKDF. See RFC 5869, section 2.2.

The Salt is the PMKID corresponding to the security association between the ISTA and RSTA. See RFC 5869, section 2.2.

A pseudorandom key (PRK) is generated using the hash function accepting IKM and Salt as inputs. See RFC 5869 section 2.2. The hash function to be employed to generate the PRK is SHA-256.

The Info field is a fixed string “EDMG Secure RTT”, which is unique to this protocol in order to guard against accidental key reuse in a different subsystem. Key reuse across different subsystems shall be avoided through careful system design, and the Secret Key shall not be visible outside of the subsystem. See RFC 5869, section 2.3.

Pseudorandom secure TRN sequences are the output of the HKDF-Expand function with the PRK, Info field, and the length of secure TRN sequences required as inputs. See RFC 5869, section 2.3. One time calculation of all pseudorandom secure TRN sequences for multiple timing measurements should be used for minimizing HKDF setup costs in case of multiple ranging attempts. The length of the Secure TRN sequences generated shall be long enough for generating Secure TRN subfields for the number of bursts expected in the session.

Furthermore, if memory is not constrained in an implementation, caching multiple Secret Keys and Salt pairs and pre-generating all secure TRN sequences for multiple timing measurements and multiple FTM sessions is allowed provided that this information is not revealed to third parties.

12.5 RSNA confidentiality and integrity protocols

12.5.3 CTR with CBC-MAC protocol (CCMP)

12.5.3.2 CCMP MPDU format

Replace Figure 12-16 with the following figure:

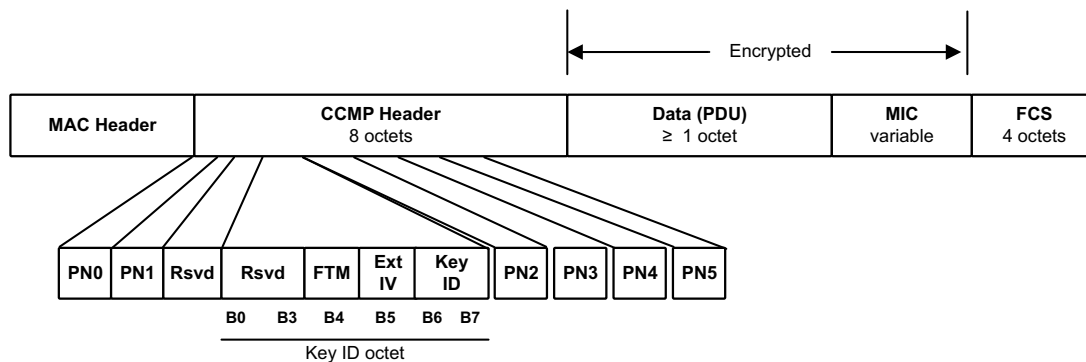


Figure 12-16—Expanded CCMP MPDU

Insert new text after the fourth paragraph (beginning “Bits 6–7 of the Key ID octet ...”) in 12.5.3.2 as follows:

In a protected unicast management Action frame, bit 4 of the Key ID octet equals 1 if the frame is a Protected Fine Timing frame—see Table 9-51. In other protected unicast frames, bit 4 is reserved.

The remaining bits of the Key ID octet are reserved.

12.5.3.4 CCMP decapsulation

12.5.3.4.4 PN and replay detection

Change the lettered list after the third paragraph of 12.5.3.4.4 as follows:

See 12.5.3.2 for a description of how the PN is encoded in the CCMP header. The following processing rules are used to detect replay:

- a) The receiver shall maintain a separate set of replay counters for each PTKSA, GTKSA, and protocol version value. The receiver initializes these replay counters to 0 when it resets the temporal key for a peer. The replay counter is set to the PN value of accepted CCMP MPDUs.
- b) For each PTKSA, GTKSA, and protocol version value, the recipient shall maintain a separate replay counter for each TID, subject to the limitation of the number of supported replay counters indicated in the RSN Capabilities field (see 9.4.2.24), and shall use the PN from a received frame to detect replayed frames. A replayed frame occurs when the PN from a received frame is less than or equal to the current replay counter value for the frame's MSDU or A-MSDU priority and frame type.
- c) If `dot11RSNAProtectedManagementFramesActivated` is true, the recipient shall maintain a single replay counter for received individually addressed robust Management frames except Protected Fine Timing frames (see 9.6.34) that are received with the To DS subfield equal to 0, and a single replay counter for received individually addressed robust PV1 Management frames except Protected Fine Timing frames (see 9.6.34) and shall use the PN from the received frame to detect replays.
- d) If `dot11RSNAProtectedManagementFramesActivated` is true and `dot11QMFActivated` is also true, the recipient shall maintain an additional replay counter for each ACI for received individually addressed robust Management frames except Protected Fine Timing frames (9.6.34) and robust PV1 Management frames except protected PV1 Protected Fine Timing frames (see 9.6.34) that are received with the To DS subfield equal to 1. The QMF receiver shall use the ACI encoded in the Sequence Number field of the received frame to select the replay counter to use for the received frame, and shall use the PN from the received frame to detect replays. A replayed frame occurs when the PN from the frame is less than or equal to the current value of the management frame replay counter that corresponds to the ACI of the frame.
- e) If `dot11RSNAProtectedManagementFramesActivated` is true, the recipient shall maintain a separate replay counter for receiving individually addressed Protected Fine Timing frames (see 9.6.34) and shall use the PN from the received frame to detect replays.
- f) ~~f)~~ The receiver shall discard any Data frame that is received with its PN less than or equal to the value of the replay counter that is associated with the TA and priority value of the received MPDU. The receiver shall discard MSDUs and MMPDUs whose constituent MPDU PN values are not incrementing in steps of 1. If `dot11RSNAProtectedManagementFramesActivated` is true, the receiver shall discard any individually addressed robust Management frame that is received with its PN less than or equal to the value of the replay counter associated with the TA of that individually addressed Management frame.
- g) ~~g)~~ When discarding a frame, the receiver shall increment by 1 `dot11RSNAStatsCCMPReplays` for Data frames or `dot11RSNAStatsRobustMgmtCCMPReplays` for robust Management frames.
- h) ~~h)~~ For MSDUs or A-MSDUs sent using the block ack feature, reordering of received MSDUs or A-MSDUs according to the block ack receiver operation is performed prior to replay detection.
- i) If the receiver performs replay detection prior to decryption, then the receiver shall check that the replay counter used to detect replays is correct and discard the frame if incorrect. In particular, the separate replay counter for individually addressed Protected Fine Timing frames shall be used if and only if the FTM subfield of CCMP Header (Figure 12-16) signals that the management PDU is a Protected Fine Timing frame. The replay counter shall not be updated unless the decryption is successful and the frame is accepted.

12.5.5 GCM protocol (GCMP)

12.5.5.2 GCMP MPDU format

Replace Figure 12-26 with the following figure:

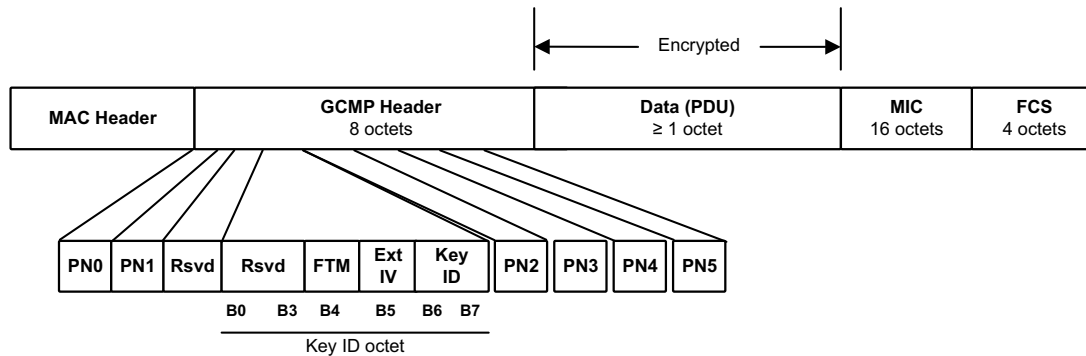


Figure 12-26—Expanded GCMP MPDU

Change the last paragraph of 12.5.5.2 as follows:

Bits 6–7 of the Key ID octet are for the Key ID subfield. ~~The remaining bits of the Key ID octet are reserved.~~

In a protected unicast management Action frame, bit 4 of the Key ID octet is set to 1 if the frame is a Protected Fine Timing frame—see Table 9-51. In other protected unicast frames, bit 4 reserved.

12.5.5.4 GCMP decapsulation

12.5.5.4.4 PN and replay detection

Change the lettered list after the first paragraph of 12.5.3.4.4 as follows:

- The receiver shall maintain a separate set of replay counters for each PTKSA and GTKSA. The receiver initializes these replay counters to 0 when it resets the temporal key for a peer. The replay counter is set to the PN value of accepted GCMP MPDUs.
- For each PTKSA and GTKSA, the recipient shall maintain a separate replay counter for each TID, subject to the limitation of the number of supported replay counters indicated in the RSN Capabilities field (see 9.4.2.24), and shall use the PN from a received frame to detect replayed frames. A replayed frame occurs when the PN from a received frame is less than or equal to the current replay counter value for the frame's MSDU or A-MSDU priority and frame type.
- If `dot11RSNAProtectedManagementFramesActivated` is true, the recipient shall maintain a single replay counter for received individually addressed robust Management frames except Protected Fine Timing frames (see 9.6.34) that are received with the To DS subfield equal to 0, and a single replay counter for received individually addressed robust PV1 Management frames except PV1 Protected Fine Timing frames (see 9.6.34) and shall use the PN from the received frame to detect replays.
- If `dot11RSNAProtectedManagementFramesActivated` is true and `dot11QMFActivated` is also true, the recipient shall maintain an additional replay counter for each ACI for received individually addressed robust Management frames except Protected Fine Timing frames (see 9.6.34) and robust PV1 Management frames except Protected Fine Timing frames (see 9.6.34) that are received with the To DS subfield equal to 1. The QMF receiver shall use the ACI encoded in the Sequence

Number field of the received frame to select the replay counter to use for the received frame, and shall use the PN from the received frame to detect replays. A replayed frame occurs when the PN from the frame is less than or equal to the current value of the management frame replay counter that corresponds to the ACI of the frame.

- e) If dot11RSNAProtectedManagementFramesActivated is true, the recipient shall maintain a separate replay counter for received individually addressed Protected Fine Timing frames (see 9.6.34) and shall use the PN from the received frame to detect replays.
- f) ~~Ⓣ~~ The receiver shall discard any Data frame that is received with its PN less than or equal to the value of the replay counter that is associated with the TA and priority value of the received MPDU. The receiver shall discard MSDUs and MMPDUs whose constituent MPDU PN values are not incrementing in steps of 1. If dot11RSNAProtectedManagementFramesActivated is true, the receiver shall discard any individually addressed robust Management frame that is received with its PN less than or equal to the value of the replay counter associated with the TA of that individually addressed Management frame.
- g) ~~Ⓣ~~ When discarding a frame, the receiver shall increment by 1 dot11RSNAStatsGCMPreplays for Data frames or dot11RSNAStatsRobustMgmtGCMPreplays for robust Management frames.
- h) ~~Ⓣ~~ For MSDUs or A-MSDUs sent using the block ack feature, reordering of received MSDUs or A-MSDUs according to the block ack receiver operation is performed prior to replay detection.

12.6 RSNA security association management

12.6.1 Security associations

12.6.1.1 Security association definitions

12.6.1.1.1 General

Change the fifth item of the dashed list after the second paragraph of 12.6.1.1.1 as follows:

- PTKSA: A result of a successful 4-way handshake, FT 4-way handshake, FT authentication sequence, ~~Ⓣ~~ FILS authentication, or PASN authentication.

12.6.1.1.6 PTKSA

Change the first paragraph of 12.6.1.1.6 as follows:

The PTKSA results from a successful 4-way handshake, FT 4-way handshake, FT protocol, FT resource request protocol, ~~Ⓣ~~ FILS authentication, or PASN authentication. This security association is also bidirectional. PTKSAs, except those established using PASN authentication, have the same lifetime as the PMKSA or PMK-R1 security Association, whichever comes first. PTKSAs for PASN authentication have a minimum of the lifetime of the PMKSA used and the timeout negotiated, if any, during PASN authentication. Because the PTKSA is tied to the PMKSA or to a PMK-R1 security association, it only has the additional information from the 4-way handshake, ~~Ⓣ~~ FT Protocol authentication, ~~Ⓣ~~ FILS authentication, or PASN authentication. For the PTKSA derived as a result of the 4-way handshake, there shall be only one PTKSA per band (see 12.6.19) with the same Supplicant and Authenticator MAC addresses. For the PTKSA derived as a result of an initial mobility domain association or fast BSS transition, there shall be only one PTKSA with the same STA's MAC address and BSSID.

12.6.7 RSNA policy selection in an MBSS

Insert new text at the end of 12.6.7 as follows:

When establishing an RSNA in an MBSS, PASN authentication shall not be used.

12.6.10 RSNA authentication in an infrastructure BSS

12.6.10.1 General

Change the first paragraph of 12.6.10.1 as follows:

When establishing an RSNA in a non-FT environment or during an FT initial mobility domain association, a STA shall use IEEE 802.11 SAE authentication, FILS authentication, or Open System authentication prior to (re)association. PASN authentication may be used without (re)association.

Insert a new note after the first paragraph of 12.6.10.1 as follows:

NOTE 1—A STA does not open 802.1X port(s) upon successful PASN authentication.

Change the note in the dashed list in 12.6.10.1 as follows:

NOTE 2—A roaming STA's IEEE 802.1X Supplicant can initiate preauthentication by sending an EAPOL-Start frame via its old AP, through the DS, to a new AP.

12.6.19 Protection of robust Management frames

Change the fourth and fifth paragraphs of 12.6.19 as follows:

A STA with dot11RSNAProtectedManagementFramesActivated equal to true and dot11RSNAUnprotectedManagementFramesAllowed equal to true shall transmit and receive protected individually addressed robust Management frames to and from any associated STA that advertised MFPC = 1, shall discard unprotected individually addressed robust Action frames and robust Action No Ack frames received from any STA that advertised MFPC = 1, and shall discard unprotected individually addressed Disassociation and Deauthentication frames received from a STA that advertised MFPC = 1 after the PTK and IGTK have been installed. The receiver shall process unprotected individually addressed Disassociation and Deauthentication frames before the PTK and IGTK are installed.

A STA with dot11RSNAProtectedManagementFramesActivated equal to true and dot11RSNAUnprotectedManagementFramesAllowed equal to false shall transmit and receive protected individually addressed robust Action frames and robust Action No Ack frames to and from any STA, shall not transmit unprotected individually addressed robust Action frames and robust Action No Ack frames to any STA, and shall discard unprotected individually addressed robust Action frames and robust Action No Ack frames received from a STA after the PTK and IGTK have been installed. The receiver shall process unprotected individually addressed Disassociation and Deauthentication frames before the PTK and IGTK are installed.

Change the last paragraph of 12.6.19 as follows:

Management frame protection cannot be applied until the PTK and IGTK has been established with the STA. A STA shall not transmit robust Action frames and robust Action No Ack frames until it has installed the PTK for the peer STA, or in the case of group addressed frames, has installed the IGTK. The STA shall discard any robust Action frames and robust Action No Ack frames received before the PTK and IGTK are installed.

12.7 Keys and key distribution

12.7.1 Key hierarchy

12.7.1.3 Pairwise key hierarchy

Change the first paragraph of 12.7.1.3 (as amended by IEEE Std 802.11ba-2021) as follows (resulting in four paragraphs in place of one):

Except when preauthentication or FILS authentication is used, the pairwise key hierarchy utilizes PRF-384, PRF-512, or PRF-704 to derive session-specific keys from a PMK, as depicted in Figure 12-30. When using AKM suite selector 00-0F-AC:12 or 00-0F-AC:15, the length of the PMK, PMK_bits, shall be 384 bits. When using AKM suite selectors for which the Authentication type column indicates FT authentication (see Table 9-151), the FT key hierarchy is used to derive session specific keys from an MPMK as defined in 12.7.1.6. With all other AKM suite selectors, the length of the PMK, PMK_bits, shall be 256 bits. The pairwise key hierarchy takes a PMK and generates a PTK.

The PTK is partitioned into KCK, KEK, a temporal key, and a KDK. A KDK is derived if and only if any of the following are true:

- ~~WUR frame protection is negotiated; otherwise the PTK is partitioned into KCK, KEK, and a temporal key.~~
- dot11SecureLTFImplemented is true and the peer STA has advertised secure HE-LTF support capability in its RSNXE (see 9.4.2.241)

The temporal key is used by the MAC to protect individually addressed communication between the Authenticator's and Supplicant's respective STAs. If WUR frame protection is negotiated, the KDK is used to derive a WTK, which is used by the MAC of the WUR AP to protect and by the MAC of the WUR non-AP STA to validate individually addressed WUR Wake-up frames. PTKs are used between a single Supplicant and a single Authenticator.

If a KDK is derived to support secure HE-LTF, it is recommended that the KDK is updated periodically for improved ranging security.

Replace Figure 12-30 with the following figure:

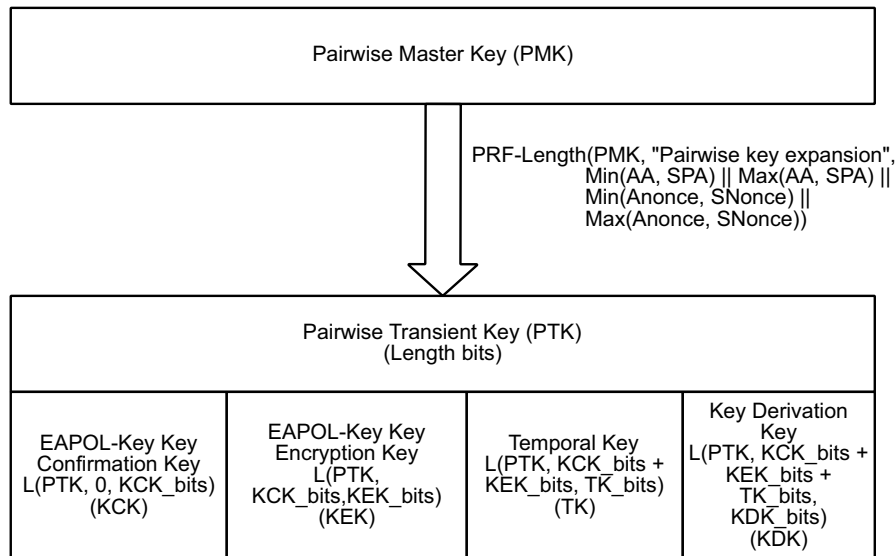


Figure 12-30—Pairwise key hierarchy

Change the third item (as amended by IEEE Std 802.11ba-2021) and the fourth item (as inserted by IEEE Std 802.11ba-2021) in the first dashed list in 12.7.1.3 as follows:

- The PTK shall be derived from the PMK by
$$\text{PTK} = \text{PRF-Length}(\text{PMK}, \text{"Pairwise key expansion"}, \text{Min}(\text{AA}, \text{SPA}) \parallel \text{Max}(\text{AA}, \text{SPA}) \parallel \text{Min}(\text{ANonce}, \text{SNonce}) \parallel \text{Max}(\text{ANonce}, \text{SNonce}))$$

where $\text{Length} = \text{KCK_bits} + \text{KEK_bits} + \text{TK_bits} + \text{KDK_bits}$, if WUR frame protection is being negotiated; otherwise, $\text{Length} = \text{KCK_bits} + \text{KEK_bits} + \text{TK_bits}$. The values of KCK_bits and KEK_bits are AKM suite dependent and are listed in Table 12-10. The value of TK_bits is cipher-suite dependent and is defined in Table 12-7. If a KDK is derived, the value of KDK_bits is equal to the value of PMK_bits; otherwise the value of KDK_bits shall be 0. The Min and Max operations for IEEE 802 addresses are with the address converted to a positive integer treating the first transmitted octet as the most significant octet of the integer. The nonces are encoded as specified in 9.2.2.
- ~~If WUR frame protection is being negotiated, the KDK~~ If a KDK is derived, it shall be computed as the next KDK_bits bits of the PTK:
$$\text{KDK} = \text{L}(\text{PTK}, \text{KCK_bits} + \text{KEK_bits} + \text{TK_bits}, \text{KDK_bits})$$

Otherwise, the KDK is not derived.

12.7.1.6 FT key hierarchy

12.7.1.6.5 PTK

Change the definition of “Length” following the equation after the second paragraph of 12.7.1.6.5 (as amended by IEEE Std 802.11ba-2021) as follows:

Length is the total number of bits to derive, i.e., number of bits of the PTK. The length is dependent on the negotiated cipher suites and AKM suites as defined by Table 12-7 in 12.7.2 and Table 12-10 in 12.7.3, and whether ~~WUR frame protection is negotiated~~ a KDK is derived. A KDK shall be derived if and only if any of the following are true:

- WUR frame protection is negotiated
- dot11SecureLTFImplemented is true and the peer STA has advertised secure HE-LTF support capability in its RSNXE (see 9.4.2.241)

Change the third paragraph of 12.7.1.6.5 (as amended by IEEE Std 802.11ba-2021) as follows:

Except when ~~WUR frame protection is negotiated~~ a KDK is derived, each PTK has five component keys, KCK, KEK, a temporal key, KCK2, and KEK2 derived as follows:

Change the 13th and 14th paragraphs of 12.7.1.6.5 (as inserted by IEEE Std 802.11ba-2021) as follows:

When ~~WUR frame protection is negotiated~~ a KDK is derived, each PTK has six component keys, KCK, KEK, a temporal key, KCK2, KEK2, and a KDK derived as follows:

The KCK, KEK, temporal key, KCK2, and KEK2 shall be computed in the same way as when ~~WUR frame protection is not negotiated~~ a KDK is not derived.

12.7.3 EAPOL-Key frame construction and processing

Insert a new row and associated note at the bottom of Table 12-10 as follows (unchanged rows not shown):

Table 12-10—Integrity and key wrap algorithms

AKM	Integrity algorithm	KCK_bits	Size of MIC	Key wrap algorithm	KEK_bits	KCK2_bits	KEK2_bits
...							
00-0F-AC:21	See NOTE	N/A	N/A	N/A	N/A	N/A	N/A
NOTE—Keys derived using PASN AKM are not used to protect EAPOL frames.							

12.11 Authentication for FILS

12.11.2 FILS authentication protocol

12.11.2.5 Key establishment with FILS authentication

12.11.2.5.3 PTKSA Key derivation with FILS authentication

Change the first paragraph of 12.11.2.5.3 (as amended by IEEE Std 802.11ba-2021) splitting it into two paragraphs as follows:

For PTKSA key generation, the inputs to the PRF are the PMK of the PMKSA, a constant label, and a concatenation of the STA's MAC address, the AP's BSS's BSSID, the STA's nonce, and the AP's nonce. When the negotiated AKM is 00-0F-AC:14 or 00-0F-AC:16, the length of KEK shall be 256 bits, and the length of the ICK shall be 256 bits. When the negotiated AKM is 00-0F-AC:15 or 00-0F-AC:17, the length of the KEK shall be 512 bits, and the length of ICK shall be 384 bits. When the negotiated AKM is 00-0F-AC:16, FILS-FT is 256 bits; when the negotiated AKM is 00-0F-AC:17, FILS-FT is 384 bits; otherwise, FILS-FT is not derived; when a KDK is derived ~~WUR frame protection is negotiated~~, the length of KDK is equal to the value of PMK_bits (see 12.7.1.3); ~~otherwise, the KDK is not derived.~~ a KDK shall be derived if and only if any of the following are true:

- WUR frame protection is negotiated
- dot11SecureLTFImplemented is true and the peer STA has advertised secure HE-LTF support capability in its RSNXE (see 9.4.2.241)

The total amount of bits extracted from the KDF shall therefore be 640+TK bits, 1124+TK bits, or 1408+TK bits depending on the negotiated AKM when WUR frame protection is negotiated, otherwise, shall be 512+TK bits, 896+TK bits, or 1280+TK bits depending on the negotiated AKM, where TK_bits are determined from Table 12-4:

Change the paragraph after the equation for “FILS-FT” (as inserted by IEEE Std 802.11ba-2021) in 12.11.2.5.3 as follows:

When ~~WUR frame protection is negotiated~~ a KDK is derived while doing FT initial mobility domain association using FILS authentication,

Change the paragraph after the first equation for “KDK” (as inserted by IEEE Std 802.11ba-2021) in 12.11.2.5.3 as follows:

When ~~WUR frame protection is negotiated~~ a KDK is derived while not doing FT initial mobility domain association using FILS authentication,

Change the definition of “X” (as amended by IEEE Std 802.11ba-2021) in 12.11.2.5.3 as follows:

- X is 512+TK_bits, 768+TK bits, 896+TK bits, or 1280+TK bits from Table 12-7 depending on the negotiated AKM ~~when WUR frame protection is negotiated a KDK is derived~~; otherwise, X is 512+TK_bits, 768+TK bits, 896+TK bits, or 1280+TK bits from Table 12-7 depending on the negotiated AKM

Insert 12.13 (and its subclauses) after 12.12.2 (as inserted by IEEE Std 802.11ax-2021) as follows:

12.13 Preassociation security negotiation

12.13.1 General

Preassociation security negotiation (PASN) is an RSNA authentication protocol in all cases where it relies on the existence of a PMKSA for an AKMP, termed Base AKMP for PASN. It is a non-RSNA protocol when there is no PMKSA and the corresponding Base AKMP used with it. The protocol supports PTKSA establishment with and without mutual authentication and allows for a PMKSA to be established for certain Base AKMPs by tunneling the Base AKMP protocol messages.

PASN is primarily intended for use in infrastructure networks for a STA and an AP to establish a PTKSA using a three-message authentication frame exchange. Some salient aspects of this protocol are:

- Where available, it leverages a cached PMK for a Base AKMP or already specified mechanisms for a Base AKMP to establish the PMKSA from which the PASN PTKSA is derived. Such a PTKSA provides mutual authentication.
- It leverages a cached PMK for a Base AKMP or already specified mechanisms for a Base AKMP to establish the PMKSA from which the PASN PTKSA is derived.
- It exchanges ephemeral public keys to provide PFS and derive the PTKSA keys, using a PMKSA if one exists. The PMKSA corresponds to the Base AKMP and consequently PASN AKMP shall not be used in the RSNE of an (re)association request.
- A PTKSA may be derived using ephemeral keys without mutual authentication at the discretion of the protocol peers, that is, it may derive a PTKSA without a PMKSA and is thus a non-RSNA protocol.
- Key establishment and key confirmation is accomplished using authentication frames.
- It allows a PMKSA to be established by tunneling FILS shared key and SAE protocol data as part of PASN Authentication frame contents.
- Authentication is initiated by a non-AP STA.
- PASN authentication does not derive GTKSA or IGTKSA.
- A provision to allow APs to request the peer to come back later based on resource constraints or other conditions; see 12.13.9.

The security of PASN depends on assumptions similar to those for FILS and SAE authentication. In particular:

- It assumed that both the STAs have in common at least one cyclic group from the dot11RSNAConfigDLGroupTable that is used to select the ephemeral key exchange parameters.
- Protection is provided against downgrade attacks that may alter security parameters (e.g., RSNE, RSNXE, Group, Hash Function) used in the negotiation.
- Denial of service protection is not guaranteed.
- A cryptographic RNG is available to generate ECC private keys.
- There is no protection for group addressed frames using PASN authentication. Like any PTKSA, the PTKSA established via PASN authentication shall be deleted when the Base PMKSA expires. Similarly, deauthentication (4.5.4.3) results in deletion of PASN PTKSA. For the purpose of interoperability, a STA shall support group 19, an ECC group defined over a 256-bit prime order field.

12.13.2 Discovery of a PASN capable AP

An AP indicates it is capable of performing PASN authentication by including the PASN AKMP as part of the RSNE included in Beacon and Probe Response frames. When PASN AKMP is advertised, the AP shall also

include at least one additional AKMP in the RSNE unless it allows PTKSA derivation without authentication using the ephemeral keys exchanged during PASN authentication.

12.13.3 Key establishment with PASN authentication

12.13.3.1 Overview

This subclause defines the procedure for establishing a PTKSA and the corresponding shared keys between a PASN capable STA and AP. The PASN frame sequence used is depicted in Figure 12-56. It consists of three Authentication frames with the Authentication Algorithm Number field (9.4.1.1) set to 7 (PASN authentication) and the corresponding Authentication Transaction Sequence Number field (9.4.1.2) set to 1, 2, and 3, respectively.

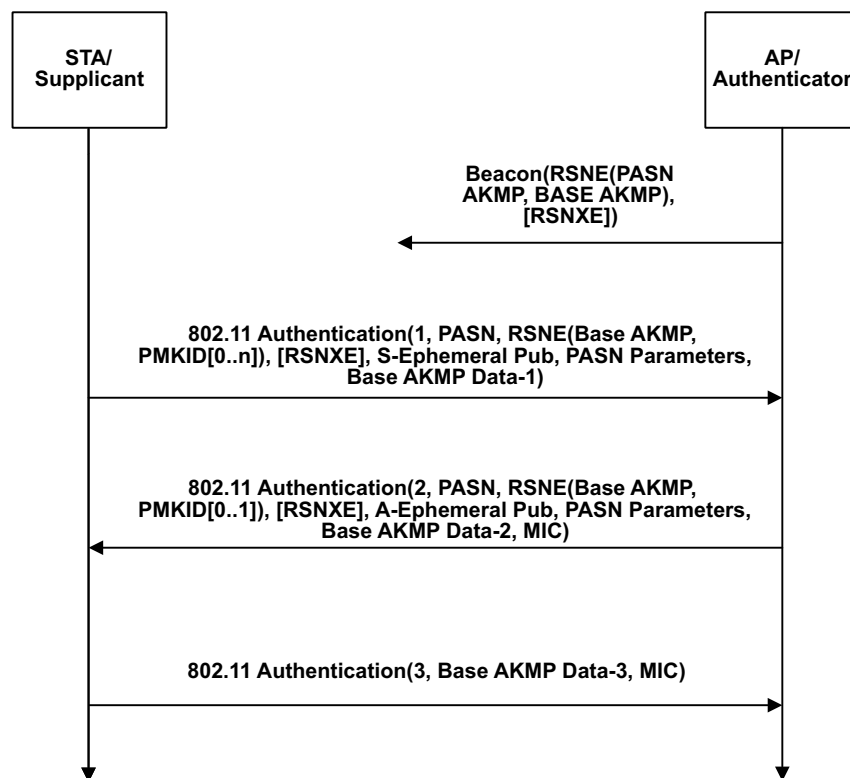


Figure 12-56—PASN authentication

A non-AP STA shall use the first PASN frame to initiate PASN authentication with a PASN capable AP by providing RSNE that contains Base AKMP parameters including a list of PMKIDs (possibly empty), STA's ephemeral public key, PASN parameters, and additional Base AKMP specific data, if any.

The PASN capable AP processes the first PASN frame and responds with a second PASN frame. The processing depends on the Base AKMP as specified in 12.13. If the processing is successful, the AP includes RSNE that contains chosen Base AKMP parameters including the chosen PMKID (possibly none), AP's ephemeral public key, PASN parameters, and additional Base AKMP specific data, if any. The frame is also integrity protected with a MIC using the Key Confirmation key (KCK) as described in 12.13.8.1. If the processing fails, the AP may return an appropriate status in the Status Code field (9.4.1.9) of the frame without including additional information or silently drop the frame.

The non-AP STA receives and processes the second PASN frame and responds with the third PASN frame. Again, the processing depends on the Base AKMP as specified later in the document. If the processing is successful, the STA responds with an integrity protected PASN third frame; otherwise, it may return an appropriate status in the Status Code field (9.4.1.9) of the frame or silently drop the frame. The frame is integrity protected with a MIC using the Key Confirmation key (KCK) as described in 12.13.8.2.

A successful PASN exchange results in establishment of PTKSA using the ephemeral keys and PMK from the Base AKMP (if any). The lifetime of the PTKSA shall be the minimum of PTKSA timeout interval(s) exchanged (if any) but shall not exceed the PMK lifetime for the PMKSA of the Base AKMP. The PASN PTKSA will be deleted by the SME of a non-AP STA or an AP when:

- Its lifetime expires.
- Upon initiation of (re)association on a non-AP STA, or successful association by the peer on an AP (see 11.3.5.2 and 11.3.5.3).

12.13.3.2 PASN frame construction and processing

If non-AP STA chooses to initiate PASN authentication, it first selects the following authentication parameters:

- Base AKMP from among AKMPs advertised by the AP if RSNA authentication is desired. Otherwise, if `dot11NoAuthPASNActivated` is true, Base AKMP chosen is the PASN AKMP, indicating that PTKSA is to be established without mutual authentication, that is, without a corresponding PMKSA.
- Pairwise cipher suite to use for the PTKSA that is being setup.
- Finite cyclic group from the `dot11RSNAConfigDLCTable` that is at least of the security strength provided by the Base AKMP and cipher suites.

With the chosen finite cyclic group, the STA generates an ephemeral (random) private key, and uses the selected group's scalar operation (see 12.4.4.1) with the private key to generate its ephemeral public key. If PMKSA caching is used, it generates a list of PMKSA identifiers, and then obtains Base AKMP specific data.

It then composes an RSNE in which:

- Chosen AKMP and pairwise cipher is included.
- MFPC and MFPR (see 9.4.2.24.4) in the RSN Capabilities field are set to 1.
- If `dot11RSNAOperatingChannelValidationActivated` is true, OCVC (see 9.4.2.24.4) in the RSN Capabilities field is set to 1.
- No Pairwise bit is set to 0.
- PMKID count and PMKID list set corresponding to PMKSA identifiers.
- Group Data Cipher Suite and Group Management Cipher Suite set to 00-0F-AC:7, indicating that group addressed traffic is not allowed.
- Extended Key ID for Individually Addressed Frames (bit B13) shall be 0.

The first PASN Authentication frame (see 9.3.3.11) of the exchange is constructed as follows:

- Authentication Algorithm Number field (9.4.1.1) set to 7 (PASN authentication).
- Authentication Transaction Sequence Number field (9.4.1.2) set to 1.
- Including the constructed RSNE.
- Including a PASN Parameters element (9.4.2.303) with the wrapped data format, chosen finite cyclic group ID, and the ephemeral public key. Comeback field shall either be absent or set to the cookie length and the cookie received from the AP if the authentication is being retried. Comeback After subfield shall not be present in the Comeback field. The Control field in the element is set appropriately to indicate the presence or absence of fields in the element.

- Including a Wrapped Data element (9.4.2.187) corresponding to Base AKMP data, if any. This element may be fragmented using the mechanism specified in 10.28.11 as necessary.
- Optionally including a Timeout Interval element (TIE) (9.4.2.48) with the Timeout Interval Value set to dot11RSNAConfigPASNPTKSATimeout and Timeout Interval Type set to 2 (key lifetime interval).
- Including RSNXE (9.4.2.241) if any subfield of the Extended RSN Capabilities field in this element, except the Field Length subfield, is nonzero.
- If dot11RSNAOperatingChannelValidationActivated is true including an OCI Element containing an OCI element as defined in 9.4.2.236.

The non-AP STA sends the first PASN frame to the AP.

Upon receiving the first PASN frame, the AP:

- Validates RSNE to ensure it is well formed and the AKMP, Pairwise Cipher, MFPC and MFPR capabilities, No Pairwise bit, and Group Ciphers fields (of length 0) are set as expected. Otherwise processing status is set appropriately, see Table 9-50, to INVALID_RSNE, INVALID_RSNE_CAPABILITIES, INVALID_AKMP, INVALID_GROUP_CIPHER, INVALID_PAIRWISE_CIPHER, UNSUPPORTED_RSNE_VERSION.
- Validates that finite cyclic group indicated in PASN Parameters element is supported (present in dot11RSNAConfigDLGroupTable). Otherwise processing status is set to UNSUPPORTED_FINITE_CYCLIC_GROUP.
- Validates PASN Parameters element. If resources or other policy does not permit PASN authentication (in general or for the peer) at this time, but will be permitted later, processing status is set to REFUSED_TEMPORARILY. Otherwise if Comeback Info field is present and AP is expecting the field, it shall validate the field. It is outside the scope of this specification as to the validation mechanism used. If the validation fails, unless a processing status of REFUSED_TEMPORARILY is being returned, the processing status is set to INVALID_PARAMETERS.
- Verifies the public key as specified in 5.6.2.3 of NIST SP 800-56A R2. If verification fails, the processing status is set to INVALID_PUBLIC_KEY. Verifies that a PMKSA named via a PMKID in the RSNE exists for the specified Base AKMP, or the Base AKMP is set to PASN AKMP or Base AKMP data exists in the frame to allow a PMK to be established. If Base AKMP is equal to PASN AKMP, verifies that dot11NoAuthPASNAActivated is set to true. Otherwise processing status is set to REFUSED.
- Extracts any Base AKMP specific data, including reassembly, (10.28.12) as necessary, and processes it according to the behavior described in a later subclause specific to the AKMP; for example, 12.13.4. This may include waiting for Base AKMP related processing to complete before advancing the PASN protocol state. If the Base AKMP related processing indicates an error or there is no PMKSA at the end of such processing, processing status is set to PASN_BASE_AKMP_FAILED.
- If dot11RSNAOperatingChannelValidationActivated is true and the peer STA's RSNE indicated OCVC capability, it validates that an OCI element is present and the Channel information in the element matches current operating channel parameters (see 12.2.9). Otherwise, if there is a mismatch, processing status is set to OCI_MISMATCH.

The AP begins the construction the second PASN frame as follows:

- Authentication Algorithm Number field (9.4.1.1) set to 7 (PASN authentication)
- Authentication Transaction Sequence Number field (9.4.1.2) set to 2
- Status code indicating the processing status

If the processing status is `REFUSED_TEMPORARILY`, the AP constructs and includes in the second PASN frame.

PASN Parameters element (9.4.2.303) with Comeback Info field with time for the peer to retry the operation. The Cookie Length subfield is set to nonzero if a cookie is being returned to the non-AP STA, otherwise it is set to 0. The Cookie subfield contains the cookie if the Cookie Length subfield is nonzero. The Control field in the element is set appropriately to indicate the presence or absence of fields in the element. Otherwise, if the validation is successful, the AP with the chosen finite cyclic group, generates an ephemeral (random) private key, and uses the selected group's scalar operation with the private key to generate its ephemeral public key.

- Derives the PTKSA; see 12.13.7.
- Perform the group's scalar-op (see 12.4.4.1) with the non-AP STA's ephemeral public key and its own ephemeral private key to produce an ephemeral Diffie-Hellman shared secret, DHss.
- Constructs and includes in the second PASN frame.
- An RSNE that contains.
- Chosen Base AKMP and Pairwise cipher.
- MFPC and MFPR (see 9.4.2.24.4) in the RSN Capabilities field are set to 1.
- If `dot11RSNAOperatingChannelValidationActivated` is true, OCVC (see 9.4.2.24.4) in the RSN Capabilities field is set to 1.
- No Pairwise bit is set to 0.
- PMKID count of 1 and PMKID list corresponding to the PMKID from the PMKSA.
- Group Data Cipher Suite and Group Management Cipher Suite set to 00-0F-AC:7, indicating that group addressed traffic is not allowed.
- Extended Key ID for individually addressed frames (bit B13) shall be 0.
- PASN Parameters element (9.4.2.303) with the wrapped data format, chosen finite cyclic group ID, and the AP's ephemeral public key. The Control field in the element is set appropriately to indicate the presence or absence of fields in the element.
- Wrapped Data element (9.4.2.187) corresponding to Base AKMP data to be returned to the non-AP STA, if any. This element may be fragmented using mechanism specified in 10.28.11 as necessary.
- Optionally including a Timeout Interval element (TIE) (9.4.2.48) with Timeout Interval Value set to `dot11RSNAConfigPASNPTKSATimeout` and Timeout Interval Type set to 2 (key lifetime interval).
- Including RSNXE (9.4.2.241) that is advertised in AP's Beacon and Probe Response frames, if any subfield of the Extended RSN Capabilities field in this element, except the Field Length subfield, is nonzero.
- If `dot11RSNAOperatingChannelValidationActivated` is true, including an OCI Element containing an OCI element as defined in 9.4.2.236, if `dot11RSNAOperatingChannelValidationActivated` is true.
- A MIC element (9.4.2.118) with MIC computed as specified in 12.13.8.1.

Once the processing complete, the AP sends the second PASN frame to the non-AP STA. If the processing status returned in the frame was not `SUCCESS`, the AP shall terminate the PASN authentication protocol. Upon receiving the second PASN frame, the non-AP STA:

- Validates the Status Code field is `SUCCESS`.
- Validates the PASN Parameters element. If the Status Code field was `REFUSED_TEMPORARILY` and Comeback Info field in the parameters specifies a comeback time, the STA may retry PASN authentication after the specified time with the specified cookie (if any).
- Validates RSNE to ensure it is well formed and the AKMP, Pairwise Cipher, MFPC and MFPR capabilities, No Pairwise bit, and Group Ciphers fields (of length 0) are set as expected. Validates that finite cyclic group indicated in PASN Parameters element is supported (present in `dot11RSNAConfigDLGroupTable`).

- Verifies that the public key as specified in 5.6.2.3 of NIST SP 800-56A R2.
- Verifies that a PMKSA named via a PMKID in the RSNE exists for the specified Base AKMP, or the Base AKMP is set to PASN AKMP or Base AKMP data exists in the frame to allow a PMK to be established. If Base AKMP is equal to PASN AKMP, verifies that dot11NoAuthPASNAActivated is set to true.
- Extracts any Base AKMP specific data and processes it according to the behavior described in a later subclause specific to the AKMP; for example, 12.13.4.
- Locates the PMKSA, which might include waiting for Base AKMP specific processing to complete.
- Perform the group's scalar-op (see 12.4.4.1) with the AP's ephemeral public key and its own ephemeral private key to produce an ephemeral Diffie-Hellman shared secret, DHss.
- Derives the PTKSA; see 12.13.7.
- If dot11RSNAOperatingChannelValidationActivated is true and the peer STA's RSNE indicated OCVC capability, it validates that an OCI element is present and the Channel information in the element matches current operating channel parameters (see 12.2.9). Otherwise, if there is a mismatch, processing status is set to OCI_MISMATCH.
- Computes the MIC as specified in 12.13.8.1 and verifies it to be the same as the MIC provided in the MIC element.

If validation fails, the non-AP STA shall discard the frame and terminate further protocol processing, but may retry PASN authentication later.

Otherwise the STA begins the construction the third PASN frame as follows:

- Authentication Algorithm Number field (9.4.1.1) set to 7 (PASN authentication).
- Authentication Transaction Sequence Number field (9.4.1.2) set to 3.
- Status code indicating success.
- Constructs and includes in the third PASN frame.
- PASN Parameters element (9.4.2.303) with the wrapped data format. Public Key information and Comeback Info fields shall not be present. The Control field in the element is set appropriately to indicate the presence or absence of fields in the element.
- Wrapped Data element (9.4.2.187) corresponding to Base AKMP data to be returned to the non-AP STA, if any. This element may be fragmented using mechanism specified in 10.28.11 as necessary.
- If dot11RSNAOperatingChannelValidationActivated is true, including an OCI Element containing an OCI element as defined in 9.4.2.236.
- A MIC element (9.4.2.118) with MIC computed as specified in 12.13.8.2.

Once the processing completes, the non-AP STA sends the third PASN frame to the AP. If a success was indicated, it installs the temporal key derived using MLME-SETKEYS.request primitive.

Upon receiving the third PASN frame, the AP:

- Validates that the Status Code field is SUCCESS.
- Validates PASN Parameters element.
- If dot11RSNAOperatingChannelValidationActivated is true and the peer STA's RSNE indicated OCVC capability, it validates that an OCI element is present and the Channel information in the element matches current operating channel parameters (see 12.2.9). Otherwise, if there is a mismatch, processing status is set to OCI_MISMATCH.
- Computes the MIC as specified in 12.13.8.2 and verifies it to be the same as the MIC provided in the MIC element.

- Extracts any Base AKMP specific data and processes it according to the behavior described in a later subclause specific to the AKMP; for example, 12.13.4. Base AKMP processing may indicate an error.

For the third PASN frame, if the MIC validation fails, the AP shall discard the frame and terminate the processing of the frame with no further effect. If other validation fails, or Base AKMP processing indicates an error, the AP shall terminate the PASN authentication. If the validation was successful, the AP installs the temporal key derived using MLME-SETKEYS.request primitive to install the new key. If the AP terminated the PASN authentication silently or by sending a failure Status Code to the non-AP STA, other than a temporary refusal, it shall delete all of the PASN authentication state.

If the non-AP STA receives an Authentication frame that does not contain Authentication Algorithm Number field set to 7 (PASN authentication) once it has sent the first PASN frame, it shall abandon the PASN authentication in progress. Likewise, if the AP receives an Authentication frame that does not contain Authentication Algorithm Number field set to 7 (PASN authentication) once it has received the first PASN frame, it shall abandon the PASN authentication in progress.

If the authentication is not successful within the AuthenticateFailureTimeout, the STA shall terminate PASN authentication and delete the PASN authentication state.

NOTE—Because of frame filtering rules (11.3.3), the protection using the key installed by PASN authentication is only used to protect management frames, as applicable.

12.13.4 PASN authentication with FILS shared key

This subclause specifies aspects of PASN authentication when FILS AKMP 00-0F-AC:14 or 00-0F-AC:15 is used as a Base AKMP when PMK Caching is not used with PASN authentication. Otherwise, when PMK Caching is used, the PMKSA identified by the PMKID in the first PASN Authentication frame is used for the PASN PTKSA derivation.

Where FILS shared key authentication without PFS is desired and there is no cached PMK, the Base AKMP data is constructed using 9.4.2.187.

In the first PASN frame, the Wrapped Data element contains the EAP-Initiate/Re-auth PPDU similar to FILS shared key processing; see 12.11.2.3.2. The EAP-Initiate/Re-auth data is forwarded to the AS by the AP and the EAP-Finish/Re-auth PPDU received from the AS is forwarded to the non-AP STA encapsulated in the Wrapped Data element in the second PASN frame. The AS returns the associated EAP-RP rMSK along with the EAP-Finish/Re-auth PPDU to the AP. Wrapped data shall be absent in the third PASN frame.

The resulting rMSK is used in PMKSA derivation (12.11.2.5.2) where:

- SNONCE is set to the non-AP STA's ephemeral public key encoded as an octet string (12.4.7.2.4)
- ANONCE is set to AP's ephemeral public key encoded as an octet string (12.4.7.2.4)
- DHss is the shared secret derived from the PASN ephemeral key exchange encoded as an octet string (12.4.7.2.2)

The PMKSA is then used in PTKSA derivation for PASN authentication. A FILS PMKSA so established shall be used only to establish PTKSAs that are negotiated via PASN authentication.

There is only a single FILS session with PASN authentication, and FILS session element is not tunneled.

12.13.5 PASN authentication with SAE

This subclause specifies aspects of PASN authentication when SAE AKMP 00-0F-AC:8 is used as the Base AKMP when PMK caching is not used. When PMK caching is used PASN authentication relies on the PMKSA already established by SAE protocol. Without PMK caching, wrapped data shall be present in first, second, and third PASN frames and is constructed as follows. When SAE is used with PASN authentication, the direct hashing technique described in 12.4.4.2.3 or 12.4.4.3.3 shall be used. A PASN capable AP shall set SAE hash-to-element bit to 1 in the Extended RSN Capabilities field of the RSNXE included in its Beacon, Probe Response, and second PASN Authentication frames. A non-AP STA initiating PASN authentication shall set SAE hash-to-element bit to 1 in the Extended RSN Capabilities field of the RSNXE in the first PASN Authentication frame.

The non-AP STA instantiates an SAE protocol instance, 12.4.8, and provides an Initiate event. The PASN authentication process receives an Authentication frame with a Commit message, 12.4.7.4, from the SAE protocol instance. It then sets the Wrapped Data in the first PASN frame to be the Authentication frame body.

The AP upon receiving the first PASN frame, delivers the frame to SAE for processing via the SAE parent process managed by the SME. If the PASN authentication process receives, from the SAE protocol instance, a failure indication, PASN authentication shall be abandoned. Otherwise PASN authentication process receives Authentication frames with a Commit message, 12.4.7.4; and a Confirm message, 12.4.7.5, in that order, as the SAE protocol instance transitions to the Confirmed state.

The AP sets the Wrapped Data in the second PASN frame, as shown in the Figure 12-57, to be the concatenation of:

- Two octet length of first Authentication frame body in little endian order
- First Authentication frame body with a Commit message
- Two octet length of second Authentication frame body in little endian order
- Second Authentication frame body with a Confirm message

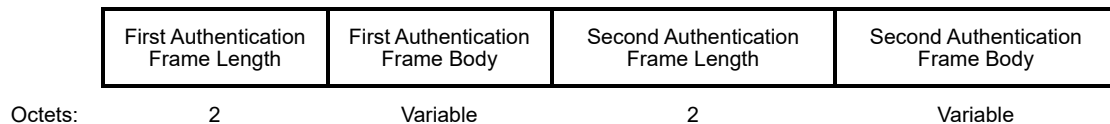


Figure 12-57—PASN Wrapped Data format with SAE

The non-AP STA upon receiving the second PASN frame, delivers the Commit and Confirm messages from the AP to the SAE protocol instance. If the PASN authentication process receives, from the SAE protocol instance, a failure indication, PASN authentication shall be abandoned. Otherwise PASN authentication process receives an Authentication frame with a Confirm message (12.4.7.5) as the SAE protocol instance transitions to the Confirmed state followed by a transition to the Accepted state (since it has received both the Commit and Confirm messages from the peer).

The non-AP STA sets the Wrapped Data in the third PASN frame to be the Authentication frame body with a Confirm message. The AP upon receiving the third PASN frame, delivers the Authentication frame from the Wrapped Data to the SAE protocol instance. Either the instance accepts the frame and transitions to the Accepted state or PASN authentication shall be abandoned and the SAE protocol instance destroyed.

Not all aspects of SAE protocol are supported by the above tunneling mechanism. In particular, the tunneled protocol may fail if the SAE state machine desires:

- Anti-Clogging Tokens
- Retries of Commit and Confirm messages

In such cases, SAE authentication may be first used to establish an SAE PMKSA.

The PMKSA is then used in PTKSA derivation for PASN authentication. If establishment of SAE PMKSA fails, PASN authentication shall be abandoned. An SAE PMKSA so established shall be used only to establish PTKSAs that are negotiated via PASN authentication.

12.13.6 PASN authentication with FT

This subclause specifies aspects of PASN authentication when one of FT AKMPs 00-0F-AC: [3, 4, 13, 19] is used as the Base AKMP.

PASN authentication with FT relies on FT key hierarchy already being established via the FT initial mobility domain association (13.4.2). PASN protocol messages carry FT PMKR1Name as the PMKID, and the PASN PTKSA is established like any other Base AKMP.

Wrapped Data shall be optionally present in PASN first frame. When present it shall contain a set of elements that include RSNE (9.4.2.24), MDE (9.4.2.46), and FTE (9.4.2.47) as specified for the first message of FT authentication sequence (13.8.2).

Wrapped Data shall be optionally present in the second PASN frames but shall be present if Wrapped Data was present in the first PASN frame. When present it shall contain a set of elements that include RSNE (9.4.2.24), MDE (9.4.2.46), and FTE (9.4.2.47) as specified for the second message of FT authentication sequence (13.8.3).

Wrapped Data shall be absent in the third PASN frame. The elements in the Wrapped Data are used for additional validation FT security parameters as being used in PASN authentication.

12.13.7 PTKSA derivation with PASN authentication

For PTKSA key derivation, the inputs to the PRF are the PMK of the PMKSA, a constant label, and a concatenation of non-AP STA's MAC address, AP's BSSID, and the DH shared secret from the ephemeral exchange.

$$\text{PTK} = \text{KDF-HASH-NNN}(\text{PMK}, \text{"PASN PTK Derivation"}, \text{SPA} \parallel \text{BSSID} \parallel \text{DHss})$$

where

PMK is the pairwise master key for the Base AKMP if the AKMP is other than PASN AKMP; see 9.4.2.24.3. Otherwise, if the Base AKMP is PASN AKMP, that is, the PASN PTKSA is being setup without mutual authentication in a non-RSN, the PMK shall be set to the string "PMKz" padded with 28 0s.

NOTE—The PMK for the derivation can come from a cached PMKSA for the AKMP or from the PMKSA established with PASN by tunneling Wrapped Data or Authentication frames.

DHss is the shared secret derived from the PASN ephemeral key exchange encoded as an octet string (12.4.7.2.2).

KDF-HASH-NNN is the key derivation function defined in 12.7.1.6.2 using the hash algorithm defined for the Base AKMP; see Table 9-151. When there is no Base AKMP, the hash algorithm is selected based on the pairwise Cipher Suite provided in the RSNE provided by the AP in the second PASN frame. SHA-256 is used as the hash algorithm, except for the ciphers 00-0F-AC:9 and 00-0F-AC:10 for which SHA-384 is used.

NNN is the Bits required for KCK, TK, and KDK depending on the pairwise cipher and whether a KDK is derived.

PTK is composed of the Key Confirmation Key (KCK), Temporal Key (TK), and the Key Derivation Key (KDK), which are derived as follows:

$$\text{KCK} = \text{L}(\text{PTK}, 0, 256)$$

KCK is the first 256 bits of the PTK.

$$\text{TK} = \text{L}(\text{PTK}, 256, \text{TK_Length_Bits})$$

TK is the transient key whose length is the same as a key for the pairwise cipher in RSNE provided by the AP in the second PASN frame. This length is 16 octets for all ciphers, except for the ciphers 00-0F-AC:9 and 00-0F-AC:10 for which it is 32 octets.

$$\text{KDK} = \text{L}(\text{PTK}, 256 + \text{TK_Length_Bits}, \text{KDK_bits})$$

The KDK is of bit length KDK_bits, which has the value 256 if a KDK is derived (see 12.7.1.3) or 0 otherwise.

KDK shall be derived if dot11SecureLTFImplemented is true and the peer STA has indicated Secure HE-LTF support capability in its advertised Extended Capabilities.

The Key ID in the PTKSA (see 12.6.1.1.6) resulting from PASN authentication shall be 3 0.

12.13.8 MIC computation with PASN authentication

12.13.8.1 MIC computation for PASN second frame

The MIC field of the MIC element in the PASN second frame is set by the AP to first MMM octets of:

$$\text{HMAC-HASH}(\text{KCK}, \text{BSSID} \parallel \text{SPA} \parallel \text{Beacon RSNE} \parallel \text{Beacon RSNXE} \parallel \text{Frame Data})$$

where

HASH	is the hash algorithm used in 12.13.7
KCK	is the key confirmation key for the PASN PTKSA
BSSID	is the BSSID for the AP's BSS
SPA	is the MAC address of the non-AP STA, the transmitter of the first PASN frame
Beacon RSNE	is the RSN element sent in the Beacons transmitted by the AP
Beacon RSNXE	is the RSNXE sent in the Beacons transmitted by the AP
Frame Data	is the body of the PASN second frame including the MIC element with the octets in the MIC field of the MIC element set to 0
MMM	is half of the output length in octets for the hash function used, that is, 16 or 24 octets for SHA-256 and SHA-384, respectively

Beacon RSNE and Beacon RSNXE are included in the MIC computation so that a downgrade attack with forged RSNE Beacons will result in a MIC mismatch and thus PASN authentication failure.

12.13.8.2 MIC computation for PASN third frame

The MIC field of the MIC element in the PASN third frame is set by the non-AP STA to first MMM octets of:

$$\text{HMAC-HASH}(\text{KCK}, \text{SPA} \parallel \text{BSSID} \parallel \text{F1 Auth} \parallel \text{Frame Data})$$

where

HASH	is the hash algorithm used in 12.13.7
KCK	is the key confirmation key for the PASN PTKSA
SPA	is the MAC address of the non-AP STA, the transmitter of the first PASN frame
BSSID	is the BSSID for the AP's BSS
F1 Auth	is the HASH of body of the PASN first frame
Frame Data	is the body of the PASN third frame including the MIC element with the octets in the MIC field of the MIC element set to 0
MMM	is half of the output length in octets for the hash function used, that is, 16 or 24 octets for SHA-256 and SHA-384, respectively

12.13.9 Comeback cookies

An AP is required to do a considerable amount of work upon receiving first PASN Authentication frame and thus may be subjected to denial of service attacks by an attacker by spraying such frames, each possibly with a different source MAC address. An AP may also have limited resources and may desire to defer authentication for new client STAs. PASN authentication allows an AP to indicate the deferral time (Comeback After field in PASN Parameters element) and optionally a cookie in the Comeback Info field of the PASN Parameters element (9.4.2.303) and refuse the authentication temporarily.

The comeback cookie is used similarly to the SAE anti-clogging token (see 12.4.6) for denial of service protection on an SAE peer. When a client STA receives the Comeback Info field with a deferral time and a cookie, the STA can reinitiate PASN authentication by providing the cookie. The AP would accept or deny authentication based on the received cookie, if one was expected. PASN comeback cookies are an opaque sequence of octets to the client. They are generated and validated by the AP by an implementation dependent scheme. An AP may start requiring clients present the comeback cookie provided based on resource or other constraints outside the scope of this specification.

Note that Clause 26 was introduced by IEEE Std 802.11ax-2021.

26. High-efficiency (HE) MAC specification

26.5 MU operation

26.5.2 UL MU operation

26.5.2.2 Rules for soliciting UL MU frames

26.5.2.2.4 Allowed settings of the Trigger frame fields and TRS Control subfield

Change the first paragraph of 26.5.2.2.4 as follows:

An AP with `dot11MultiBSSIDImplemented` equal to true shall not send a Trigger frame (other than an NFRP Trigger frame or Ranging Trigger frame) with the TA field set to the transmitted BSSID to a non-AP STA that is associated with ~~an AP corresponding to~~ a nontransmitted BSSID in the multiple BSSID set, ~~set~~ unless the AP has received an HE Capabilities element from non-AP STA with the Rx Control Frame To MultiBSS subfield in the HE MAC Capabilities Information field equal to 1. An AP with `dot11MultiBSSIDImplemented` equal to true may send an NFRP Trigger frame with the TA field set to the transmitted BSSID to a non-AP STA that is associated with ~~an AP corresponding to~~ a nontransmitted BSSID in a multiple BSSID set. An AP with `dot11MultiBSSIDImplemented` equal to true ~~shall not send an individually addressed~~ may send a Ranging Trigger frame, with the TA field set to ~~the~~ a transmitted BSSID, to a non-AP STA associated with ~~an AP corresponding to an ISTA during a TB ranging measurement exchange (11.21.6.4.3) when the FTM session was established using~~ a nontransmitted BSSID in the multiple BSSID set {of, associated with} the transmitted BSSID.

26.5.2.5 UL MU CS mechanism

Insert new text at the end of 26.5.2.5 as follows:

An RSTA that transmits a Ranging Trigger frame shall set the CS Required subfield to 1 unless one of the following conditions is met:

- The Ranging Trigger frame is of subvariant Poll, Sounding, Secure Sounding, or Passive Sounding.
- The Ranging Trigger frame is of subvariant Report and the UL Length subfield in the Common Info field of the Trigger frame is less than or equal to 418.

26.12 HE PPDU post-FEC padding and packet extension

Change the note after the seventh paragraph of 26.12 as follows:

NOTE 1—If the pre-FEC padding factor is 4, then the value of nominal T_{PE} is equal to the nominal packet padding (see Table 27-46).

Insert new text before the last paragraph of 26.12 as follows:

A STA A shall set the value of the TXVECTOR parameter `NOMINAL_PACKET_PADDING` to 16 μ s when transmitting to a STA B if the STA A has not received a frame including the HE Capabilities element from the STA B.

NOTE 2—One such situation is an AP transmitting to an unassociated STA. Another such situation is an unassociated STA transmitting to an AP without having received a management frame including an HE Capabilities element from the AP, such as a Beacon or Probe Response frame.

26.15 PPDU format, BW, MCS, NSS, and DCM selection rules

26.15.2 PPDU format selection

Change the last paragraph of 26.15.2 as follows:

During an EDCA FTM session, an HE STA should send an Ack frame that is the response to a Fine Timing Measurement FTM frame carried in a VHT PPDU or HT PPDU in the same PPDU format as the PPDU carrying the Fine Timing Measurement FTM frame. It shall also send an Ack frame in an HE SU PPDU in response to an FTM frame carried in an HE SU PPDU format; see 26.17.2.

26.17 HE BSS operation

26.17.2 HE BSS operation in the 6 GHz band

26.17.2.1 General

Insert new text at the end of 26.17.2.1 as follows:

A 6 GHz STA that negotiates an EDCA based FTM session, as defined in 11.21.6, shall set the Format And Bandwidth subfield, in the FTM Parameters field, to a value that corresponds to an EDCA based HE format, see Table 9-282, in the IFTMR frame for the ISTA, and in the IFTM frame for the RSTA.

A 6 GHz STA that negotiates an EDCA FTM session shall transmit FTM frames in an HE SU PPDU and FTM Request frames in either a non-HT PPDU, or an HE SU PPDU. The HE SU PPDU shall use 2x HE-LTF and 1.6 μ s GI for both the FTM frame and corresponding the acknowledgement frame.

Note that Clause 27 was introduced by IEEE Std 802.11ax-2021.

27. High-efficiency (HE) PHY specification

27.1 Introduction

27.1.1 Introduction to the HE PHY

Change the 12th paragraph of 27.1.1 as follows:

An HE STA may support the following features:

- HE-MCSs 8 to 11 (transmit and receive).
- Two or more spatial streams (transmit and receive).
- DCM (transmit and receive).
- HE SU PPDU and HE ER SU PPDU with a 1x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols (transmit and receive).
- HE SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on the HE-LTF and Data field OFDM symbols if the STA does not support HE ER SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on both the HE-LTF and Data field OFDM symbols (transmit and receive).
- HE ER SU PPDU with a 4x HE-LTF and 0.8 μ s GI duration on both the HE-LTF and Data field OFDM symbols (transmit and receive).
- LDPC coding (transmit and receive) if the maximum number of spatial streams the STA is capable of transmitting or receiving in an HE SU PPDU is less than or equal to 4.
- Single spatial stream HE-MCS 0 in the higher frequency 106-tone RU of the primary 20 MHz channel for an HE ER SU PPDU.
- STBC (transmit and receive).
- HE Ranging NDP (transmit and receive).
- HE TB Ranging NDP (transmit and receive).

NOTE—Support for HE Ranging NDP and HE TB Ranging NDP are indicated in the Extended Capabilities element.

27.2 HE PHY service interface

27.2.2 TXVECTOR and RXVECTOR parameters

Change the existing rows for parameters “APEP_LENGTH”, “PSDU_LENGTH”, and “NUM_USERS” and insert rows for seven new parameters at the end of Table 27-1 (but before the notes) as follows (unchanged rows not shown):

Table 27-1—TXVECTOR and RXVECTOR parameters

Parameter	Condition	Value	TXVECTOR	RXVECTOR
..				
APEP_LENGTH	FORMAT is HE_SU or HE_ER_SU	Integer.	Y	O
	FORMAT is HE_MU or HE_TB	If 0 and FORMAT is HE_SU, indicates an HE sounding NDP, or <u>HE Ranging NDP</u> . If 0 and FORMAT is HE_TB, indicates an <u>HE TB Ranging NDP</u> . Otherwise, indicates the number of octets in the range 1 to <i>aPSDUMaxLength</i> in the A-MPDU pre-EOF padding (see Table 27-54) that is carried in the PSDU.	M U	O
	Otherwise	See corresponding entry in Table 19-1 or Table 21-1.		
PSDU_LENGTH	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB	Indicates the number of octets in the PSDU in the range of 0 to <i>aPSDUMaxLength</i> octets (see Table 27-54). A value of 0 indicates an HE sounding NDP, <u>HE Ranging NDP</u> , or <u>HE TB Ranging NDP</u> .	N	Y
	Otherwise	See corresponding entry in Table 21-1.		
..				
NUM_USERS	<u>FORMAT is HE_SU and RANGING_FLAG is present</u>	<u>If SECURE_LTF_FLAG is 0, set to 1.</u> <u>If SECURE_LTF_FLAG is 1, indicates the number of users of the HE Ranging NDP with secure HE-LTF.</u> <u>If NUM_USERS is larger than 1, NUM_STS, LTF_REP, and LTF_KEY are arrays with number of entries equal to NUM_USERS.</u>	<u>Y</u>	<u>N</u>
	FORMAT is HE_SU, HE_MU, HE_ER_SU, or HE_TB, <u>and RANGING_FLAG is not present</u>	Not present NOTE—The number of users for an HE SU PPDU, HE ER SU PPDU, or HE TB PPDU is always 1. The number of users in an RU in an HE MU PPDU is determined by RU_ALLOCATION and STA_ID parameters for that RU.	N	N
	Otherwise	See corresponding entry in Table 21-1.		
...				

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
TIME_OF_DEPARTURE_REQUESTED	FORMAT is HE_SU or (HE_TB and RANGING_FLAG is present)	Enumerated type: True indicates that the MAC entity requests that the PHY entity measures and reports time of departure parameters corresponding to the time when the first frame energy is sent by the transmitting port. False indicates that the MAC entity requests that the PHY entity neither measures nor reports time of departure parameters.	O	N
	FORMAT is HE_ER_SU or HE_MU	Not present.	N	N
	Otherwise	See corresponding entry in Table 21-1 (TXVECTOR and RXVECTOR parameters).		
LTF_KEY	FORMAT is either HE_SU or HE_TB and RANGING_FLAG is present and SECURE_LTF_FLAG is 1	Contains the <i>rsta-ltf-key</i> or <i>ista-ltf-key</i> (see 11.21.6.4.5.4) when the secure HE-LTFs are used (see 11.21.6.4.5).	Y	N
	Otherwise	Not present.		
LTF_IV	FORMAT is either HE_SU or HE_TB and RANGING_FLAG is present and SECURE_LTF_FLAG is 1	Contains the <i>ltf-iv</i> (see 11.21.6.4.5.4) used to generate the secure HE-LTFs.	Y	N
	Otherwise	Not present.		
LTF_REP	FORMAT is either HE_SU or HE_TB and RANGING_FLAG is present	Indicates the number of HE-LTF repetitions.	Y	N
	Otherwise	Not present.		
RANGING_FLAG	FORMAT is HE_SU	If present, indicates that the PPDU is an HE Ranging NDP. Not present otherwise.	O	N
	FORMAT is HE_TB	If present, indicates that the PPDU is an HE TB Ranging NDP. Not present otherwise.	O	N
	Otherwise	Not present.	N	N
SECURE_LTF_FLAG	FORMAT is either HE_SU or HE_TB and the RANGING_FLAG is present.	Set to 1 to indicate that the HE Ranging NDP or HE TB Ranging NDP will use secure HE-LTF. Set to 0 otherwise.	Y	N
	Otherwise	Not present.		

Table 27-1—TXVECTOR and RXVECTOR parameters (continued)

Parameter	Condition	Value	TXVECTOR	RXVECTOR
TX_WINDOW_FLAG	FORMAT is either HE_SU or HE_TB and RANGING_FLAG is present and SECURE_LTF_FLAG is 1	Set to 1 to indicate that the secure HE-LTF of an HE Ranging NDP or HE TB Ranging NDP will use the optional frequency domain Tx window. Set to 0 otherwise.	Y	N
	Otherwise	Not present.		
⋮				

Insert 27.2.3a (including Table 27-2a) after 27.2.3 as follows:

27.2.3a LTFVECTOR parameters

The LTFVECTOR is carried in a PHY-RXLTFSEQUENCE.request for the PHY of a STA to receive an HE Ranging NDP or an HE TB Ranging NDP. The parameters in Table 27-2a are defined as part of the LTFVECTOR parameter list in the PHY-RXLTFSEQUENCE.request primitive.

Table 27-2a—LTFVECTOR parameters

Parameter	Value
LTF_NSTS	Indicates the number of space-time streams in the following HE Ranging NDP or the following HE TB Ranging NDP.
LTF_REP	Indicates the number of HE-LTF repetitions in the following HE Ranging NDP or the following HE TB Ranging NDP.
SECURE_LTF_FLAG	Set to 1 to indicate that the HE Ranging NDP or HE TB Ranging NDP uses secure HE-LTF.
LTF_KEY	Present when SECURE_LTF_FLAG is set to 1. Contains the <i>rsta-ltf-key</i> or <i>ista-ltf-key</i> (see 11.21.6.4.5.4) when receiving the secure HE-LTFs.
LTF_IV	Present when SECURE_LTF_FLAG is set to 1. Contains the <i>ltf-iv</i> (see 11.21.6.4.5.4) for secure HE-LTFs or null otherwise.
LTF_OFFSET	Present when SECURE_LTF_FLAG is set to 1. Indicates the number of HE-LTF to skip before beginning to process the HE-LTF symbols.
TX_WINDOW_FLAG	Present when SECURE_LTF_FLAG is set to 1. Set to 1 to indicate that the secure HE-LTF of an HE Ranging NDP or HE TB Ranging NDP will use the optional frequency domain Tx Window.

27.3 HE PHY

27.3.4 HE PPDU formats

Change the first paragraph of 27.3.4 as follows:

Four HE PPDU formats are defined: HE SU PPDU, HE MU PPDU, HE ER SU PPDU, and HE TB PPDU. The HE sounding NDP and HE Ranging NDP ~~is a variant~~ are variants of the HE SU PPDU and are defined in 27.3.17 and 27.3.18a.1, respectively. The HE TB ~~NDP feedback PPDU NDP is a variant~~ and the HE TB Ranging NDP are variants of the HE TB PPDU and are defined in 27.3.18 and 27.3.18a.2, respectively.

27.3.11 HE preamble

27.3.11.1 Introduction

Insert a new paragraph at the end of 27.3.11.1 as follows:

See 27.3.18b for the description of the HE-LTF field using secure HE-LTF in the preamble of HE Ranging NDP and HE TB Ranging NDP.

Insert 27.3.18a and 27.3.18b (and their subclauses) after 27.3.18 as follows:

27.3.18a HE Ranging NDP and HE TB Ranging NDP

This subclause describes the HE Ranging NDP and HE TB Ranging NDP.

27.3.18a.1 HE Ranging NDP

The format of an HE Ranging NDP is shown in Figure 27-46a.

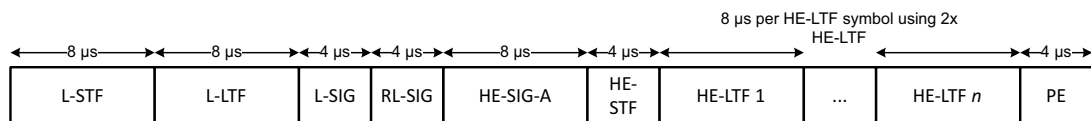


Figure 27-46a—HE Ranging NDP format

The HE Ranging NDP has the following properties:

- Uses the HE SU PPDU format but without the Data field.
- The L-STF and L-LTF fields in an HE Ranging NDP are the same as the L-STF and L-LTF fields in an HE SU PPDU. The subfields in L-SIG, RL-SIG and HE-SIG-A have the same bit assignments and definitions as the subfields in a HE SU PPDU unless specified otherwise.
- The HE-STF field in an HE Ranging NDP is the same as the HE-STF field in an HE SU PPDU.
- Uses HE-LTFs or secure HE-LTFs when the TXVECTOR parameter SECURE_LTF_FLAG is equal to 0 or 1, respectively.
- Secure HE-LTFs use randomized HE-LTF sequences, pseudorandom and deterministic per stream phase rotation and, when the TXVECTOR parameter TX_WINDOW_FLAG is equal to 1, a frequency domain flat top window is used instead of the frequency domain rectangular window; see 27.3.18b.6.
- Uses HE-LTF repetitions, if indicated in the TXVECTOR parameter LTF_REP by values larger than one.

- The HE-LTF field consists of one or more HE-LTF User Blocks; each HE-LTF User Block contains one or more HE-LTF Repetition Blocks, where the number of HE-LTF Repetition Blocks is equal to LTF_REP. An HE-LTF Repetition Block in an HE-LTF User Block comprises one or more HE-LTF symbols, NHE-LTF, calculated using the number of space-time streams NUM_STS for this user, as defined in Table 21-13.
- When the TXVECTOR parameter SECURE_LTF_FLAG is equal to 0, the number of HE-LTF User Blocks is equal to 1, and the same HE-LTF User Block is applied to all users. When the TXVECTOR parameter SECURE_LTF_FLAG is equal to 1, an HE-LTF User Block contains all HE-LTF symbols for one user, and the number of HE-LTF User Blocks in the HE-LTF field is equal to the number of Users, NUM_USERS, and each HE-LTF User Block contains all the HE-LTF symbols for one user.
- No beamforming steering matrix is applied to the waveform. The Beamformed field in HE-SIG-A of an HE Ranging NDP is always set to 0.
- For transmission of HE-STFs and HE-LTFs, if NSTS = NTx, the Q matrix shall be an Identity matrix, and if NSTS < NTx, the Q matrix shall be based on an antenna selection matrix with no antenna swapping. The Q matrix becomes an Identity matrix when all 0 rows are removed.
- Has a Packet Extension (PE) field that is 4 μ s in duration. No energy is transmitted during the first 1.6 μ s of the PE field if the HE-LTF field is using the secure HE-LTF, similar to no energy being transmitted during the GI of HE-LTF symbols.
- For decoding the HE-LTF fields, a PHY-RXLTFSEQUENCE.request primitive issued from the MAC provides the LTF_REP, LTF_NSTS, and LTF_OFFSET parameters, which are not encoded in the HE-SIG-A, but included in the preceding Ranging NDP Announcement frame. The LTF_OFFSET parameter indicates the number of secure HE-LTF symbols to skip for receiving the corresponding user's HE-LTF User Block.
- When the TXVECTOR parameter NUM_USERS is greater than 1, the TXVECTOR parameter NUM_STS[1] is used to encode the NSTS And Mid-amble Periodicity field of the HE-SIG-A1. Otherwise, the TXVECTOR parameter NUM_STS is used to encode the NSTS And Mid-amble Periodicity field of the HE-SIG-A1.
- The only supported mode is 2x HE-LTF with 1.6 μ s GI. The other combinations of HE-LTF modes and GI duration are disallowed. No energy is transmitted during the GI of the HE-LTF symbols when secure HE-LTF are used, which is referred to as a zero-power GI.
- The number of HE-LTF symbols in the HE-LTF field in an HE Ranging NDP depends on the number of space-time streams NUM_STS, the number of HE-LTF repetitions, LTF-REP, and when secure HE-LTFs are used, the number of users NUM_USERS.

NOTE 1—See examples in Figure 27-46b and Figure 27-46c.

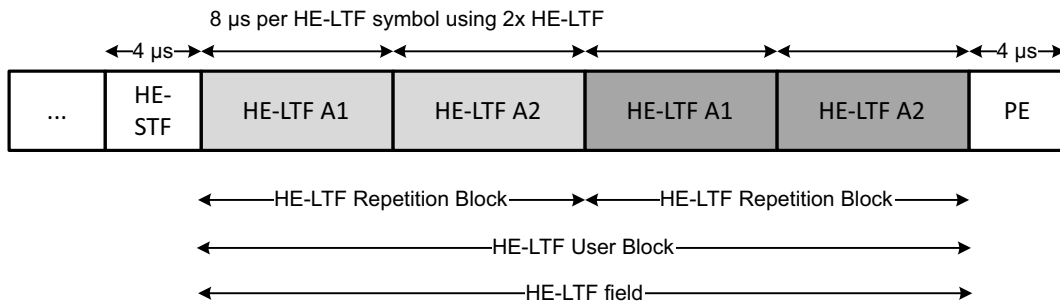


Figure 27-46b—An example of HE-LTFs in an HE Ranging NDP with NUM_USERS = 1, NUM_STS = 2, and N_LTF_REP = 2

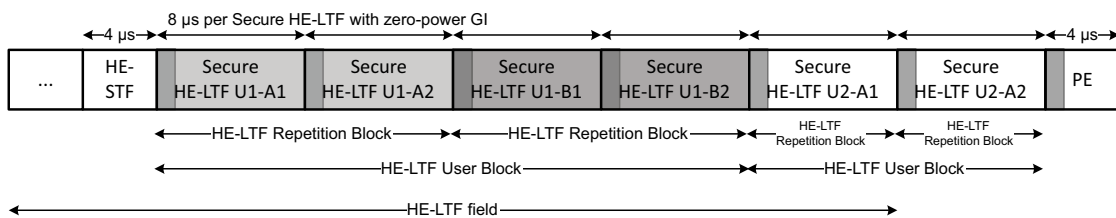


Figure 27-46c—An example of secure HE-LTFs with NUM_USERS = 2, NUM_STS = [2,1], and LTF_REP = [2,2]

When the TXVECTOR parameter SECURE_LTF_FLAG is equal to 0, HE-LTFs as defined in 27.3.11.10 are used in the HE Ranging NDP. There is a single HE-LTF User Block in an HE-LTF field, and the total number of HE-LTF symbols in the HE-LTF field is the product of the number of symbols in an HE-LTF Repetition Block, NHE-LTF, and the number of HE-LTF repetitions, given in LTF_REP.

The construction of the HE-LTFs in an HE Ranging NDP is done by repeating the steps in 27.3.6.9 LTF_REP times, that is, a value of LTF_REP equal to 1 indicates a single HE-LTF Repetition Block, and a value of LTF_REP greater than 1 indicates the use of repetitions, that is, multiple HE-LTF Repetition Blocks are included in an HE-LTF User Block. If the TXVECTOR parameter SECURE_LTF_FLAG is set to 0, the TXVECTOR parameter NUM_USERS is not present, which is then assumed to be 1, and all the HE-LTF symbols belong to a single HE-LTF User Block.

When the TXVECTOR parameter SECURE_LTF_FLAG is equal to 1, secure HE-LTFs as defined in 27.3.18b.6, are used and the Packet Extension field will be partially replaced by a zero power GI in its first 1.6 μs, see Figure 27-46d. For the secure HE-LTF symbol or Packet Extension field with zero power GI, the time domain signal has zero power during the period of the GI. The TXVECTOR parameters LTF_KEY, NUM_STS, and LTF_REP will be in array form with NUM_USERS entries. The repetitions of the HE-LTF symbols are repetitions of the HE-LTF Repetition Block. The randomized HE-LTF sequences are different in each of the HE-LTF Repetition Blocks. The total number of HE-LTF symbols in an HE-LTF User Block is the product of the number of symbols in an HE-LTF Repetition Block, NHE-LTF, and the number of HE-LTF repetitions given in LTF_REP.

For secure HE-LTF transmissions, the number of HE-LTF repetitions given in LTF_REP shall be greater than 1, and there are a minimum of two HE-LTF Repetition Blocks in each HE-LTF User Block.

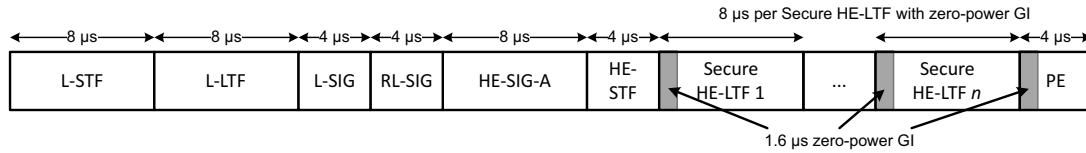


Figure 27-46d—HE Ranging NDP format with secure HE-LTFs

NOTE 2—The intended receiver of the Secure HE-LTFs can use implementation specific methods to detect the occurrence of an attack. It can explore the HE-LTF repetitions to check the consistency of the channel estimation or the signal to interference ratio (SIR) drop from impact of an attack. Please see Annex AE for more details on SIR drop due to attack.

The secure HE-LTFs of each User Block are concatenated one after another and form the HE-LTF field of an HE Ranging NDP; the total number of HE-LTF symbols in an HE-LTF field shall not exceed a maximum of 64 secure HE-LTF symbols over all users, all space-time streams, and number of HE-LTF repetitions.

In each HE-LTF User Block within the HE-LTF field, the number of Tx antennas are the same as the number indicated in NUM_STS for the corresponding HE-LTF User Block and may vary from one HE-LTF User Block to another. Within the HE-STF Field, the number of Tx antennas should match the first HE-LTF User Block. In the pre-HE modulated fields, the number of Tx antennas used shall be no less than the minimum number of Tx antennas used in the HE modulated fields. The sum of the Tx power across all Tx antennas shall remain constant throughout the entire HE Ranging NDP PPDU.

27.3.18a.2 HE TB Ranging NDP

The format of an HE TB Ranging NDP is shown in Figure 27-46e.

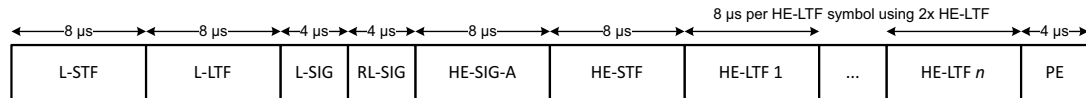


Figure 27-46e—HE TB Ranging NDP format

The HE TB Ranging NDP has the following properties:

- Uses the HE TB PPDU format but without the Data field.
- The HE-STF field in HE TB Ranging NDP is the same as the HE-STF field in a HE TB PPDU.
- Uses HE-LTFs or secure HE-LTFs when the TXVECTOR parameter SECURE_LTF_FLAG is equal to 0 or 1, respectively.
- Secure HE-LTFs use randomized HE-LTF sequences, pseudorandom and deterministic per stream phase rotation and, when the TXVECTOR parameter TX_WINDOW_FLAG is equal to 1, a frequency domain flat top window is used instead of the frequency domain rectangular window; see 27.3.18b.6.
- Uses HE-LTF repetitions, if indicated in the TXVECTOR parameter LTF_REP by values larger than one.
- The HE-LTF field of an HE TB Ranging NDP consists of a single HE-LTF User Block. The HE-LTF User Block contains one or more HE-LTF Repetition Blocks, and the number of HE-LTF Repetition Blocks is equal to LTF_REP. Each HE-LTF Repetition Block in the HE-LTF User Block comprises one or more HE-LTF symbols, $N_{\text{HE-LTF}}$ specified in the Common Info field within the Sounding Ranging Trigger frame.

- Has a Packet Extension (PE) field that is 4 μ s in duration. No energy is transmitted during the first 1.6 μ s of the PE field if the HE-LTF field is using the secure HE-LTF, similar to no energy being transmitted during the GI of HE-LTF symbols.
- No beamforming steering matrix is applied to the waveform.
- For transmission of HE-LTFs, if NSTS = NTx, the Q matrix shall be an Identity matrix, and if NSTS < NTx, the Q matrix shall be an antenna selection matrix with no antenna swapping. The Q matrix becomes an Identity matrix when all 0 rows are removed.
- The only supported mode is the 2x HE-LTF with 1.6 μ s GI, with HE single stream pilot HE-LTF mode. The other combinations of HE-LTF modes and GI duration are disallowed.

The number of HE-LTF symbols in an HE TB Ranging NDP is the product of the number of HE-LTF symbols in an HE-LTF Repetition Block, $N_{\text{HE-LTF}}$, and the number of HE-LTF repetitions, given in LTF_REP. A value of LTF_REP equal to 1 indicates no repetition, that is, a single HE-LTF Repetition Block is included in an HE-LTF User Block, and a value of LTF_REP greater than 1 indicates the use of repetitions, that is, multiple HE-LTF Repetition Blocks are included in the HE-LTF User Block. The sum of Tx power shall remain constant throughout the entire HE TB Ranging NDP PPDU.

When the TXVECTOR parameter SECURE_LTF_FLAG is equal to 0, HE-LTFs as defined in 27.3.11.10 are used in each HE-LTF Repetition Block.

When the TXVECTOR parameter SECURE_LTF_FLAG is equal to 1, secure HE-LTFs as defined in 27.3.18b are used in the HE-LTF Repetition Blocks, and the Packet Extension field will be partially replaced by a zero power GI in its first 1.6 μ s; see Figure 27-46f. The repetitions of the HE-LTF symbols form the HE-LTF Repetition Blocks. The randomized HE-LTF sequences are different in each HE-LTF Repetition Block.

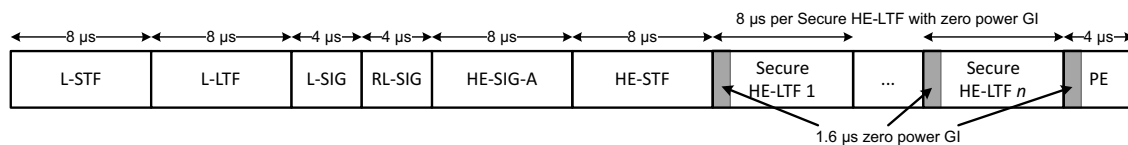


Figure 27-46f—HE TB Ranging NDP with secure HE-LTFs

27.3.18b HE-LTF field using secure HE-LTF

27.3.18b.1 Introduction

The HE-LTF field using secure HE-LTF is similar to the HE-LTF field, see 27.3.11.10, the main differences are as follows:

- The HE-LTF sequence is replaced by the randomized LTF sequence described in 27.3.18b.2.
- The conventional GI is replaced by a zero power GI.
- There are no single stream pilot subcarriers in the secure HE-LTFs, all subcarriers are mapped using the $P_{\text{HE-LTF}}$ matrix.
- No CSD is applied to the space-time streams.
- Each spatial stream has a per stream pseudorandom and deterministic phase rotation applied to all the subcarriers.
- A frequency domain flat top window is applied to the secure HE-LTF when configured.
- Only 2x HE-LTF and 1.6 μ s GI is supported.
- No beamforming is applied; Q is a square identity matrix.

27.3.18b.2 Generation of a randomized secure HE-LTF sequence

The secure HE-LTF sequence is constructed using pseudorandom 64-QAM modulation. Pseudorandom octets defined in 11.21.6.4.5.4 are used in the construction of the pseudorandom 64-QAM values.

The first seven pseudorandom octets ($Octet_0$ – $Octet_6$) in the secure NDP are used for per stream phase rotations see 27.3.18b.3. Starting with $Octet_7$, these pseudorandom octets are used for construction of pseudorandom 64-QAM values in the secure HE-LTF sequences.

27.3.18b.2.1 Randomized LTF sequence for 20 MHz secure NDP

There are 122 nonzero entries in the 20 MHz secure 2x HE-LTF sequence. This subclause describes the mapping of pseudorandom octets to the nonzero entries of the 20 MHz secure 2x HE-LTF sequence, and then the construction of the 64-QAM values for each nonzero entry of the secure HE-LTF sequence.

The indices of the nonzero entries of the 20 MHz secure 2x HE-LTF sequence are those of the nonzero entries in Equation (27-42). The indices for the nonzero entries are provided in Equation (27-126a).

$$NZ_{20\text{ MHz}} = \{-122, -120, -118, -116, -114, -112, -110, -108, -106, -104, -102, -100, -98, -96, -94, -92, -90, -88, -86, -84, -82, -80, -78, -76, -74, -72, -70, -68, -66, -64, -62, -60, -58, -56, -54, -52, -50, -48, -46, -44, -42, -40, -38, -36, -34, -32, -30, -28, -26, -24, -22, -20, -18, -16, -14, -12, -10, -8, -6, -4, -2, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122\} \quad (27-126a)$$

There are up to 64 secure HE-LTF sequences in an NDP. For notational convenience we indicate the LTF sequence number with the integer n , which is an integer between 1 and 64. Since each secure HE-LTF sequence is used to generate each of the LTF symbols, n also indicates the LTF symbol number. Table 27-47a provides the pseudorandom octet index for each nonzero subcarrier index in the n th 20 MHz secure HE-LTF sequence.

Table 27-47a—Pseudorandom octet index for each nonzero subcarrier index in the n th 20 MHz secure HE-LTF sequence

Secure HE-LTF tone index	Pseudorandom octet index
–122	$7 + (n - 1) \times 122$
–120	$8 + (n - 1) \times 122$
–118	$9 + (n - 1) \times 122$
$\vdots$	$\vdots$
–4	$66 + (n - 1) \times 122$
–2	$67 + (n - 1) \times 122$
2	$68 + (n - 1) \times 122$

Table 27-47a—Pseudorandom octet index for each nonzero subcarrier index in the n th 20 MHz secure HE-LTF sequence (*continued*)

Secure HE-LTF tone index	Pseudorandom octet index
4	$69 + (n - 1) \times 122$
$\vdots$	$\vdots$
118	$126 + (n - 1) \times 122$
120	$127 + (n - 1) \times 122$
122	$128 + (n - 1) \times 122$

All entries in the 20 MHz secure HE-LTF sequence other than the entries with indices defined in Equation (27-126a) shall be set to 0.

The six least significant bits ($B_0, B_1, B_2, B_3, B_4, B_5$) of an octet are used in the construction of the 64-QAM value, as specified in Table 17-15.

27.3.18b.2 Randomized LTF sequence for 40 MHz secure NDP

There are 242 nonzero entries in the 40 MHz secure 2x HE-LTF sequence. This subclause describes the mapping of pseudorandom octets to the nonzero entries of the 40 MHz secure 2x HE-LTF sequence, and then the construction of the 64-QAM values for each nonzero entry of the secure HE-LTF sequence.

The indices of the nonzero entries of the 40 MHz secure 2x HE-LTF sequence are those of the nonzero entries in Equation (27-45). The indices for the nonzero entries are provided in Equation (27-126b).

$$\begin{aligned}
 NZ_{40 \text{ MHz}} = \{ & -244, -242, -240, -238, -236, -234, -232, -230, -228, -226, -224, \\
 & -222, -220, -218, -216, -214, -212, -210, -208, -206, -204, -202, -200, -198, \\
 & -196, -194, -192, -190, -188, -186, -184, -182, -180, -178, -176, -174, -172, \\
 & -170, -168, -166, -164, -162, -160, -158, -156, -154, -152, -150, -148, -146, \\
 & -144, -142, -140, -138, -136, -134, -132, -130, -128, -126, -124, -122, -120, \\
 & -118, -116, -114, -112, -110, -108, -106, -104, -102, -100, -98, -96, -94, -92, \\
 & -90, -88, -86, -84, -82, -80, -78, -76, -74, -72, -70, -68, -66, -64, -62, -60, -58, \\
 & -56, -54, -52, -50, -48, -46, -44, -42, -40, -38, -36, -34, -32, -30, -28, -26, -24, \\
 & -22, -20, -18, -16, -14, -12, -10, -8, -6, -4, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, \\
 & 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, \\
 & 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, \\
 & 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, \\
 & 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, \\
 & 182, 184, 186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, \\
 & 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244 \}
 \end{aligned} \tag{27-126b}$$

There are up to 64 secure HE-LTF sequences in an NDP. For notational convenience we indicate the LTF sequence number with the integer n , which is an integer between 1 and 64. Since each secure HE-LTF sequence is used to generate each of the LTF symbols, n also indicates the LTF symbol number. Table 27-47b provides the pseudorandom octet index for each nonzero subcarrier index in the n th 40 MHz secure HE-LTF sequence.

All entries in the 40 MHz secure HE-LTF sequence other than the entries with indices defined in Equation (27-126b) shall be set to 0.

Table 27-47b—Pseudorandom octet index for each nonzero subcarrier index in the n th 40 MHz secure HE-LTF sequence

Secure HE-LTF tone index	Pseudorandom octet index
−244	$7 + (n - 1) \times 242$
−242	$8 + (n - 1) \times 242$
−240	$9 + (n - 1) \times 242$
$\vdots$	$\vdots$
−6	$126 + (n - 1) \times 242$
−4	$127 + (n - 1) \times 242$
4	$128 + (n - 1) \times 242$
6	$129 + (n - 1) \times 242$
$\vdots$	$\vdots$
240	$246 + (n - 1) \times 242$
242	$247 + (n - 1) \times 242$
244	$248 + (n - 1) \times 242$

The six least significant bits ($B_0, B_1, B_2, B_3, B_4, B_5$) of an octet are used in the construction of the 64-QAM value, as specified in Table 17-15.

27.3.18b.2.3 Randomized LTF sequence for 80 MHz secure NDP

There are 498 nonzero entries in the 80 MHz secure 2x HE-LTF sequence. This subclause describes the mapping of pseudorandom octets to the nonzero entries of the 80 MHz secure 2x HE-LTF sequence, and then the construction of the 64-QAM values for each nonzero entry of the secure HE-LTF sequence.

The indices of the nonzero entries of the 80 MHz secure 2x HE-LTF sequence are those of the nonzero entries in Equation (27-48). The indices for the nonzero entries are provided in Equation (27-126c).

$$NZ_{80 \text{ MHz}} = \{-500, -498, -496, -494, -492, -490, -488, -486, -484, -482, -480, -478, -476, -474, -472, -470, -468, -466, -464, -462, -460, -458, -456, -454, -452, -450, -448, -446, -444, -442, -440, -438, -436, -434, -432, -430, -428, -426, -424, -422, -420, -418, -416, -414, -412, -410, -408, -406, -404, -402, -400, -398, -396, -394, -392, -390, -388, -386, -384, -382, -380, -378, -376, -374, -372, -370, -368, -366, -364, -362, -360, -358, -356, -354, -352, -350, -348, -346, -344, -342, -340, -338, -336, -334, -332, -330, -328, -326, -324, -322, -320, -318, -316, -314, -312, -310, -308, -306, -304, -302, -300, -298, -296, -294, -292, -290, -288, -286, -284, -282, -280, -278, -276, -274, -272, -270, -268, -266, -264, -262, -260, -258, -256, -254, -252, -250, -248, -246, -244, -242, -240, -238, -236, -234, -232, -230, -228, -226, -224, -222, -220, -218, -216, -214, -212, -210, -208, -206, -204, -202, -200, -198, -196, -194, -192, -190, -188, -186, -184, -182, -180, -178, -176, -174, -172, -170, -168, -166, -164, -162, -160, -158, -156, -154, -152, -150, -148, -146, -144, -142, -140, -138, -136, -134, -132, -130, -128, -126, -124, -122, -120, -118, -116, -114, -112, -110, -108, -106, -104, -102, -100, -98, -96, -94, -92, -90, -88, -86, -84, -82, -80, -78, -76, -74, -72, -70, -68, -66, -64, -62, -60, -58, -56, -54, -52, -50, -48, -46, -44, -42, -40, -38, -36, -34, -32, -30, -28, -26, -24, -22, -20, -18, -16, -14, -12, -10, -8, -6, -4, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176, 178, 180, 182, 184, 186, 188, 190, 192, 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 270, 272, 274, 276, 278, 280, 282, 284, 286, 288, 290, 292, 294, 296, 298, 300, 302, 304, 306, 308, 310, 312, 314, 316, 318, 320, 322, 324, 326, 328, 330, 332, 334, 336, 338, 340, 342, 344, 346, 348, 350, 352, 354, 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, 384, 386, 388, 390, 392, 394, 396, 398, 400, 402, 404, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478, 480, 482, 484, 486, 488, 490, 492, 494, 496, 498, 500\}$$

(27-126c)

There are up to 64 secure HE-LTF sequences in an NDP. For notational convenience we indicate the LTF sequence number with the integer n , which is an integer between 1 and 64. Since each secure HE-LTF sequence is used to generate each of the LTF symbols, n also indicates the LTF symbol number. Table 27-47c provides the pseudorandom octet index for each nonzero subcarrier index in the n th 80 MHz secure HE-LTF sequence.

All entries in the 80 MHz secure HE-LTF sequence other than the entries with indices defined in Equation (27-126c) shall be set to 0.

The six least significant bits ($B_0, B_1, B_2, B_3, B_4, B_5$) of an octet are used in the construction of the 64-QAM value, as specified in Table 17-15.

Table 27-47c—Pseudorandom octet index for each nonzero subcarrier index in the n th 80 MHz secure HE-LTF sequence

Secure HE-LTF tone index	Pseudorandom octet index
−500	$7 + (n - 1) \times 498$
−498	$8 + (n - 1) \times 498$
−496	$9 + (n - 1) \times 498$
$\vdots$	$\vdots$
−6	$254 + (n - 1) \times 498$
−4	$255 + (n - 1) \times 498$
4	$256 + (n - 1) \times 498$
6	$257 + (n - 1) \times 498$
$\vdots$	$\vdots$
496	$502 + (n - 1) \times 498$
498	$503 + (n - 1) \times 498$
500	$504 + (n - 1) \times 498$

27.3.18b.2.4 Randomized LTF sequence for the 160 MHz secure NDP and for the 80+80 MHz secure NDP

This subclause describes the mapping of pseudorandom octets to the nonzero entries of either the 160 MHz secure 2x HE-LTF sequence, or the 80+80 MHz secure 2x LTF sequence, and then the construction of the 64-QAM values for each nonzero entry of the secure HE-LTF sequence.

The construction of either the 160 MHz secure HE-LTF sequence, or the 80+80 MHz secure HE-LTF sequence, uses a segment parser to divide the pseudorandom octets between the sequence for the lower 80 MHz segment and the sequence for the upper 80 MHz segment. Figure 27-46g illustrates the segment parser distribution of pseudorandom octets between the sequence for the lower 80 MHz segment and the sequence for the upper 80 MHz segment.

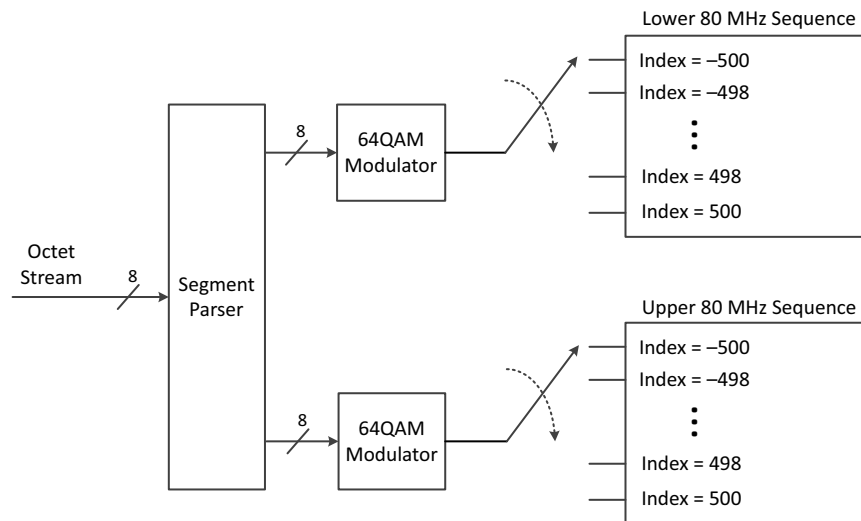


Figure 27-46g—Segment parser distributing pseudorandom octets to the sequences for the lower and upper 80 MHz segments in either the 160 MHz secure HE-LTF, or the 80+80 MHz secure HE-LTF sequence

There are up to 64 secure HE-LTF sequences in an NDP. For notational convenience we indicate the LTF sequence number with the integer n , which is an integer between 1 and 64. Since each secure HE-LTF sequence is used to generate each of the LTF symbols, n also indicates the LTF symbol number. Table 27-47d provides the pseudorandom octet index for each nonzero subcarrier index for the n th pair of lower and upper 80 MHz segments.

Table 27-47d—Pseudorandom octet index for each nonzero subcarrier index in the n th pair of lower and upper 80 MHz segments

80 MHz Segment	Secure HE-LTF tone index	Pseudorandom octet index
Lower	–500	$7 + (n - 1) \times 996$
Upper	–500	$8 + (n - 1) \times 996$
Lower	–498	$9 + (n - 1) \times 996$
Upper	–498	$10 + (n - 1) \times 996$
Lower	–496	$11 + (n - 1) \times 996$
Upper	–496	$12 + (n - 1) \times 996$
⋮	⋮	⋮
Lower	–6	$501 + (n - 1) \times 996$
Upper	–6	$502 + (n - 1) \times 996$
Lower	–4	$503 + (n - 1) \times 996$
Upper	–4	$504 + (n - 1) \times 996$
Lower	4	$505 + (n - 1) \times 996$
Upper	4	$506 + (n - 1) \times 996$

Table 27-47d—Pseudorandom octet index for each nonzero subcarrier index in the n th pair of lower and upper 80 MHz segments (*continued*)

80 MHz Segment	Secure HE-LTF tone index	Pseudorandom octet index
Lower	6	$507 + (n - 1) \times 996$
Upper	6	$508 + (n - 1) \times 996$
$\vdots$	$\vdots$	$\vdots$
Lower	496	$997 + (n - 1) \times 996$
Upper	496	$998 + (n - 1) \times 996$
Lower	498	$999 + (n - 1) \times 996$
Upper	489	$1000 + (n - 1) \times 996$
Lower	500	$1001 + (n - 1) \times 996$
Upper	500	$1002 + (n - 1) \times 996$

All entries in the 160 MHz secure HE-LTF sequence, or the 80+80 secure HE-LTF sequence, other than the nonzero entries shall be set to 0.

The six least significant bits ($B_0, B_1, B_2, B_3, B_4, B_5$) of an octet are used in the construction of the 64-QAM value, as specified in Table 17-15.

27.3.18b.3 Pseudorandom and deterministic per spatial stream phase rotations

In each repetition of the secure HE-LTF symbols, each spatial stream has a phase rotation applied to all the subcarriers. The same phase rotation is applied to all subcarriers. The phase rotation can vary in different repetitions of the secure HE-LTF symbols.

The phase rotation for a given spatial stream consists of two components: a pseudorandom phase rotation and a deterministic phase rotation. For each spatial stream, the pseudorandom phase rotation is the same for all the repetitions of HE-LTF symbols within the secure NDP. As specified in 11.21.6.4.5.4, a stream of pseudorandom octets are provided by the AES-128 CTR Mode. The first seven pseudorandom octets are used in the construction of the pseudorandom phase rotations.

The deterministic phase rotation is a function of the repetition number within the secure NDP and the spatial stream number.

The total phase rotation applied to spatial stream s within repetition r is given by Equation (27-126d).

$$\Theta_{r,s} = \alpha_s + \beta_{r,s} \quad (27-126d)$$

In Equation (27-126d) the term α_s is the pseudorandom phase rotation that is the same for the entire secure NDP and the term $\beta_{r,s}$ is the deterministic phase rotation that depends on the rotation value r .

The pseudorandom phase α_s is constructed using one of the pseudorandom octets defined in 11.21.6.4.5.4. The index of spatial streams, s , is a number between one and eight. For $s = 1$ the phase rotation is zero: $\alpha_1 = 0$. The first octet ($Octet_0$) is used to construct α_2 , the second octet ($Octet_1$) is used to construct α_3 .

Table 27-47e provides a summary of pseudorandom octets used in construction of each of the pseudorandom phase rotations.

Table 27-47e—Pseudorandom octets for generating per stream pseudorandom phase rotations

Pseudorandom octet	Spatial stream pseudorandom phase rotations
$Octet_0$	α_2
$Octet_1$	α_3
$Octet_2$	α_4
$Octet_3$	α_5
$Octet_4$	α_6
$Octet_5$	α_7
$Octet_6$	α_8

For STAs with fewer than eight spatial streams, not all of the pseudorandom octets up to $Octet_6$ are used.

The three most significant bits of the pseudorandom octet are used to construct the pseudorandom phase rotation. The integer k in Equation (27-126e) is derived from the three MSBs of the pseudorandom octet.

$$k = (4 \times B_5) + (2 \times B_6) + B_7 \quad (27-126e)$$

Equation (27-126f) specifies the pseudorandom phase rotation as a function of the integer k .

$$\alpha_s = k \times \frac{\pi}{4} \quad (27-126f)$$

The deterministic phase rotation $\beta_{r,s}$, which depends on both the spatial stream number s and the repetition number r , is provided in Table 27-47f.

The steps in the process of the construction of the secure HE-LTF, including the application of the spatial stream phase rotation, are described in 27.3.18b.6.

27.3.18b.4 Frequency domain windowing in HE-LTF field using secure HE-LTF

In an HE Ranging NDP or HE TB Ranging NDP using secure HE-LTF a frequency domain windowing function $w_{FD}(k)$ is applied to the subcarriers modulated with the secure HE-LTF sequence $X_{k,n}^m$.

There are two frequency domain windowing functions defined, the rectangle window $w_{FD}(k) = 1$ and the flat top window defined by Equation (27-126g) and Equation (27-126h).

$$w_{FD}(k) = \sum_{n=-10}^{10} p(n) \exp\left(\frac{-j\pi kn}{N_{FD}}\right), \text{ where } k \in K_s \quad (27-126g)$$

Table 27-47f—Deterministic phase rotation $\beta_{r,s}$ (radians)

	$s = 1$	$s = 2$	$s = 3$	$s = 4$	$s = 5$	$s = 6$	$s = 7$	$s = 8$
$r = 1$	0	0	0	0	0	0	0	0
$r = 2$	0	$\frac{2\pi}{4}$	$\frac{3\pi}{4}$	$\frac{5\pi}{4}$	$\frac{4\pi}{4}$	$\frac{6\pi}{4}$	$\frac{7\pi}{4}$	$\frac{\pi}{4}$
$r = 3$	0	$\frac{3\pi}{4}$	$\frac{5\pi}{4}$	$\frac{\pi}{4}$	$\frac{4\pi}{4}$	$\frac{7\pi}{4}$	$\frac{2\pi}{4}$	$\frac{6\pi}{4}$
$r = 4$	0	$\frac{7\pi}{4}$	$\frac{6\pi}{4}$	$\frac{4\pi}{4}$	$\frac{\pi}{4}$	$\frac{5\pi}{4}$	$\frac{3\pi}{4}$	$\frac{2\pi}{4}$
$r = 5$	0	$\frac{\pi}{4}$	$\frac{4\pi}{4}$	$\frac{6\pi}{4}$	$\frac{7\pi}{4}$	$\frac{5\pi}{4}$	$\frac{3\pi}{4}$	$\frac{2\pi}{4}$
$r = 6$	0	$\frac{6\pi}{4}$	$\frac{\pi}{4}$	$\frac{2\pi}{4}$	$\frac{7\pi}{4}$	$\frac{4\pi}{4}$	$\frac{5\pi}{4}$	$\frac{3\pi}{4}$
$r = 7$	0	$\frac{4\pi}{4}$	$\frac{7\pi}{4}$	$\frac{3\pi}{4}$	$\frac{2\pi}{4}$	$\frac{\pi}{4}$	$\frac{5\pi}{4}$	$\frac{6\pi}{4}$
$r = 8$	0	$\frac{5\pi}{4}$	$\frac{2\pi}{4}$	$\frac{7\pi}{4}$	$\frac{\pi}{4}$	$\frac{4\pi}{4}$	$\frac{6\pi}{4}$	$\frac{3\pi}{4}$

$$N_{FD} = \begin{cases} 512 & \text{for 20 MHz} \\ 1024 & \text{for 40 MHz} \\ 2048 & \text{for 80 MHz} \\ 4096 & \text{for 160 MHz} \end{cases} \quad (27-126h)$$

where

$K_s = NZ_{BW}$ is the set of nonzero subcarriers for a given bandwidth defined in 27.3.18b.2
 $p(n)$ is the impulse response given by Equation (27-126i)

$$p(n) = \sum_{m=0}^4 a_m \cos\left(\frac{2m\pi(n+10)}{N_{WinFT}}\right) \text{ where } n = -10, \dots, 10 \quad (27-126i)$$

where

$$\begin{aligned} a_0 &= 0.21557895 \\ a_1 &= -0.41663158 \\ a_2 &= 0.277263158 \\ a_3 &= -0.083578947 \\ a_4 &= 0.006947368 \\ N_{WinFT} &= 20 \end{aligned}$$

The $w_{FD}(k)$ shall be normalized to have unit RMS power.

27.3.18b.5 Mathematical description of HE-LTF field using secure HE-LTF

In an HE Ranging NDP or HE TB Ranging NDP using secure HE-LTF, the time domain representation of the waveform of user u , transmitted on frequency segment i_{Seg} , and transmit chain i_{TX} shall be as described by Equation (27-126j).

$$r_{HE-LTF, u}^{(i_{Seg}, i_{TX})}(t) = \frac{1}{\sqrt{|K_s|N_{STS, u}}} \sum_{r=0}^{N_{rep, u}-1} \sum_{n=0}^{N_{HE-LTF, u}-1} w_{T_{HE-LTF}}(t - (n + rN_{HE-LTF, u})T_{HE-LTF-SYM} - T_{GI, HE-LTF}) \\ \times \sum_{k \in K_s} \sum_{m=1}^{N_{STS, u}} \left([Q]_{i_{TX}, m} [P_{HE-LTF}]_{m, (n+1)} w_{FD}(k) X_{k, (n+rN_{HE-LTF, u})}^u \exp(j\Theta_{r+1, m}^u) \right) \exp(j2\pi k \Delta_{F, HE}(t - (n + rN_{HE-LTF, u})T_{HE-LTF-SYM} - T_{GI, HE-LTF})) \quad (27-126j)$$

where

$K_s = NZ_{BW}$	is the set of nonzero subcarriers for a given bandwidth defined in 27.3.18b.2
$N_{STS, u}$	is the number of spatial streams for user u
$N_{rep, u}$	is the number of HE-LTF repetitions for user u
$N_{HE-LTF, u}$	is the number of HE-LTF symbols N_{HE-LTF} in one Repetition Block of user u
$w_{T_{HE-LTF}}(t)$	is the zero power GI windowing function, that avoids cyclic extension of the signal outside of $[0, T_{HE-LTF}]$
T_{HE-LTF}	is always set to $T_{HE-LTF-2X} = 6.4 \mu s$
$T_{GI, HE-LTF}$	is always set to $1.6 \mu s$, accordingly $T_{HE-LTF-SYM} = 8.0 \mu s$
Q	is an identity matrix, that is, $Q = I_{N_{TX}}$
P_{HE-LTF}	is defined in Equation (27-57)
$w_{FD}(k)$	is the frequency domain windowing function
$X_{k, (n+rN_{HE-LTF, u})}^u$	is the k th subcarrier of the 64-QAM modulated $(n + rN_{HE-LTF, u})$ secure HE-LTF sequence for user u
$\Theta_{r+1, m}^u$	is the total phase rotation applied to spatial stream m within repetition $r + 1$ for user u , defined in Equation (27-126j)

The time domain representation of the waveform including all users, transmitted on frequency segment i_{Seg} , and transmit chain i_{TX} shall be as described by Equation (27-126k).

$$r_{HE-LTF}^{(i_{Seg}, i_{TX})}(t) = \sum_{u=1}^{N_u} r_{HE-LTF, u}^{(i_{Seg}, i_{TX})}(t - T_{HE-LTF, u}) \quad (27-126k)$$

where

N_u	is the total number of users (NUM_USERS)
$T_{HE-LTF, u}$	is the start time of the secure HE-LTF symbols for user u as defined by Equation (27-126l) and Equation (27-126m)

$$T_{\text{HE-LTF}, u} = \sum_{k=1}^{u-1} N_{\text{rep}, k} N_{\text{HE-LTF}, k} T_{\text{HE-LTF-SYM}} \quad (27-126l)$$

$$T_{\text{HE-LTF}, 1} = 0 \quad (27-126m)$$

27.3.18b.6 Construction of secure HE-LTF symbols

The construction of the n th secure HE-LTF symbol is achieved as follows:

- Sequence generation: Construct the n th randomized HE-LTF sequence $X_{k,r,n}^u$ in the frequency domain over the bandwidth indicated by CH_BANDWIDTH as described in 27.3.18b.2.
- Apply frequency domain window function $w_{FD}(k)$ to all the tones of the secure HE-LTF sequence. The frequency domain window can be the Rectangular window or flat top window, when the TXVECTOR parameter TX_WINDOW_FLAG is equal to 0 or 1, respectively.
- Apply the $P_{\text{HE-LTF}}$ matrix to all tones of the secure HE-LTF sequence.
- Apply per spatial stream phase rotation: Generate the pseudorandom phase rotation for each spatial stream. Apply the pseudorandom phase rotation along with the deterministic phase rotation to the spatial streams as described in 27.3.18b.3.
- Apply a zero CSD on each space-time stream, which is equivalent to no CSD per space-time stream.
- There is no spatial mapping, the Q matrix is a block identity matrix.
- IDFT: Compute the inverse discrete Fourier transform.
- Insert zero power GI: Prepend values of zero, of length indicated by the TXVECTOR parameter GI_TYPE.
- Analog and RF: Upconvert the resulting complex baseband waveform associated with each transmit chain to an RF signal according to the center frequency of the desired channel and transmit. Refer to 27.3.10.

The generation of the time domain secure HE-LTF symbol per repetition for repetition r , symbol n , and tone index k is shown in Figure 27-46h.

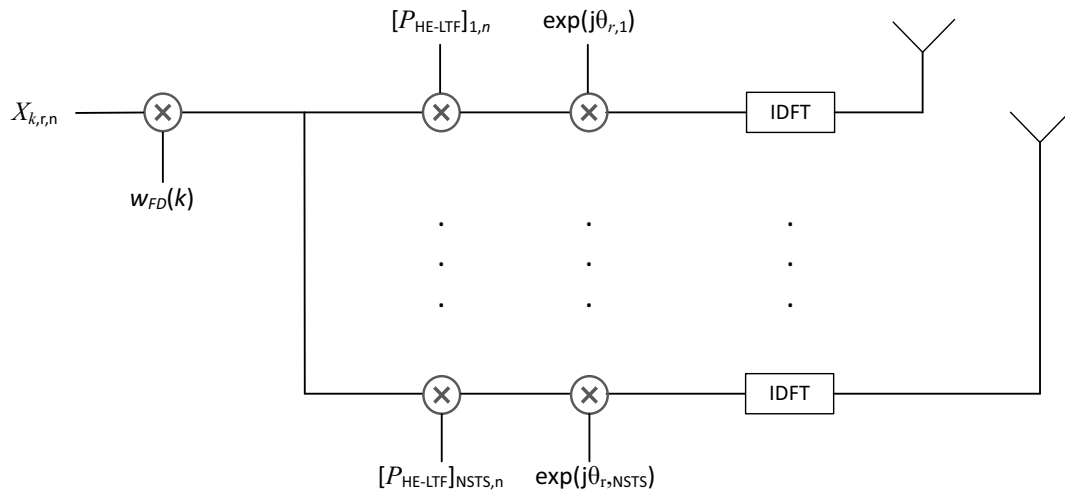


Figure 27-46h—Generation of secure HE-LTF symbols per repetition in a HE Ranging NDP PPDU

27.3.19 Transmit specification

Insert 27.3.19.5 after 27.3.19.4.4 as follows:

27.3.19.5 Time of departure accuracy

The time of departure accuracy test evaluates TIME_OF_DEPARTURE against aTxPHYTxStartRMS and aTxPHYTxStartRMS against TIME_OF_DEPARTURE_ACCURACY_TEST_THRESH as defined in Annex P with the following test parameters:

- MULTICHANNEL_SAMPLING_RATE is:

$$20 \times 10^6 \left(1 + \left\lceil \frac{f_H - f_L}{20 \text{ MHz}} \right\rceil \right) \text{ sample/s, for a CH_BANDWIDTH parameter equal to CBW20}$$

$$40 \times 10^6 \left(1 + \left\lceil \frac{f_H - f_L}{40 \text{ MHz}} \right\rceil \right) \text{ sample/s, for a CH_BANDWIDTH parameter equal to CBW40}$$

$$80 \times 10^6 \left(1 + \left\lceil \frac{f_H - f_L}{80 \text{ MHz}} \right\rceil \right) \text{ sample/s, for a CH_BANDWIDTH parameter equal to CBW80}$$

$$160 \times 10^6 \left(1 + \left\lceil \frac{f_H - f_L}{160 \text{ MHz}} \right\rceil \right) \text{ sample/s, for a CH_BANDWIDTH parameter equal to CBW160 or CBW80+80}$$

where

f_H is the nominal center frequency in Hz of the highest channel in the channel set
 f_L is the nominal center frequency in Hz of the lowest channel in the channel set
the channel set is the set of channels upon which frames providing measurements are transmitted, the channel set comprises channels uniformly spaced across

- FIRST_TRANSITION_FIELD is L-STF
- SECOND_TRANSITION_FIELD is L-LTF
- TRAINING_FIELD is L-LTF windowed in a manner which should approximate the windowing described in 17.3.2.5 with TTR = 100 ns
- TIME_OF_DEPARTURE_ACCURACY_TEST_THRESH is 80 ns

NOTE—The indicated windowing applies to the time of departure accuracy test equipment, and not the transmitter or receiver.

27.3.21 HE transmit procedure

Insert new text after the fourth paragraph of 27.3.21 as follows:

Transmission of the PHY preamble may start if TIME_OF_DEPARTURE_REQUESTED is false and shall start immediately if TIME_OF_DEPARTURE_REQUESTED is true, based on the parameters passed in the PHY-TXSTART.request primitive.

If all of the following conditions are met:

- If dot11TODImplemented and dot11TODActivated are true or if dot11TimingMsmtActivated is true
- The TXVECTOR parameter TIME_OF_DEPARTURE_REQUESTED is true

then the PHY shall issue a PHY-TXSTART.confirm(TXSTATUS) primitive to the MAC, forwarding the TIME_OF_DEPARTURE corresponding to the time when the first frame energy is sent by the transmitting port and TIME_OF_DEPARTURE_ClockRate parameter within the TXSTATUS vector. If dot11TimingMsmtActivated is true, then the PHY shall forward the value of TX_START_OF_FRAME_OFFSET in the TXSTATUS vector.

27.3.22 HE receive procedure

Change the third paragraph of 27.3.22 as follows:

This receive procedure and state machine do not describe the operation of optional features, such as DCM₁, HE Ranging NDP, and HE TB Ranging NDP. If the detected format indicates a non-HT PPDU, refer to the receive procedure and state machine in Clause 15, Clause 16, Clause 17, and Clause 18. If the detected format indicates an HT PPDU format, refer to the receive procedure and state machine in Clause 19. If the detected format indicates a VHT PPDU format, refer to the receive procedure and state machine in Clause 21. Through station management (via the PLME), the PHY is set to the appropriate frequency, as specified in 27.4. The PHY has also been configured with BSS identification information and STA identification information (i.e., BSS color value and STA-ID in the BSS) so that it can receive data intended for the STA in the specific cell. Other receive parameters, such as RSSI and indicated DATARATE, may be accessed via the PHY-SAP.

27.4 HE PLME

27.4.3 TXTIME and PSDU_LENGTH calculation

Insert new text after the second paragraph of 27.4.3 (beginning “For an HE sounding NDP and ...”) as follows:

For an HE TB Ranging NDP, TXTIME is defined by Equation (27-136a).

$$\text{TXTIME} = 40 + T_{\text{HE-PREAMBLE}} + 8N_{\text{LTF-REP}}N_{\text{HE-LTF}} + T_{PE} + \text{SignalExtension} \quad (27-136a)$$

where

$N_{\text{LTF-REP}}$ is given by the TXVECTOR parameter LTF_REP

$N_{\text{HE-LTF}}$ is given in Table 27-15

NOTE 1—The value 40 corresponds to the duration of L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, and HE-STF in μs . T_{PE} is 4 μs in an HE TB Ranging NDP (see 27.3.18a.2).

For an HE Ranging NDP with the TXVECTOR parameter SECURE_LTF_FLAG equal to 0 or SECURE_LTF_FLAG equal to 1 and the TXVECTOR parameter NUM_USERS is equal to 1, TXTIME is defined by Equation (27-136b).

$$\text{TXTIME} = 36 + 8N_{\text{LTF-REP}}N_{\text{HE-LTF}} + T_{PE} + \text{SignalExtension} \quad (27-136b)$$

NOTE 2—The value 36 corresponds to the duration of L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, and HE-STF in μs . T_{PE} is 4 μs in an HE Ranging NDP (see 27.3.18a.1).

For an HE Ranging NDP with the TXVECTOR parameter SECURE_LTF_FLAG equal to 1 and the TXVECTOR parameter NUM_USERS greater than 1, TXTIME is defined by Equation (27-136c).

$$TXTIME = 36 + \sum_{n=1}^{NUM_USERS} 8N_{LTF_REP}(n)N_{HE-LTF}(n) + T_{PE} + SignalExtension \quad (27-136c)$$

NOTE 3—The value 36 corresponds to the duration of L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, and HE-STF in μ s. T_{PE} is 4 μ s in an HE Ranging NDP (see 27.3.18a.1).

Note that Clause 28 was introduced by IEEE Std 802.11ay-2021.

28. Enhanced directional multi-gigabit (EDMG) PHY specification

28.1 Introduction

28.1.1 Introduction to EDMG PHY

Insert a new item at the end of the dashed list at the end of 28.1.1 as follows:

- 2.16 GHz, 4.32 GHz, 6.48 GHz, and 8.64 GHz EDMG secure ranging PPDU using EDMG SC mode (transmit and receive)

28.2 EDMG PHY service interface

28.2.2 TXVECTOR and RXVECTOR parameters

Insert rows for three new parameters at the end of Table 28-1 (but before the notes) as follows (unchanged rows not shown):

Table 28-1—TXVECTOR and RXVECTOR parameters

Parameter	Condition	Value	TXVECTOR	RXVECTOR
SECURED_TRN	FORMAT is EDMG, EDMG_MODULATION is EDMG_SC_MODE, NUM_USERS is 1, NUM_STS is 1.	Indicates whether the TRN field, if present in the PPDU, contains Secure TRN sequences. Enumerated type: SECURED_TRN NON_SECURED_TRN	Y	Y
	Otherwise.	Not present.	N	N
SECURED_TRN_SEQUENCES	FORMAT is EDMG, EDMG_MODULATION is EDMG_SC_MODE, NUM_USERS is 1, NUM_STS is 1.	Provides the secure TRN sequences used in the EDMG secure ranging PPDU.	Y	N
	Otherwise.	Not present.	N	N
FIRST_PATH_AWV_TRN	FORMAT is EDMG, EDMG_MODULATION is EDMG_SC_MODE, NUM_USERS is 1, NUM_STS is 1.	Enumerated Type FIRST_PATH_AWV_ON_TRN BEST_PATH_AWV_ON_TRN Indicates whether the TRN field of the PPDU is transmitted using the first path AWV or the best path AWV.	Y	N
	Otherwise.	Not present.	N	N

28.3 Common parameters

28.3.2 EDMG PPDU format

28.3.2.1 General

Insert a new note after Table 28-3 as follows:

NOTE—For EDMG secure ranging PPDU, the TRN field contains secure TRN sequences.

28.3.3 EDMG preamble

28.3.3.3 EDMG portion of EDMG format preamble

28.3.3.3.2 EDMG-Header-A definition

28.3.3.3.2.3 Definition for EDMG SC mode and EDMG OFDM mode PPDUs

Insert two new rows above the “Reserved” row and change the “Reserved” row in Table 28-13 as follows (unchanged rows not shown):

Table 28-13—EDMG-MCS field definition when the Number of SS field is 0

Subfield	Number of bits	Start bit	Description
...			
Secure TRN	1	7	Corresponds to the TXVECTOR parameter SECURED_TRN. When set to 1 (TXVECTOR parameter SECURED_TRN equals SECURED_TRN), indicates that the TRN field contains secure TRN sequences specified in 28.9.3.5. Otherwise, (TXVECTOR parameter SECURED_TRN equals NON_SECURED_TRN) the TRN field uses the format specified in 28.9.2.2.5. This field is reserved when the EDMG TRN Length field is equal to 0.
First Path AWV TRN	1	8	When set to 1, indicates that the TRN field is transmitted (and shall be received) using the first path AWV antenna setting. Otherwise the TRN field is transmitted and shall be received using the best path AWV antenna setting.
Reserved	14	79	

28.5 EDMG and non-EDMG SC mode

28.5.9 Data field

28.5.9.2 Symbol blocking and guard insertion

28.5.9.2.2 SU PPDU transmission

28.5.9.2.2.2 SU PPDU transmission over a 2.16 GHz or a 2.16+2.16 GHz channel with $N_{STS} = 1$

Insert a new paragraph at the end of 28.5.9.2.2.2 as follows:

A Secure TRN field (see 28.9.3.5) shall be appended to an SU PPDU for EDMG secure ranging PPDU.

28.9 EDMG beamforming

Insert 28.9.3 (and its subclauses) after 28.9.2.2.8 as follows:

28.9.3 EDMG secure ranging PPDU

28.9.3.1 General

EDMG secure ranging PPDU are used for secure RTT measurement. The EDMG secure ranging PPDU is defined for single space-time stream ($i_{STS} = 1$) PPDU only.

28.9.3.2 EDMG secure ranging PPDU structure

An EDMG secure ranging PPDU shall be composed of a non-EDMG portion containing an L-STF, an L-CEF, and L-Header, and of an EDMG portion containing an EDMG-Header-A, a Data field, and a TRN field that contains Secure TRN subfields only. An EDMG secure ranging PPDU may include an EDMG-STF and an EDMG-CEF.

If beam refinement has been performed on a 4.32 GHz, 6.48 GHz, or 8.64 GHz channel, the Secure TRN subfields in the TRN field of EDMG secure ranging PPDU shall be transmitted over the entire signal bandwidth of the channel.

28.9.3.3 EDMG secure ranging PPDU header fields

An EDMG secure ranging PPDU is indicated by setting the Secure TRN field to 1 in EDMG-Header A.

28.9.3.4 EDMG secure ranging PPDU duration

The duration of an EDMG secure ranging PPDU follows the same calculation procedure as an EDMG SU PPDU with TRN field appended as described in 28.9.2.2.4. Each Secure TRN subfield is of the same duration as each TRN subfield as described in Subclause 28.9.3.6.

28.9.3.5 TRN field definition with Secure TRN subfields

The Secure TRN subfields enable secure ranging measurements by an EDMG STA.

28.9.3.5.1 TRN field structure with Secure TRN subfields

The TRN field structure containing the Secure TRN subfields in EDMG secure ranging PPDU is shown in Figure 28-45. The header fields related to TRN should be set according to the Table 28-74a.

Table 28-74a—EDMG-A header fields setting for secure DMG TRNs

Header field	Value
PPDU Type	1
EDMG Beam Tracking Request	0
EDMG TRN Length	1
RX TRN-Units per Each TX TRN-Unit	0
EDMG TRN-Unit P	0
EDMG TRN-Unit M	3
EDMG TRN-Unit N	3
TRN Subfield Sequence Length	0

EDMG secure ranging PPDU are EDMG BRP-TX PPDU. In such PPDU, all TRN subfields of all TRN-Units shall be transmitted using the same AWV as the preamble and data field of the PPDU. The EDMG TRN-Unit M field in the EDMG-A header shall be set to 3 so that each TRN-Unit shall have 4 secure TRN subfields that contain Secure TRN sequences.

28.9.3.6 Secure TRN subfield definition for EDMG secure ranging PPDU

An EDMG secure ranging PPDU transmission over a 2.16 GHz, 4.32 GHz, 6.48 GHz, and 8.64 GHz channel is defined at the $NCB \times 1.76$ GHz chip rate. The symbol blocking structure for the normal GI is as shown in Figure 27-48. An EDMG STA that supports secure TRN shall support the TRN field structure with normal GI as shown in Figure 27-48.

$GIe^1_{64 \times N_{CB}}$	Zero prefix	Secure ranging field	Zero postfix	$GIe^1_{64 \times N_{CB}}$
$64 \times N_{CB}$	$128 \times N_{CB}$	$384 \times N_{CB}$	$128 \times N_{CB}$	$64 \times N_{CB}$

Figure 27-48—DMG secure ranging TRN subfield structure over 2.16 GHz, 4.32 GHz, 6.48 GHz, and 8.64 GHz channel

An EDMG secure ranging PPDU shall have a single spatial stream ($i_{TS} = 1$) mapped to a single transmit chain ($i_{TX} = 1$) as defined in 28.5.10.2. The single transmit chain is chosen by the first path beamforming training procedure in 10.4.2.10.6. All fields of EDMG secure ranging PPDU shall be transmitted with the same single transmit chain and AWV chosen by the first path beamforming training procedure.

As shown in Figure 27-48, each secure TRN subfield shall consist of five consecutive segments:

- 1) GI: The $GIe^1_{64 \times N_{CB}}$ as defined in 28.5.9.1.
- 2) Zero prefix: A prefix of $128 \times N_{CB}$ zero channel symbols.
- 3) Secure Ranging field: A secure ranging waveform composed of $384 \times N_{CB}$ $\pi/2$ -BPSK modulated channel symbols. The modulated symbols are based on bit sequences of length $384 \times N_{CB}$ which are

taken from the binary pseudorandom sequence `SECURE_TRN_SEQUENCE` in the `TXVECTOR` parameters as defined in Table 28-1. Each group of $384 \times N_{CB}$ bits is taken consecutively without overlap from the sequence. The constellation mapper maps the sequence of bits to constellation points; see 20.5.3.2.4.2 ($\pi/2$ -BPSK modulation).

- 4) Zero postfix: A postfix of $128 \times N_{CB}$ zero channel symbols.
- 5) GI: The $G_{le}^{1_{64 \times N_{CB}}}$ as defined in 28.5.9.1.

The overall length of each Secure TRN subfield is the same as each TRN subfield defined in 28.9.2.2.6.

28.9.3.7 Transmission of an EDMG secure ranging PPDU

Transmission of an EDMG secure ranging PPDU follows the same procedure as EDMG SU PPDU transmission in 28.5.10.4 with the TRN field containing secure TRN sequences.

Annex A

(informative)

Bibliography

Insert the following bibliography entry after the entry for “IETF RFC 5139” [B45]:

[B45a] IETF RFC 5903, Elliptic Curve Groups modulo a Prime (ECP Groups) for IKE and IKEv2, June 2010.⁷

Insert the following bibliography entry after the entry for “NENA 08-002” [B57]:

[B57a] NIST Special Publication 800-90A R1, Recommendation for Random Number Generation Using Deterministic Random Bit Generators, June 2015.⁸

⁷ IETF RFCs are available from the Internet Engineering Task Force (<https://www.ietf.org/>).

⁸ NIST publications are available from the National Institute of Standards and Technology (<https://csrc.nist.gov/>).

Annex B

(normative)

Protocol Implementation Conformance Statement (PICS) proforma

B.4 PICS proforma—IEEE Std 802.11-2020⁹

Insert ten new rows at the end of the table in B.4.3 as follows (unchanged rows not shown):

B.4.3 IUT configuration

Item	IUT configuration	References	Status	Support
...				
CFRSTA	FTMRSTA operation	11.21.6	WNM23:O	Yes ☐ No ☐ N/A ☐
CFISTA	FTM ISTA operation	11.21.6	WNM23:O	Yes ☐ No ☐ N/A ☐
CFPLISTA	FTM passive TB ranging ISTA	11.21.6.4.8	WNM23:O	Yes ☐ No ☐ N/A ☐
CFPLRSTA	FTM passive TB ranging RSTA	11.21.6.4.8	WNM23:O	Yes ☐ No ☐ N/A ☐
CFPLPSTA	FTM passive STA	11.21.6.4.8	WNM23:O	Yes ☐ No ☐ N/A ☐
CFTB	Support for FTM TB sounding	11.21.6.4.3	CFHE:O	Yes ☐ No ☐ N/A ☐
CFNTB	Support for FTM non-TB sounding	11.21.6.4.4	CFHE:O	Yes ☐ No ☐ N/A ☐
CFPTB	Support for Passive Sounding	11.21.6.4.8	CFHE:O	Yes ☐ No ☐ N/A ☐
CFPASN	Support for PASN	12.13	PC34:O	Yes ☐ No ☐ N/A ☐
CFPSEC	Support for PHY security, ability to transmit and receive secure HE-LTF waveforms for TB and non-TB operation	11.21.6.4.5	PC34:O CFHE:O	Yes ☐ No ☐ N/A ☐

B.4.4 MAC protocol

Insert new rows at the end of the table in B.4.4.1 as follows (unchanged rows not shown):

B.4.4.1 MAC protocol capabilities

Item	Protocol capability	References	Status	Support
...				
PC47	Support for TB ranging sounding exchange sequence	11.21.6.4.3	CFTB:M	Yes ☐ No ☐ N/A ☐
PC48	Support for non-TB ranging sounding exchange sequence	11.21.6.4.4	CFNTB:M	Yes ☐ No ☐ N/A ☐
PC49	Support for passive TB ranging sounding exchange sequence	11.21.6.4.8	CFPTB:M	Yes ☐ No ☐ N/A ☐

⁹Copyright release for PICS proforma: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Insert new rows after the row for “FT65” (as inserted by IEEE Std 802.11ba-2021) in the table in B.4.4.2 as follows (unchanged rows not shown):

B.4.4.2 MAC frames

Item	MAC frame	References	Status	Support
...				
FT66	FTM Request	11.21.6.3	CFISTA:M (CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
FT67	FTM	11.21.6.3	CFRSTA:M (CFTB OR CFNTB OR CFPTB):M	Yes ☐ No ☐ N/A ☐
FT68	LMR	9.6.7.49, 11.21.6.4.3, 11.21.6.4.4	(CFRSTA OR CFISTA):M (CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
FT69	Transmission of TF type Ranging using FTM TB ranging operation	9.3.1.22.10, 11.21.6.4.3	CFRSTA:M CFTB:M	Yes ☐ No ☐ N/A ☐
FT70	Transmission of Ranging NDP Announcement frame using FTM TB ranging operation	9.3.1.22.10, 11.21.6.4.3	CFRSTA:M CFTB:M	Yes ☐ No ☐ N/A ☐
FT71	Transmission of Ranging NDP Announcement frame using FTM non-TB ranging operation	9.3.1.22.10, 11.21.6.4.4	CFISTA:M CFNTB:M	Yes ☐ No ☐ N/A ☐
FT72	Transmission of TF subvariant Passive Sounding	9.3.1.22a.5	(CFRSTA AND CFPTB): M	Yes ☐ No ☐ N/A ☐
FT73	Transmission of Ranging NDP Announcement frame	9.3.1.19	(CFRSTA AND CFPTB): M	Yes ☐ No ☐ N/A ☐
...				

Insert new rows after the row for “FR66” (as inserted by IEEE Std 802.11ba-2021) in the table in B.4.4.2 as follows (unchanged rows not shown):

B.4.4.2 MAC frames

Item	MAC frame	References	Status	Support
...				
FR67	FTM Request	11.21.6.3	CFRSTA:M (CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
FR68	FTM	11.21.6.3	CFISTA:M (CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
FR69	LMR	9.3.1.22.10, 11.21.6.4.3	(CFRSTA OR CFISTA):M (CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
FR70	Receive of TF of type Ranging for TB ranging operation	9.3.1.22.10, 11.21.6.4.3	CFISTA:M CFTB:M	Yes ☐ No ☐ N/A ☐

B.4.4.2 MAC frames *(continued)*

Item	MAC frame	References	Status	Support
FR71	Receive of Ranging NDP Announcement frame for TB ranging operation	9.3.1.22.10, 11.21.6.4.3	CFISTA:M CFTB:M	Yes ☐ No ☐ N/A ☐
FR72	Receive of Ranging NDP Announcement frame for non-TB ranging operation	9.3.1.22.10, 11.21.6.4.4	CFRSTA:M CFNTB:M	Yes ☐ No ☐ N/A ☐
FR73	Reception of TF subvariant Passive Sounding	9.3.1.22a.5	(CFISTA AND CFPTB):M CFPLPSTA:M	Yes ☐ No ☐ N/A ☐
FR74	Reception of Ranging NDP Announcement frame	9.3.1.19	(CFISTA AND CFPTB):M CFPLPSTA:M	Yes ☐ No ☐ N/A ☐
...				

Insert B.4.37 after B.4.36 (as inserted by IEEE Std 802.11ba-2021) as follows:

B.4.37 Enhanced positioning (Ranging) features

B.4.37.1 EP MAC features

Item	Protocol capability	References	Status	Support
	Are the following MAC protocol features supported?			
EP7	Passive TB ranging operation			
EP7.1	Passive TB ranging sequence	11.21.6.4.8	CFPTB:M CFHE:O	Yes ☐ No ☐ N/A ☐
EPM1	EP capability signaling			
EPM1.1	EP FTM support advertisement	11.21.6.2, 11.21.6.3.1	(CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
EPM1.2	EP FTM capability negotiation	11.21.6.3.1	(CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
EPM1.3	EP secured MAC operation negotiation	12.13, 11.21.6.4.5	CFPASN:M (CFTB OR CFNTB):M	Yes ☐ No ☐ N/A ☐
EPM2	TB ranging operation			
EPM2.1	TB measurement sequence	11.21.6.4.3	CFTB:M	Yes ☐ No ☐ N/A ☐
EPM2.2	Protected LMR exchange in TB ranging exchange	12.13, 11.21.6.4.5	(CFTB AND CFPASN): M CFPSEC:M	Yes ☐ No ☐ N/A ☐
EPM3	MAC support of PHY security			
EPM3.1	SAC exchange for TB operation	12.13, 11.21.6.4.5	CFTB:O CFPSEC:M	Yes ☐ No ☐ N/A ☐
EPM4	Non-TB ranging operation			
EPM4.1	non-TB ranging exchange	11.21.6.4.4	CFNTB:M	Yes ☐ No ☐ N/A ☐

B.4.37.1 EP MAC features *(continued)*

Item	Protocol capability	References	Status	Support
EPM4.2	Protected LMR exchange in non-TB ranging exchange	12.13, 11.21.6.4.5	(CFNTB AND CFPASN):M CFPSEC:M	Yes ☐ No ☐ N/A ☐
EPM4.3	SAC exchange for non-TB ranging operation	12.13, 11.21.6.4.5	CFNTB:O CFPSEC:M	Yes ☐ No ☐ N/A ☐
EPM5	EDMG/DMG operation	11.21.6.4.2.2	O	Yes ☐ No ☐
EPM5.1	EDMG/DMG direction measurement as ISTA	11.21.6.4.2.3	CFISTA: O	Yes ☐ No ☐ N/A ☐
EPM5.2	EDMG/DMG direction measurement as RSTA	11.21.6.4.2.3	CFRSTA: O	Yes ☐ No ☐ N/A ☐
EPM5.3	EDMG FTM measurement with first arrival path as ISTA	11.21.6.4.2.6	(CFISTA and EMDG-M16.7): M	Yes ☐ No ☐ N/A ☐
EPM5.4	EDMG FTM measurement with first arrival path as RSTA	11.21.6.4.2.6	(CFRSTA and EMDG-M16.7): M	Yes ☐ No ☐ N/A ☐
EPM5.5	EDMG direction measurement with first arrival path as ISTA	11.21.6.4.2.3	(CFISTA and EMDG-M16.7): O	Yes ☐ No ☐ N/A ☐
EPM5.6	EDMG direction measurement with first arrival path as RSTA	11.21.6.4.2.3	(CFRSTA and EMDG-M16.7): O	Yes ☐ No ☐ N/A ☐
EPM5.7	EDMG LOS assessment as ISTA	11.21.6.4.2.5	(CFISTA and EMDG-M16.7 and EDMG-M16.10) :O	Yes ☐ No ☐ N/A ☐
EPM5.8	EDMG LOS assessment as RSTA	11.21.6.4.2.5	(CFISTA and EMDG-M16.7 and EDMG-M16.10) :O	Yes ☐ No ☐ N/A ☐
EPM6	Secure EDMG FTM operation	11.21.6.4.8	(CFEDMG and (CFISTA or CFRSTA) and EPP4 and EPM1.3):	Yes ☐ No ☐ N/A ☐

B.4.37.2 EP PHY features

Item	Protocol capability	References	Status	Support
	Are the following PHY protocol features supported?			
EPP1	EP TB and non-TB measurement exchange waveforms	27.3		
EPP1.1	HE ranging NDP	27.3.18a.1	(CFTB OR CFNTB OR CFPTB):M	Yes ☐ No ☐ N/A ☐
EPP1.2	HE ranging NDP with secure HE-LTFs	27.3.18a.1	(CFTB OR CFNTB):M CFPASN:M CFPSEC:M	Yes ☐ No ☐ N/A ☐
EPP1.3	HE TB ranging NDP	27.3.18a.2	CFTB:M	Yes ☐ No ☐ N/A ☐
EPP1.4	HE TB ranging NDP with secure HE-LTFs	27.3.18a.2	CFPASN:M CFPSEC:M CFTB:M	Yes ☐ No ☐ N/A ☐
EPP2	MIMO operation			
EPP2.1	Multi transmit streams	11.21.6.4.3, 11.21.6.4.4	(CFTB or CFNTB):M	Yes ☐ No ☐ N/A ☐
EPP2.2	Multi receive streams	11.21.6.4.3, 11.21.6.4.4	(CFTB or CFNTB):M	Yes ☐ No ☐ N/A ☐
EPP3	EDMG frame formats			
EPP4	Secure ranging EDMG TRN field	28.9.3.5, 28.9.3.6, 28.9.3.7		Yes ☐ No ☐ N/A ☐

Annex C

(normative)

ASN.1 encoding of the MAC and PHY MIB

C.3 MIB detail

Change the item “dot11UnsolicitedBAActivated” (as inserted by IEEE Std 802.3ay-2021) at the end of the list in the object definition that begins with “Dot11StationConfigEntry ::=” in C.3 as follows:

dot11UnsolicitedBAActivated	TruthValue,
dot11PASNAActivated	TruthValue,
<u>dot11NoAuthPASNAActivated</u>	<u>TruthValue</u>

Insert new text after the object definition that begins with “dot11ColocatedRNRImplemented OBJECT-TYPE” (as inserted by IEEE Std 802.3ax-2021) in C.3 as follows:

```
dot11PASNAActivated OBJECT-TYPE
  SYNTAX TruthValue
  MAX-ACCESS read-write
  STATUS current
  DESCRIPTION
    "This is a control variable. It is written by an external management
    entity or the SME. Changes take effect for the next MLME-START.request
    primitive or MLME JOIN.request primitive. This attribute indicates whether
    or not PASN authentication is enabled."
  DEFVAL {false}
  ::= { dot11StationConfigEntry 201 }

dot11NoAuthPASNAActivated OBJECT-TYPE
  SYNTAX TruthValue
  MAX-ACCESS read-write
  STATUS current
  DESCRIPTION
    "This is a control variable. It is written by an external management
    entity or the SME. Changes take effect for the next MLME-START.request
    primitive or MLME JOIN.request primitive. This attribute indicates whether
    or not PASN without mutual authentication is allowed."
  DEFVAL {false}
  ::= { dot11StationConfigEntry 202 }
```

Change the item “dot11RSNAStatsRobustMgmtGCMPReplays” at the end of the list in the object definition that begins with “Dot11RSNAConfigEntry ::=” in C.3 as follows:

dot11RSNAStatsRobustMgmtGCMPReplays	Unsigned32,
<u>dot11RSNAConfigPASNPTKSATimeout</u>	<u>Unsigned32</u>

Insert new text after the object definition that begins with “dot11RSNASESync OBJECT-TYPE” in C.3 as follows:

```
dot11RSNAConfigPASNPTKSATimeout OBJECT-TYPE
  SYNTAX Unsigned32 (1..4294967295)
  UNITS "seconds"
  MAX-ACCESS read-write
  STATUS current
```

DESCRIPTION

"This is a control variable. It is written by an external management entity. Changes take effect as soon as practical in the implementation. The time for which the derived PTKSA derived from PASN authentication is valid."

DEFVAL { 3600 }

::= { dot11RSNAConfigEntry 43 }

Change the item “dot11RMCivicConfigured” at the end of the list in the object definition that begins with “Dot11WirelessMgmtOptionsEntry ::=” in C.3 as follows:

<u>dot11RMCivicConfigured</u>	<u>TruthValue,</u>
<u>dot11SecureLTFImplemented</u>	<u>TruthValue,</u>
<u>dot11TriggerBasedRangingRespImplemented</u>	<u>TruthValue,</u>
<u>dot11NonTriggerBasedRangingRespImplemented</u>	<u>TruthValue,</u>
<u>dot11RSTARequiresPMFActivated</u>	<u>INTEGER,</u>
<u>dot11PassiveTBRangingResponderImplemented</u>	<u>TruthValue,</u>
<u>dot11PassiveTBRangingInitiatorImplemented</u>	<u>TruthValue,</u>
<u>dot11AOAMeasurementImplemented</u>	<u>TruthValue,</u>
<u>dot11I2RLMRFeedbackPolicy</u>	<u>TruthValue,</u>
<u>dot11LOSAssessmentTXImplemented</u>	<u>TruthValue,</u>
<u>dot11LOSAssessmentRXImplemented</u>	<u>TruthValue,</u>
<u>dot11PassiveTBRangingAODImplemented</u>	<u>TruthValue,</u>
<u>dot11PhaseShiftFeedbackImplemented</u>	<u>TruthValue</u>

Insert new text after the object definition that begins with “dot11RMCivicConfigured OBJECT-TYPE” in C.3 as follows:

dot11SecureLTFImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.

Its value is determined by device capabilities.

This attribute, when true, indicates that a secure HE-LTF measurement exchange protocol; see 11.21.6.4.5 is implemented. The capability is disabled otherwise."

::= { dot11WirelessMgmtOptionsEntry 54 }

dot11TriggerBasedRangingRespImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable. Its value is determined by device capabilities. This attribute, when true, indicates that support for negotiating and executing TB ranging measurement exchange as an RSTA (see 11.21.6) is implemented. The capability is disabled otherwise."

::= { dot11WirelessMgmtOptionsEntry 55 }

dot11NonTriggerBasedRangingRespImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.

Its value is determined by device capabilities.

This attribute, when true, indicates that support for negotiating and executing non-TB ranging measurement exchange as an RSTA (see 11.21.6) is implemented. The capability is disabled otherwise."

```
 ::= { dot11WirelessMgmtOptionsEntry 56 }

dot11RSTARequiresPMFActivated OBJECT-TYPE
    SYNTAX INTEGER {Inactive (0), Required-X20M (1), Required (2)}
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME.

        Changes take effect at the next occurrence of an MLME-START.request or
        MLME-JOIN.request primitive.

        The attribute applies only to preassociation ranging behavior.

        When set to Required (2), indicates that the station requires Management
        Frame Protection for all management frames exchanged during the
        negotiation and measurement reporting.

        When set to Required-X20M (1), indicates that the station Management Frame
        Protection for all management frames exchanged during the negotiation and
        measurement reporting except those using a 20 MHz bandwidth for
        measurements.

        Otherwise, when set to Inactive (0), indicates that Management Frame
        Protection is not required for ranging.

        See 11.21.6.3.1 and range measurement procedure, see 11.21.6.4.3,
        11.21.6.4.4, and 11.21.6.4.5 to successfully negotiate a range measurement
        session, see 11.21.6.3.1."
    DEFVAL { Inactive }
 ::= { dot11WirelessMgmtOptionsEntry 57 }

dot11PassiveTBRangingResponderImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that support for passive TB ranging
        acting as a responder is implemented (see 11.21.6.4.8). The capability is
        disabled otherwise."
 ::= { dot11WirelessMgmtOptionsEntry 58 }

dot11PassiveTBRangingInitiatorImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities.

        This attribute, when true, indicates that support for passive TB ranging
        acting as an initiator is implemented (see 11.21.6.4.8). The capability is
        disabled otherwise."
 ::= { dot11WirelessMgmtOptionsEntry 59 }

dot11AOAMeasurementImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
```


"This is a capability variable.
Its value is determined by device capabilities.

This attribute, when true, indicates that the STA supports Angle-of-Arrival (AOA) measurements and feedback of these measurements is available via the LMR frame. When false, there is no such feedback available in the LMR frame."

::= { dot11WirelessMgmtOptionsEntry 60 }

dot11I2RLMRFeedbackPolicy OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"This is a control variable.

It is written by an external management entity or the SME.

Changes take effect at the next occurrence of an MLME-START.request or MLME-JOIN.request primitive.

This attribute, when true, indicates that the station (RSTA) does not require any initiating stations to support the capability to generate and transmit I2R LMRs (see 11.21.6.3.3).

False indicates that the stations shall negotiate the transmission of I2R LMR."

DEFVAL { true }

::= { dot11WirelessMgmtOptionsEntry 61 }

dot11LOSAssessmentTXImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.

Its value is determined by device capabilities.

This attribute, when true, indicates that the station capability for participation in LOS assessment FTM exchange by transmitting a Loss Assessment PPDU. It is set to false otherwise."

::= { dot11WirelessMgmtOptionsEntry 62 }

dot11LOSAssessmentRXImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.

Its value is determined by device capabilities.

This attribute, when true, indicates that the station has capability for participation in LOS assessment FTM exchange by switching polarization on the TRN field when a Loss Assessment PPDU is received. It is set to false otherwise."

::= { dot11WirelessMgmtOptionsEntry 63 }

dot11PassiveTBRangingAODEnablementImplemented OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This is a capability variable.

Its value is determined by device capabilities.

This attribute, when true, indicates that the station, when performing passive TB ranging, is capable of estimating the AOD measurement by the client STA that locates itself through passive TB ranging (see 11.21.6.4.8).

False indicates that the station, when performing passive TB ranging, is not capable of estimating the AOD measurement by the client STA that locates itself through passive TB ranging. (See 11.21.6.4.8)."
 ::= { dot11WirelessMgmtOptionsEntry 64 }

dot11PhaseShiftFeedbackImplemented OBJECT-TYPE
 SYNTAX TruthValue
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "This is a capability variable.
 Its value is determined by device capabilities. This attribute, when true, indicates that the STA supports reporting of time of arrival estimated using the Phase Shift feedback method and including these measurements in the corresponding LMR frame. When false, the time of arrival estimates Reported in the LMR frame are not based on the Phase Shift feedback method."
 ::= { dot11WirelessMgmtOptionsEntry 65 }

Change the object definition that begins with “dot11FineTimingMeasurement OBJECT-GROUP” in C.3 as follows:

dot11FineTimingMeasurement OBJECT-GROUP
 OBJECTS {
 dot11WirelessManagementImplemented,
 dot11FineTimingMsmtRespActivated,
 dot11FineTimingMsmtInitActivated,
 dot11LciCivicInNeighborReport,
 dot11RMFineTimingMsmtRangeRepImplemented,
 dot11RMFineTimingMsmtRangeRepActivated,
 dot11RMLCIMEasurementActivated,
 dot11RMLCIConfigured,
 dot11RMCivicMeasurementActivated,
 dot11RMCivicConfigured,
 dot11PASNAActivated,
 dot11NoAuthPASNAActivated,
 dot11SecureLTFImplemented,
 dot11TriggerBasedRangingRespImplemented,
 dot11NonTriggerBasedRangingRespImplemented,
 dot11RSTARequiresPMFAActivated,
 dot11PassiveTBRangingResponderImplemented,
 dot11PassiveTBRangingInitiatorImplemented,
 dot11AOAMEasurementImplemented,
 dot11I2RLMRFeedbackPolicy,
 dot11RSNAConfigPASNPTKSATimeout,
 dot11LOSAssessmentRXImplemented,
 dot11LOSAssessmentTXImplemented,
 dot11PhaseShiftFeedbackImplemented,
 dot11PassiveTBRangingAODImplemented
 }
 STATUS current
 DESCRIPTION
 "Attributes that configure the Fine Timing Measurement feature for IEEE Std 802.11."
 ::= { dot11Groups 93 }

Annex J

(informative)

RSNA reference implementations and test vectors

Insert J.12, J.13, and J.14 after J.11.2 as follows:

J.12 PASN Test Vectors

The Diffie-Hellman exchange to establish a shared secret used in key derivation for PASN and FILS with PFS is a well known algorithm. See IETF RFC 5903 [B45a] for test vectors when using ECP groups. Only the x-coordinate of the result of scalar multiplication of a private key with the received public key is used as the shared secret.

Sample PTKSA derivation with PASN Authentication

$KCK \parallel TK \parallel KDK = KDF\text{-}HASH\text{-}NNN (PMK, \text{“PASN PTK Derivation”}, SPA \parallel BSSID \parallel DHss)$

Hash: SHA-256
 NNN: 640
 PMK: de f4 3e 55 67 e0 1c a6 64 92 65 f1 9a 29 0e ef
 f8 bd 88 8f 6c 1d 9c c9 d1 0f 04 bd 37 8f 3c ad
 SPA: 00 90 4c 01 c1 07
 BSSID: c0 ff d4 a8 db c1
 DHss: Sample using NIST P256 (Group 19)
 f8 7b 20 8e 7e d2 b7 37 af db c2 e1 3e ae 78 da
 30 01 23 d4 d8 4b a8 b0 ea fe 90 c4 8c df 1f 93
 KCK: 7b b8 21 ac 0a a5 90 9d d6 54 a5 60 65 ad 7c 77
 eb 88 9c be 29 05 bb f0 5a bb 1e ea c8 8b a3 06
 TK: 67 3e ab 46 b8 32 d5 a8 0c bc 02 43 01 6e 20 7e
 KDK: 2d 0f 0e 82 c7 0d d2 6b 79 06 1a 46 81 e8 db b2
 ea 83 be a3 99 84 4b d5 89 4e b3 20 f6 9d 7d d6

J.13 KDK Test Vectors when PASN authentication is not used

$KCK \parallel KEK \parallel TK \parallel KDK = KDF\text{-}Hash\text{-}Length(PMK, \text{“Pairwise key expansion”}, \text{Min}(AA, SPA) \parallel \text{Max}(AA, SPA) \parallel \text{Min}(ANonce, SNonce) \parallel \text{Max}(ANonce, SNonce))$

Hash: SHA-256
 Length: 640
 PMK: de f4 3e 55 67 e0 1c a6 64 92 65 f1 9a 29 0e ef
 f8 bd 88 8f 6c 1d 9c c9 d1 0f 04 bd 37 8f 3c ad
 AA: c0 ff d4 a8 db c1
 SPA: 00 90 4c 01 c1 07
 ANonce: be 7a 1c a2 84 34 7b 5b d6 7d bd 2d fd b4 d9 9f
 1a fa e0 b8 8b a1 8e 00 87 18 41 7e 4b 27 ef 5f

SNonce: 40 4b 01 2f fb 43 ed 0f b4 3e a1 f2 87 c9 1f 25
 06 d2 1b 4a 92 d7 4b 5e a5 0c 94 33 50 ce 86 71
 KCK: cd 7b 9e 75 55 36 2d f0 b6 35 68 48 4a 81 12 f5
 KEK: 99 ca d3 58 8d a0 f1 e6 3f d1 90 19 10 39 bb 4b
 TK: 9e 2e 93 77 e7 53 2e 73 7a 1b c2 50 fe 19 4a 03
 KDK: 6c 7f b9 7c eb 55 b0 1a cf f0 0f 07 09 42 bd f5
 29 1f eb 4b ee 38 e0 36 5b 25 a2 50 bb 2a c9 ff

J.14 LTF Sequence Generation Test Vectors

As defined in 11.21.6.3.4, Secure-LTF-Key-Seed is derived from KDK as follows:

Secure-LTF-Key-Seed = HMAC-Hash(KDK, “Secure HE-LTF key seed”)

Hash: SHA-256
 KDK: 6c 7f b9 7c eb 55 b0 1a cf f0 0f 07 09 42 bd f5
 29 1f eb 4b ee 38 e0 36 5b 25 a2 50 bb 2a c9 ff
 Secure-LTF-Key-Seed: 07 60 6f 7b 0d 98 ca 03 ec 2d 61 e1 7c 6b df d3
 0e 2f 20 30 e3 47 02 22 55 1a 05 ec 55 d1 35 b9

SAC || ista-ltf-key || rsta-ltf-key = KDF-Hash-Length(Secure-LTF-Key-Seed, “Secure HE-LTF Expansion”, Secure-LTF-Counter)

Hash: SHA-256
 Length: 272 (bits)
 Secure-LTF-Key-Seed: 07 60 6f 7b 0d 98 ca 03 ec 2d 61 e1 7c 6b df d3
 0e 2f 20 30 e3 47 02 22 55 1a 05 ec 55 d1 35 b9
 Secure-LTF-Counter: 0x000000000100
 SAC: 23 cf
 ista-ltf-key: d2 a8 a2 b7 6c 3c 29 2d 81 e1 82 a4 69 fd e8 3c
 rsta-ltf-key: 65 02 7a 83 8d 58 59 3c 57 b9 41 6f 17 24 e6 c4
 Transmitter MAC address: 00 10 18 32 76 54

LTF octet stream generator inputs and outputs:

Octet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
LTF_KEY	d2	a8	a2	b7	6c	3c	29	2d	81	e1	82	a4	69	fd	e8	3c
LTF_IV	00	10	18	32	76	54	00	00	00	00	01	00	00	00	00	00
AES counter [0]	00	10	18	32	76	54	00	00	00	00	01	00	00	00	00	00
Output block [0]	aa	f6	2c	30	6b	cd	8a	5d	89	80	8b	03	8e	da	43	f1
AES counter [1]	00	10	18	32	76	54	00	00	00	00	01	00	00	00	00	01
Output block [1]	54	15	f0	5c	7f	c7	ee	f5	9b	c4	58	d2	f4	6b	5b	5a
...																

Insert Annex AD and Annex AE after Annex AC (inserted by IEEE Std 802.11ba-2021) as follows:

Annex AD

(informative)

Phase shift feedback calculation

Assume there is a single Tx and Rx RF chain between the ISTA and RSTA and denote the frequency domain channel estimation of I2R NDP and R2I NDP as a column vector $h = [h_1, h_2, \dots, h_N]^T$ and N is the number of subcarriers, then the corresponding phase shift might be calculated from Equation (AD-1).

$$PS = -\frac{\text{angle}(h_0^H h_K)}{2\pi K \Delta_{F, \text{HE-LTF}}} + t_{\text{HE-LTF, DFT boundary}} - T_{\text{GI, HE-LTF}} - T_{\text{HE-STF}} - T_{\text{Pre-HE portion}} \quad (\text{AD-1})$$

where

h_0	is $[h_1, h_2, \dots, h_{N-K}]^T$
h_K	is $[h_{K+1}, h_{K+2}, \dots, h_N]^T$
$(\dots)^T$	denotes the transpose of a vector
$(\dots)^H$	represents the Hermitian transpose of a vector
$\text{angle}(\dots)$	is the phase angle of a complex number
K	is implementation specific
$\Delta_{F, \text{HE-LTF}}$	is the frequency subcarrier spacing for the HE-LTF field
$t_{\text{HE-LTF, DFT boundary}}$	is the timing boundary of the DFT window for the HE-LTF symbol expressed in the receiver's local clock
$T_{\text{GI, HE-LTF}}$	is the guard interval duration for the HE-LTF portion
$T_{\text{HE-STF}}$	is the HE-STF field duration in Ranging NDP or TB Ranging NDP
$T_{\text{Pre-HE portion}}$	is the duration of the pre-HE portion in the preamble of a Ranging NDP or TB Ranging NDP

Channel estimates might be filtered for noise reduction and ranging accuracy improvement before they are applied in Equation (AD-1).

When the ISTA and RSTA have multiple Tx and Rx chains, the frequency domain channel estimation vector between the i th Tx spatial stream and j th Rx antenna is denoted as $h_{i,j} = [h_{i,j,1}, h_{i,j,2}, \dots, h_{i,j,N}]^T$, and the phase shift is calculated from Equation (AD-2).

$$PS = -\frac{\text{angle}\left(\sum_i \sum_j h_{i,j,0}^H h_{i,j,K}\right)}{2\pi K \Delta_{F, \text{HE-LTF}}} + t_{\text{HE-LTF, DFT boundary}} - T_{\text{GI, HE-LTF}} - T_{\text{HE-STF}} - T_{\text{Pre-HE portion}} \quad (\text{AD-2})$$

where

$h_{i,j,0}$	is $[h_{i,j,1}, h_{i,j,2}, \dots, h_{i,j,N-K}]^T$
$h_{i,j,K}$	is $[h_{i,j,K+1}, h_{i,j,K+2}, \dots, h_{i,j,N}]^T$

Annex AE

(informative)

SIR-based attack detection

The presence of an attack signal will cause the receiver SINR to drop. Given a measured noise floor, the SINR can be mapped to a signal to interference ratio (SIR). In this annex, an analysis is shown to illustrate the impact of an attack signal on the SIR drop.

Assume that an attacker observes the Secure HE-LTF symbol for a duration of T_{ob} , referred to as the observation period; see Figure AE-1 (Attacking secure HE-LTF symbols). The attacker can use these observations to generate an attack signal over the duration T_c , referred to as the computation period. Finally, the attacker will transmit the attack signal during the attack period, T_a , to create a false first path of arrival that has an earlier arrival time than the true first path of arrival.

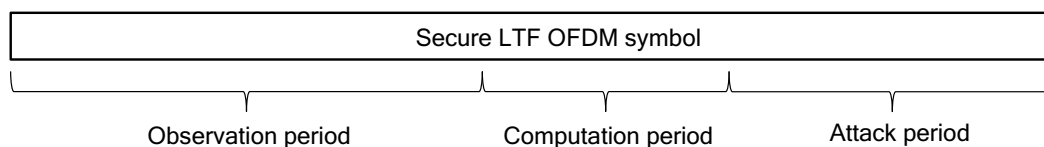


Figure AE-1—Attacking secure HE-LTF symbols

A side effect of creating the false first path of arrival is that the attacker will also generate interference at the intended receiver. If we assume that the attack signal has a correlation of ρ with the true Secure HE-LTF signal and creates a false first path of arrival with a relative power level of FFP (dB), then the signal-to-interference ratio (SIR) that is seen by the intended receiver will be:

$$SIR(\text{dB}) = -FFP(\text{dB}) + 10\log_{10}(\rho^2 T_a) - 10\log_{10}(1 - \rho^2 T_a)$$

When an attack signal is present, the interference introduced by the attack signal will lead to an SIR drop and the value will depend on the relative power level of FFP , correlation, and attack period.

For example, if $FFP = -10$ dB, then the expected SIR levels for different values of ρ and T_a are shown in Figure AE-2 (Received SIR after attack).

From this figure, we see that as the correlation of the attack signal decreases or the attack duration decreases, the SIR will drop even further.

Therefore, the presence of an attack signal can be detected by looking for drops in the SIR over a single, or multiple, HE-LTF repetitions. The exact details of the algorithm are left to the implementer.

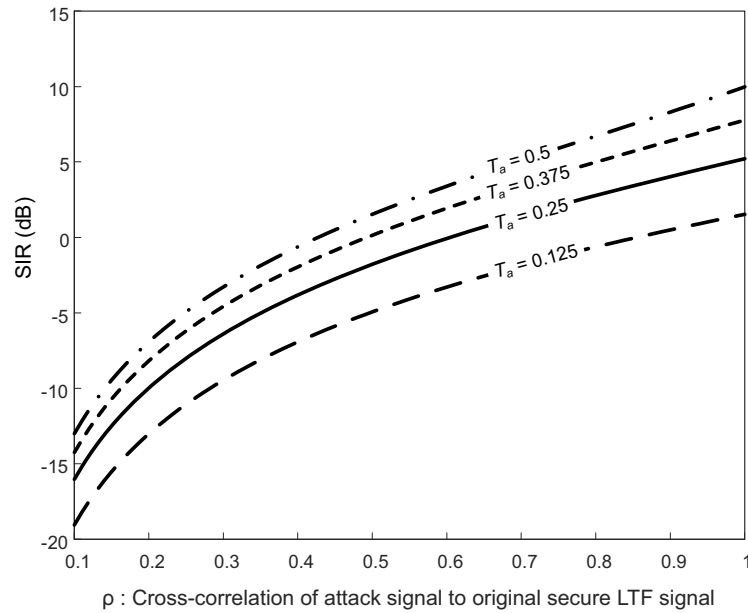


Figure AE-2—Received SIR after attack

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